Principles of Systematics

Module# 1

Principles of Systematics

Basic Terminologies

Basic Terminologies

Taxonomy

Theory and practice of classification of organisms

- Taxonomy is derived from Greek words “Taxis” which means arrangements and “Nomos” means laws /methods

Basic Terminologies

Taxonomy

Carolus Linnaeus (1707-1778)

- Carolus Linnaeus, the Swedish botanist, Zoologist and Physician
- Regarded as the father of modern taxonomy.

Basic Terminologies

Contribution of Carolus Linnaeus

His two important contributions to taxonomy were:

- A hierarchical classification system
- The system of binomial nomenclature

Basic Terminologies

Carolus Linnaeus

He proposed that there are three broad groups called Kingdoms into which whole nature could fit.

These kingdoms were:

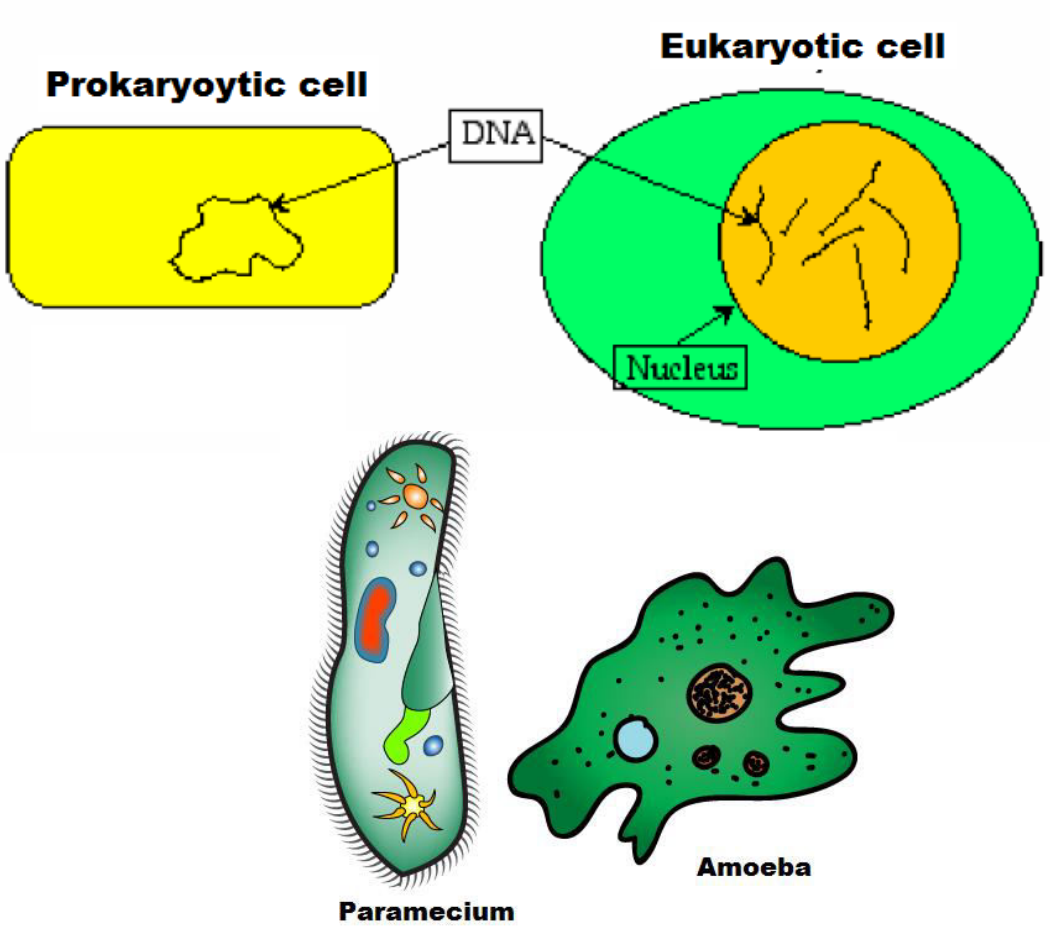
- Animals
- Plants and
- Minerals

Basic Terminologies

Classification

- It is a systematic method of arranging organisms into different groups and subgroups, based on their similarities and their differences.
- It makes the study of organisms very easy and convenient.

Basic Terminologies

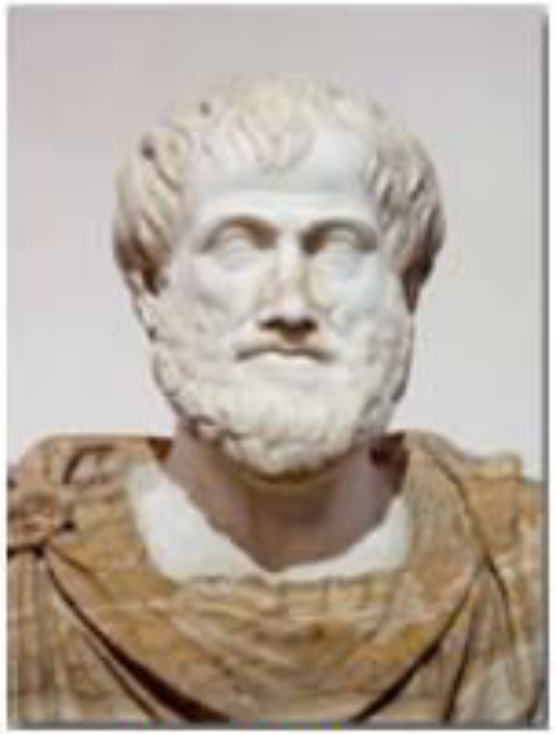


Classification

- Organisms can be classified on the basis of several different factors:

- Cell nature –Are cells Prokaryotic or Eukaryotic?
- Cells – Do they occur Singly or in Clusters?
- Mode of Nutrition
- Level of Body Organization

Basic Terminologies

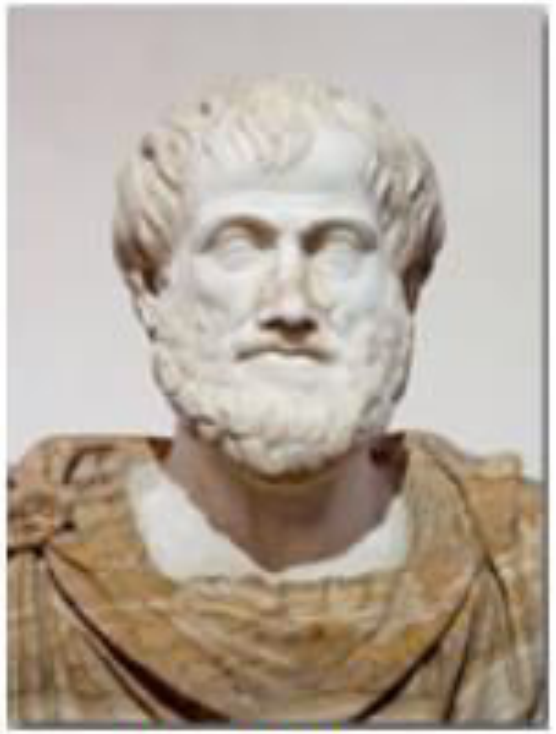


Aristotle
Greek Philosopher
384-322 BC

Aristotle-Father of Biological Classification

- To understand anything, one must classify it according to its parts
- He established numerous collective categories or genera using differentiating characteristics

Basic Terminologies



Aristotle
Greek Philosopher
384-322 BC

Aristotle-Father of Biological Classification

- Blooded and bloodless
- Two-footed vs four footed
- Hairy vs feathered
- With or without an outer shell
- Note: He failed to provide orderly and fully consistent classification of animals.

Principles of Systematics

Module# 2

Principles of Systematics

Basic Terminologies

Basic Terminologies

Simpson

(1902-1884)

Systematics

According to Simpson (1961)

- Scientific study of kinds and diversity of organisms and of any or all relationship among them
- Broad term which include Taxonomy, Nomenclature and Classification etc.
- Professor of Biology at San Diego State University, Colombia

Basic Terminologies

Aims of systematics

- Invent worlds kinds of organisms (flora and fauna)
- Provide method for identification and communication
- To produce coherent and universal system of classification and
- Demonstrate evolutionary implications of biodiversity

Basic Terminologies

Component fields of systematics

- Component fields of systematics are:
  - Biodiversity
  - Taxonomy
  - Classification
  - Nomenclature
  - Biogeography
  - Evolutionary biology

Basic Terminologies

Taxon

- According to Simpson (1961) taxon is a group of real organisms recognized as formal unit at any level of hierarchic classification

- Blue birds, song birds, vertebrate refer to groups of organisms.

Basic Terminologies

Taxon

- Taxon always refer to concrete Zoological object
- Species of Robins is not a taxon but category
- Robin is a Taxon

END

Principles of Systematics

Module# 3

Contribution of Systematics in Biology

Contribution of Systematics in Biology

Systematics contribution

The contribution of systematic to biology can be studied under two heads:-

- Theoretical Biology
- Applied Biology

Contribution of Systematics in Biology

Theoratical Biology

- Evolution
- Ecology
- Molecular Biology
- Evolutionary Biology

Contribution of Systematics in Biology

Evolution

- Evolution is a process by which plants and animals that exist today have been evolved from plants and animal of past.
- Before the rise of genetics, the study of evolution was carried out entirely by taxonomists.
- Virtually all major evolutionary problems were first pointed out and often solved by systematics.

Contribution of Systematics in Biology

Evolution

- Taxonomists helps us categorize organisms so we can more easily communicate biological information.
- No other branch of biology has made greater contribution to our understanding than systematics.

Contribution of Systematics in Biology

Ecology

- No through ecological surveys can be conducted without the identification of all species that are of ecological significance.
- Elton (1947) acknowledged the progress in ecology depends upon accurate identifications of species.

Contribution of Systematics in Biology

Molecular Biology

- Molecular biologist use genetic data for establishing relationships among different organisms.
- However, they need assistance of traditional taxonomist for accurate results.
- Examples: DNA barcoding and taxonomist

END

Principles of Systematics

Module# 4

Contribution of Systematics in Biology

Contribution of Systematics in Biology

Applied Biology

- Agriculture and forestry
- Biological control
- Public health
- Quarantine
- Wild life management

Contribution of Systematics in Biology

Applied Biology

- Mineral prospecting
- National defense
- Environmental problem
- Soil fertility
- In commerce

Contribution of Systematics in Biology

Agriculture and Forestry

- In order to control and minimize pest attack on crops and diseases caused by vector
- It is necessary to know about correct names of pests
- Taxonomists give us correct identification of pests
- Identification is vital for control

Contribution of Systematics in Biology

Biological Control

- Involved in designing biological control programs of pests

- Determination of exact country of origin and their parasite and parasitoids is compulsory.

Contribution of Systematics in Biology

Asian Citrus psylid *Tamaria radiata*

(Hoddle & Pandey, 2014)

- Asian citrus psylid is serious pest of citrus orchards
- Its origin is Asia
- *Tamarixia radiata,* a parasitic wasp was introduced from Pakistan to California to control this pest.
- Best example of classical biological control.

Contribution of Systematics in Biology

Public Health

- Number of diseases are spread by Arthropods which are controlled by targeting specific species.
- For example: All *Anopheles maculipennis* are not responsible for transmitting malaria.
- Out six siblings some are involved in transmission of malaria.

Contribution of Systematics in Biology

Public Health

- Although the species are virtually indistinguishable morphologically, they are isolated reproductively, in part because they breed in different habitats.
- Taxonomists help to identify this particular sibling species for effective control at minimum cost.

END

Principles of Systematics

Module# 5

Contribution of Systematics in Biology

Contribution of Systematics in Biology

Quarantine

- Many diseases are spread from one country to other via transportation.
- Respective Governments established quarantine labs at aerodromes and ports etc
- Taxonomists help in prompt identification of these diseases

Contribution of Systematics in Biology

Wild Life management

- The indiscriminate killing and felling of trees create disturbance in natural environment.
- It help environmental protectors by identifying economically and ecologically important wild life.
- It is important for preservation and protection of biodiversity.

Contribution of Systematics in Biology

Mineral Prospecting

- The identification of fauna and flora by paleontologists in sedimentary rocks show the sequence of geological events
- Which help in search for fuels and mineral deposits.
- Example: It attain great success in American industrialization

Contribution of Systematics in Biology

National Defense

- During World War II (On November 3, 1944), **Japan** released **balloon bombs**, into the Pacific jet stream
- Created havoc in the forest of north east America.
- Balloons covered with sand contained a large number of shells of micro-organisms.
- Bombing of this beach area destroyed the balloon launching site.

Contribution of Systematics in Biology

Detection of Environmental Problems

- Phytoplankton and zooplankton are sensitive to changes in water quality.
- They are good indicators for changes in water quality as they respond quickly to changes in nutrient input.
- Taxonomists help to detect such problem by identifying specific species.

Contribution of Systematics in Biology

Soil Fertility

- Taxonomist help to detect organisms that enhance soil fertility
- For example bacteria, earthworms etc.

Contribution of Systematics in Biology

In commerce

- Systematics play role in increasing and improving the qualities of commercially used animal products such as:
    - Honey
    - Silk
    - Lac
    - Dyes etc

END

Principles of Systematics

Module# 6

Principles of Systematics

Binomial Nomenclature

Binomial Nomenclature

- Scientific names of species is a combination of two names:
    - Generic name
    - Species name

Binomial Nomenclature

- Spider *Oxyopes javanus* is found in agro-ecosystems.
- The first part of a scientific name, *Oxyopes, is called the genus.*
- The second part of a scientific name, *javanus* is in this example, is the species name

Binomial Nomenclature

- *Hottentotta tamulus*, the Indian red scorpion also known as the eastern Indian scorpion, is a species of scorpion of the family Buthidae.
- It occurs in most of India, eastern Pakistan and the eastern lowlands of Nepal, and recently from Sri Lanka.

Binomial Nomenclature

https://en.wikipedia.org/wiki/Hoplobatrachus_tigerinus#/media/File:HoplobatrachusTigerinus.jpg

- *Rana tagrina* is an amphibian which is solitary and nocturnal in nature.
- They inhabit holes and bushes near water source.
- They do not stay in water for longer time.
- They belong to order anura.

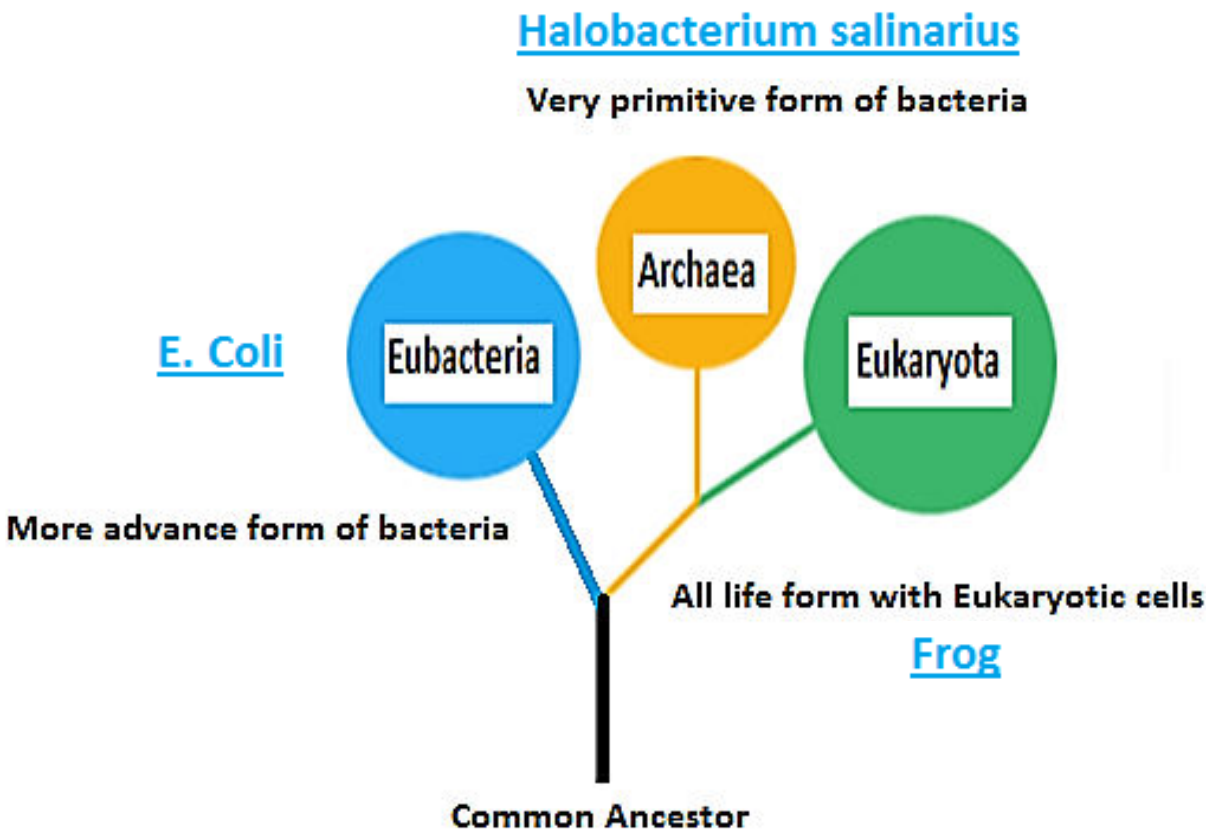
Principles of Systematics

Module# 7

Three Domain system

Archaea
Eubacteria
Eukaryota

Three Domain system



- A domain is a broadest taxonomic category.
- *A*ll organism belong to one of the three domains, depending upon their characteristics.
- A single domain can contain one or more kingdoms.
- Introduced by Charl Woese et al. In 1990.

Three Domain system

Domain of Viruses

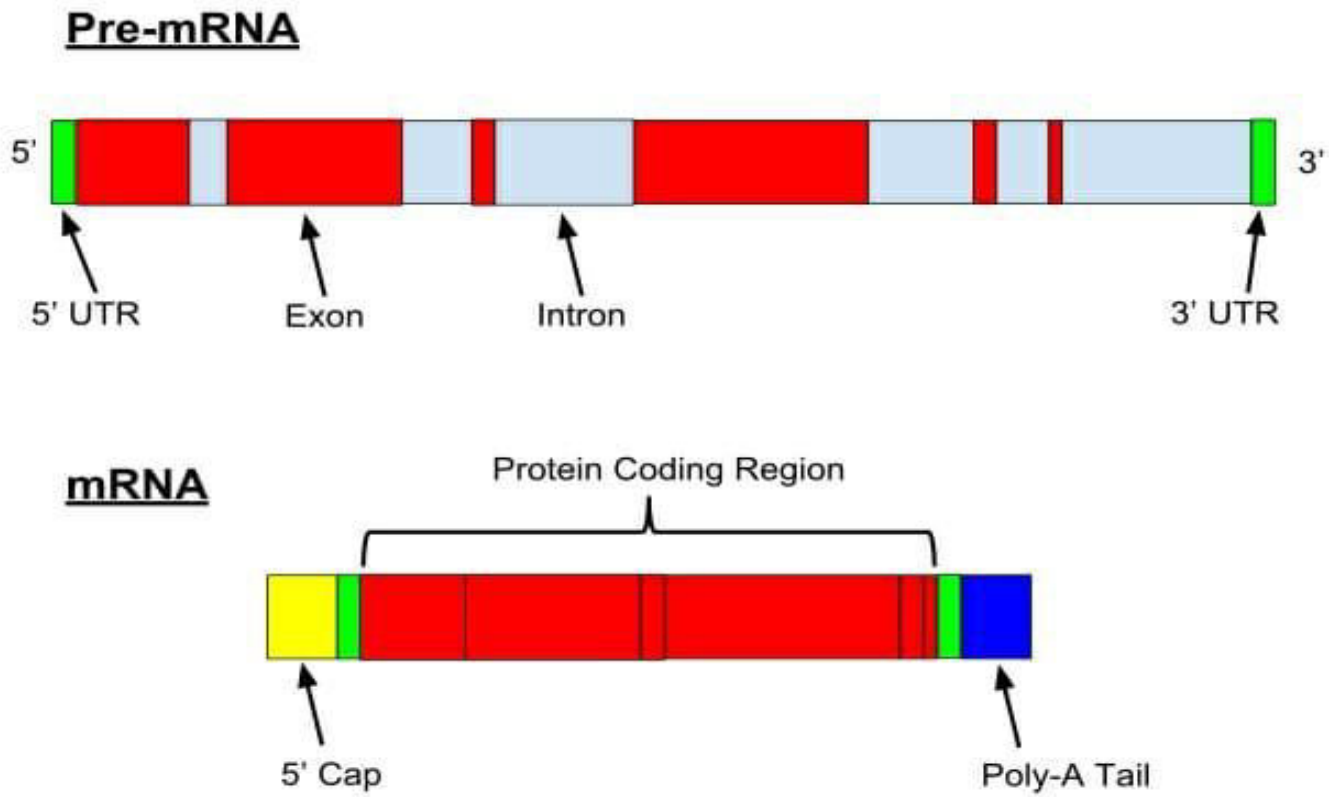
- Viruses are DNA and RNA shielded by protein coat called capsid.
- Viruses do not have a domain

Three Domain system

Similarities and differences of Eubacteria and Archaea

Characteristics	Eubacteria	Archaea
Cell type	Prokaryotic	Prokaryotic
Cell wall	Contain peptidoglycan	Carbohydrates, proteins or No cell wall
Interons	No interons	Interons present
Ribosomal proteins	No ribosomal proteins	Ribosomal proteins present
Diseases	Can be parasitic and cause diseases	None are parasitic
Gene resemblance	No resemblance	Eukaryotes

Three Domain system



https://biologydictionary.net/intron/

Principles of Systematics

Module# 8

Principles of Systematics

Five Kingdom Classification

Five Kingdom Classification

Two Kingdom classification

- Plants (Autotrophs)
- Animals (Hetrotrophs)

Problem

- Euglena

Solution by Ernst Hackel (1866)

- Protista

Five Kingdom Classification

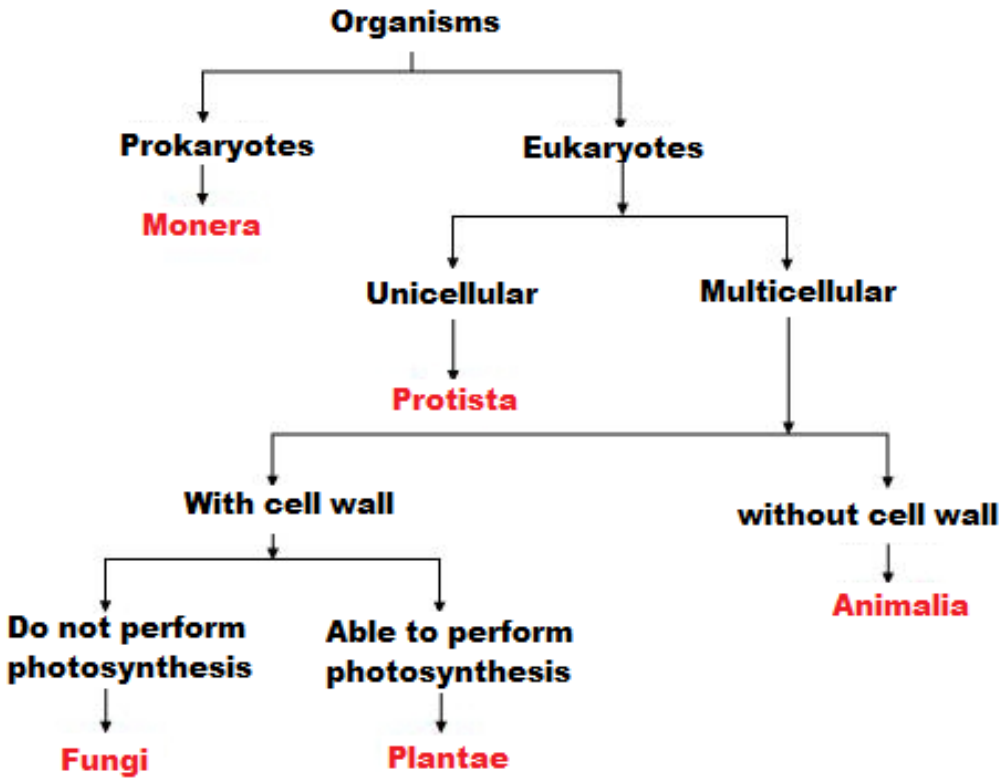
- E- Chatton in 1937 introduced the term “Procariotique” to describe bacteria and “Eucariotique” to describe animals and plant cells.

- Five kingdom classification was proposed by Robert Whittaker in 1967.

Five Kingdom Classification

- Morgulis and Schwatz in 1988 modified R. Whittaker classification.
- They considered genetics along with cellular organization and mode of nutrition in classification.

Five Kingdom Classification



- Five kingdoms proposed by Robert Whittaker include:
  - Monera
  - Protista
  - Fungi
  - Plantae
  - Animalia

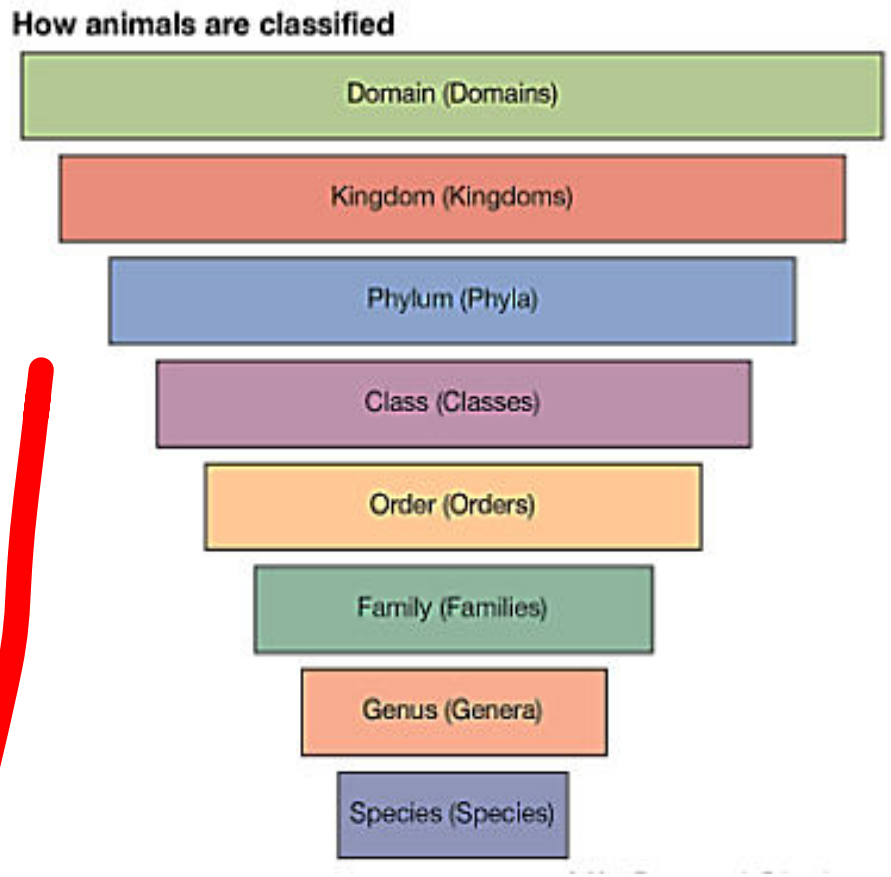
Principles of Systematics

Module# 9

Principles of Systematics

History of Taxonomy

History of Taxonomy



https://Britannica.com/science/taxonomy

- Taxonomy is as old as the language skill of mankind.
- Taxonomy is the science of naming, describing and classifying organisms including all plants, animals and microorganisms of the world.
- Taxonomy identifies and enumerates the components of biological diversity.

History of Taxonomy

- Native tribes used to call the species on the basis of their morphological characters
- Development of scientific theory of classification ….a remarkably recent phenomenon

History of Taxonomy

George Gaylord Simpson (1902-1984)

https://www.google.com/search?q=george+gaylord+simpson+pic&sxsrf=ALeKk016UQ5865EIGFEM0Awryp5IHsxm-Q:1613054410489&source=lnms&tbm=isch&sa=X&ved=2ahUKEwilsI_5hLuAhUJnlkKHYD5DkgQ_AUoAXoECA8QAw&biw=1366&bih=625#imgrc=siNionRQ1ItxWM

- Simpson in 1961 gives a valuable survey of History of Taxonomy.
- Simpson was the most influential paleontologists of the twentieth century.
- He published extensively on the taxonomy of fossil and extinct mammals.

History of Taxonomy

Frederick Smith 1805-1879

https://www.google.com/search?q=frederick+smith+pic+pic+taxonomy&tbm=isch&ved=2ahUKEwiSPCJiuLuAhUP_BoKHfOBCwsQ2cCegQIABAA&oq=frederick+smith+pic+pic+taxonomy&gs_lcp=CgNpbWcQA1CQEFj1KWCxLWgAcAB4AIABjQOIAegXkgEFMi0xLjiYAQCgAQGqAQtnd3Mtd2l6LWltZ8ABAQ&sclient=img&ei=BkQlYJLdHo_4a_ODrlg&bih=625&biw=1366#imgrc=k331uXCLcv1lsM

- Frederick Smith worked in the zoology department of British Museum from 1849.
- He published from 1851 to 1870 on ant taxonomy.
- During the 28 years of his activity he describe 25 genera, 702 species and subspecies of whic hare.

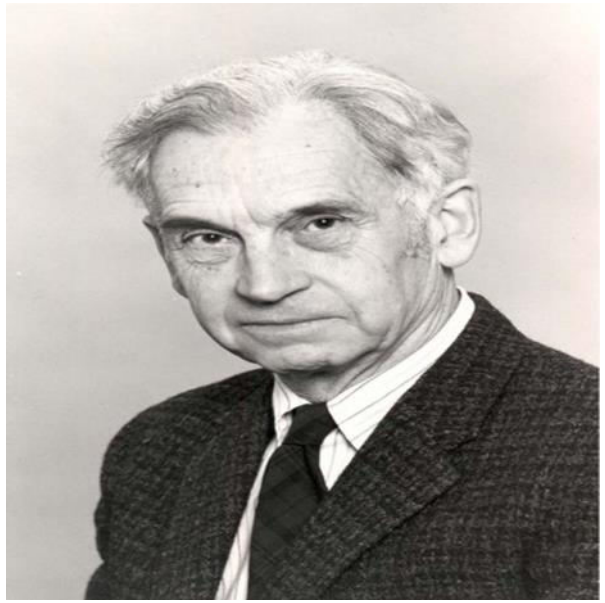
History of Taxonomy

Thomas E. Mittler (1928-2012)

https://www.google.com/search?q=thomas+E.Mittler+pic+pic+taxonomy&tbm=isch&ved=2ahUKEwiSPCJiuLuAhUP_BoKHfOBCwsQ2cCegQIABAA&oq=frederick+smith+pic+pic+taxonomy&gs_lcp=CgNpbWcQA1CQEFj1KWCxLWgAcAB4AIABjQOIAegXkgEFMi0xLjiYAQCgAQGqAQtnd3Mtd2l6LWltZ8ABAQ&sclient=img&ei=BkQlYJLdHo_4a_ODrlg&bih=625&biw=1366#imgrc=k331uXCLcv1lsM

- Thomas E.Mittler was an Austrian entomologist.
- He worked on history of higher taxonomy and also on the classification of insects

History of Taxonomy



Ernst Mayr (1904- 2005)

https://www.google.com/search?q=ernst+mayr+pic+taxonomy+&tbm=isch&ved=2ahUKEwjs6e2JjuLuAhXBwIUKHfhfAU0Q2cCegQIABAA&oq=ernst+mayr+pic+taxonomy+&gs_lcp=CgNpbWcQA1D6IlibLmD8MmgAcAB4AIABjgSIAcgXkgEHMy0zLjMuMZgBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=OEglYOztC8GBlwT4v4XoBA&bih=625&biw=1366#imgrc=PdnA5xdNdrOdwM//

- Mayr in 1982 provided a details of various periods of History of Taxonomy
- His work contributed to the conceptual revolution that led to modern evolutionary synthesis of taxonomy and systematics

Principles of Systematics

Module# 10

Principles of Systematics

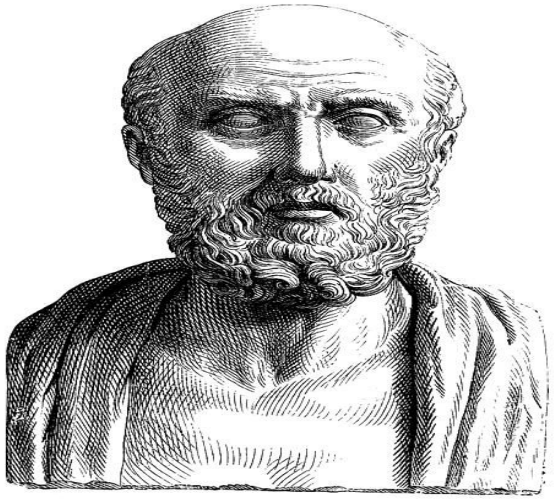
History of Taxonomy

History of Taxonomy

- Mayr described the History of Taxonomy into four periods:

- 1st Period
- 2nd Period
- 3rd Period
- 4th Period

History of Taxonomy



Hippocrates (460-377 B.C.)

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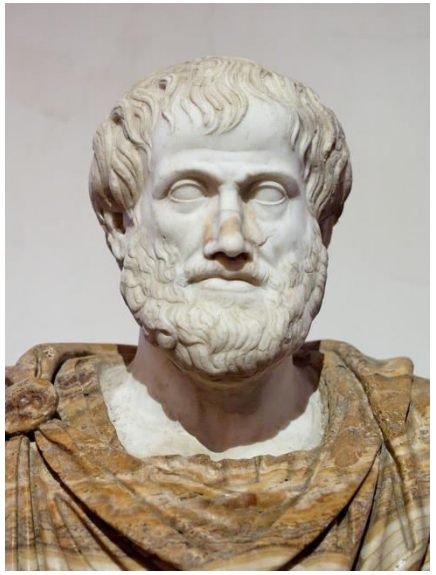
1st Period (Study of local fauna)

- Several early Greek scholars notably Hippocrates enumerated the types of animals But he did not classify.

History of Taxonomy



Aristotle (384-322 B.C.)

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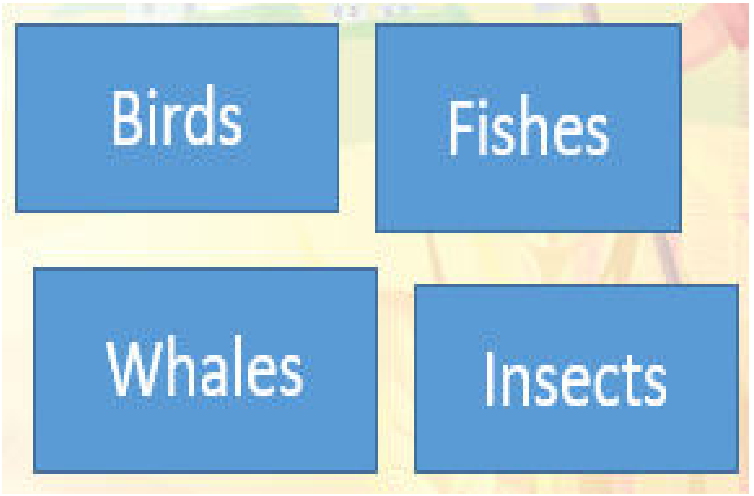
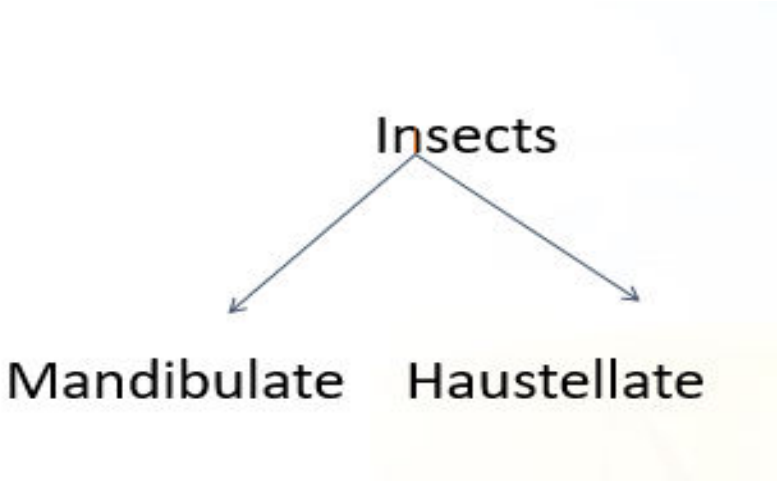
1st Period (Study of local fauna)

- Aristotle known as father of biological classification, referred to the major groups of animals.

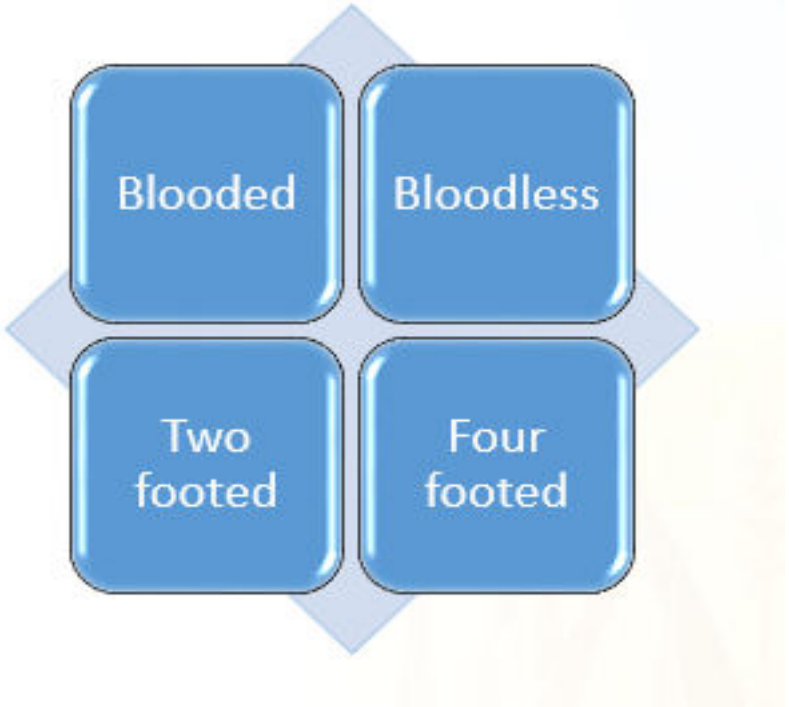


History of Taxonomy



- Aristotle also made distinction in insects
- He first used the following terms: Coleoptera and Diptera

History of Taxonomy



- He established numerous collective categories, or genera using differentiating characters

History of Taxonomy

- Aristotle made further categories of animals
- His thinking dominated for next two thousand years
- After his death no system of classification were recognized

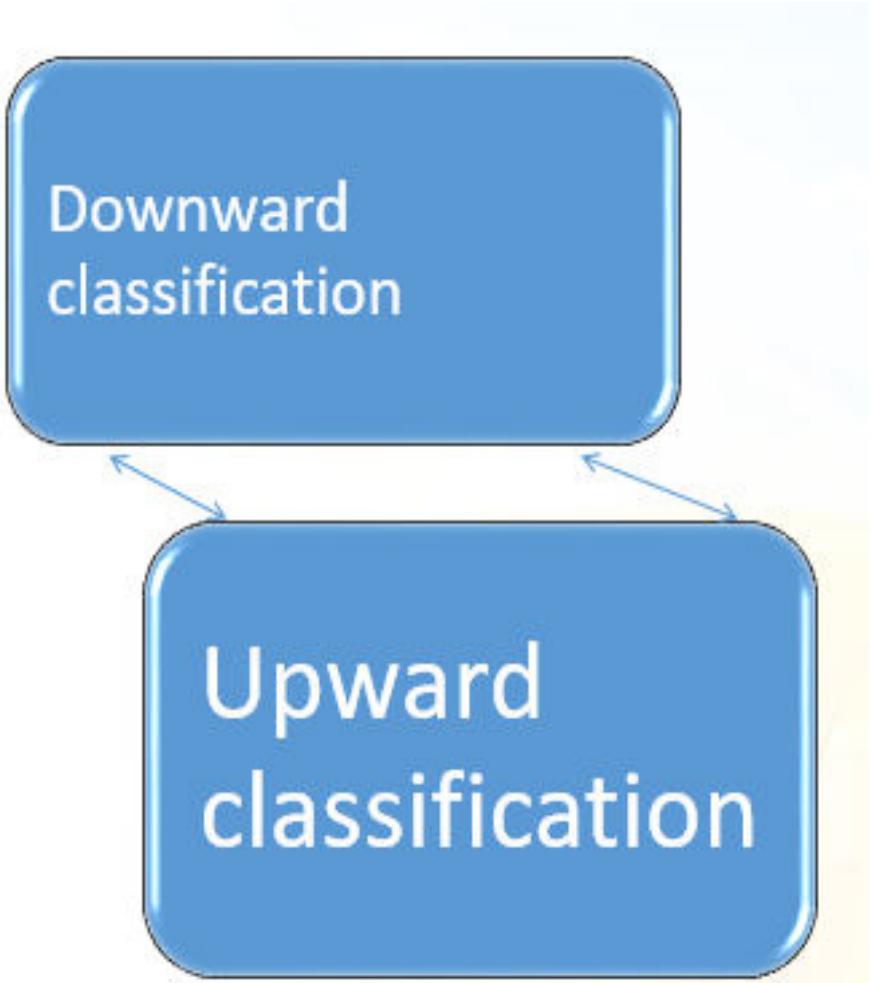
Principles of Systematics

Module# 11

Principles of Systematics

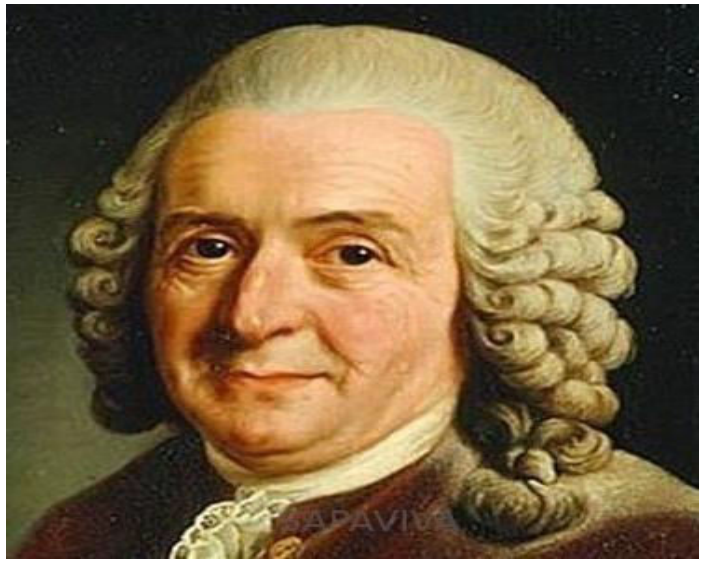
History of Taxonomy

History of Taxonomy



- From 1551 rapid progress started in knowledge about animals.
- There are two major forms of classification in first period

History of Taxonomy



Carl Linnaeus (1707-1778)

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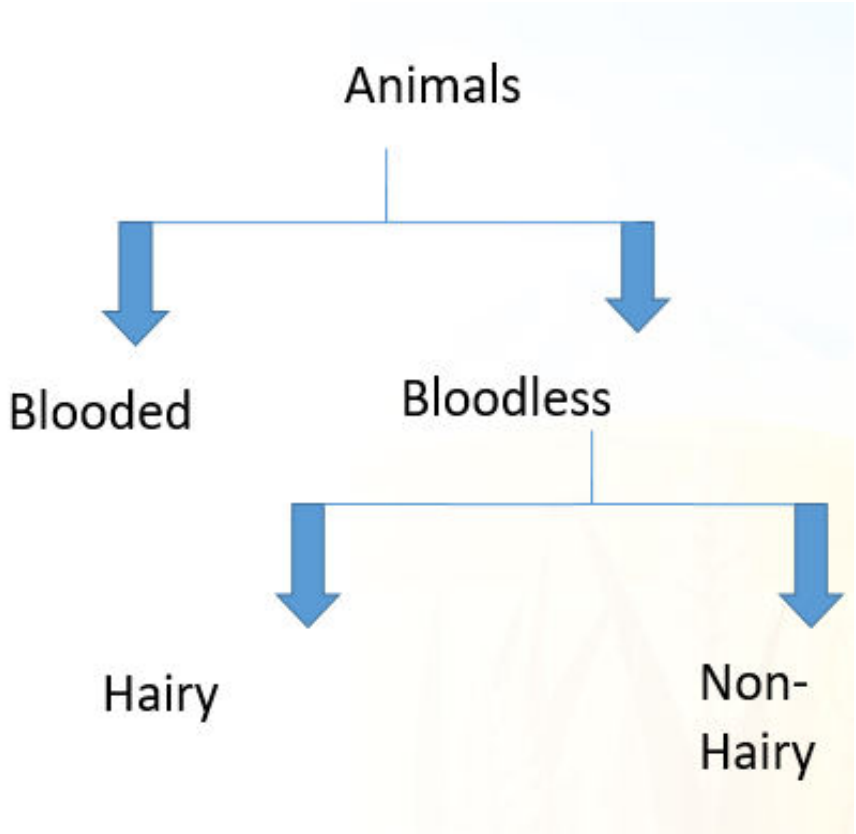
- Taxonomy based on local fauna reached its peak by the work of Linnaeus (1707-1778)

History of Taxonomy



- Downward classification are the principles of logical division which consist of dividing the larger groups based on dichotomy into smaller groups. e.g,

Animals

History of Taxonomy

- Linnaeus adopted the principle of downward classification in 15th edition of his book (Systema naturae)
- He for the first time applied binomial nomenclature to animal kingdom.

History of Taxonomy

- Linnaeus's idea was largely acceptable for the classification of insects.
- Not acceptable for the classification of birds, amphibians, and lower vertebrates (Vermes)

History of Taxonomy

Comte de Buffon (1707-1778)

https//www.google.com/search?q=comte+deBuffon+taxonomy&tbm=isch&ved=2ahUKEwi19KNjuLuAhUEPhoKHY92BQQQ2cCegQIABAA&oq=hippocratespic+taxonomy&gs_lcp=CgNpbWcQAzoGCAAQCBAeUM_LJliV9CZgy_cmaABwAHgCgAGHBYgBvTySAQczLTYuNy40mAEAoAEBqgELZ3dzLXdpei1pbWfAAQE&sclient=img&ei=QEglYLXbCIT8aI_tlSA&bih=625&biw=1366#imgrc=77dKLgexqooJoM

- Buffon prepared the way for the biological species concept, by using sterility barriers instead of degree of morphological differences as the species criterion

Principles of Systematics

Module# 12

Principles of Systematics

History of Taxonomy

History of Taxonomy

- By the mid of 18th century shortcomings of downward classification system were recognized.
- Gradually replaced by upward classification system

History of Taxonomy

- It consists of assembling species by inspection into groups of similar and related species and forming hierarchy by again grouping of similar taxa of next lower rank.

History of Taxonomy

Comte de Buffon (1707-1778)

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- Buffon in 1749 supported this system.
- He said that the only way to design a method is to group together the things that resemble each other and separate the things that differ from each other

History of Taxonomy

James Anderson (1738– 1809)

https//www.google.com/search?q=comte+deBuffon+taxonomy&tbm=isch&ved=2ahUKEwi19KNjuLuAhUEPhoKHY92BQQQ2cCegQIABAA&oq=hippocratespic+taxonomy&gs_lcp=CgNpbWcQAzoGCAAQCBAeUM_LJliV9CZgy_cmaABwAHgCgAGHBYgBvTySAQczLTYuNy40mAEAoAEBqgELZ3dzLXdpei1pbWfAAQE&sclient=img&ei=QEglYLXbCIT8aI_tlSA&bih=625&biw=1366#imgrc=77dKLgexqooJoM

- Anderson 1763 (Botanist) applied this design for the classification of plants

History of Taxonomy

- Cuvier recognized five phyla

Georges Cuvier (1769-1832)

https://www.google.com/search?q=george+cuvier+1769+to+1832&tbm=isch&ved=2ahUKEwjQx9O-kOLuAhUMEhoKHf1eCH4Q2-cCegQIABAA&oq=george+cuvier+1769+to+1832&gs_lcp=CgNpbWcQAzoECCMQJzoECAAQQzoHCCMQ6gIQJzoFCAAQsQM6AggAOggIABCxAxCDAToECAAQAzoGCAAQBRAeOgQIABAeUNkRWJ6PAWCskgFoAXAAeACAAbsDiAHCbpIBBjMtMzQuNpgBAKABAaoBC2d3cy13aXotaW1nsAEKwAEB&sclient=img&ei=v0olYJDsL4ykaP29ofAH&bih=625&biw=1366#imgrc=_rEsvqaiDKV5CM

History of Taxonomy

Karl Ernst von Baer (1892-1876)

Richard Owen (1804-1892)

- Van Baer, Owen divided all species into limited no. of groups each representing a distinct type

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History of Taxonomy

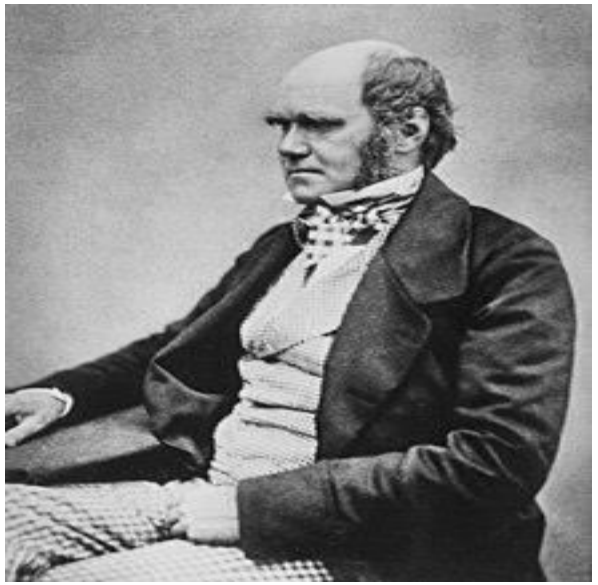
Developments during Linnaeus and Darwin Period

1. Specialization became more pronounced. Many authors identified birds, beetles, or butterflies
2. Classification became more hierarchical and soon the categories family and phylum were added
3. Philosophical guidelines were rejected
4. Search for a natural system was intensified

History of Taxonomy

- This thought was applied by a botanist in 1873 and later by nearly all zoologists.
- This shift from downward classification to upward classification was a philosophical change

History of Taxonomy

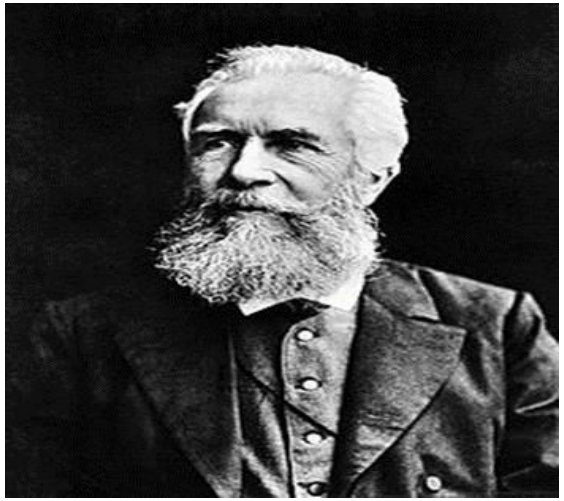


Charles Darwin (1809-1882)

https://www.google.com/search?q=charles+darwin&tbm=isch&ved=2ahUKEwi5vpykneLuAhUN-YUKHSv2Dr4Q2-cCegQIABAA&oq=charles+darwin&gs_lcp=CgNpbWcQAzIECAAQQzICCAAyAggAMgIIADICCAAyAggAMgIIADICCAAyAggAMgIIADoHCCMQ6gIQJzoECCMQJzoFCAAQsQM6CAgAELEDEIMBOgcIABCxAxBDULLGJFid-SRgzvskaAFwAHgDgAHWBYgB4kOSAQkzLTYuNi42LjGYAQCgAQGqAQtnd3Mtd2l6LWltZ7ABCsABAQ&sclient=img&ei=KlglYLmBAY3ylwSr7LvwCw&bih=625&biw=1366#imgrc=mrcMmxqef7ryCM

- Evolutionary thought was widespread in 18th century
- Darwin believed in fixity of species but after observing natural fauna in Galapagos island, he begin to believe in the plasticity of species

History of Taxonomy



Ernest Haeckel (1834-1919)

https://https://www.google.com/search?q=ernst+hackel&tbm=isch&ved=2ahUKEwiH3uDFn-LuAhUt5IUKHQxDDdoQ2-cCegQIABAA&oq=ernst+hackel&gs_lcp=CgNpbWcQAzICCAAyBggAEAoQGDIGCAAQChAYMgYIABAKEBgyBggAEAoQGDIGCAAQChAYMgYIABAKEBg6BAgjECc6BAgAEEM6BwgjEOoCECc6BQgAELEDOgQIABAeOgQIABAYULWsDFjy5wxgpu0MaAFwAHgAgAGCBogBrUWSAQkzLTguMy42LjKYAQCgAQGqAQtnd3Mtd2l6LWltZ7ABCsABAQ&sclient=img&ei=iVolYMf1DK3IlwSMhrXQDQ&bih=625&biw=1366#imgrc=cIElbdaRJgSkeM

- Empirical taxonomists were greatly influenced by Darwin's idea.
- The phylogenetic tree of Ernest Haeckel also stimulated the empirical workers.
- Large number of new species were discovered and described.

History of Taxonomy

- The greatest development of modern taxonomy started around 1930, when the workers realized that the Linnaeus species based on one or two specimens are not as perfect as those which are based on population

History of Taxonomy

- Due to this Mayr in 1942 considered species as a group of interbreeding natural population.
- This idea of population taxonomy was useful in developing the polytypic concepts.
- New terms like new systematics and biosystematics were added to extend the taxonomic theory.

History of Taxonomy

- The publication of the book ‘new systematics’
- Become a landmark in the History of Taxonomy.
- Thus taxonomy got a new label of biological taxonomy
- By 1955 taxonomy reach such a status that it was called taxonomic explosion

History of Taxonomy

- New systematic deal exclusively with the species level.
- The present day taxonomic work include all available differences and similarities.

History of Taxonomy

- Instead of morphology as an original base, phylogenetic adaptations, embryological patterns, biochemical variations, genetic variabilities and similarities, behavioral characteristics are all applicable to taxonomic study

END

Principles of Systematics

Module# 13

Principles of Systematics

Basic Definitions

Basic Definitions

Sibling Species

- Two or more than two closely related species which are morphologically alike but behaviorally or reproductively isolated from each other.
- Examples are *Drosophila persimilis* and *D. pseudoobscura*.

Basic Definitions

Sibling Species

- The mosquito *Anopheles maculipennis* complex consists of several subspecies, of which a few are vector of malaria and the rest are harmless.

Basic Definitions

Monotypic Species

- species with a single subspecies, called monotypic species
- *Example: Bardala labarda*

Polytypic Species

- A species contains two or more subspecies
- Examples are tiger, *Panthera tigris* which has several subspecies

Basic Definitions

Endemic species

- The species which are found in a particular region, called endemic species.

Example

- The Darwinian finches are the endemic species of Galapagos Islands
- Indus dolphin in Pakistan

END

Principles of Systematics

Module# 14

Principles of Systematics

Basic Definitions

Basic Definitions

Deme

- A deme is a sub-population (a smaller group within the population) that can freely interbreed.
- A population is usually made of several demes (groups) that are partially isolated from each other.

Basic Definitions

Deme

https://www.pathwayz.org

- Often a geographical barrier limits gene flow between demes, but not sufficiently to result in speciation.
- Enough gene flow occurs between demes ensuring that all of it's members remain the same species.

Basic Definitions

https://www.pathwayz.org

Cline

- A cline is the gradual change in the phenotypes and genotypes of a species across a geographical gradient
-
- Allele frequencies and traits might vary slightly across the geographical gradient because of different selection pressures.

Basic Definitions

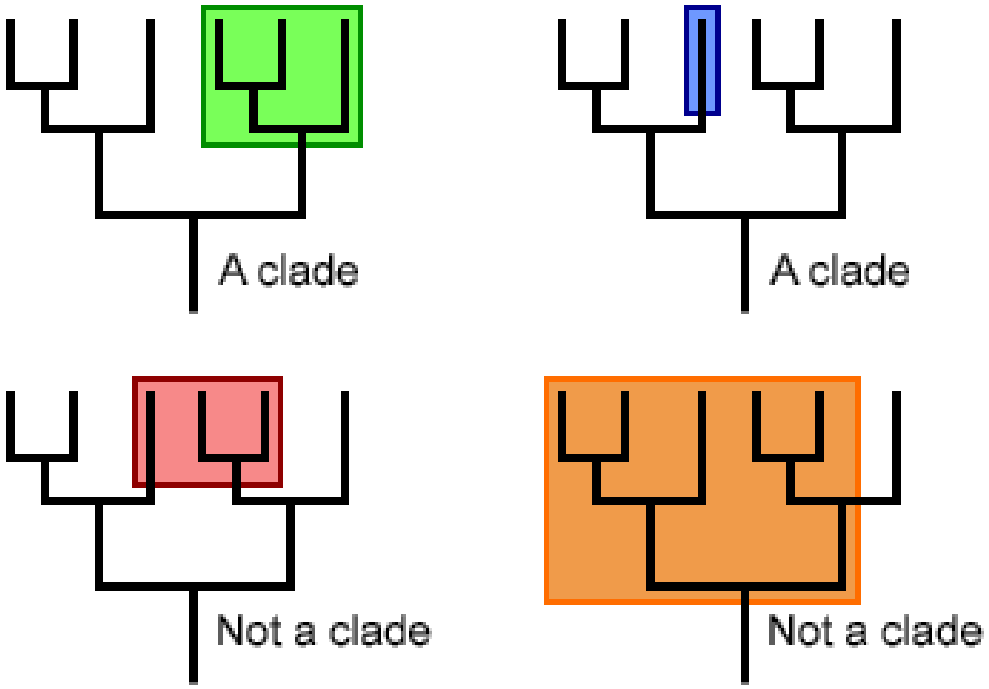
https://www.pathwayz.org

Types of cline

- Smooth cline
- Step cline

Basic Definitions

Clade



- A clade is a group of organisms that evolved from a common ancestor

- Eutheria is one mammalian clade and the other is Metatheria, which includes marsupials.
- Another example of a clade could be birds: they all also descended from a common ancestor.

Principles of Systematics

Module# 15

Concept of Speciation and Causes

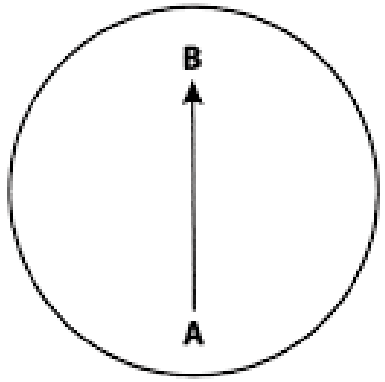
Concept of Speciation and Causes

Speciation

Speciation refers to the creation a new species.

Through this process, the earliest groups of similar organisms were able branch out and populate the world with millions of different varieties of life

Basic Definitions



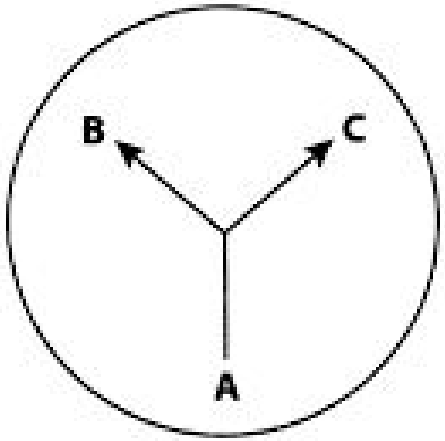
Anagenesis

Types of speciation

Anagenesis

- A single species may change over time into a new form that is different enough to be considered a new species.
- One species replaces other
- No net change in number of species

Basic Definitions



Cladogenesis

Types of speciation

Cladogenesis

- A species may become split into two groups that no longer share the same gene pool.
- Net increase in number of species
- Increase in diversity

Concept of Speciation and Causes

Cusses of speciation

- Geographical Isolation
- Behavioural isolation
- Reproductive isolation

END

Principles of Systematics

Module# 16

Concept of Speciation and Causes

Types of speciation

Concept of Speciation and Causes

Types of speciation

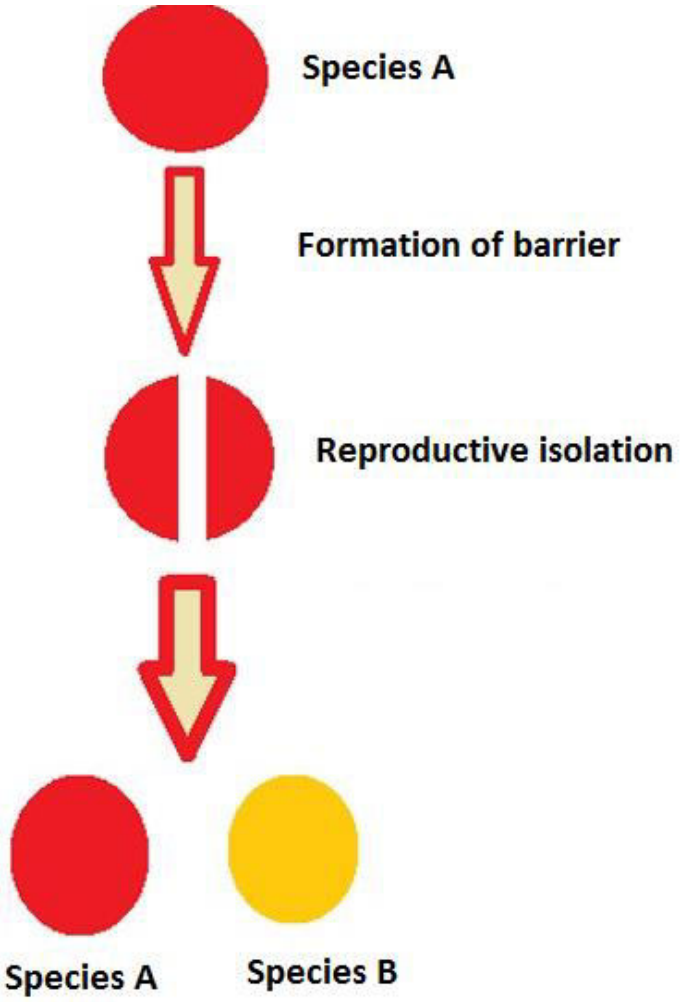
- Allopatric speciation
- Peripatric speciation
- Sympatric speciation
- Parapatric speciation

Concept of Speciation and Causes

Allopatric Speciation

- The most common form of speciation
- Occurs when populations of a species become geographically isolated.

Concept of Speciation and Causes



Allopatric Speciation

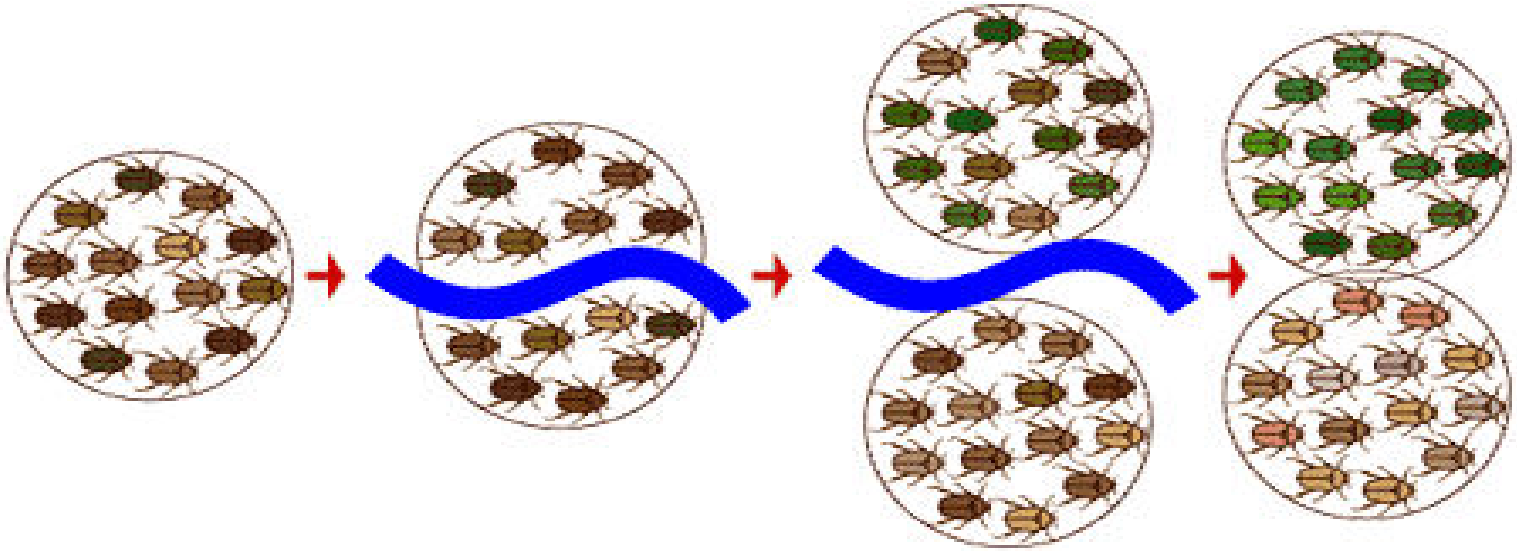
- When populations become separated, gene flow between them ceases.
- Over time, the populations may become genetically different

Concept of Speciation and Causes

Examples

- Speciation is the Galápagos finch
- Different species of these birds live on different islands.
- The finches are isolated from one another by the ocean.
- Over millions of years, each species of finch developed a unique beak that is especially adapted to the kinds of food it eats.

Concept of Speciation and Causes



https://evolution.berkeley.edu

Principles of Systematics

Module# 17

Principles of Systematics

Sympatric Speciation

Sympatric Speciation

- A speciation in which a species is evolved from a single common ancestral species while inhabiting the same geographical range.
- Share the same habitat but become reproductively isolated from each other.
- More common in plants

Sympatric Speciation

Popyploidy in plants

- Plants produce offspring that are polyploid. Hence, the offspring live in the same environment as their parents but are reproductively isolated.

Sympatric Speciation

https://images.app.goo.gl/5Q15k3p4fVkKcktA7

Example

- 200 year ago ancestor of maggot flies were used to lay eggs on hawthorn.
- Now they lay eggs on hawthorn (native to America) and domestic apples (introduced)
- Host shift/genetic differences (Coyne, 2007)

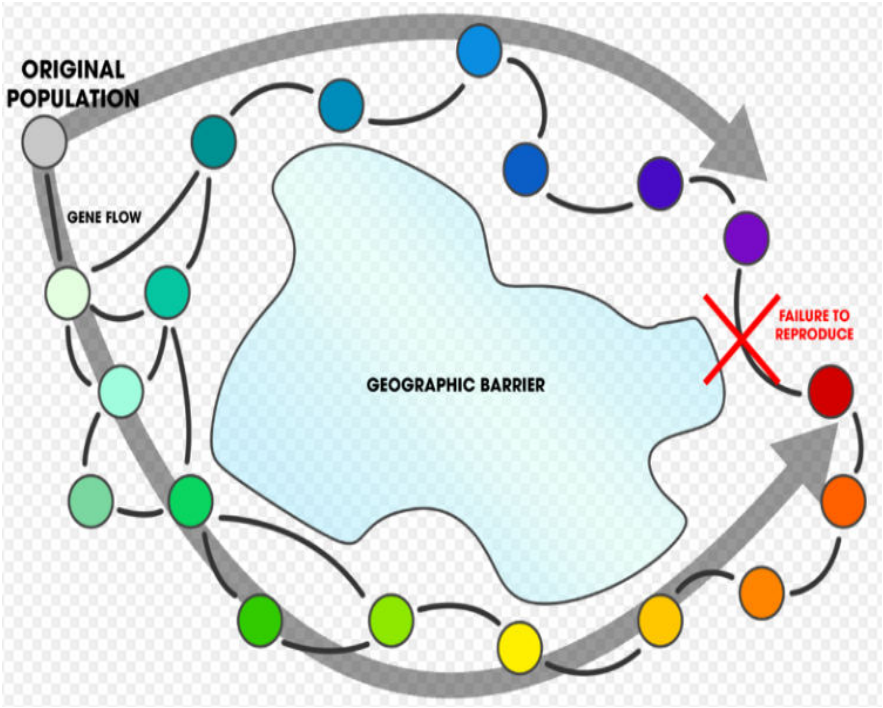
Principles of Systematics

Module# 18

Principles of Systematics

Parapatric Speciation

Parapatric Speciation



https://en.wikipedia.org/wiki/Parapatric_speciation#/media/File:Ring_Species_(gene_flow_around_a_barrier).png

Parapatric speciation

- It occurs when populations are separated not by a geographical barrier, such as a body of water, but by an extreme change in habitat.

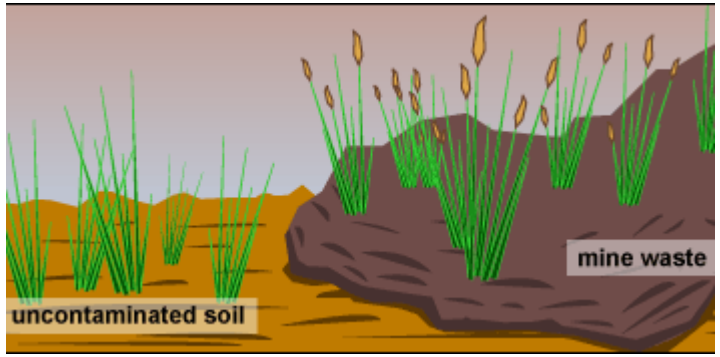
- Populations in these area may interbreed.
- Charles Darwin was the first to propose this mode of speciation.

Parapatric Speciation

Parapatric speciation

- Although interbreed but reduced gene flow.
- They often develop distinct characteristics and lifestyles.
- Reproductive isolation is rather temporal or behavioral.
- It is extremely rare.

Parapatric Speciation



https://images.app.goo.gl/iMY6DE1PcER2Dx1k7

Example

- Population of grass plant is present near mining area
- Soil near mining area is infused with metals due to which plants grow slowly than those that grows away from mines.
- As a result, two populations of grasses arise having different flowering time.

Parapatric Speciation

https://www.google.com/search?q=cave+snails&tbm

Example

- A study of tropical cave snails *(Georissa saulae)* found that cave-dwelling population descended from the above-ground population, likely speciating in parapatry

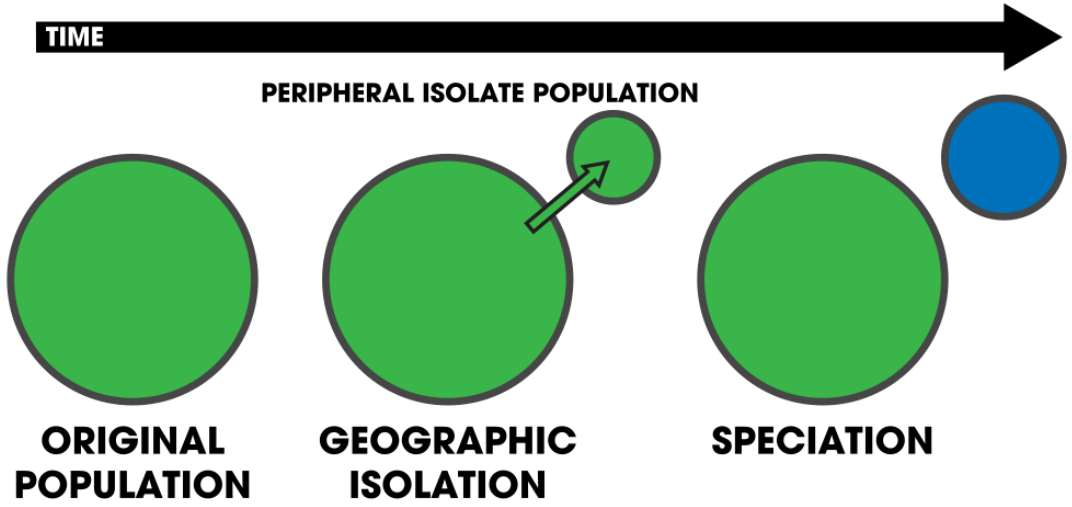
Principles of Systematics

Module# 19

Principles of Systematics

Paripatric Speciation

Paripatric Speciation

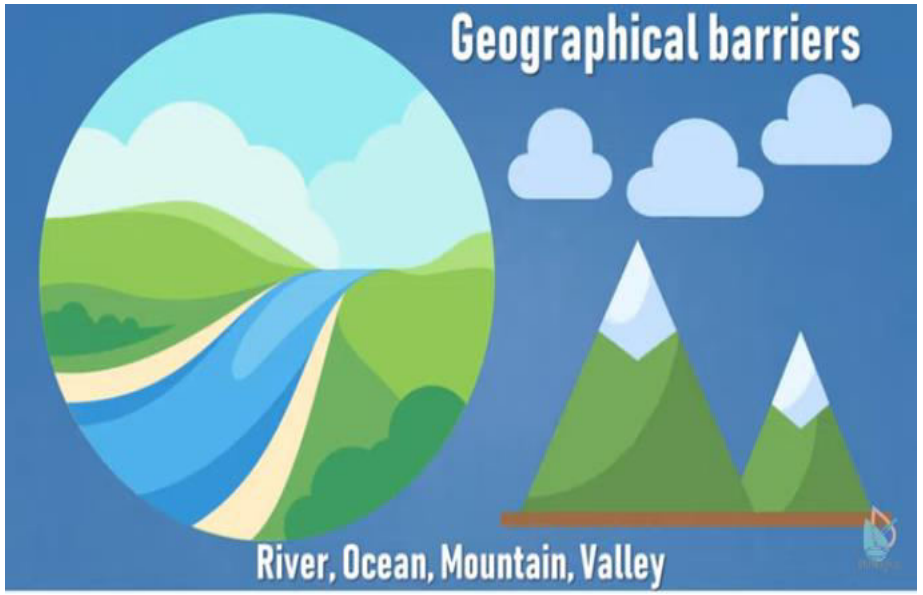


https://en.wikipedia.org/wiki/Peripatric_speciation#/media/File:Peripatric_Speciation_Schematic.svg

Paripatric speciation

- Peripatric speciation is a mode of speciation in which a new species is formed from an isolated peripheral population.
- Resemble alloprtraic speciation/barriers
- Isolating population is small

Paripatric Speciation



https://www.youtube.com/watch?v=7TNN-WGYWks

Paripatric speciation

- Peripatric speciation is a mode of speciation in which a new species is formed from an isolated peripheral population.
- Resemble alloprtraic speciation/barriers
- Isolating population is small

Paripatric Speciation

https://www.google.com/search?q=london+underground+mosquito&hl

Example

- An **example** of this is the London Underground mosquito, a variant of the mosquito *Culex pipiens*, which entered in the London Undgerground in 19th century.

- It is found in the London Underground railway system

Paripatric Speciation

Example

- Evidence for its **speciation** includes genetic divergence, behavioral differences, and difficulty in mating.

END

Principles of Systematics

Module# 20

Typological Species Concept

Typological Species Concept

- Species is the basic unit of classification according to the International Commission of Zoological Nomenclature.

- Survey of taxonomic literature shows that there are more than 20 species concepts.

Typological Species Concept

Typological species concept

- This concept was proposed by Aristotle and Plato

According to this concept

"A certain pattern of features is found in each species of organisms and all members of species show maximum equality in this format"

Example: certain pattern in dogs

Typological Species Concept

Typological species concept

- Monotypic species

- Polytypic species

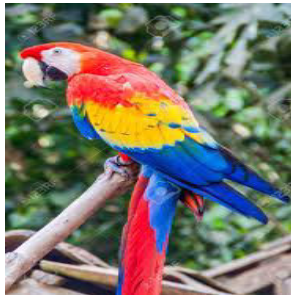
Typological Species Concept

Postulates of Typological species concept

- Species consist of similar individuals sharing the same essence.
- Each species is separated from all others by sharp discontinuity.
- Each species is completely constant through time.
- There are strict possible limits to variations within any one species.

Typological Species Concept

Ara ararauna



Ara macao

sterile hybrid
Catalina macaw

Example

- Two macaw species, blue-and-yellow macaw ***Ara ararauna*** and the scarlet macaw ***Ara macao*** are considered different species due to different coloration.
- In nature, their habitats overlap but they not produce hybrids except in captivity.

Typological Species Concept

Anchura substriata

https://www.google.com/search?q=macaw&tbm=isch&ved=2ahUKEwiQyL

Strengths

- Applicable to living or extinct species.
- Doesn't depend on complex genetic analysis.
- Morphology can be readily observed, without harming the organisms.

Typological Species Concept

Problems with Typological species concept

- Polymorphism within populations.
- Sibling or cryptic species.
- Geographical variations among populations

END

Principles of Systematics

Module# 21

Principles of Systematics

Biological Species Concept

Biological Species Concept

- According to Biological species concept:

- Species are groups of actually or potentially interbreeding natural populations, which are reproductively isolated from other such groups.” Ernst Mayr(1942,1963)

Biological Species Concept

- Can successfully interbreed and produce fertile offspring.
- A species' integrity is maintained by interbreeding within a species.

Biological Species Concept

Why members of species resemble

- This concept explains why members of species resemble and differ from other species.
- When organisms breed they shuffle their gene pool which gives the species its identity.
- Genes are not shared with other species so they look different.

Biological Species Concept

Biological species concept

Members of species constitute:

- Ecological unit
- Reproductive unit
- Genetic unit

END

Principles of Systematics

Module# 22

Principles of Systematics

Biological Species Concept

Biological Species Concept

Biological species concept

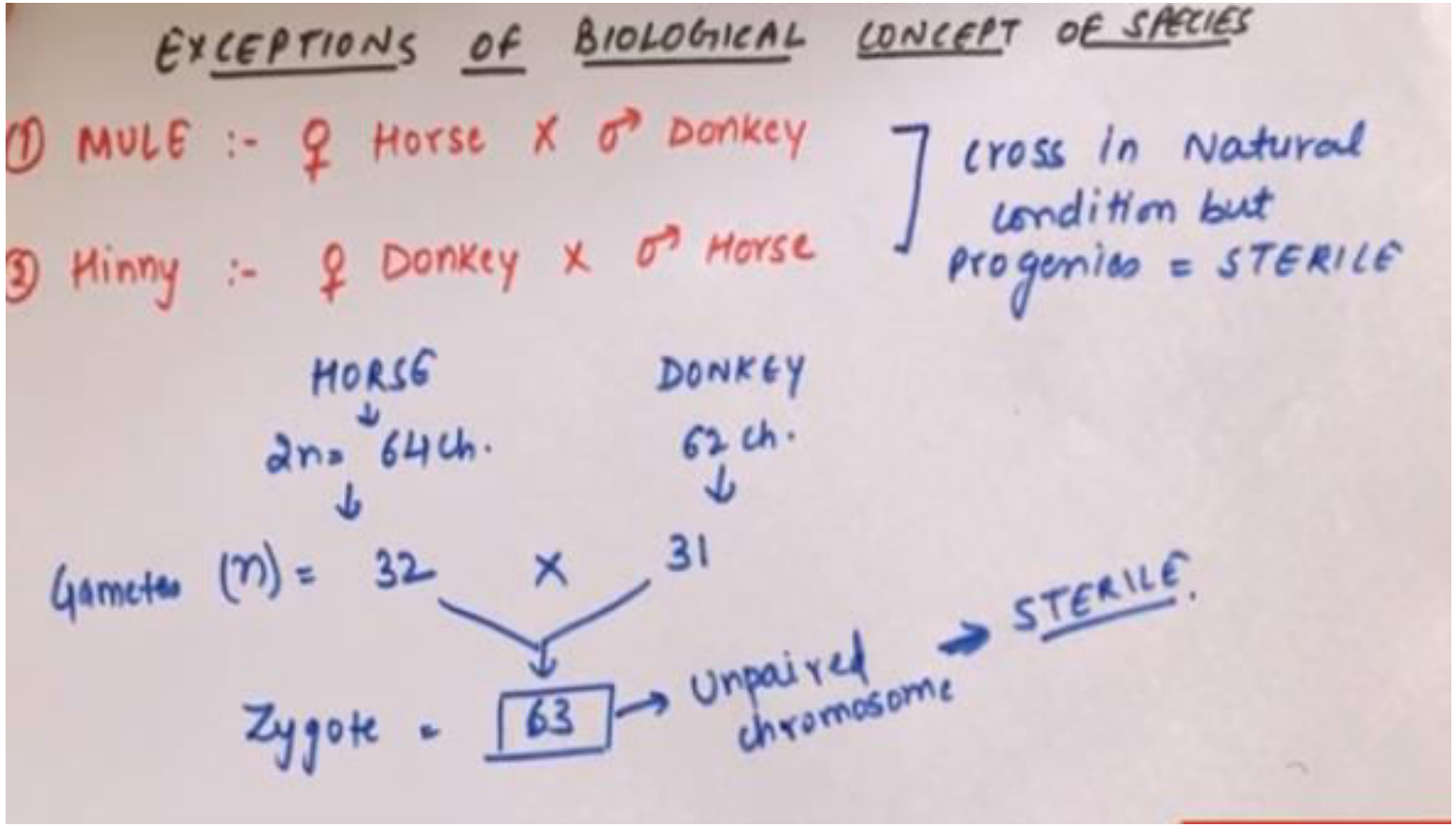
- Ecological unit
- Reproductive unit
- Genetic unit
- ." Ernst Mayr (1942, Darwin of 20th century)

Biological Species Concept

Biological species concept

- Most widely accepted species concept.
- Dobzhansky (1951) and Hanson (1981) accepted the definition of biological Species concept because of two reasons:
- Gene pool
- Reproductive isolation

Biological Species Concept



https://www.youtube.com/watch?v=3Chz-XJHvNE

Principles of Systematics

Module# 23

Principles of Systematics

Drawbacks of Biological Species Concept

Drawbacks of Biological Species Concept

Drawbacks

- Asexual reproducing organisms.
- Fossils
- Sibling or cryptic species
- Distinguishing between species on the basis of reproductive separation is problematic because reproductive isolation is often incomplete .

Drawbacks of Biological Species Concept

Asexual reproduction/Uniparental reproduction

- Not applicable on apomictic species
- Do not fulfill interbreeding criterion **(Most important in BSC)**
- Apomictic groups (parthenogenesis, apomixis, budding etc)

Drawbacks of Biological Species Concept

Asexual reproduction/Uniparental reproduction

- Uniparental reproduction is Common in lower invertebrates and lower vertebrate
- Decedents are called agamospecies, binoms or paraspecies
- According to Mayr (1988a) these are not species

Drawbacks of Biological Species Concept

Sibling species/Cryptic species

- As they are morphological similar but reproductively isolated

Incompleteness of Speciation/fossil record

- Evolution is gradual and continuous process. However, fossil record is not complete

END

Principles of Systematics

Module# 24

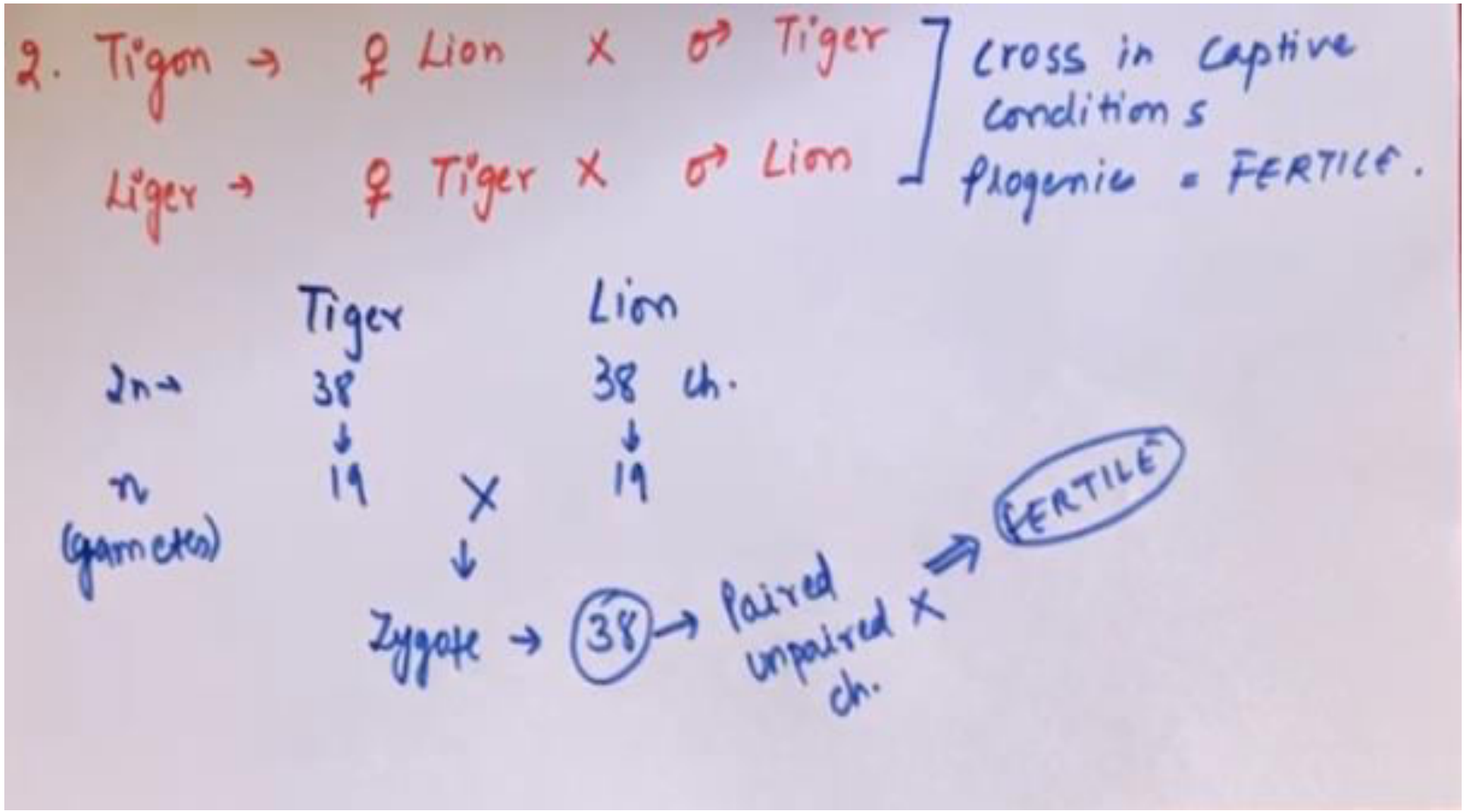
Drawbacks of Biological Species Concept

Drawbacks of Biological Species Concept

Interbreeding between two different species

- According to BSC two different species can not breed and produce fertile offsprings
- However, there are exceptions

Drawbacks of Biological Species Concept



https://www.youtube.com/watch?v=3Chz-XJHvNE

Principles of Systematics

Module# 25

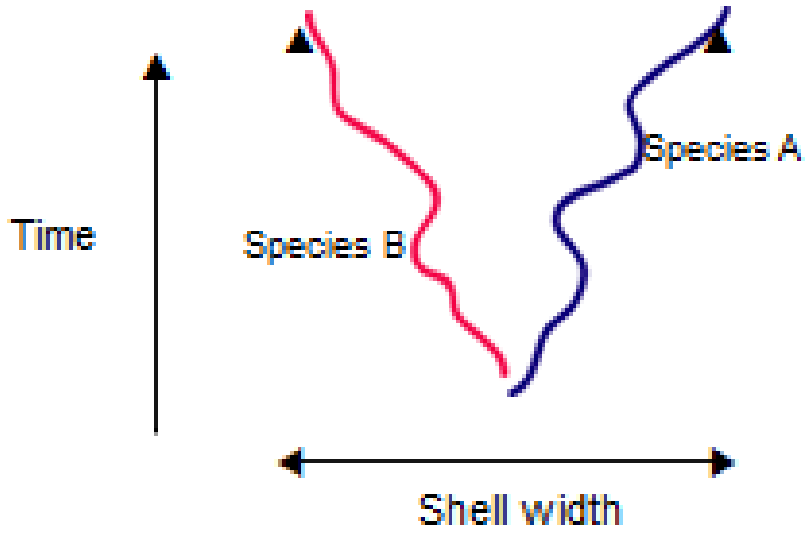
Evolutionary Species Concept

Evolutionary Species Concept

Evolutionary species concept

- Simpson (1940s)
- An **evolutionary species** "is a single lineage of ancestor-descendant populations of organisms which maintains its:
- Identity from other such lineages [in space and time]
- Own **evolutionary** tendencies
- historical fate" (Wiley, 1981)

Evolutionary Species Concept



Evolutionary species concept

- Initially developed to define fossil species.
- Applicable to both sexually and asexually reproducing species and emphasizes common descent.

Evolutionary Species Concept

Drawbacks

- Arbitrary: how to define independent roles and tendencies.
- Does not provide mechanism.

END

Principles of Systematics

Module# 26

Principles of Systematics

Taxonomic Characters

Taxonomic Characters

Character

- It is any attribute of member of taxon by which it may differ from member of other taxon.
- A heritable trait possessed by an organism
- Characters are usually described in terms of character states.

Taxonomic Characters

Character state

- One of the variant conditions of a trait.
- **Binary states:** For example: "hair present" vs. "hair absent," where "hair" is the **character**, and "present" and "absent" are its **states**
- **Multistate characters:** Yellow, white or pink flowers or AGCT for nucleotide sequence data.

Taxonomic Characters

Qualities of key characters

- Apply to all member of taxon
- Fairly stable and durable
- Qualitative and absolute
- Detectable without special equipment's, dissection or histological preparations

Taxonomic Characters

Taxonomic Characters are illustrated as:

- Morphological character
- Physiological character
- Behavioral character
- Ecological character
- Geographic character
- Molecular and biochemistry character
- Ontogenetic character

END

Principles of Systematics

Module# 27

Principles of Systematics

Morphological Characters

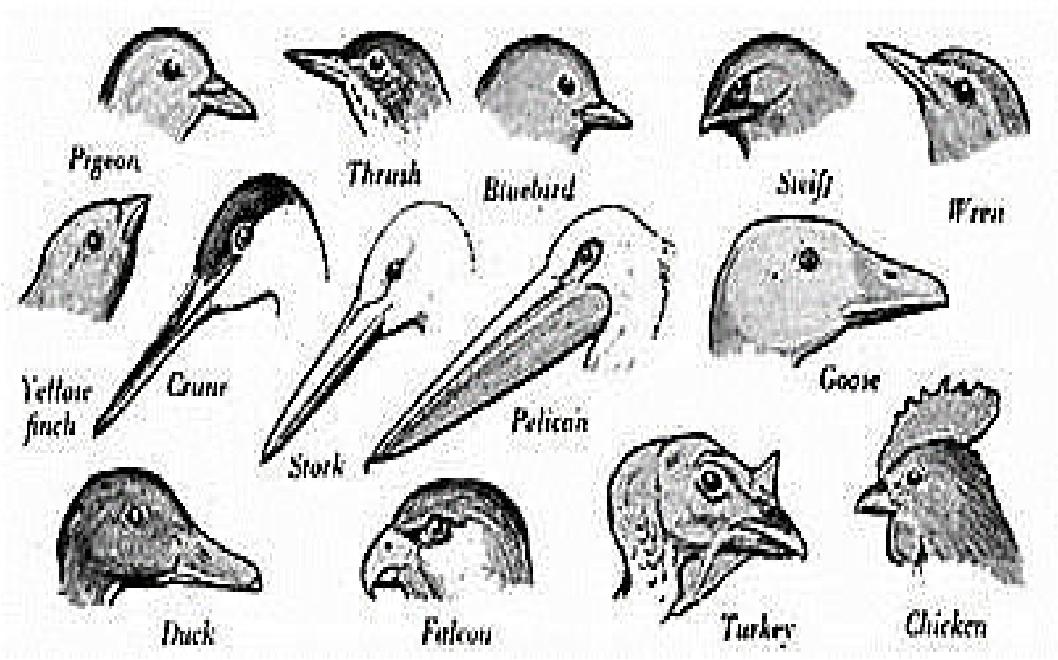
Morphological Characters

Morphological Characters

Morphology describe the structural features. Morphological characters include:

- External morphology
- Genetelic structure
- Internal morphology
- Embryology
- Karyology and other cytological differences

Morphological Characters



https://tse2.mm.bing.net/th?id=OIP.k_dPDFg2fI7OVNpR7Y72-gHaDa&pid=Api&P=0&w=

General external morphology

The apparent features are easy to detect and observe.

e.g. Plumage of birds, scales of fish and reptiles, pelage of mammals, phylogenetically significan sutures and sclerites of arthropod body etc.

Morphological Characters

https://www.researchgate.net/profile/Aidas_Saldaitis/publication/321343778/figure/fig2/AS:568147356336130@1512468293710/Male-genitalia-and-leg-structures-of-the-C-katsumii-holotype_Q320.jpg

Genetalic structures

- The genital structure of each specie are different to depict the reproductive isolation.
- Used to delimit species
- Key and Lock model
- Highly effective insects
- Genetalic structures appear to be among the first to change in the course of speciation.

Morphological Characters

Coloration

- Color pattern and other aspects of coloration are convenient characters in certain group of animals.
- Bird and coloration except few genera with sibling species.
- Same is true or certain reef fishes and butterflies
- Many subspecies are indentified on the basis of colour

END

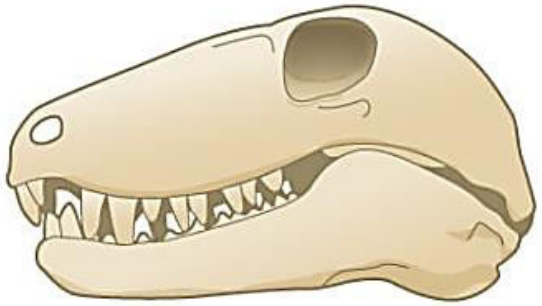
Principles of Systematics

Module# 28

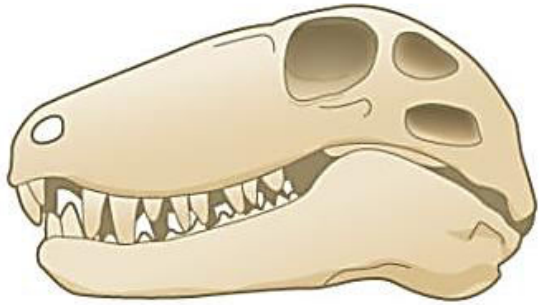
Principles of Systematics

Morphological Characters

Morphological Characters



Anapsid skull (Turtles)



Diapsid skull (Reptiles including birds)

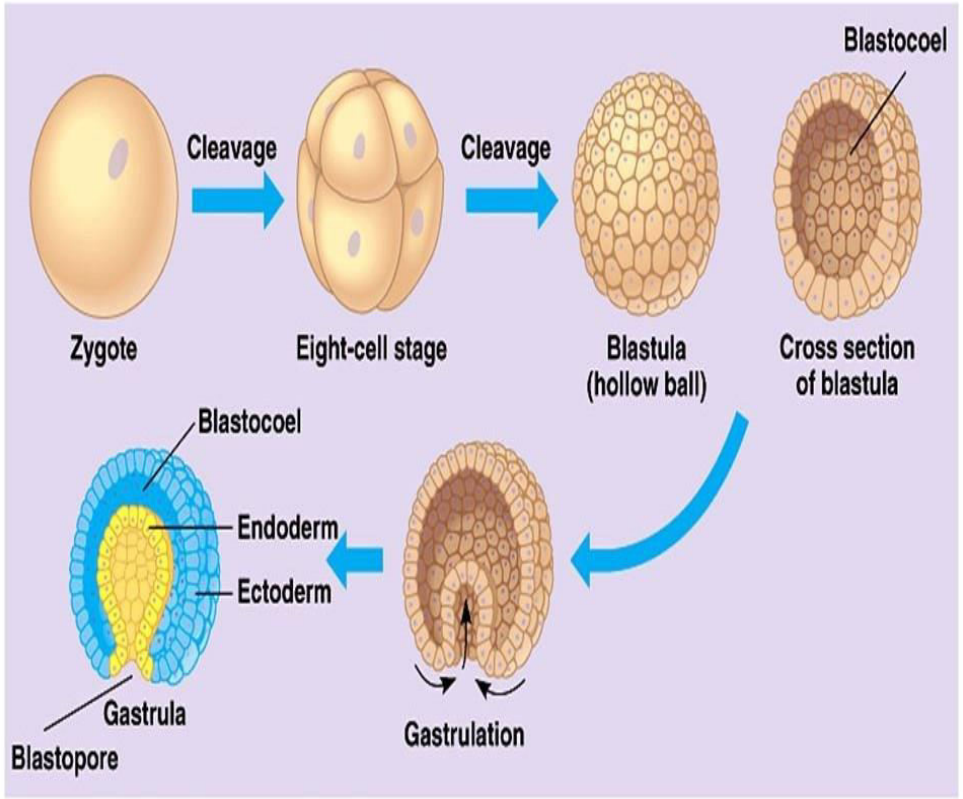
Synapsid skull (Mammals)

https://www.google.com/search?q=evolution+in+skulls+of+vertebrates

Internal morphology

- Internal anatomy provides Taxonomic Characters for all groups of higher animals.
- The extend to use vary from group to group.
- The skull of routinely used in classification of higher mammals.
- Teeth and bones of fossils and other hard parts.

Morphological Characters



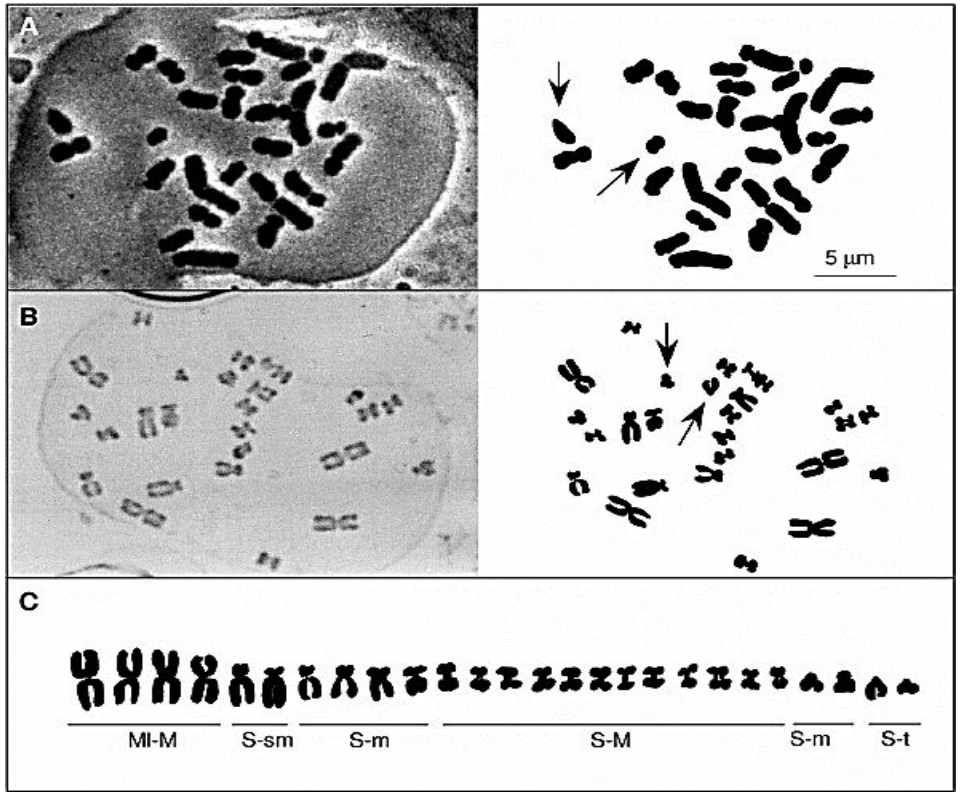
https://i1.wp.com/orbitbiotech.com/wp-content/uploads/2018/04/embryonic-development-Orbit-Biotech-Training.jpg?ssl=1

Embryology

Various larval stages, embryology and even eggs may provide taxonomic information.

- Sibling species of *Anopheles maculipenis------Egg structure*
- Identification of white flies are based on pupae
- Comparative embyological stages like cleavage, blastulation, gastrulation etc are helpful

Morphological Characters



https://www.liberaldictionary.com/wp-content/uploads/2019/02/karyology-4570.jpg

Karyology and other cytological factors

- Karyology is description of size, shape and number of chromosomes
- The simplest cytological character is chromosomal number (species specific)
- Comparison of closely related species (sibling species)
- Establishing phyletic lineage

Principles of Systematics

Module# 29

Principles of Systematics

Physiological Characters

Physiological Characters

Physiological Characters

Physiological characters are low weight characters and are variable:

- All structures are product of physiological process and are thus physiological characters.
- This character can not be studied in preserved animals.

Physiological Characters

Physiological Characters

- Metabolic activities
- Body secretion
- Serological protein and other biochemical differences
- Genic sterility factor

Physiological Characters

Metabolic Factors

- Metabolic factors/enzymes help to find the relations in the closely related species

e.g. Gram +ive and Gram –ive bacteria

Physiological Characters

https://thecotyledon.files.wordpress.com/2010/10/scale-honey.jpg

Body secretion

Body secretions also have the consistent pattern

- Links can be created by finding the number
- Size and
- Structure of the glands

e.g. waxy secretion of scale insects

Principles of Systematics

Module# 30

Principles of Systematics

Physiological Characters

Physiological Characters

Serological protein and other biochemical differences

- Serology is concerned with the nature and interaction of antigens and antibodies.
- Antigens are the substances capable for inducing the formation of antibodies when introduced into the blood stream of animals.

Physiological Characters

- According to antibodies produced in response to the introduction of foreign antigen

Bovden (1943)

- Protein antigens are conservative hereditary trait.
- Proteins of one organism react more strongly with the antibodies of closely related organism than more distantly related.

Physiological Characters

- Good precipitin technique are reveal the relative degree of biochemical similarity of protein antigens among organisms.

Physiological Characters

Genic sterility factors

- Reproductively incompatible species cannot reproduce new young one
- Only closely related species can hybridize successfully.
- As a result give genetically sterile species.
- Cross fertility in Dobkey and Hourse- Mule
- Cross fertility in ducts (species and genera)

END

Principles of Systematics

Module# 31

Principles of Systematics

Behavioral Characters

Behavioral Characters

Behavioral characters

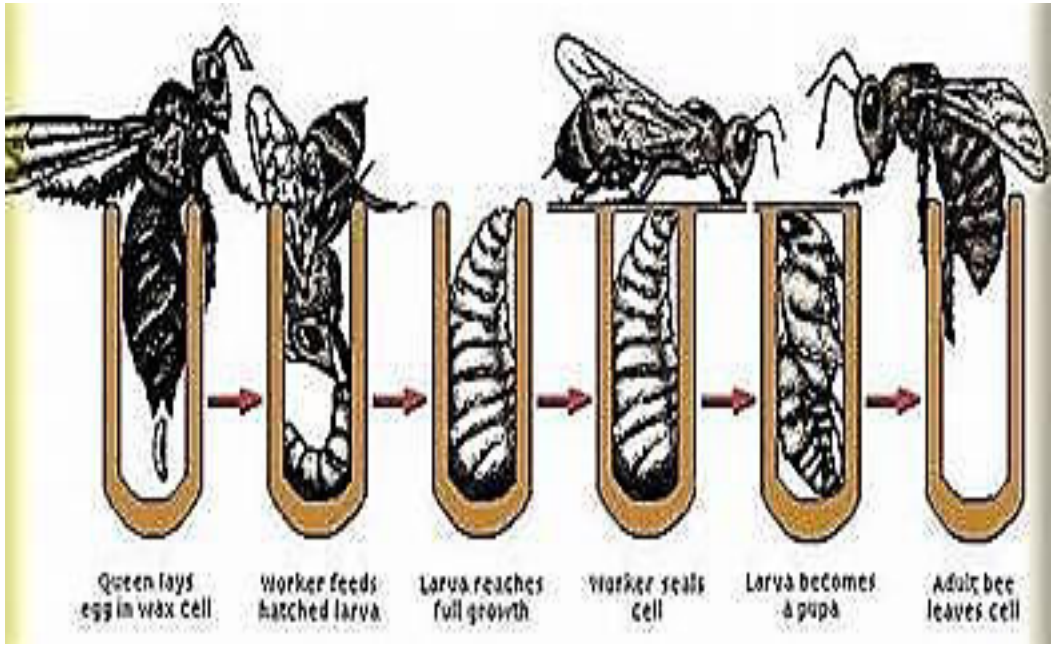
- Behaviour is one of the most important source of Taxonomic Characters.
- They are more superior than morphological characters in the study of closely related species.
- Especially in sibling species
- Mating behaviour, egg laying, nesting behaviour, drinking habit of birds etc.

Behavioral Characters

Behavioral characters

- Mating calls of frogs, toads and crickets are used in species delimitation.
- Webs pattern in spiders can be at various level in classification.

Behavioral Characters



https://tse3.mm.bing.net/th?id=OIP.eubyf9OPir29CdF7wsFukQHaDO&pid=Api&P=0&w=392&h=171

Reproduction

Mating and reproduction behavior of species help in taxonomic classification and also help in determining reproductive isolation and speciation.

E.g. slugs

Behavioral Characters

https://media.buzzle.com/media/images-en/photos/birds/peacocks/1200-77866817-peacock-drinking-water.jpg

Other behavior pattern

The living, eating and nesting behavior of the animals also depicts the taxonomic character

e.g. drinking habit of pigeon and sand grouse, use of mud in nest building

Behavioral Characters

Drawbacks

- Can not be studied in preserved material
- It is intermittent even in living animal.
- Some behaviours are seasonal
- Change within 24 hours

Principles of Systematics

Module# 32

Principles of Systematics

Ecological Character

Ecological Character

Ecological characters

- Each species has specific niche
- Types of niches
- Habitats and hosts environment
- Food sources
- Seasonal variation
- Parasite and host interaction reaction

Ecological Character

Habitats and host environment

- Habitat of the organisms is not same
- Species cannot coexist at same place or in same environment or partitioned their resource to avoid competition.

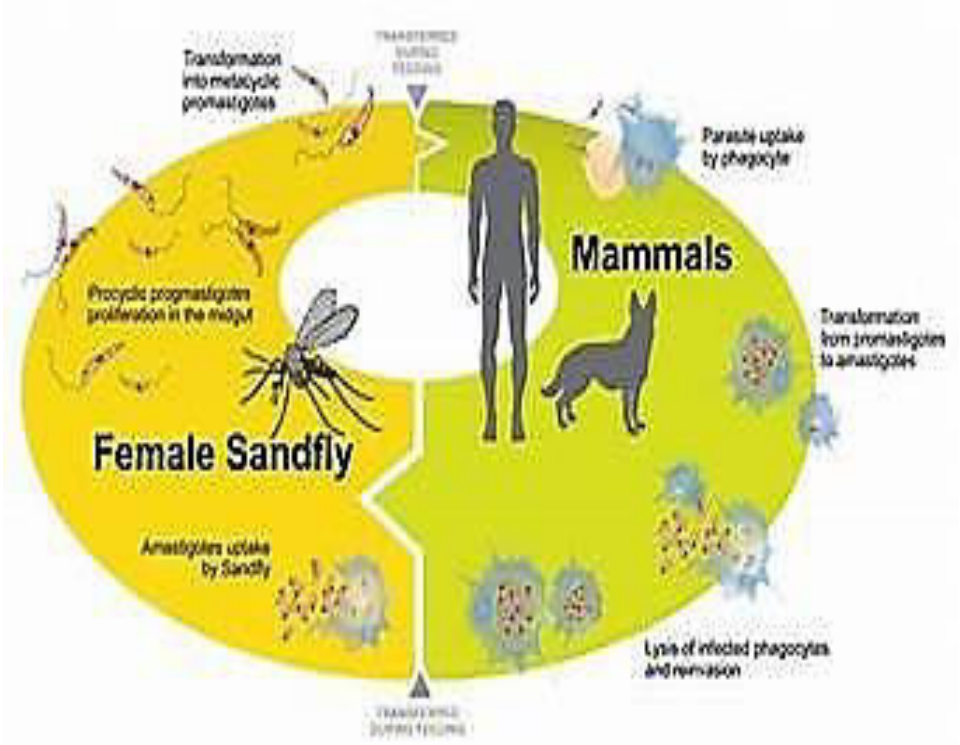
Ecological Character

Food and seasonal variation

- Food sources also varies like in parasites and many others.
- Characteristics of organism also vary depending on season

e.g. plumage of birds and breeding coloration

Ecological Character



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Parasite and host interaction reaction

- Structural changes can occur in host after parasitic attack
- Various responses of body like distortion, swelling etc.

Principles of Systematics

Module# 33

Principles of Systematics

Ecological Character

Ecological Character

Ecological Character

- Every species has its own ecological niche in nature differing from its close relatives in (spider example)
- Food preference
- Breeding season
- Tolerance to physical factors
- Resistance to predators
- Compotators and pathogens

Ecological Character

Ecological Character

- Larvae of both *Drosophila mulleri* and *Drosophila aldriachi* live simultaneously in decaying pulp of the fruits of cactus *Opuntia lindheimeri.*
- Both species are specialized in using certain types of yeast and bacteria.

Ecological Character

- Similarly tapeworms of man and rodents and Ascaris of man and pigs are host specific.
- Each species of Galapagos finch is also specialized according its habitat/ environments

END

Principles of Systematics

Module# 34

Principles of Systematics

Geographical Characters

Geographical Character

Geographical Character

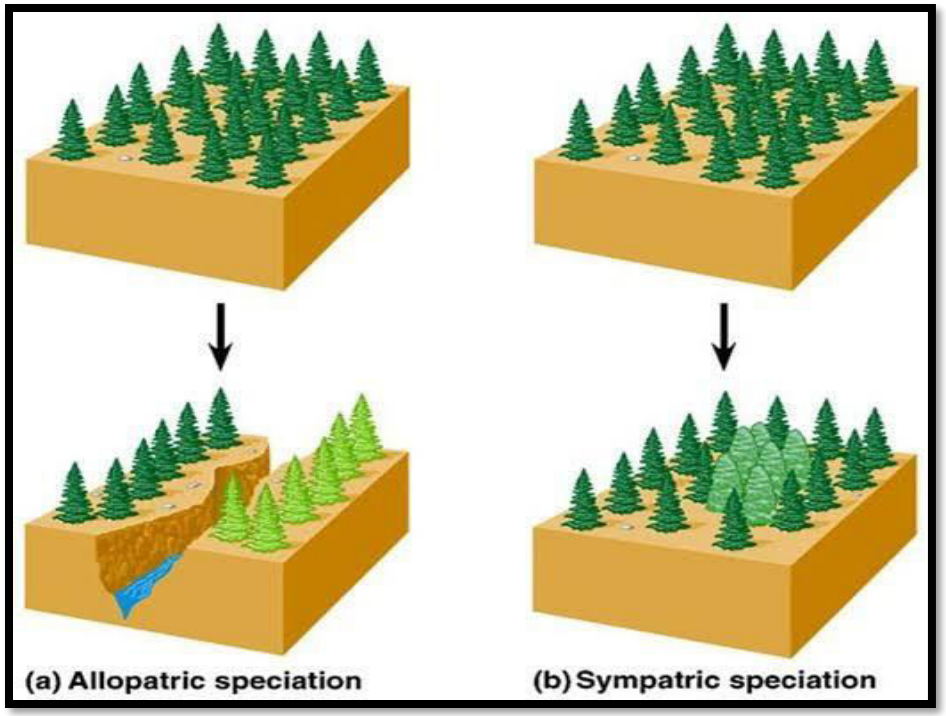
- Each taxon has a geographical range
- Range changes as the taxon evolve
- Distribution of taxa provide past history
- Geographic characters are important at two levels
- Microtaxonomy
- Macrotaxonomy

Geographical Character

Geographical Character

- Taxonomists are important in two types of geographical characters:
- General biogeogrphical patterns, which are especially useful in arrangement of higher Texa
- Allopatric-sympatric relationship, which is helpful in determining wether or not two species are conspecific.

Geographical Character



https://s-media-cache-ak0.pinimg.com/736x/18/49/1c/18491ced4bf91ceac2f85a3fdee59a5f.jpg

Sympatric-allopatric relationship of population

These isolation pattern that occur in the population also determine the links in population

Principles of Systematics

Module# 35

Molecular and biochemistry character

Molecular and Biochemistry Character

Molecular and biochemistry character

This character as taxonomic character is also an approach for the evolutionary links and relations determines if classical taxonomy is defined.

Molecular and Biochemistry Character

Molecular and biochemistry character

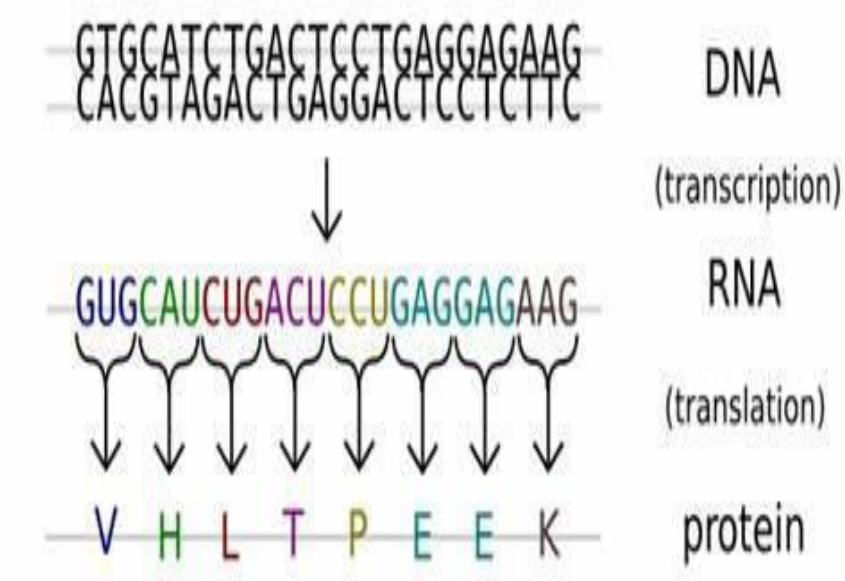
- Immunological distances
- Electrophoretic differences
- Amino acid sequence of protein
- DNA hybridization
- DNA and RNA sequence

Molecular and Biochemistry Character

Amino acid sequences

- Amino acid sequences help to determine the links.
- This approach as a taxononmic character is more absolute.

Molecular and Biochemistry Character



https://tse2.mm.bing.net/th?id=OIP.Uo3B4dsPEz-PEzHf3uJ4OwHaCz&pid=Api&P=0&w=517&h=197

DNA and RNA sequences

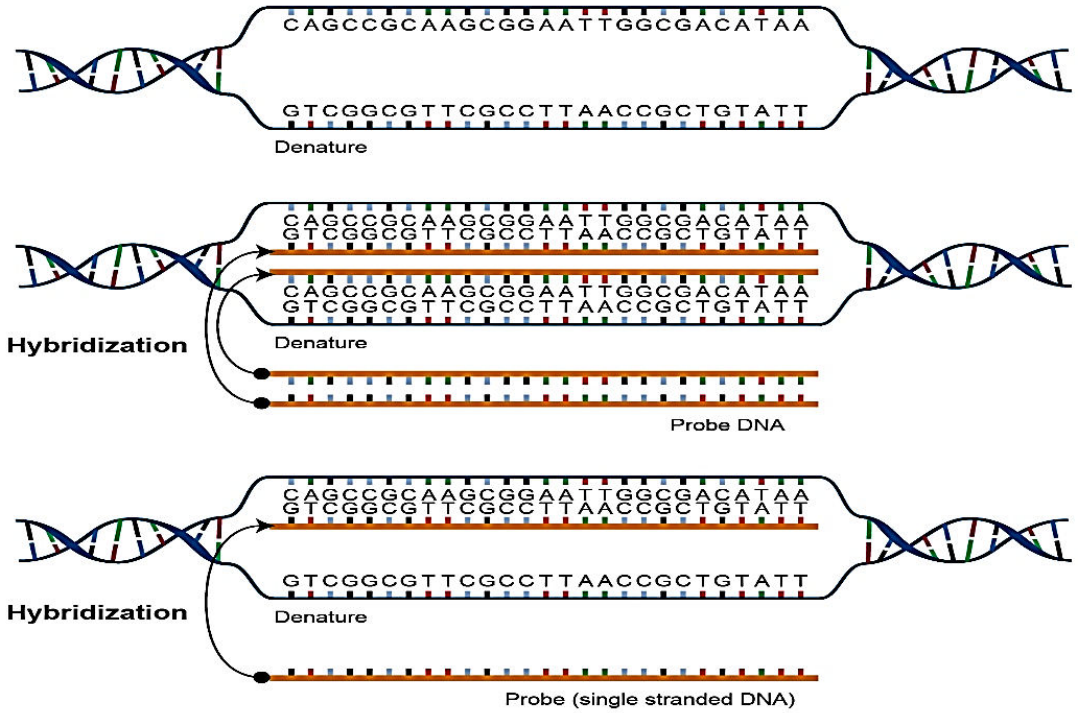
- These sequences help to determine the direct links on the basis of sequences
- But these sequences can also undergo some changes under the evolutionary pressure

Molecular and Biochemistry Character

Electrophoretic differences

- This method determine the difference in the isozyme and allozyme.
- Can help to determine share character among taxa.

Molecular and Biochemistry Character



https://www.genome.gov/sites/default/files/tg/en/illustration/hybridization.jpg

DNA hybridization

- This method is used to determine the genetic distance between two organisms
- Mainly measure the genetic similarity

Principles of Systematics

Module# 36

Principles of Systematics

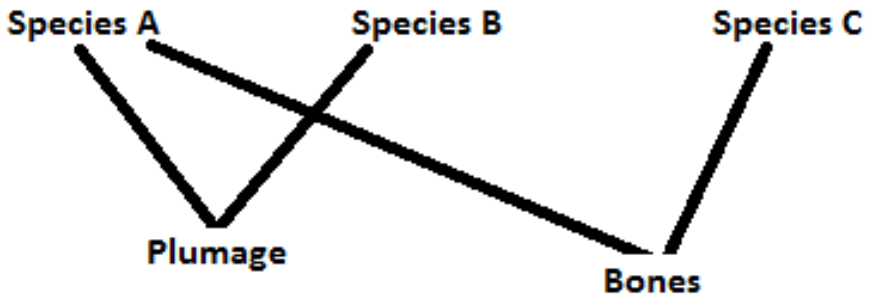
Weighing of Characters

Weighing of Characters

- Any attribute by which member of a taxon differ
- Raw material of classification
- Amount of phyletic information contents in a given character
- More phyletic information---- high weigh

- Low phyletic information---low weight

Weighing of Characters



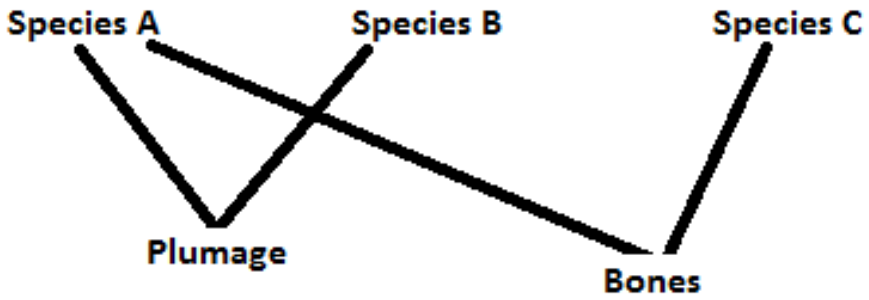
Discordent characters: difficult to verify

- Animal character may of high weight ----mostly used by taxonomist
- Low weight---rarely used

- Also characters of intermediate weight

- Challenge for taxonomists which discordant character should be truested?

Weighing of Characters



Discordent characters: difficult to verify

- Process to determine that which character is more reliable is called *weighting.*

Weighing of Characters

1. **Priori (before/without phyletic information)**
   - Proposed by Artistotle

2. **Posteriori**
   - Proposed by Adenson (after phyletic information collection)
   - All classification is based on this model.

END

Principles of Systematics

Module# 37

Principles of Systematics

Correlated Characters

Correlated Characters

- Characters which are functionally correlated have very low weight
- They have low phylogenetic content
- Do not deserve to be treated as separate character
- Analogues organs birds wings, bats and insects (analogus)/convergent evolution
- Redundant

Correlated Characters

- Phyletic correlation
- High weight
- Higher genetic contents
- Diagnostic characters of Deuterostomes and Chordates (concordance)----phyletic correlation—evolutionary related

END

Principles of Systematics

Module# 38

Principles of Systematics

High Weight Characters

High Weight Characters

High weight characters

- Complexity
- Constancy
- Consistency
- Not affected by ecological shift
- Apomorphy
- Not a specific adaptation
- e.g. Morphological and behavioral character

High Weight Characters

Complexity

- Complexity-Genetalia
- Deep genetic character
- Higher complexity-higher weight age

Constancy

- Character which is constant in a large group of species than variable character (Darwin 1859)
- According to Farris (1969) Low phenotypic variability be given high weight

High Weight Characters

Consistency

- A character which is consistently present in one group and consistently absent in related groups should have high weight

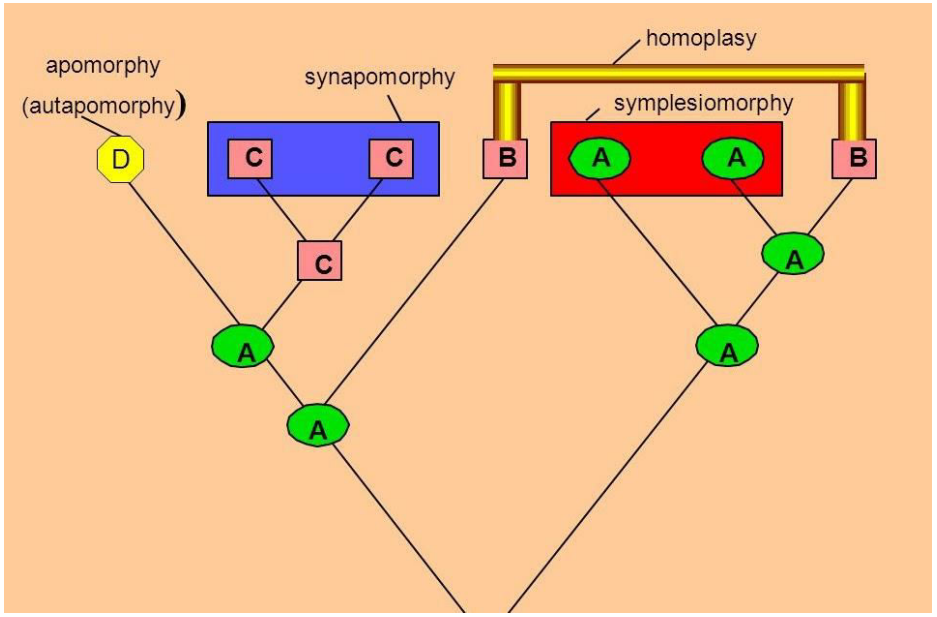
High Weight Characters

https://www.google.com/search?q=mergansers&sxsrf=ALeKk03jIyrLE0TeIgIBxO6mWo9hBiogvg:1603685582863&source=lnms&tbm=isch&sa=X&ved=2ahUKEwjdtv2rstHsAhXvDmMBHZtJBOcQ_AUoAXoECCYQAw&biw=1360&bih=657

Not affected by ecological shift

- A character that no change with ecology
- A bird feeding on fish in one habitat continue to feed on fish in another habitat
- An example is adaptation of mergansers (Mergus) for fish eating

High Weight Characters



https://www.google.com/search?q=apomorphy&sxsrf=ALeKk03Y_tzp73qJK0Em_Bfingu_YHOxiw:1603614144234&source=lnms&tbm=isch&sa=X&ved=2ahUKEwjbjLGbqM_sAhWzShUIHdkiB1oQ_AUoAXoECCAQAw&biw=1360&bih=600#imgrc=GCYTyU7VcBY0LM

Apomorphy

- A novel evolutionary trait that is unique to a particular species and all its descendants and which can be used as a defining character for a species or group in phylogenetic terms.
- Hence, the possession of feathers is unique to birds and defines all members of the class Aves.

Principles of Systematics

Module# 39

Principles of Systematics

Low Weight Characters

Low Weight Characters

- Characters having less taxonomic importance
- Less phylogenetic content
- Not reliable
- Poor indicators of relationships

Low Weight Characters

- Highly variable Characters (hairs, branching pattern of arteries)
- Monogenic and oligogenic characters (albinism) varies independently of other characters

Low Weight Characters

- Regressive characters (loss of eyes, loss of wings, wing veins in insects, loss of teeth in mammals, loss of segments in segmented animals are unnatural.
- Loss may be due to environmental conditions (eye loss in cave, loss of digestive systems in parasitic tapeworms)
- Narrow specialization

  e.g. Physiological character

Low Weight Characters

- Not constant
- Not consistent
- Change with ecology
- Less complex

  e.g. Physiological character

END

Principles of Systematics

Module# 40

Principles of Systematics

Non-Genetic Variations

Non-Genetic Variations

Types of Non-Genetic variations

- Individual variations in Time
- Social Variations
- Ecological Variations
- Traumatic variation
- Post-morten variations

Non-Genetic Variations

Individual variations in Time

- Age variations
- Seasonal variations of individuals
- Seasonal variations of generations

Non-Genetic Variations

https://www.google.com/search?q=eel&sxsrf=ALeKk02QiXShlrI7phddXt7HPFtE0QOvmA:1603309550565&source=lnms&tbm=isch&sa=X&ved=2ahUKEwihuefBucbsAhWPTN8KHUuJCQsQ_AUoAXoECDUQAw&biw=1360&bih=600#imgrc=eWSxWZoTy-WYyM

Age Variations

- Animals pass through large numbers of juvenile or larvae which are quite different from adult.
- Immature stage of eel (*Anguilla anguilla*) were originally described as *Leptocephalus brevirosteris in 1856*
- Larval stages different from adult (Caterpillar and butterfly)

Non-Genetic Variations

Seasonal Variations of individuals

- Same animal may have different appearance in different parts of the year
- In arctic and sub-arctic birds there may be change from white winter dress to normal colour summer dress.

Non-Genetic Variations

https://www.google.com/search?q=ptarmigan&sxsrf=ALeKk015nxwTbGZOdxCyoPYsK-TETw_DOA:1603517715139&source=lnms&tbm=isch&sa=X&ved=2ahUKEwjH3bT-wMzsAhVoQEEAHabtBjEQ_AUoAXoECCcQAw&biw=1360&bih=600#imgrc=sK318okbBMjJj

Rock ptarmigians (*Lagopus muta*)

- Gray and brown upper parts in summer
- White plumage in winter

Principles of Systematics

Module# 41

Principles of Systematics

Individual Variations

Individual Variations

https://www.google.com/search?q=european+starling+in+october&hl=en&sxsrf=ALeKk00lEQwoUE0nvDEAwcEeS_aFlcXkJg:1603518950740&source=lnms&tbm=isch&sa=X&ved=2ahUKEwil7MvLxczsAhV4UhUIHTdZA0QQ_AUoAXoECBYQAw&biw=1360&bih=600#imgrc=nUnihAZz9S-NkM

European Starling (seasonal variation)

- Freshly molted bird of October is covered with white spots and all feathers show whitish margins
- In winter edges of feathers wear off

Individual Variations

https://www.google.com/search?q=european+starling+in+spring%2C+glossy+black&tbm=isch&ved=2ahUKEwi6xauzxszsAhVC-4UKHZh3BakQ2-cCegQIABAA&oq=european+starling+in

European Starling

- In spring (beginning of breeding season) whole bird is glossy black without the mold of single feather

Individual Variations

https://www.google.com/search?q=bicycles+anynana+in+wet+season&tbm=isch&ved=2ahUKEwjQsK3DxszsAhULcxoKHfjYAKMQ2-cCegQIABAA&oq=bicycles+anynana+in+wet+season&gs_lcp=CgNpbWcQAzoECCMQJzoHCAAQsQMQQzoECAAQQzoFCAAQsQM6AggAOgYIABAIEB5QyY0uWN6XL2CLmi9oAHAAeACAAegDiAGKRJIBCDItMjEuOC4xmAEAoAEBqgELZ3dzLXdpei1pbWfAAQE&sclient=img&ei=4cKTX5C8Novmafixg5gK&bih=600&biw=1360&hl=en#imgrc=zxMXuCnjPwLPSM

Seasonal variations of consecutive generations

- Seasonal plasticity
- These variations are called polyhenism.
- African butterfly *Bicyclus anynana*
- Eyespots on margins (ventral side)---- in wet season
- Very small or No eyespot ---Dry season

Principles of Systematics

Module# 42

Principles of Systematics

Social Variations

Social Variations

- In social insects there are definite groups of individuals in a colony.
- Each type of individual in a colony perform specific function.
- For examples in a colony of Honey bee:

- Queen, males (drones) and workers, soldiers

Social Variations

Social variations

- **workers:** All workers are females, product of fertilized eggs.
- They are in majority in the colony.
- May be from 10000 to 50,000 per colony.
- development.
- Projection of colony, food collection, feeding of queen, take care of young ones, cleaning of hive etc

END

Principles of Systematics

Module# 43

Principles of Systematics

Social Variations

Social Variations

Social variation in ants

- Ants live in complex social colonies.
- **Queen:** founder and leader of the colony.
- Primary goal is to populate colony .
- Lay thousands of eggs
- Queen chambers are deep within the colony for protection.
- Long life in some species upto 30 years.

Social Variations

Social variation in ants

- **Drones:** Only function is to fertilize female
- Die after fertilization
- Rarely encountered outside the colony
- **Workers:** Foraging and protection
- Sometime: queen lay eggs, winged male and females-**Alates or reproductive.**

Social Variations

Social variation in ants

- After maturation leave the colony in swarms.
- Establish new colony

END

Principles of Systematics

Module# 44

Principles of Systematics

Ecological Variations

Ecological Variations

A. Habitat Variation
B. Variations induced by climatic conditions
C. Host determined variations
D. Density dependent variations
E. Allometric variations
F. Neurogenic color variations

Ecological Variations

A. Habitat variation

- Populations of single species in different habitat are visibly different.

Oyster in still water

- Round or broader

Oyster in strong currents

- Narrow, elongated and usually straight

Ecological Variations

A. Habitat variation

Oyster in clean water on pebble

- Lower valve is deep and more or less sharply ribbed to acquire strength
- Not needed attachment is with perfectly flat surface

Oyster in muddy area

- Valves are less commonly ribbed

END

Principles of Systematics

Module# 45

Principles of Systematics

Ecological Variations

Ecological Variations

B. Variations induced by climatic conditions

- Animals with phenotypic plasticity
- Variations with temperature, draught, cold etc.

Ecological Variations

https://www.google.com/search?q=apenteles+flaviconchae&tbm=isch&ved=2ahUKEwjkg7Ob_czsAhUD8IUKHQs4BLsQ2-

C. Host determined variations

- Braconid wasp, *Apanteles flaviconchae* spin white cocoons if reared from blue green caterpillars of *Choliase philodice*

- Golden cocoons if reared from yellow green caterpillars of same species.

Principles of Systematics

Module# 46

Principles of Systematics

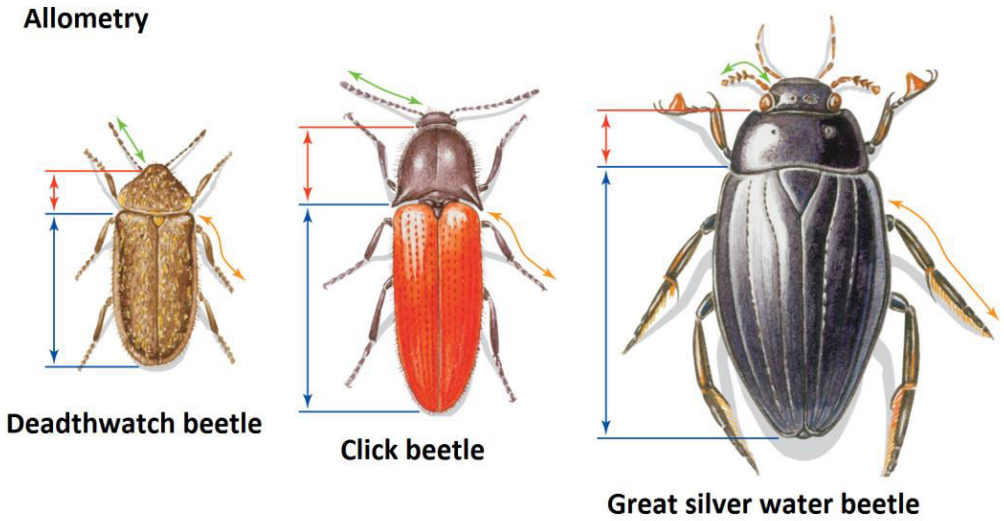
Ecological Variations

Ecological Variations

D. Density-Dependent variations

- When newly hatched nymphs of locust are reared under crowded conditions----Gregarious phase
- Less crowded condition---Transitional phase
- Separate rearing---Solitary phase
- These phase differ in anatomy, colour and behavioral characteristics----distinct species

Ecological Variations



https://www.google.com/search?q=allometric+variations&tbm=isch&ved=2ahUKEwjjrbyKg83sAhXC4YUKHRKVBOwQ2-cCegQIABAA&oq=allometric+variations&gs

E. Allometric variation

- Allometry results in misappropriate size of some structure in relation to rest of the body.
- This results in animals of different sizes----Allometric variability

Ecological Variations

https://www.google.com/search?q=chameleon&sxsrf=ALeKk03w9hngHVfxaApti0b4bszoZ9g-

F. Neurogenic variation

- It is colour change in response to environment.
- Chameleons have Chromatophores in three layers below their transparent skin.
- Xenthophors (Yellow) and erythrophores (red) in upper layer
- Iridophores or guanophores (blue or white) 2nd layer
- Melanophores (dark colour) in deep layer

Ecological Variations

https://www.google.com/search?q=chameleon&sxsrf=ALeKk03w9hngHVfxaApti0b4bszoZ9g-

F. Neurogenic variation

- Chameleons change their colours in response to temperature and to chamouflage.

Principles of Systematics

Module# 47

Principles of Systematics

Traumatic Variations

Traumatic Variations

- **Teratology** is the study of abnormalities of physical development.
- **Taratogen** are the toxic substances which cause abnormal development.
- Mercury, alcohol, lead, thalidomide
- Teratological variations are common in animals.

Traumatic Variations

- In those forms which undergo metamorphosis injury to an earlier stage may produce abnormalities which are not easily recognized.
- Important if abnormality is in characters of taxonomic importance.
- Modification in wing pattern

END

Principles of Systematics

Module# 48

Principles of Systematics

Post-Mortem Variations

Post-Mortem Variations

https://www.google.com/search?q=twelve+wire+bird+of+paradise&sxsrf=ALeKk00Z6PMYPqESiQ3ZTFbrIC6T6hsR6Q:1603611950298&tbm=isch&source

- It is impossible to prevent post-mortem changes in some animals.
- Deep yellow plumage of twelve wire bird of paradise----white

Post-Mortem Variations

https://www.google.com/search?q=Chinese+jay+bird%2C+green&tbm=isch&ved=2ahUKEwiMgpaQocsAhUCYBoKHRVPDI0Q2-

- Chinese jay plumage changes from green to blue in collection
- Wasps over-expressed to cyanide turn bright red
- Insects
- Spiders

Principles of Systematics

Module# 49

Principles of Systematics

Genetic Variation

Genetic Variations

- Difference in the individual members of a population
- Genetic variations are important for survival and adaptation of species
- Evolution
- Natural selection

Causes of evolution

- Mutations
- Genetic recombination
- Genetic drift
- Migration

Genetic Variations

- Population s with small variation
- Population with large variability among individual
- Variability in characters
- Variability biological fitness

Major causes variability

- Mutations
- Genetic recombination
- Mutations are not always bad (Sickle cell disease—present Malaria) END

END

Principles of Systematics

Module# 50

Types of Genetic Variation

Type of Genetic Variation

Sexual Dimorphism

Many of the variants in a population are sex-associated.

- They may be sex limited (express themselves in one sex only) or otherwise associated with one sex or the others
- **Primary Sex differences (PSD)** Difference in primary sex organs such as gonads genetalia etc.)
- When two sexes are similar morphologically—PSD helpful

Type of Genetic Variation

Sexual Dimorphism

- **Secondly Sex differences (SSD)**
- Striking differences in male in females e.g bird of paradise, humming bird and ducks
- Due to differences different sexes were considered different species until painstaking works by naturalist established their relationship.
- Celebrated case of King parrot, ***Eclectus roratus*** (Muller)

Type of Genetic Variation

https://www.google.com/search?q=King+parrot+Eclectus+roratus+,+male+female&sxsrf=ALeKk02h0vVHJDTAOjHTEAE5PQBiERJ9_Q:1604040706059&source=lnms&tbm=isch&sa=X&ved=2ahUKEwirmfWj3dvsAhXBDGMBHZgZCBEQ_AUoAXoECBkQAw&biw=1360&bih=657#imgrc=kLDkWlLeXUZ6cM

King parrot *Eclectus roratus*
(Muller) of Papuan region

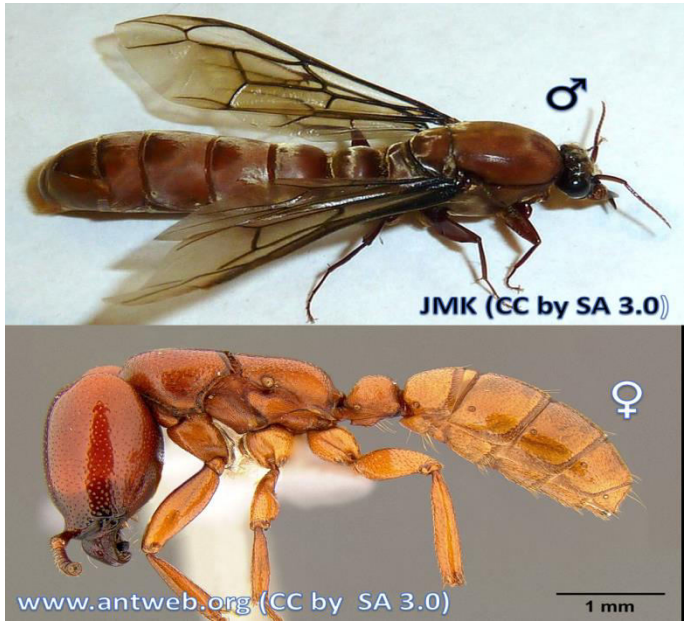
- Male is green with orange bill
- Female is red and blue with black bill.
- Described in 1776, two considered different for nearly 100 years
- Until naturalist proved conclusively in 1873 that they belonged togather.

Principles of Systematics

Module# 51

Types of Genetic Variation

Type of Genetic Variation



https://www.google.com/search?q=African+Ant+Dorylus+male+females&tbm=isch&ved=2ahUKEwiajc-y39vsAhUU_RoKHbZ2CW0Q2-cCegQIABAA&oq=African+Ant+Dorylus+male+females&gs_lcp=CgNpbWcQA1DywQFY1egBYJ_qAWgAcAB4AIAB_QOIAfQakgELMC4xLjkuMi4wLjGYAQCgAQGqAQtnd3Mtd2l6LWltZ8ABAQ&sclient=img&ei=ObqbX9qMLpT6a7btpegG&bih=657&biw=1360#imgrc=-C_HA25vlwfHRM&imgdii=kDy2tGctcC0wiM

Sexual Dimorphism

Striking sexual dimorphism in Hymenoptera.

- Males of African Ant *Dorylus* are so unlike others ants that they were not recognized as ants and considered belong to different family.
- In certain families of wasps small wingless females and large winged males are so different that some taxonomists use different nomenclature for both.

Type of Genetic Variation

Sexual Dimorphism

Some time males and females are so different that it is very difficult to decide which male belong to which female:

- Field study by ecologist
- Which male is attracted to which female

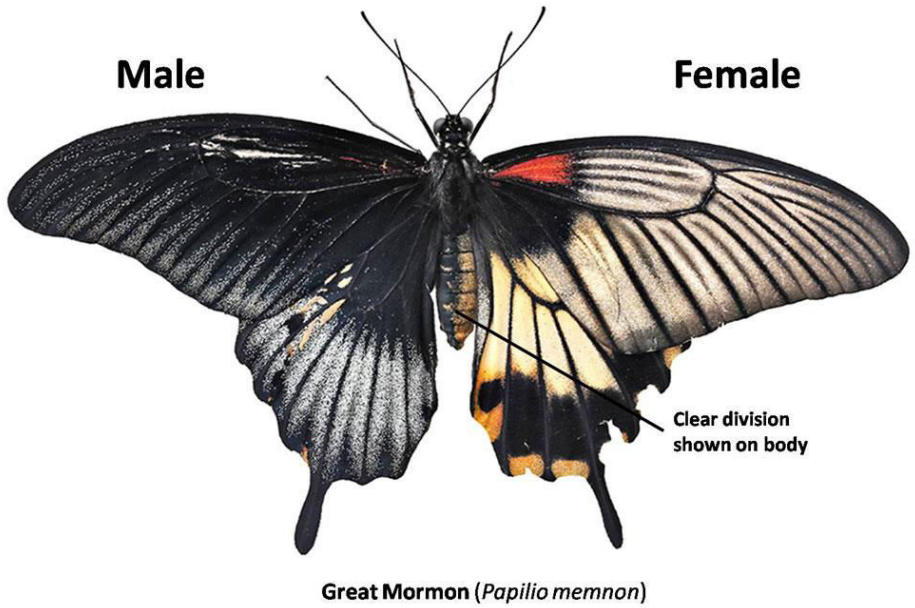
END

Principles of Systematics

Module# 52

Types of Genetic Variation

Type of Genetic Variation



Great Mormon (*Papilio memnon*)

https://www.google.com/search?q=gynandromorph+of+papilo+adardanus&tbm=isch&ved=2ahUKEwiGuK-l5tvsAhWO0YUKHdbFCn4Q2-cCegQIABAA&oq=gynandromorph+of+papilo+adardanus&gs_lcp=CgNpbWcQAzoECAAQQzoCCABQoa0BWImWAmC8mQJoAHAAeAGAAYYJiAGNPpIBEDAuMi4xMS4yLjAuMS4yLjKYAQCgAQGqAQtnd3Mtd2l6LWltZ8ABAQ&sclient=img&ei=dcGbX8apAY6jlwTWi6vwBw&bih=657&biw=1360#imgrc=sCA1PSyDaBTAiM

Gynandromorphs and intersexes

Individuals that shoe male characters in one part and female characters in other parts of body.

- Two halves of the body may be of opposite sexes.
- The division male be transverse or the sex characters may be scattered in mosaic.
- In later case symmetrical variants may be produced.
- Produced due to unequal distribution of chromosomes/SC

Type of Genetic Variation

https://www.google.com/search?q=intersexes,+Example:+Lymantria&hl=en&sxsrf=ALeKk01diF3YZHp0xEbXUCJVibS92jJ0-w:1604043782080&source=lnms&tbm=isch&sa=X&ved=2ahUKEwi_udbe6NvsAhXiA2MBHQRuCRYQ_AUoAXoECAsQAw&biw=1360&bih=657#imgrc=_xPgprPy1k57CM&imgdii=Uia9hvBTpb27TM

Intersexes

Blending of male female characters. body. Intersexes are produced due to:

- Upset in balance between male-tendency and female-tendency genes
- May result from irregularities in fertilization or mitosis
- Physiological disturbances associated with parasitism.
- Example: Gipsy moth, Lymantaria

Principles of Systematics

Module# 53

Types of Genetic Variation

Type of Genetic Variation

https://www.google.com/search?q=agemic+generation+gall+wasp&hl=en&sxsrf=ALeKk02oVJis8aalDyE2Y0wVwXJoaodWcQ:1604054434479&source=lnms&tbm=isch&sa=X&ved=2ahUKEwiTu5G2kNzsAhXVh1wKHZ_ZBjoQ_AUoAXoECBkQAw&biw=1360&bih=657#imgrc=SO7OXhMqVG7A9M

Reproductively different generations

- In many insects thesre is an alternation of generations that is very confusing to the taxonomist.
- In cynips (gall wasp) agemic generation is different from bisexual one.
- In aphids parthenogenetic wingless females are different from winged females of sexual generation.

Type of Genetic Variation

https://www.google.com/search?q=Slugs,+Arion&hl=en&sxsrf=ALeKk02zw1UrLr9HBnEK7giReGZMJt64pA:1604055019292&source=lnms&tbm=isch&sa=X&ved=2ahUKEwiOuv_MktzsAhWjnVwKHTUaB1MQ_AUoAXoECCUQAw&biw=1360&bih=657#imgrc=AKvOHe8xqdKJVM

Sexual and uniparental strains

- In some species of slugs (Arion) several strains are found which differ in their sexuality.
- Sexually reproducing strains are highly polymorphic in their enzymes
- While self fertilizing hermaphroditic strains are highly homozygous.

Type of Genetic Variation

Discontinuous variation (Polymorphism)

- Certain members of population have certain conspicuous characters.
- Such discontinuous individual variations is called polymorphism.
- Frequently controlled by single gene.
- Follow simple Mendelian inheritance.

Type of Genetic Variation

https://www.google.com/search?q=laddy+beeles%2C+spotted&tbm=isch&ved=2ahUKEwi0gs_PktzsAhVRlxoKHdwxAYwQ2-

- Spotting in lady beetles
- Industrial melanism in moths
- Understanding of polymorphism is helpful to taxonomists.
- More than 100 birds species names were given to morphs.

Principles of Systematics

Module# 54

Types of Genetic Variation

Type of Genetic Variation

Continuous variations

- Most common type of individual variations is due to slight genetic differences among individuals.
- No two individuals (except monozygotic twins) in a population of sexually reproducing animals are exactly alike
- Morphologically or genetically
- Small difference/special techniques.

Type of Genetic Variation

Continuous variations

- Each characters may show a different degree of variability within a single population.
- Different degree of variability among related species.
- Why one species is highly variable while another is NOT?
- Answer this question required lot of background knowledge and research.

Type of Genetic Variation

Continuous variations

- Some early taxonomists vastly underrated individual variations in many genera of animals.
- Species of the snail genus *Melania* (fresh water and brackish water) have been described largely on the basis of shell characters as presence or absence of spines and diagonal versus spiral ribs.
- Revision showed 114 species were just because of individual variation.

Principles of Systematics

Module# 55

Principles of Systematics

Systematic Collection

Systematic Collection

Systematic collection

Systematic collection is a method of collecting valuable biological specimens and preserving them along with specimen documentations such as labels and notations. It possess various attributes:

- Raw material of classification
- Document earth's biodiversity
- Initially additional specimens of species—duplicates-exchange purpose

Systematic Collection

Systematic collection

- As there is variability in members of species
- Adequate sample of every population should be collected and preserved.

What is adequate?

- More material is needed in s species with strong individual and geographical variation than in uniform species.

Systematic Collection

Objective of study?

Simple study or genetic variability

- When population analysis is attempted dealing with size, proportions, coloration, or polymorphism large samples from numerous locations are needed.
- For example in fishes where each population of a species may vary in some metric characters (fin ray, vertebrae), a single specimen per locality would tell us little.

Systematic Collection

- Facilitate the process of researching relationships among and within different groups of organisms biodiversity
- We can use systematics to find out how insects are related to the rest of the invertebrates.

Systematic Collection

Preservation of collection samples

The collected samples are preserved by;

- Wet method and
- Dry method

Systematic Collection

Wet method

- Wet method of preservation involves the use of liquid preservatives to prevent deterioration.
- Chemicals like alcohols, formalin etc. are commonly used for specimen preservation. Animals belonging to phylum porifera, annelida, nematodaare mostly preserved by wet method.

Systematic Collection

Dry method

- In this method, specimens are preserved in dry state is a traditional method to place several kinds of insects in a box for an indefinite period allowing them to become dry.
- Not advisable for soft bodied insects.
- The process of pinning insect specimens is a dry method to preserve and display collections.

END

Principles of Systematics

Module# 56

Principles of Systematics

Systematic Collection

Systematic Collection

Importance of systematic collection

- Safe storage of historical material such as fossils
- Rarely found specimens
- Specimens of geographically inaccessible areas
- Extinct species specimens to place in museums for documentation
- Provide permanent record of faunas and floras from natural catastrophes

Systematic Collection

Importance of systematic collection

- Help in the verification of original data or for renewed study from new knowledge and technique
- The systematic method assumes the size of the population is available
- Aid in identification of organisms through comparison
- Teaching/Public Outreach: stimulate interest in earth's

Systematic Collection

- Facilitate the process of researching relationships among and within different groups of organisms biodiversity
- We can use systematics to find out how insects are related to the rest of the invertebrates.

END

Principles of Systematics

Module# 57

Principles of Systematics

Collection Methods

Collection Methods

Collection methods

There are various collection methods to collect specimens

- Black light trap
- Sweep net
- Suction device
- De-Vac
- Pit fall method
- Aspirator
- Beating sheet
- Trawling
- Dipper
- Hand picking

Collection Methods

http://mississippientomologicalmuseum.org.msstate.edu/images/traps/blacklight.sheet.

Black light trap

- Entomologists often collect night flying insects using a black light and sheet.
- The black light is suspended in front of a white sheet.
- Insects attracted to ultraviolet light fly toward the light, and alight on the sheet.

Collection Methods

Black light trap

- **A:** Tying the rope between two trees
- **B:** Sheet and light are hung
- **C:** A quite night of collecting

https://mississippientomologicalmuseum.org.msstate.edu/collecting.preparation.methods/Blacklight.traps.htm

Collection Methods

https://ffs.ipm-info.org/wp-content/uploads/2015/05/Collecting-insects-with-a-sweep-net.

Sweep Net

- Sweep nets are used to collect arthropods from long grass and other types of vegetation.
- This net is with round mouth and canvas bag fixed with a long rod.
- Its advantages lie in its simplicity and speed, and its ability to collect relatively dispersed insect species on top of the vegetation.
- It cannot be used on very short vegetation.

Collection Methods

https://www.google.com/search?q=suction+device+for+insect+collection%2F+devac&tbm=isch&ved=2ahUKEwikjtP_hN7sAhUKahoKHTraDt0Q2

Suction device/De-Vac

- Used for collection of arthropods from vegetation, long grasses, forest, orchards etc.
- Engine operated.
- Large scale insect collection.

Principles of Systematics

Module# 58

Principles of Systematics

Collection Methods

Collection Methods

https://upload.wikimedia.org/wikipedia/commons/b/b2/Barber_pitfall_trap

Pit Fall Method

- In this method, the insect falls into a pit, a container that is buried in the soil.
- The pitfall trap catches ground-dwelling insects.
- Glass jar is buried in soil with its rims at the level of ground.
- Filled with formalin/alcohol
- Detergent drops to beak surface tension
- May be covered

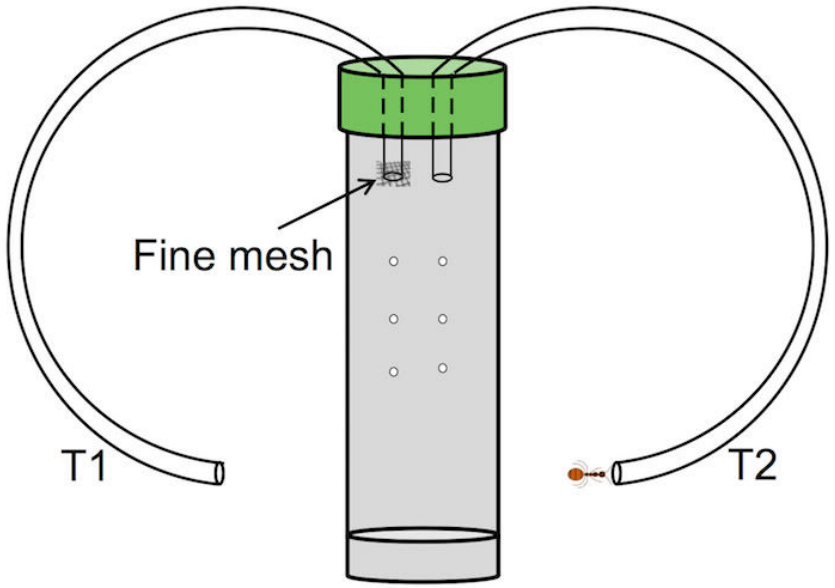
Collection Methods

https://bioquipinc.com/wp-content/uploads/2017/05/%C3%92Pooter%C3%93-Style-Aspirator-4

Aspirator

- Small insects or insects in hard to reach places, can be collected using an aspirator.
- The aspirator is a vial with two pieces of tubing, one with a fine screen material over it.
- By sucking on one tube, you draw the insect into the vial through the other.

Collection Methods



https://www.google.com/search?q=aspirator%2C+insect+colection&tbm=isch&ved=2ahUKEwiNwOKFiN7sAhUI7xoKHS47DLQQ2-cCegQIABAA#imgrc=7omrJPNCN0eAqM

Aspirator

- The screen/fine mesh prevents the insect or anything else unpleasant from getting drawn into your mouth.

Principles of Systematics

Module# 59

Principles of Systematics

Collection Methods

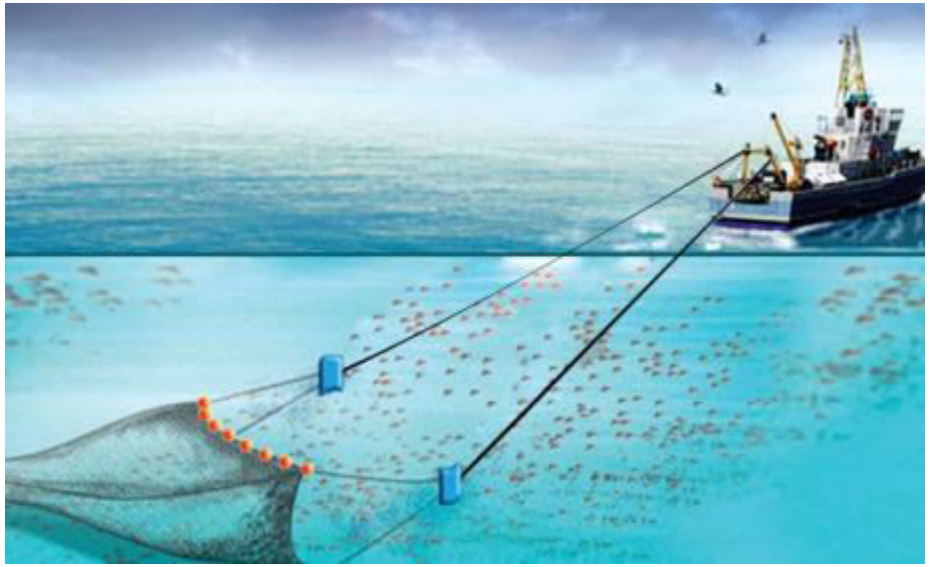
Collection Methods

http://www.nzffa.org.nz/images/design/beatingbig.jpg

Beating Sheet

- To study insects that live on branches and leaves, like caterpillars, a beating sheet is a tool to use.
- Stretch a white or light-colored sheet below the tree branches. With a pole or stick, beat the branches above.
- Insects feeding on the foliage and twigs will fall down onto the sheet, where they can be collected.

Collection Methods



https://bharatabharati.files.wordpress.com/2015/03/bottom-trawling.jpg?w=460&h=260

Trawling

- Trawling is a method of fishing that involves pulling a fishing net through the water behind one or more boats.
- The boats that are used for trawling are called trawlers or draggers
- In Trawling baited fishing lines are drawn through the water.
- Trawling is commonly used as a scientific sampling, or survey method for marine specimens.

Collection Methods

https://www.researchgate.net/profile/Richard_Nisbett/publication/265226839/figure/download/fig15/AS:669421387382787@1536613904116/Dipper-for-the-collection-of-mosquito-larvae-41.jpg

Dipper

- The most commonly used larval collection method is the "standard dipper.
- Estimates of **larval** density can be carried out by counting the number of mosquito **larvae** per **dip**, using a standard 1-pint **dipper**.
- A minimum of three to five dips should be completed at each site.

E.g. collection of mosquito larvae

Principles of Systematics

Module# 60

Principles of Systematics

Curation of Collection

Curation of Collection

- A **curator** is a person who manages, administers or organizes a collection for exhibition.
- A museum curator is custodian of collection of museum and is responsible for their management.
- **Curation** is process of identification and organization animal collection.

Curation of Collection

- All types of museums have at least one curator.
- A large museum may employee several curators
- Each have specialization in particular fields.
- American Museum of Natural History (AMNH), New York
- Housing of collection in fire free buildings that are dust proof
- Air-conditioned Building

Curation of Collection

Cataloging

- Museum number
- Original field number
- Scientific Name (at least generic name)
- Exact locality
- Date of collection
- Name of Collector
- Remarks
- Card systems/electronic data processing
- Exchange of materials

END

Principles of Systematics

Module# 61

Important Properties of Texa Names

Important Properties of Texa Names

Three important properties

- Uniqueness
- Universality
- Stability

Important Properties of Texa Names

Uniqueness

- Names of an animal gives immediate access to all known information about that taxon.
- Every names has to be unique
- If several names---clear cut method should be there to determine valid name.
- Principle of priority usually decides in case of conflict.

Important Properties of Texa Names

Universality

- Scientific communication would be very difficult if there were only vernacular names for animals.
- Specialist would have to learn the name of texa in different languages in order to communicate with each other.
- Zoologist use single language

Important Properties of Texa Names

Stability

- A name should be stable
- Any change of a well-established name is likely to produce confusion and impede information retrieval.
- Even a principle of priority can be set aside by International Commission on Zoological Nomenclature in cases where stability is threatened.

END

Principles of Systematics

Module# 62

Principles of Systematics

Type Specimen

Type Specimen

- A type is a specimen (or a group of specimens) which serves to illustrate the defining characters of a species or genus.
- A type is a specimen serve as a reference which can be used by later scholars to verify the identity of species being described.

Type Specimen

- Precise collecting locality and other relevant data on the labels of specimen.
- Sex
- Developmental stage or form to which the type is referable
- In the case of parasites, the name of host species

Type Specimen

- Name of collector
- Specimens numbered
- Altitude of the type locality or depth in meters below sea level at which the holotype was taken
- In the case of fossils species, geological horizon

Type Specimen

- Type designation or fixation should always be completed before publication
- Type designation should be clear and unambiguous
- The deposition and museum number of the type always be recorded

END

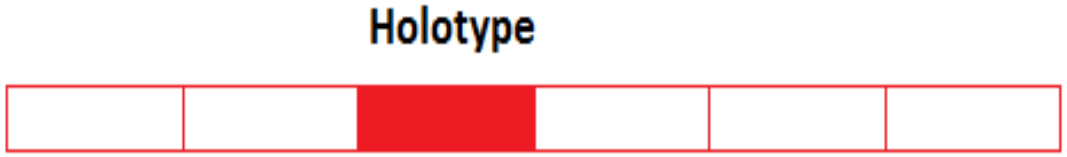
Principles of Systematics

Module# 63

Principles of Systematics

Kinds of Type Specimen

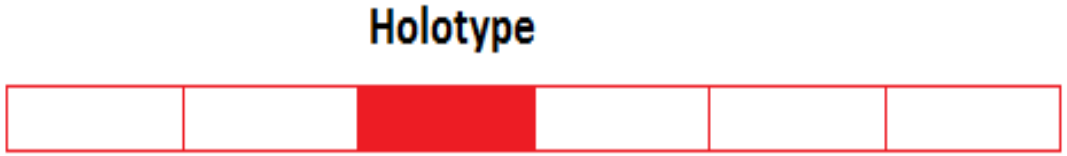
Kind of Type Specimens



Holotype

- A single specimen designated or indicated as the type by the original author at the time of publication of the original description of a species”
- If no specimen designated or indicated as the type by the original author but it is clear that only one specimen was used for description then it will be considered as Holotype.

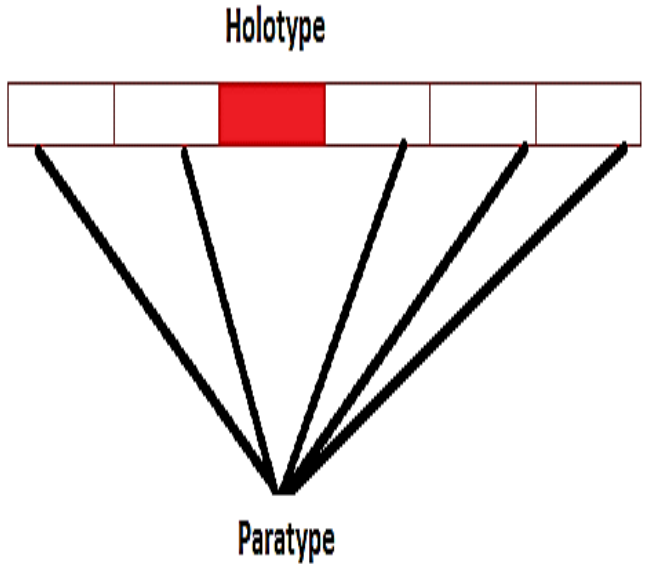
Kind of Type Specimens



Holotype

- For a species described after 1999, holotype must be designated.

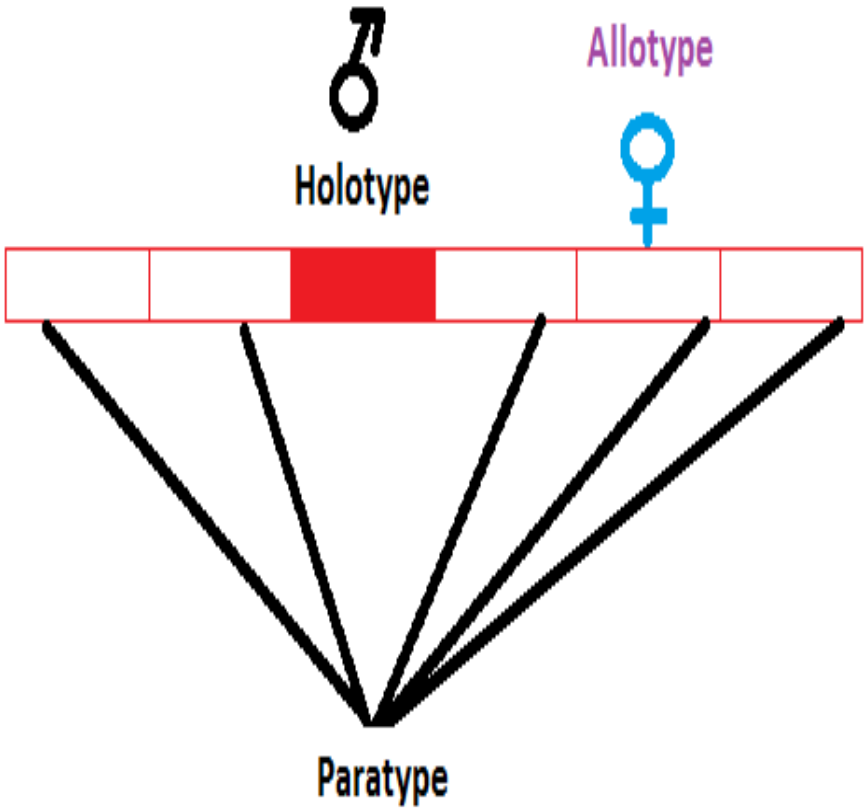
Kind of Type Specimens



Paratype

- The specimen other than the holotype, which has been used in the description.

Kind of Type Specimens



Allotype

- A specimen of opposite sex as the holotype, which has been used in the description.
- Its is often used to describe a morphological character not shown in the Holotype

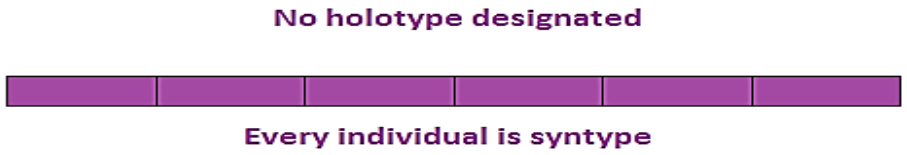
Principles of Systematics

Module# 64

Principles of Systematics

Kinds of Type Specimen

Kind of Type Specimens



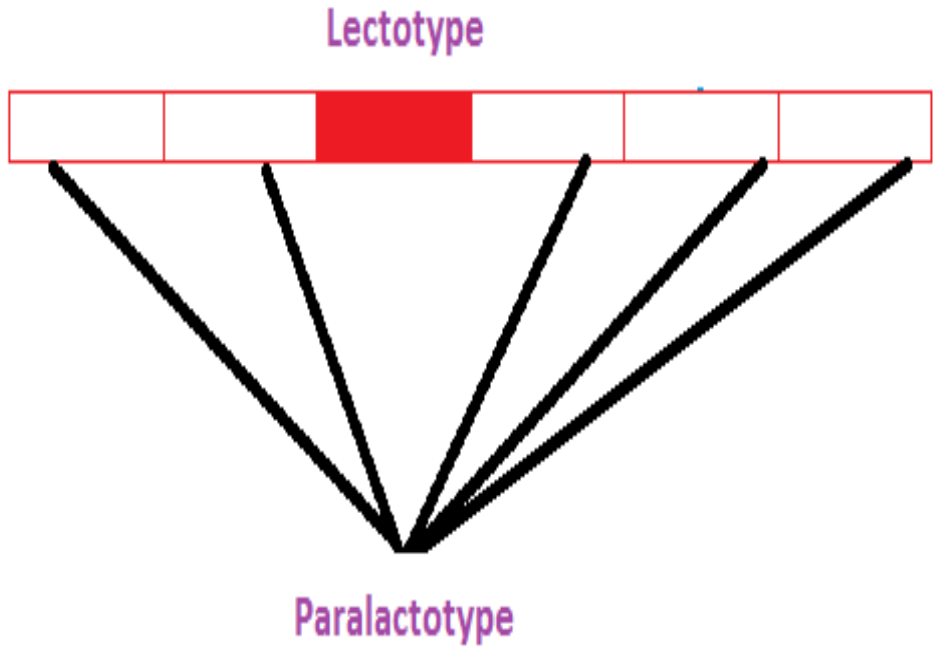
Syntype

- When description is based on two or more specimens, no holotype was designated, every specimen in a type series will be syntype

Kind of Type Specimens

Lectotype/Paralectotype

e



Kind of Type Specimens

Neotype

- A specimen selected as a type subsequent to the original description in the cases where the original types are known to have been destroyed or suppressed by commission

Kind of Type Specimens

Isotype

- A specimen that is a duplicate of holotype or very similar to the type specimen collected at the same time but not from same place as holotypes, paratype or syntype and not included in any of these.
- Can be used as a reference specimen if the type specimen is lost.

Kind of Type Specimens

Monotype

- When holotype of the species is the only specimen known to science, it is known as monotype.

END

Principles of Systematics

Module# 65

Principles of Systematics

Type Locality

Type Locality

- Place of collection of Holotype and lectotype.
- Paratype can be from different locality to holotype.
- Specimens collected at type locality are called **Topotypes.**
- Population that occur at type locality is called **Topotypical population.**

Type Locality

- In the case of parasites, the term type host is used to indicate the host organism from which the type specimen was obtained.
- Importance of type locality

END

Principles of Systematics

Module# 66

Principles of Systematics

Type Locality

Type Locality

- According to Article 72h type locality is the place from where type specimen was taken.
- Specimens collected from type locality are called topotype
- Population that occur at type locality is topotypical population
- Preserved specimen/type locality

Type Locality

- Damage holotype
- Information about environment
- Temperature/Humidity/soil type
- Missing type locality
- Article 72H contains advice on designations or restriction
- Designations or restriction not done routinely but by specialists by their revisionary work

END

Principles of Systematics

Module# 67

Principles of Systematics

Concept of Priority

Concept of Priority

- Fundamental principal of Zoological nomenclature
- Essentially it is a principle of recognizing first valid name of a species.
- A principle in taxonomy: the first properly published name of a species or genus takes precedence over any subsequently published
- (Article 19)

Concept of Priority

- The name which was use first time should be valid name
- Once name has been used, no subsequent publication of that name for other texon.
- There are formal provisions for making exceptions to this principle.

Concept of Priority

Limitations

- Priority starts from May 01, 1753
- Apply only to family level
- Not according to Zoological code

Concept of Priority

- If an archaic or obscure prior name is discovered for an established taxon, the current name can be declared conserved name (zoology), and so conserved against the prior name.
- Conservation can may be avoided in Zoology as these names fall in formal category of ***Nomen oblitum.***

Concept of Priority

Nomen oblitum

- In taxonomy, a forgotten name; i.e. the name of a senior synonym that has not been used in the zoological literature for at least 50 years.
- Such names should not be used if they upset a better-known name, but should preferably be suppressed by the ICZN.

Concept of Priority

- If a current name for a taxon have archaic or obscure homonym (same name to different taxon), the current name can be declared as ***Nomen protectum*** (available name).

END

Principles of Systematics

Module# 68

Principles of Systematics

Concept of priority

Concept of Priority

https://www.google.com/search?q=leopard+shark+&tbm=isch&ved=2ahUKEwj5i_2yg_PsAhWU0oUKHUviByEQ2-cCegQIABAA&oq=leopard+shark+&gs_lcp=CgNpbWcQAzICCAAyAggAMgIIADICCAAyAggAMgIIADICCAAyAggAMgIIADICCABQq52IEVirnYgRYKmmiBFoAHAAeACAAdgBiAHYAZIBAzItMZgBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=_O6nX_mdCpSllwTLxJ-IAg&bih=600&biw=1360#imgrc=uTp_hTV4HeDvJM

What does priority means?

- The first Scientific name applied to the leopard shark was ***Triakis californica***
- Proposed by British Zoologist Edward Gray (1851)
- He did not provided proper description
- In 1854 ***Mustelus felis*** *by william Orville published*
- *In 1855* ***Triakis semifasciata*** *by C F Girad*

Concept of Priority

Error in date recording

- ***Triakis semifasciata***
- ***Nomen protectum*** (available name)
- *Mustelis felis* was invalidated (as a ***nomen oblitum***).

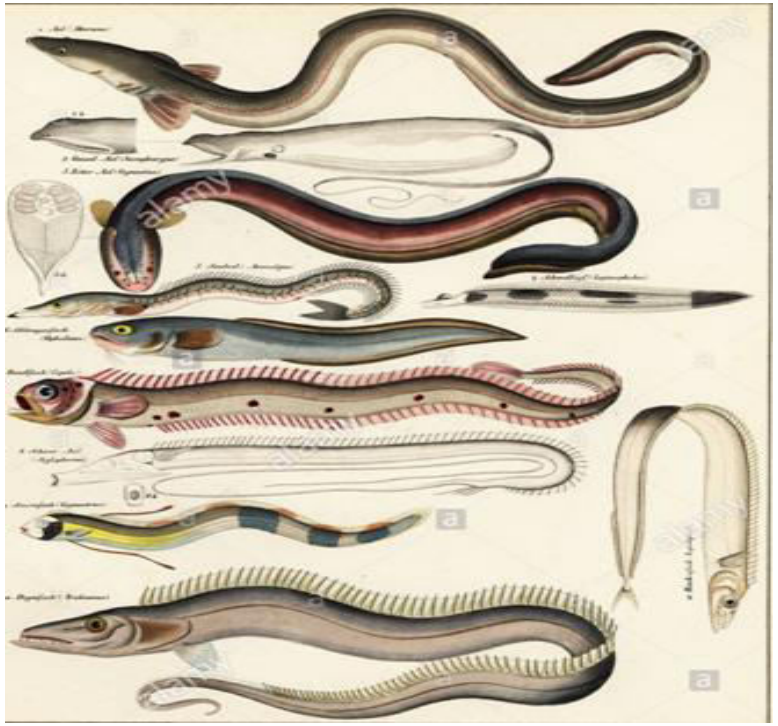
Concept of Priority

https://www.google.com/search?q=pronghorn&sxsrf=ALeKk016hNi2hIH42LJ1FojE0bD517_3Tw:1604565941448&source=lnms&tbm=isch&sa=X&ved=2ahUKEwj_pNX3gevsAhWMYcAKHUJyCboQ_AUoAXoECDQQAw&biw=1360&bih=600#imgrc=zCJQlNSIekozdM

Example

- In 1855 John Edward Gray published the name of Pronghorn *(Antilocarpa anteflexa)*
- Based on pairs of horns
- 1815 George Ordin Published *Antilocarpa americana* for the same species.
- So the older name *Antilocarpa americana* takes the priority over *the Antilocarpa anteflexa*

Concept of Priority



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Example

- In 1856 Johann Jakob Kaup published the name *Leptocephalus brevirosteris* for new species of eel.
- In 1893 same name was given to juvenile form of European eel whose name was *Muraena anguilla* given by Carl Linnaeus in 1758.
- Old name was retained.

Principles of Systematics

Module# 69

Principles of Systematics

Concept of Availability

Concept of Availability

- Taxonomic name which should be used is called Valid (Accepted, correct) name.
- The name must have been published according to requirements detailed in Articles 8-20.
- An available name can be valid or not
- An unavailable name is never valid.
- Availability criteria vary depending on nomenclature

Concept of Availability

- The term available means “legitimate”
- A name may be available in this technical sense even though as Junior homonym
- It is automatically invalid at the time of publication (Article 53 and 59a).
- Need to differentiate between **available** and **valid names**

Concept of Availability

- Available **(properly published).**
- Valid **(not preoccupied by senior synonym or homonym)**
- Article 11-19 describe prerequisite for making a specific name available.
- Particularly article **13a** is very important.

Concept of Availability

- According to article **13a** names published after 1930 must be
- Accompanied by description that states the characters that are purported to differentiate the Taxon
- It is good taxonomic practice to include in the original description not only such diagnostic information but also differential diagnosis (comparison with other related species)

END

Principles of Systematics

Module# 70

Principles of Systematics

Concept of Availability

Concept of Availability

Nomen nudum

- A name published without satisfying the condition of availability is called nomen nudum.
- Especially if it fails to satisfy the conditions of Article 12-16.
- A nomen nudum has no standing in Zoological nomenclature

Concept of Availability

Nomen nudum

- If editors of journals were more familiar with the provisions of the Code, they would not permit publications of so many unavailable names because they violate one or another of Article 8-20.

Concept of Availability

Differential Diagnosis

- Description of characters after 1930 (Article 13a)
- Comparison with close relative

- Description of diagnostic differences (NOT mandatory)

Concept of Availability

Names Given to Hybrids

- Hybrids are normally individuals not population
- Not considered as Texa
- Article 1b (3) states clearly that a name given to an animal that is later found hybrid remain available for the purposes of homonymy, not for purposes of Synonymy
- A name given to hybrid can not be applied to either of parental species.

Concept of Availability

Synonyms

- Different names for same thing
- Earliest published name **senior synonym**
- Later published name **junior synonym**

Concept of Availability

Types of Synonyms

Objective synonym

- The name that objectively refer to the same thing/name based on the same specimen or illustration (at Genera level)

Subjective synonym

- Name based on different type material (*Antelocarpa anteflexa* 1855 vs *Antelocarpa americana*)
- *Antelocarpa anteflexa* was unusual specimen of *Antelocarpa Americana)*

END

Principles of Systematics

Module# 71

Principles of Systematics

International Code for Zoological Nomenclature

International Code for Zoological Nomenclature

Zoological Nomenclature

- System of scientific names applied to taxonomic units of animals
- The term nomenclature comes from Latin words
- **Nomen** which means Names
- **Calare** mean to call

- Call by names

International Code for Zoological Nomenclature

Important requisite of taxonomic names

- **Uniqueness**
- **Universality**
- **Stability**

International Code for Zoological Nomenclature

ICZC

- Set of rules for naming animals and resolution of nomenclatural problems
- Authoritative document that contain valid rule of Zoological nomenclature
- Original codes and its revision were the work of International Congress of Zoology

International Code for Zoological Nomenclature

ICZC

- Now these congresses are defunct
- Role have been taken over by **General Assemblies of International Union of Biological Sciences** (IUBS)
- International Congresses of Zoology have appointed a judicial body (**International Commission on Zoological Nomenclature**) to interpret the rules and publish opinions on controversial issues.

International Code for Zoological Nomenclature

ICZC

- Article 76 through 82 specify the duties of the commission.
- Latest edition of code was published in February 1985
- Can be ordered from International Trust for Zoological Nomenclature 56, Queen Gate London or
- University of California Press, Berkeley and Los Angeles.

International Code for Zoological Nomenclature

ICZC

- Code consists of three parts
- The code proper
- Appendices (A section giving additional information at the end of Book)
- Glossary (a list of technical or special words)

International Code for Zoological Nomenclature

Code proper

- **Preamble:** Introduction to book or written documents
- 90 articles grouped in 18 chapters
- Each article is composed of one or more mandatory rules
- Followed by recommendations and illustrative examples
- Use of recommendation is not mandatory

International Code for Zoological Nomenclature

Appendices

- Appendices are like recommendations not to be followed strictly as article.
- These are five in number and are designated as A-E.

Glossary

- The terms used in the text are clearly defined in the Glossary.

END

Principles of Systematics

Module# 72

Principles of Systematics

Molecular Systematics

Molecular Systematics

- Classification of organisms on the basis of distribution and composition of chemical substances in them.
- Comparisons of Molecular components (DNA, RNA and Proteins) help to establish relationships among different animal texa.
- **Molecular Phylogenetics-** relationships based on molecular data and statistical analysis.

Molecular Systematics

- First phylogenetic tree based on molecular data 1967 **Fitch and Margoliash**
- Researcher realized that although traditional taxonomic methods are good but Molecular evidences could be final confirmatory
- Phylogenetic studies help to assess the historical processes which affect relationships

Molecular Systematics

- Phylogeographic studies help to understand the distribution
- Phylogenetic and Phylogeographic studies started with the introduction of mDNA markers in population genetic studies.
- The goal is to provide insight into the history of groups of organisms and the evolutionary processes that create diversity among species.

Molecular Systematics

- The use of amino-acid or nucleotide-sequence data in determining the evolutionary relationships of different organisms.
- Essentially it involves comparing the sequences of functionally homologous molecules from each organism being studied, and determining the number of differences between them.

Molecular Systematics

- The greater the number of differences, the more distantly related the organisms are likely to be.
- Genetic distances
- Cluster analysis
- Gel electrophoresis
- Agarose/PAGA

END

Principles of Systematics

Module# 73

Principles of Systematics

Molecular Systematics

Molecular Systematics

Applications

- Solving taxonomic problems
- Reorganization of tree of life
- I identifications of immature stages
- I identification of cryptic species
- Identification of disease vectors
- Testing food authenticity
- Monitoring of illegal trade
- Monitoring quality of water
- Identification of damage specimens

Molecular Systematics

Applications

- Identification of sibling species and sub-species

END

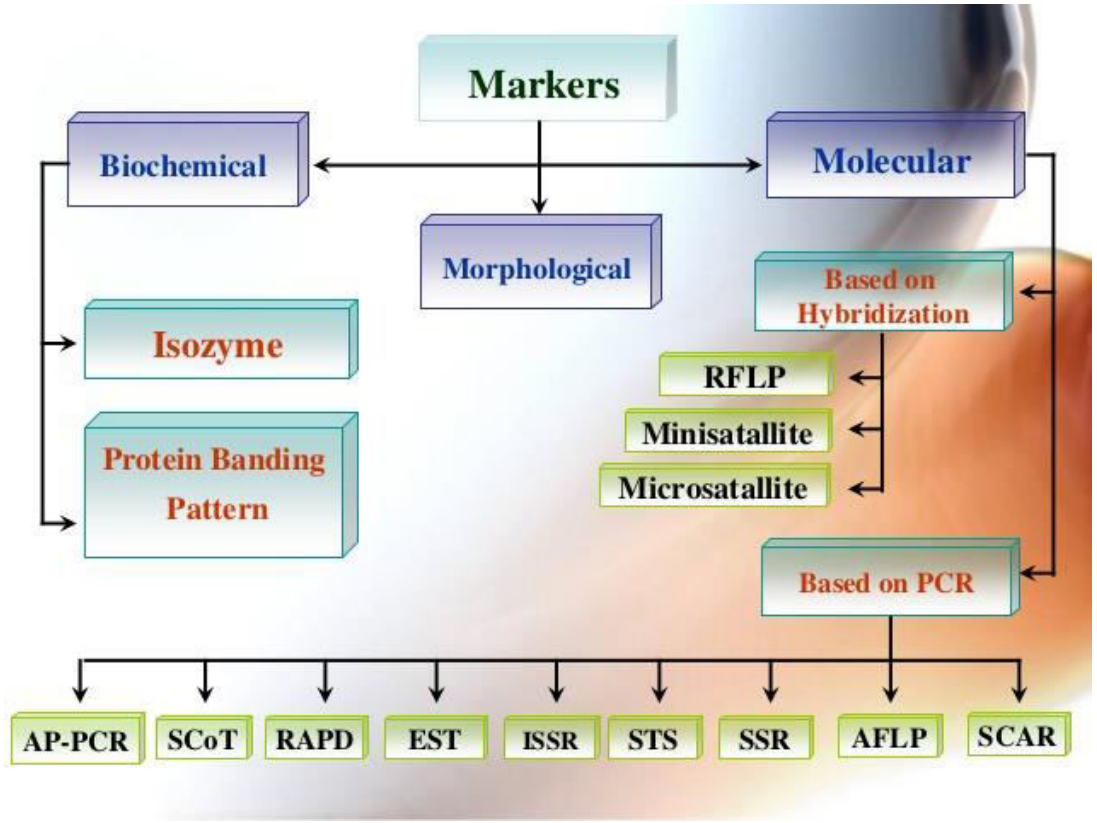
Principles of Systematics

Module# 74

Principles of Systematics

Molecular Markers

Molecular Markers

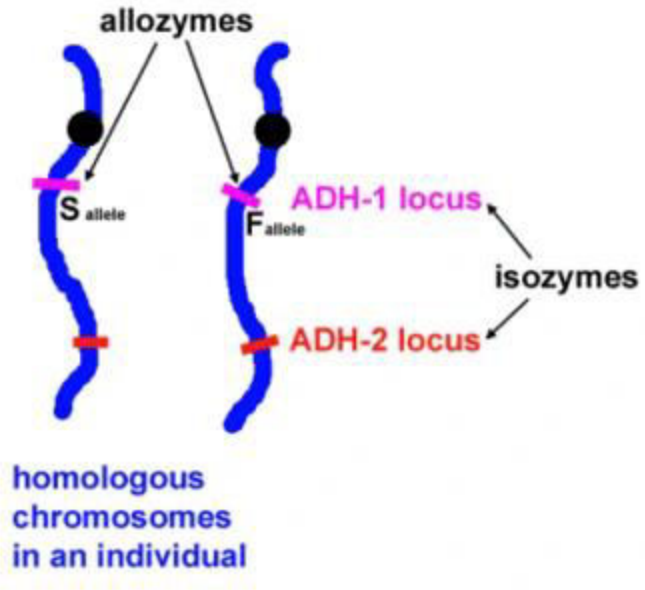


https://www.slideshare.net/FAOoftheUN/molecular-markers-types-and-applications-55688001

Types of Markers

- **Type I:** Associated with genes of known function. Alloenzymes are type I markers as the protein they encode are associated with genes of known function.
- **Type II:** Associated with genes of unknown function. Microsatellite and other neutral markers are type II markers.

Molecular Markers



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Alloenzymes

- Variant form of an enzyme
- Different structurally but not functionally from other alloenzyme coded by different allele present at same locus.
- Codominant markers that express in hetrogygous in MF

Isoenzymes

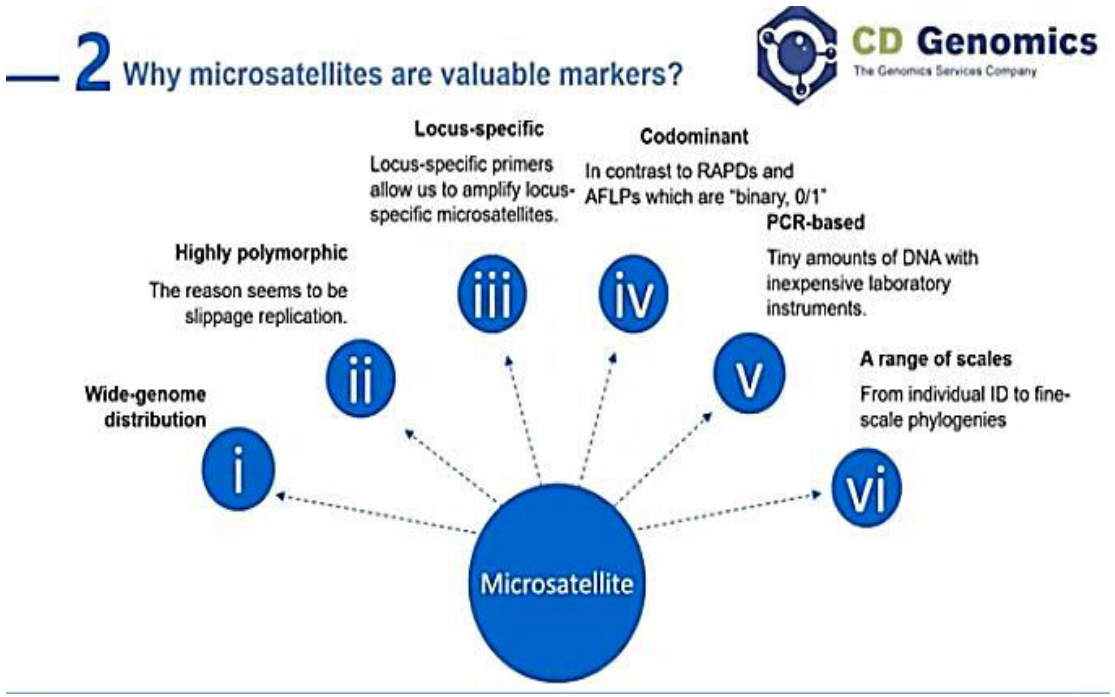
- The enzymes that perform the same function, but which are coded by genes located at different loci

Molecular Markers

Mitochondrial DNA Markers

- Non-Nuclear DNA
- Present in Mitochondrial
- Maternally inherited as haploid genome
- Entire genome undergo transcription as single unit
- No recombination
- Homologous markers

Molecular Markers



https://www.youtube.com/watch?v=r-8SXTHzJFU

Microsatellite

- Simple DNA sequence
- Repeated several times across various points in DNA of an organism
- 2-6 nucleotides
- 5-50 times
- Present at thousands of locations in an organism
- Microsatellite sequences are composed of non-coding DNA and are not parts of genes.

Principles of Systematics

Module# 75

Principles of Systematics

Molecular Markers

Molecular Markers

SNPs

- Single Nucleotide substitution
- Single Nucleotide insertion/deletion
- These point mutations give rise to various alleles with alternate bases at particular nucleotide position
- Most abundant polymorphism in genome of an organism
- Can be detected by PCR, microchip array or fluorescence technology

Molecular Markers

DNA barcoding

- CO1 gene
- 658 base pairs
- Species specific
- Highly conserved
- High mutation rate
- Widely used in identification of arthropods
- Universal primers
- Reliable and quick method
- Hebert Paul 2003

END

Principles of Systematics

Module# 76

Principles of Systematics

Molecular Markers

Molecular Markers

Protein (Biochemical markers)

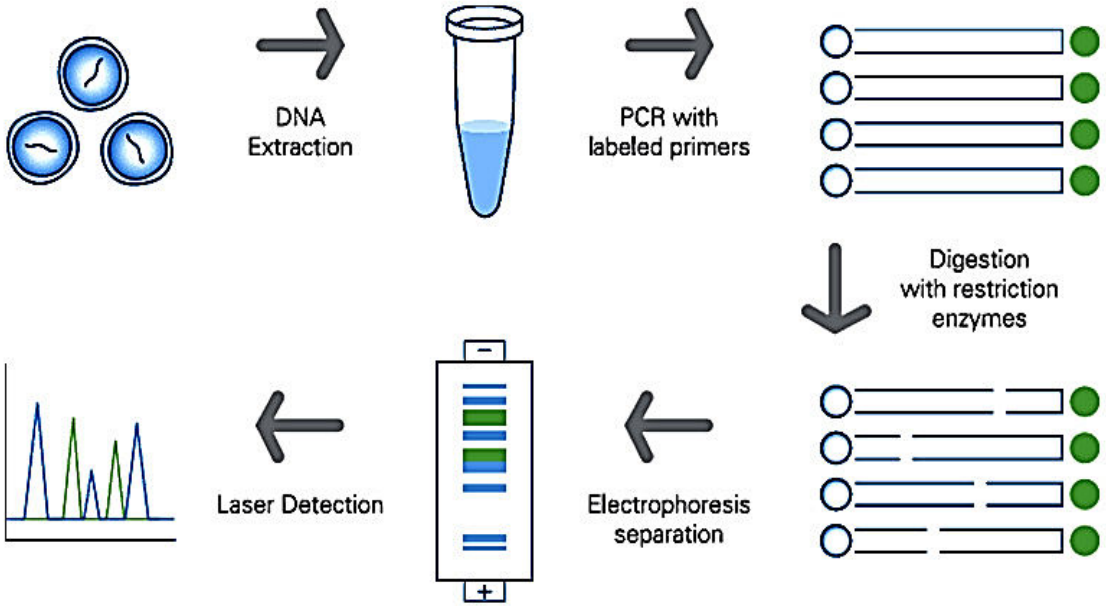
- Available since 1950s
- Related to variation in protein and amino acid bending pattern
- Can be studied using Gel electrophoresis e.g esterase, Acid phosphatase, peroxidase etc.
- Advantage
- Require simple equipment
- A vigorous complement to the morphological assessment of variation

Molecular Markers

Protein (Biochemical markers) Disadvantages

- Subject to environmental influences
- Limited in numbers

Molecular Markers



Hybridization based techniques (Non-PCR)

- RFLPs (Restriction Fragment Length polymorphism) Analysis
- Introduced by Bostein et al. 1980
- Relatively slow process
- Use of radioisotopes has limited use of RFLPs in certified laboratories

Molecular Markers

PCR based techniques

- RAPD (Random Amplified polymorphic DNA)
- ISSR (Inter-Simple Sequence Repeat)
- SSR (Simple Sequence Repeat)
- AFLP (Amplified Fragment length Polymorphism)
- EST (Expressed sequence Tags
- SCoT (Stop Codon targeted marker)

END

Principles of Systematics

Module# 77

Principles of Systematics

Molecular Markers

Molecular Markers

Advantages of Molecular Markers

- Ubiquitous
- Stably inherited
- Multiple alleles for each marker
- Devoid of pleiotrpic effects
- Detectable in all tissues, all ages
- Long shelf life of DNA samples

Molecular Markers

Application of Molecular Markers

- Fingerprinting
- Diversity studies
- Genetic maps
- Gene tagging
- F1 identification
- Seed testing
- Comparative maps

END

Principles of Systematics

Module# 78

Principles of Systematics

Methods of Animal Preservation

Methods of Animal Preservation

Animal Preservation

- Long term animal preservation in best possible condition
- May be assessed as reference collection in future for scientific purposes
- Many methods are used for preservation of vertebrate and invertebrates

Methods of Animal Preservation

Reasons of Animal Preservation

- Taxonomic reasons
- For detail examination
- For morphological examination of particular animal as each and every animals can not be in researcher's vicinity
- For zoological museum record

Methods of Animal Preservation

Steps for Animal Preservation

- Killing and relaxing of animals
- Fixation (stop cellular respiration, kill bacteria within organism, a good penetrating ability)
- Storage in bottle, vials, jars, bottles etc.

Methods of Animal Preservation

Type of specimens

- Entire fluid preserved specimens (for study of histology and anatomy, entirely change fur colour)
- Study skins with skulls /partial skeleton (Some bones in skin) for studying colour, hair quality and moulting pattern
- Mounted skins
- Entire skeletons (Anatomy, age determination, geographic variations)

END

Principles of Systematics

Module# 79

Principles of Systematics

Methods of Animal Preservation

Methods of Animal Preservation

Formalin as Preservative

- It is used for vertebrate only
- Not used for long term storage
- Acidic and difficult to handle
- Used when colour of animal is important
- Alcohols dissolve most colour immediately
- Penetrate more easily and internal organs remain in better condition

Methods of Animal Preservation

Side effects of Formalin

- Fumes of formalin are harmful
- Extreme discomfort of eyes and nose
- Severe irritation if contact with body
- Can cause cancer
- Hand should be rinsed after use
- Should be placed at safe place
- Water-tight

Methods of Animal Preservation

Industrial Alcohol (For fixing and storage)

- Not used for killing and fixing of vertebrates.
- For long term storage
- Colour of sample is lost quickly
- Glycerin in alcohol help to preserve colours and to keep specimens flexible
- 70-75% alcohol is used for long term storage
- 95% of DNA studies

Methods of Animal Preservation

Side effects

- Skin irritation with prolong contact.
- Industrial alcohol is toxic and should not be drunk
- Highly flammable
- Rapid evaporation
- Should be kept covered not in open jars.

END

Principles of Systematics

Module# 80

Principles of Systematics

Methods of Animal Preservation

Methods of Animal Preservation

Iropropyl Alcohol

- Cheep
- Easy to obtain
- Available strengths (90%, 70%)
- For use dilute upto 40%
- Can be hard to specimens and specimens tends to brittle over the time
- Can be buffered with few drops of glycerin
- Pinch of calcium carbonate tablet (crush form of tablet)

Methods of Animal Preservation

Ethyl Alcohol

- Better solution for long term storage of invertebrate specimens
- 80% ethyl alcohol
- Denatured alcohol
- Painting supplies
- Should be buffered with glycerin/antacid tablets

Methods of Animal Preservation

Fish preservation

- After capture fishes are placed in 10% formalin for quick killing.
- This is highly painful
- Fishes dies with its parts well spread off and body straight and well stretched.
- Examination and counting of rays, fins etc. very easy.
- 30 cm fish, Fish in formalin for one week, in water for one day (leech out formalin), use alcohol for long term preservation

END

Principles of Systematics

Module# 81

Principles of Systematics

Methods of Animal Preservation

Methods of Animal Preservation

Herptiles

- Individually kept in plastic jars
- Killed by freezing or chloroform
- Shallow trays are used for herptiles
- Snakes should be coiled
- Frogs are placed on their bellies with their limbs set at right angles to their bodies.
- Small salamanders fixing time 2 days
- Month for alligator, snapping turtle

Methods of Animal Preservation

Herptiles

- Add water for day or two
- 75% alcohol for long term storage.

Methods of Animal Preservation

Birds an Mamamls

- Skinned
- Skulls or skeletal mounts
- Remove flesh
- Boiling or dermested beetle
- Placed defleshed bones in bleach or hydrogen peroxide solution to whiten
- Allow to dry
- Place in bag with complete label

END

Principles of Systematics

Module# 82

Principles of Systematics

Methods of Animal Preservation

Methods of Animal Preservation

Preservation of invertebrates

- Use of alcohol is the easiest way
- One should be aware about the type of alcohol
- Each specimen type require different concentration of alcohol
- Preserved in jars, tubes or bottles

Methods of Animal Preservation

Coelenterates, platyhelminthes and echinoderms

Coelenterates are difficult to preserve

- Preserve in 70% alcohol or 5% formalin
- Hydra can be quickly fixed in warm Bouin's warm not hot.

Platyhelminthes (flatworms)

- Bouin solution for fixation
- Paraffin is best long term preserving and storage medium for all flatworms.

Methods of Animal Preservation

- **Echinoderms** are narcotized by addition of magnesium sulphate or methanol to the sea water in which they live
- When completely insensitive to stimulus such as pricking, 70% alcohol for preservation and storage

Methods of Animal Preservation

- **Arthropods** are easy to process as killed immediately and preserved and store in alcohol
- Crabs and prawn may be killed in formalin but renders their joints hard and brittle.
- Larger invertebrate need to be injected 10% formalin to prevent from rotting
- Industrial alcohol is used for most arthropods.

Methods of Animal Preservation

- Insects, crustaceans and arachnids can be simply dropped in alcohol for immediate preservation.
- Industrial alcohol is used for most arthropods.

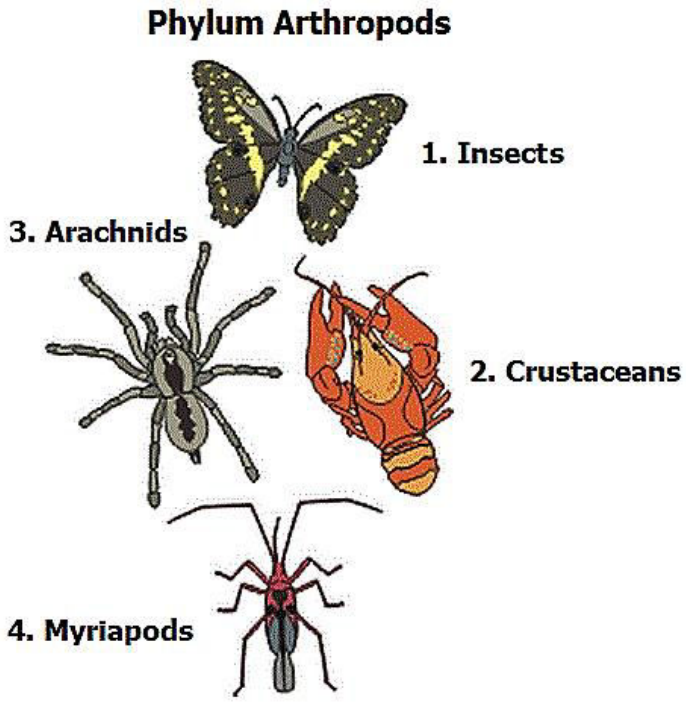
END

Principles of Systematics

Module# 83

Morphological Identifications of Model Arthropod

Morphological Identifications of Model Arthropods



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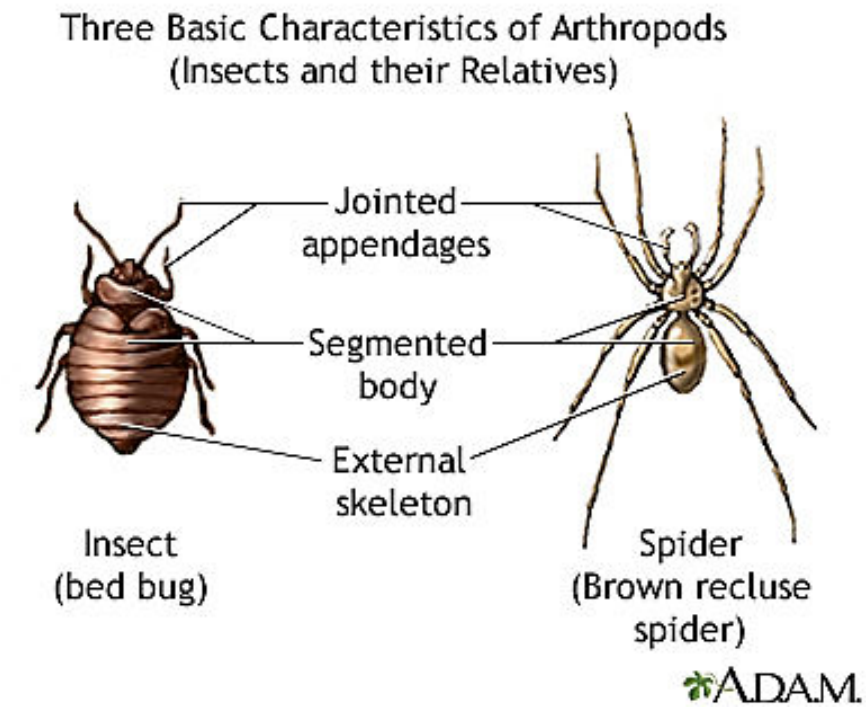
What is an Arthropod?

- Any member of Phylum Arthropoda
- Include lobster, prawn, mites, insects, spiders, centipedes, millipedes
- Represented in every habitat of earth
- Show great diversity of adaptations

Morphological Identifications of Model Arthropods



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- Some are aquatic, others are terrestrial, some are free living, other are parasitic
- All arthropods have a body supported by hard exoskeleton
- Legs, antennae and mouth are jointed
- Tube like heart
- Overall body bilateral symmetrical

Morphological Identifications of Model Arthropods

- Seven million species
- Arthropods are the most successful of all animal phyla based on diversity, distribution, and numbers.
- Nearly one million species identified so far, mostly insects.
- The exoskeleton, or cuticle, is composed of protein and chitin.

Morphological Identifications of Model Arthropods

- Molting of the cuticle is called ecdysis
- Extensive cephalization
- Open circulatory systems in which a heart pumps hemolymph through short arteries and into open spaces (sinuses)
- Aquatic members- gills for gas exchange; terrestrial members- tracheal system of branched tubes leading from surface throughout body

END

Principles of Systematics

Module# 84

Morphological Identifications of Model Arthropod

Morphological Identifications of Model Arthropods

- Appendages in arthropods are modified for a number of functions:
- Walking legs
- Mouth parts
- Antennae
- Claws
- Organs for mating
- Holding eggs

Morphological Identifications of Model Arthropods

- Segmentation is an ancestral characteristic that carries over from the Annelids
- Note the similarity of the insect caterpillar to polychaete worms
- The exoskeleton provides rigid support and strong points for muscle attachment
- The exoskeleton also forms a protective shell

Morphological Identifications of Model Arthropods

- The exoskeleton prevents drying out, but also prevents respiration through the skin
- The exoskeleton provides rigid support, but prevents growth – requiring molting
- ***Striated muscle*** associated with the exoskeleton, and ***smooth muscle*** associated with the gut

Morphological Identifications of Model Arthropods

- Reduced body cavity, filled with blood, forming an ***open circulatory system***
- Specialized ***excretory glands*** (coxal, antennal or maxillary glands; malpighian tubules)
- Specialized ***respiratory organs*** (gills, tracheae, book lungs)
- Dorsal brain with fused ventral nerve cords

Morphological Identifications of Model Arthropods

Four lineages of arthropods

- Trilobites – extinct
- Chelicerates – horseshoe crabs, spiders
- Uniramians – centipedes, millipedes, insects
- Crustaceans – crabs, lobsters, barnacles

From these lineages arose five major classes of arthropods.

END

Principles of Systematics

Module# 85

Principles of Systematics

Identification Keys and catalogues

Identification Keys and Catalogues

- Identifications key aids in the identification of biological identities:
- Plants
- Animals
- Fossils
- Microorganisms
- Pollens

Identification Keys and Catalogues

- Keys are constructed so that user is presented with series of choices about the characteristics of unknown organisms
- By making correct choice at each step of key user is ultimately let to identity of unknown specimen
- At each step use must answer a question about one of more character of gentility to be identified.

Identification Keys and Catalogues

- If at each step there are only two alternative, the key is said to be **DICHOTOMOUS (s**equential key, analytical key, pathway key)

- **POLYTOMUS** key

Identification Keys and Catalogues

Methods of using taxonomic keys

- Read introductory comments, format of the key, abbreviations etc before using key
- Read both choices at each step of the key
- Use glossary to find out the definition of any term your do not understood
- Use a ruler if measurements are needed, do not guess
- Examine several specimens of same species

END

Principles of Systematics

Module# 86

Principles of Systematics

Identification Keys and catalogues

Identification Keys and Catalogues

Dichotomous keys

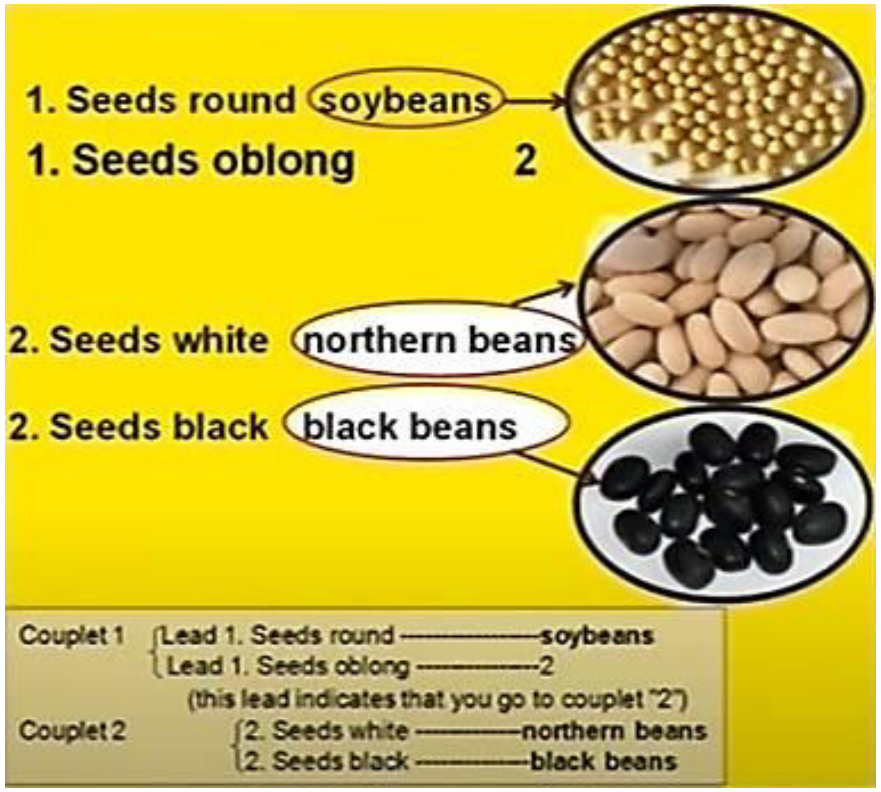
- If each step in a key has only two alternative the key will be dichotomous
- A dichotomous key is a device for identifying unknown organisms to some taxonomic levels such as species, genus or family level.
- These most commonly published keys were first published by Jean B. Lamark 1778

Identification Keys and Catalogues

Dichotomous keys

- Consists of series of paired statements, called couplets.
- Couplets describes some features of organism
- Each statements in a couplet is called a lead
- The statements or leads are in contract (Mutually exclusive)

Identification Keys and Catalogues

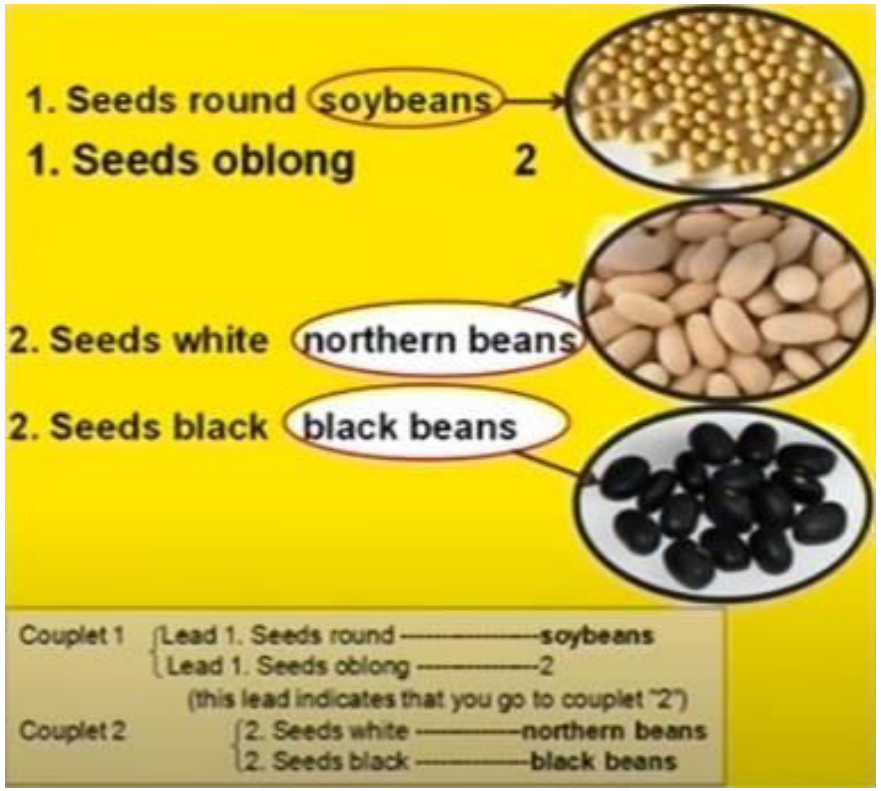


https://www.youtube.com/watch?v=srJ4aab5RYc

Dichotomous keys

- The use of key begins with first couplet and select the statement which best fit the organism to be identified
- The leads typically begins with board characteristics and become narrower as more choicely are required.
- We move through key-Single choice remains

Identification Keys and Catalogues



https://www.youtube.com/watch?v=srJ4aab5RYc

Dichotomous keys

- If we continue to select correct option of couplets for unknown organism, correct identification of animals will be achieved.

Principles of Systematics

Module# 87

Principles of Systematics

Identification Keys and catalogues

Identification Keys and Catalogues

Catalogue

- No study in science is complete until it is made available to others
- The best way is to made it available through circulation
- In taxonomy publication may be in the form of Books, pamphlets, journal articles, catalogues, checklist etc.

Identification Keys and Catalogues

Catalogue

- Catalogue is mainly an index to the texa arrange in such a way that as to give a vivid picture of references for both zoological and nomenclatural purpose.
- The **Catalogue** currently compiles data from 168 peer-reviewed **taxonomic** databases, that are maintained by specialist institutions around the world.

Identification Keys and Catalogues

Catalogue of life

- The **Catalogue of Life** is an online database
- It provides the most comprehensive and authoritative index of known species of organisms .
- The Catalogue interface is available in twelve languages and is used by research scientists, citizen scientists, educators, and policy makers

Identification Keys and Catalogues

Catalogue of life

- June 2001; 19 years ago
- Current status is active
- Registration not required
- Not commercial
- Available in English

Identification Keys and Catalogues

Catalogue of life

- The Catalogue listed 300,000 species by 2003, 500,000 species by 2005, and over 800,000 species by 2006. As of 2019, the Catalogue lists 1.9M extant and extinct species.
- There are an estimated 14M mainly unpublished species.

END

Principles of Systematics

Module# 88

Principles of Systematics

Construction of Identification Key

Construction of Identification Key

Key /Taxonomic aids

- Key is a list of statements
- Used for identification
- Similarities and dissimilarities
- Used pair of contrasting characters “couplet”
- Each statement in couplet is “Lead”

Construction of Identification Key

- Taxonomist have different opinions about the design and format of the key
- Some use numbers to separate the couplets
- Other use letters
- Some taxonomist place each couplets together
- Others may seprate couplets

Construction of Identification Key

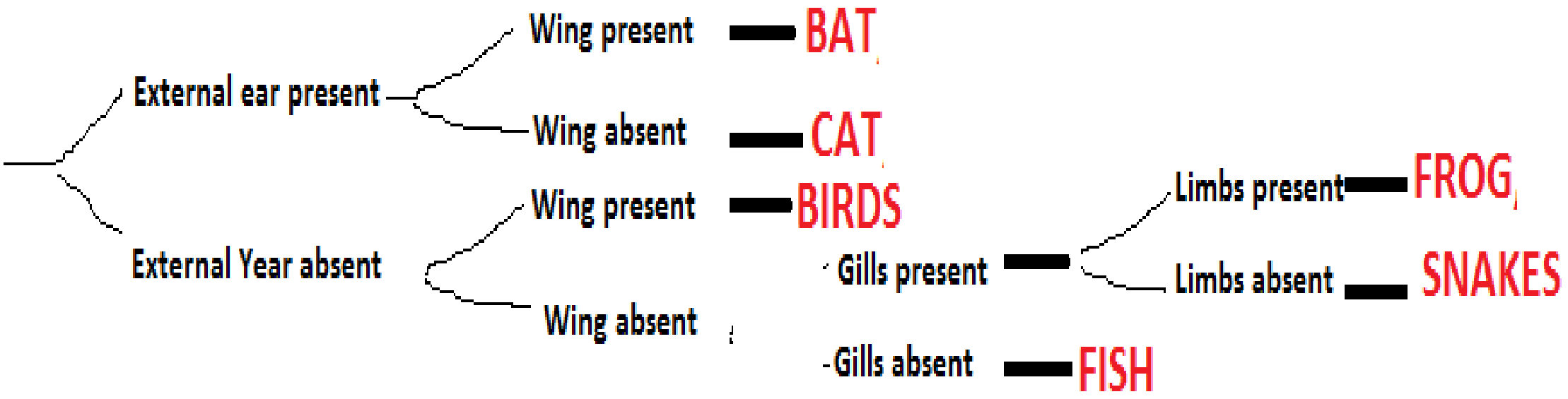
Types of Keys

- Indented Key
- Bracket Key
- Pictoral Key
- Polyclave Key

Construction of Identification Key

In indentd key we have sequences of choices between two or more statements

For example: BAT, CAT, FISH, FROG, SNAKES, BIRDS



Principles of Systematics

Module# 89

Principles of Systematics

Construction of Identification Key

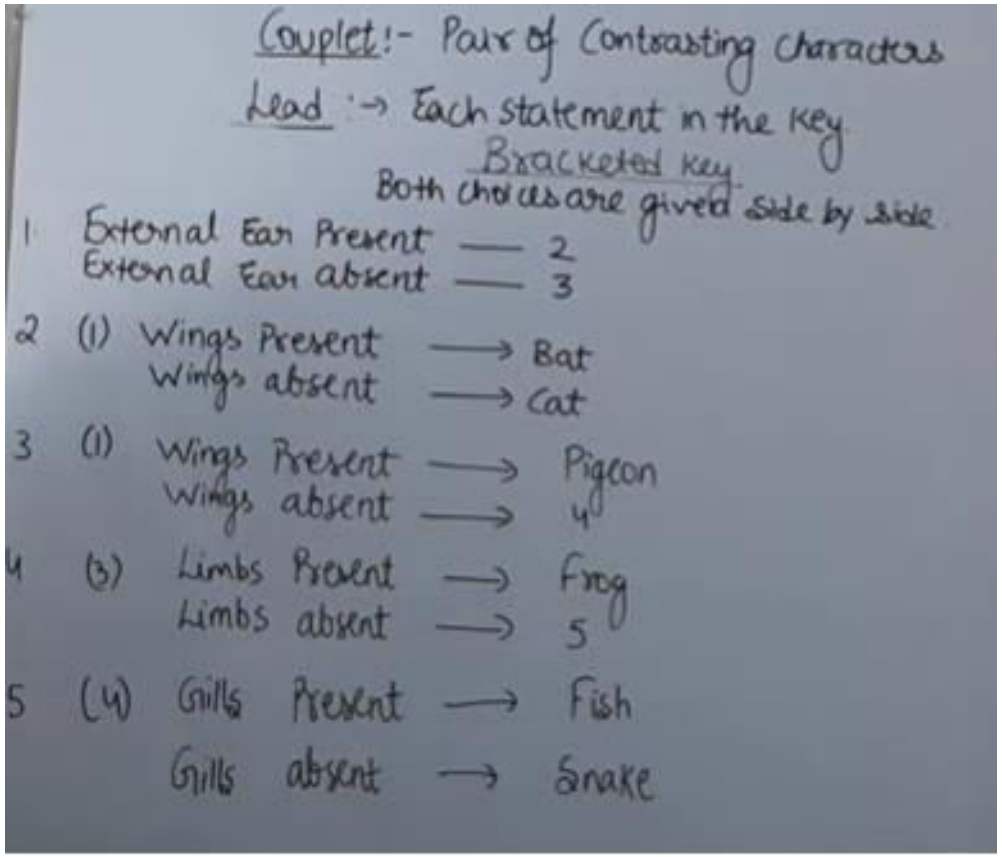
Identification Keys and Catalogues

Characters	Vertebrates
(1) External ear present	2
(1) External ear absent	3
(2) Wings present	BAT
(2) Wings absent	CAT
(3) Wings present	BIRDS
(3) Wing absent	4
(4) Gills present	FISH
(4) Gills absent	5
(5) Limbs Present	FROG
(5) Limbs absent	SNAKE

Bracketed Key

- Pair of contrasting characters are taken in brackets
- Simple to use
- It is most commonly used in the filed of taxonomic study.

Identification Keys and Catalogues



https://www.youtube.com/watch?v=Uqbm-8ZIqwo

Bracketed Key

- Pair of contrasting characters are taken in brackets
- Simple to use
- It is most commonly used in the filed of taxonomic study.

Identification Keys and Catalogues

Pectoral Keys

- For non-scientists
- Simple to use
- Key with pictures
- Jungle or desert/easy for identification of plants and animals

Identification Keys and Catalogues

Polyclave Key

- Based on computer program
- Simple to use
- Plant identification
- Take photos/Google/identification

END

Principles of Systematics

Module# 90

Morphological Identifications of Model Arthropod

Morphological Identifications of Model Arthropods

- Spiders (Arthropoda: Arachnida)Most
- Most abundant natural predators
- Present in all types of habitat
- Highly diverse groups of Arthropod
- More than 46000 species
- Ranked 7th in animal Kingdom

Morphological Identifications of Model Arthropods

- Present on every continent
- Excellent predators
- Types of spiders
- Can survive without food for several months
- Low metabolic rate
- Prey switching ability

Morphological Identifications of Model Arthropods

- Prey switching
- Four pairs of legs
- Two body parts
- No antennae

END

Principles of Systematics

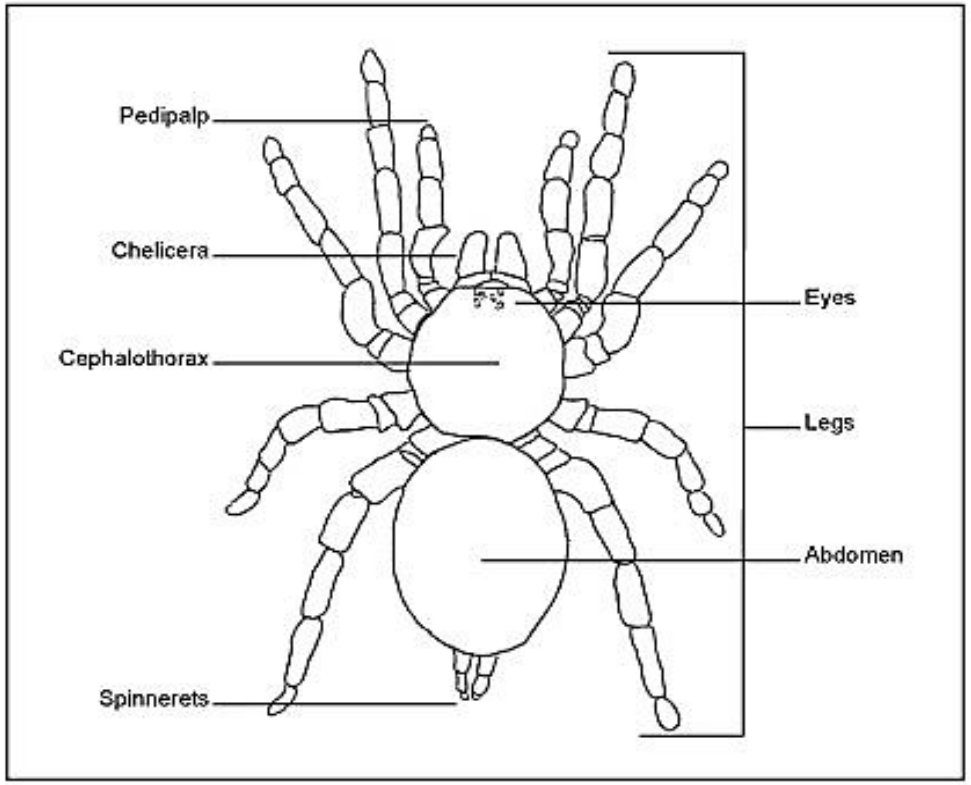
Module# 91

Morphological Identifications of Model Arthropod

Morphological Identifications of Model Arthropods

- Spiders have two body segments
- Front segment is called **Cephalothorax**
- It contain eyes, mouth parts, fangs, brain, venom gland etc.
- Legs are also connected to this parts
- Four pairs of legs/Chelicera/padipalp

Morphological Identifications of Model Arthropods



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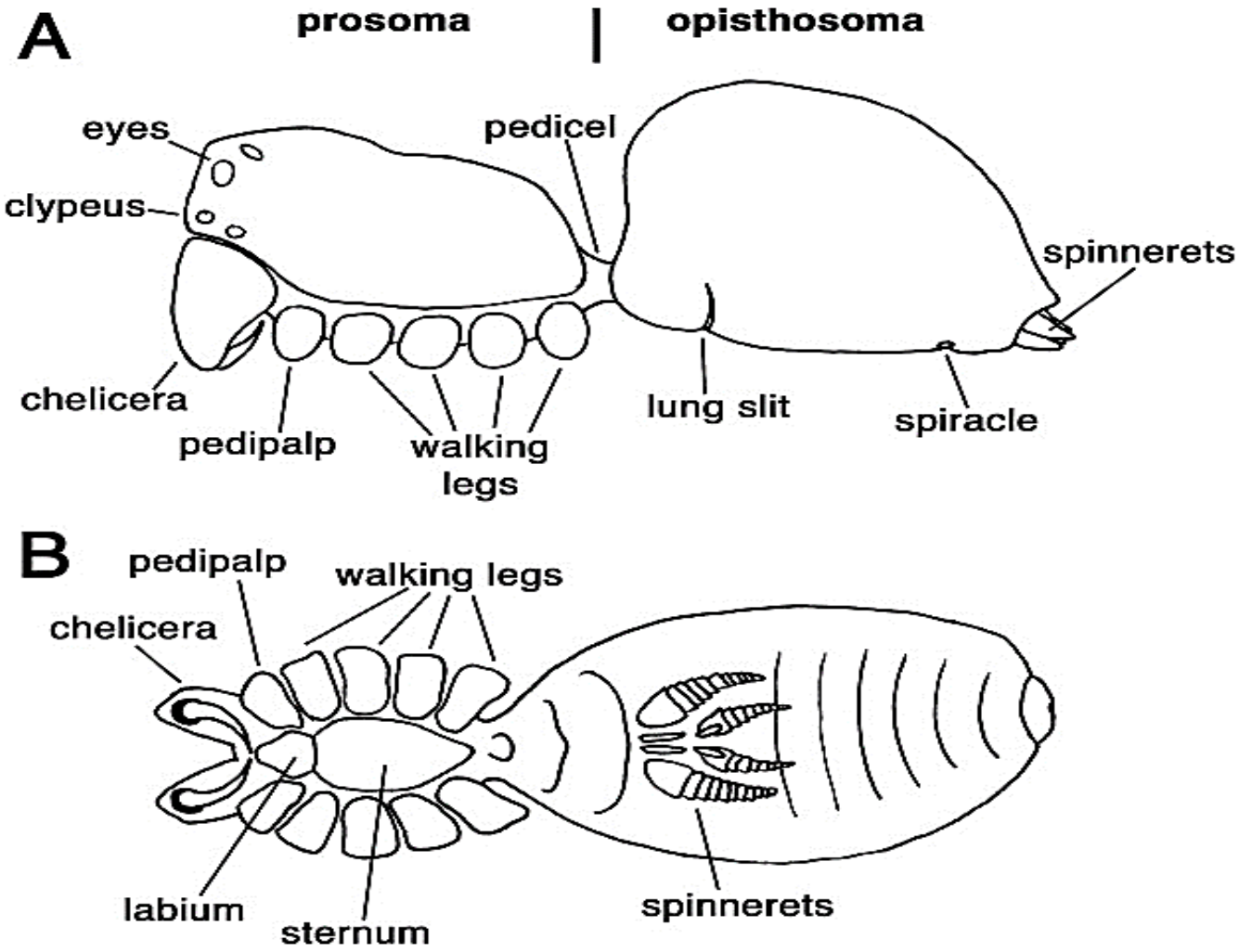
- 2nd part is abdomen
- Spinneret are attached at the end of this part
- Silk glands
- Legs are covered with many hairs
- Hairs pick up vibrations and smell from air
- Claws 2-3 at the end of legs

Morphological Identifications of Model Arthropods

- 2nd part is abdomen
- No skeleton inside their bodies
- Hard outer shell called exoskeleton
- Molting
- Males are smaller than females

Morphological Identifications of Model Arthropods



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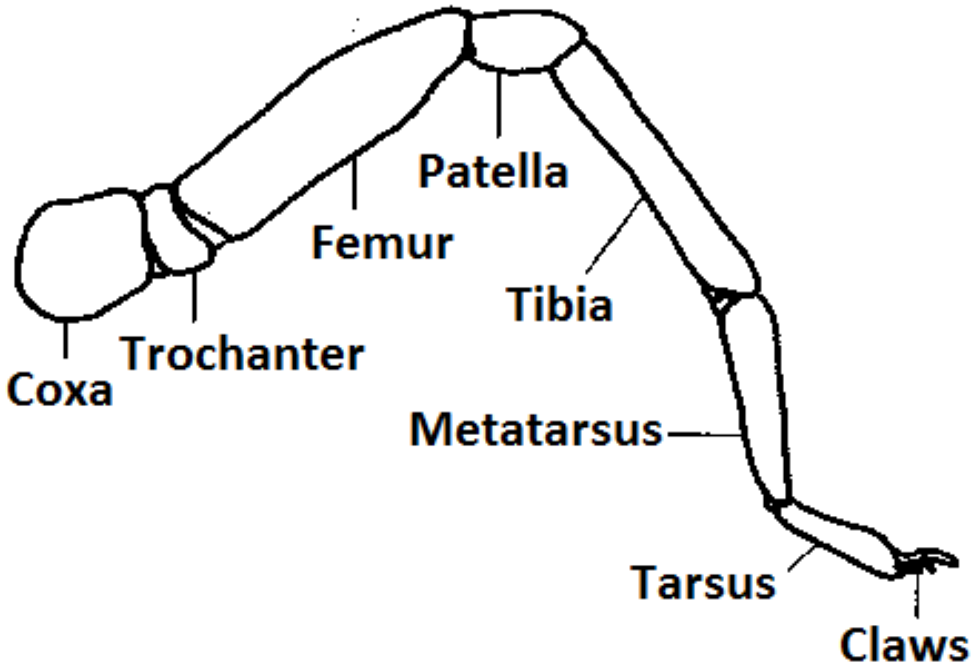
Principles of Systematics

Module# 92

Morphological Identifications of Model Arthropod

Morphological Identifications of Model Arthropods



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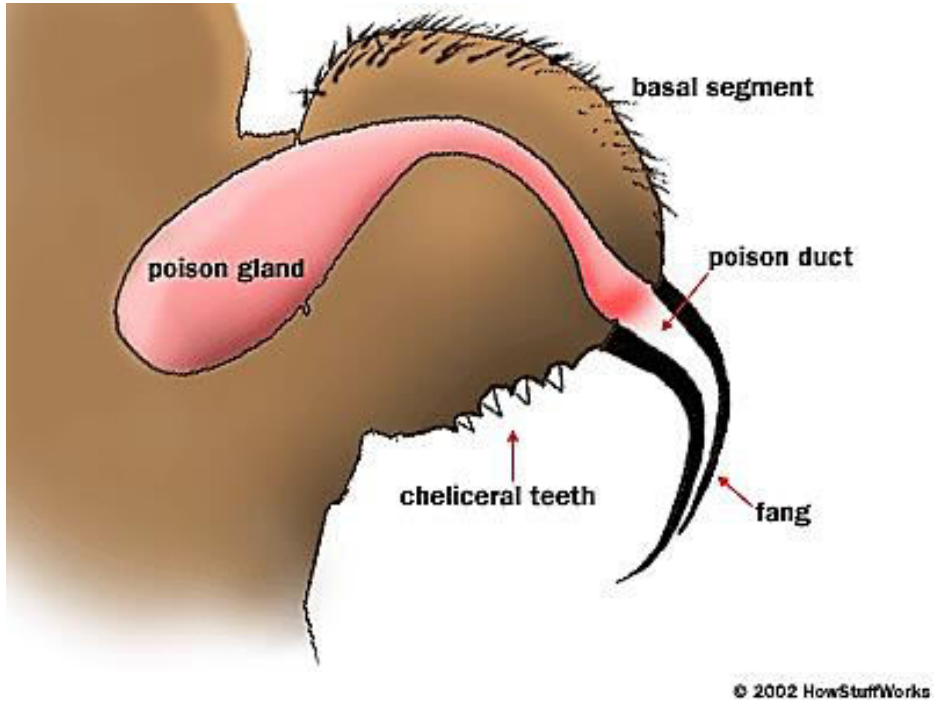
Parts of Legs

- Coxa
- Traochanter
- Femur
- Patella
- Tibia
- Metatarsus
- Tarsus
- Claws

Morphological Identifications of Model Arthropods

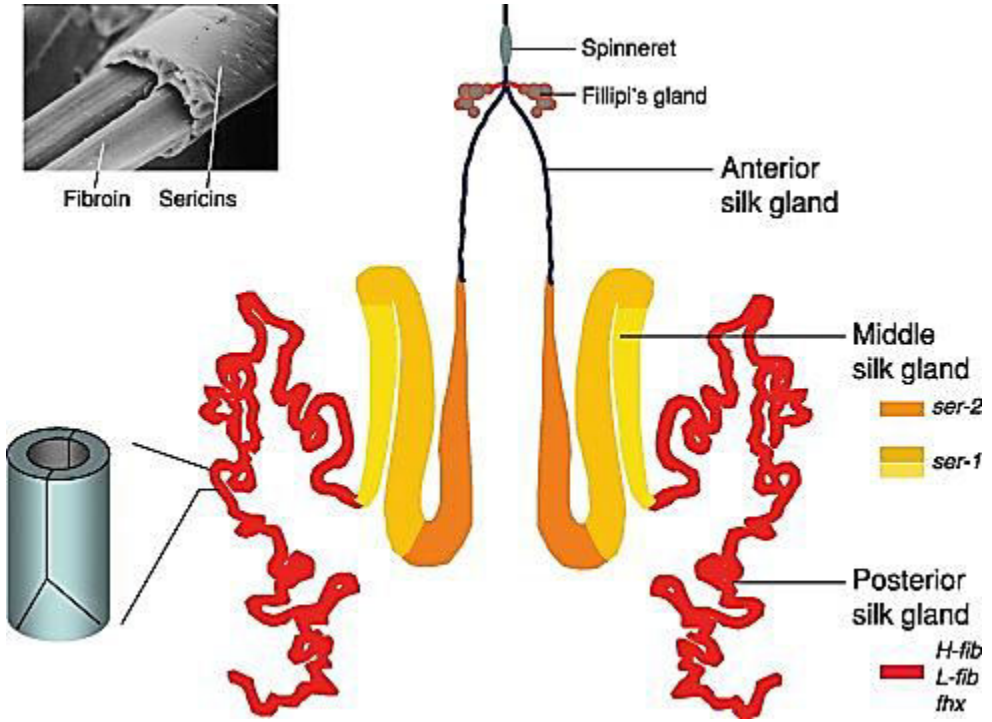


https://www.google.com/search?q=venom+gland+of+spiders&sxsrf=ALeKk03L0hhd-Hg2f_BL_MYyBTdPVy7X5w:1606355341110&source=lnms&tbm=isch&sa=X&ved=2ahUKEwjWm_X7i5_tAhWCThUIHVHSDmEQ_AUoAXoECBEQAw&biw=1360&bih=657#imgrc=JnmQ_pY9IUUA_M

Parts of Venom gland and use

- Poison gland
- Poison Duct
- Fang
- Absent in the family Uloboridae. The **glands** are located either in the chelicerae or under the carapace.

Morphological Identifications of Model Arthropods



https://www.google.com/search?q=silk+gland+of+spiders&hl=en&sxsrf=ALeKk000m9KED5vBpgr86HJjqrtJTcCQeg:1606355636107&source=lnms&tbm=isch&sa=X&ved=2ahUKEwjlmMqIjZ_tAhVtRBUIHbNdB34Q_AUoAXoECBIQAw&biw=1360&bih=657#imgrc=nLaQ7GALGB51TM

Parts of Silk gland and use

- For capturing prey
- For food storage
- Travelling
- Ballooning

Principles of Systematics

Module# 93

Morphological Identifications of Model Arthropod

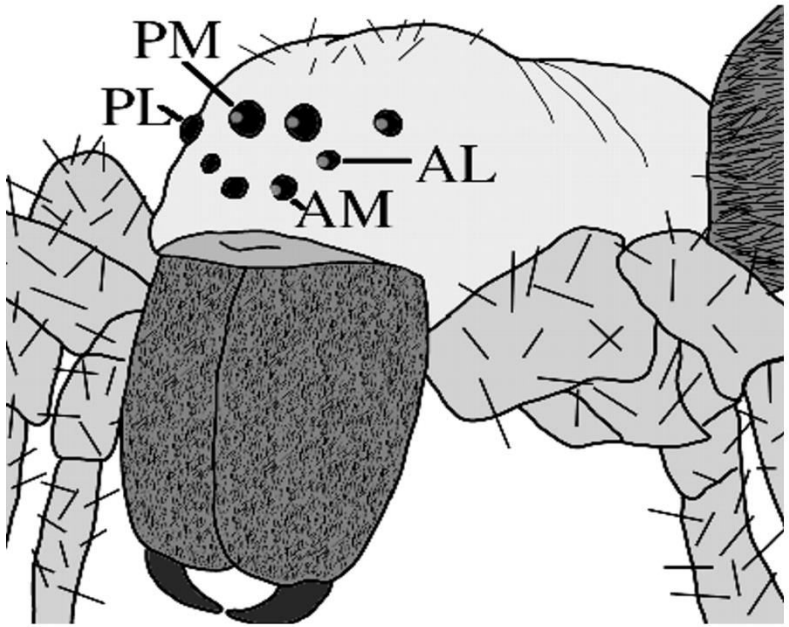
Morphological Identifications of Model Arthropods

https://www.google.com/search?q=eyes+of+spiders&hl=en&sxsrf=ALeKk03MPwHBA8hK0MNcLPNSAuHBGFOVoQ:1606355925079&source=lnms&tbm=isch&sa=X&ved=2ahUKEwj89K-Sjp_tAhU1rHEKHfiCALgQ_AUoAXoECBEQAw&biw=1360&bih=657

Eyes in spiders

- Most spiders have eight eyes. Some species have six or fewer eyes, but they always come in an even number.
- Some species of spiders, such as those that live in caves or under the soil, have no eyes at all. Even those species with eight eyes don't usually see very well.

Morphological Identifications of Model Arthropods

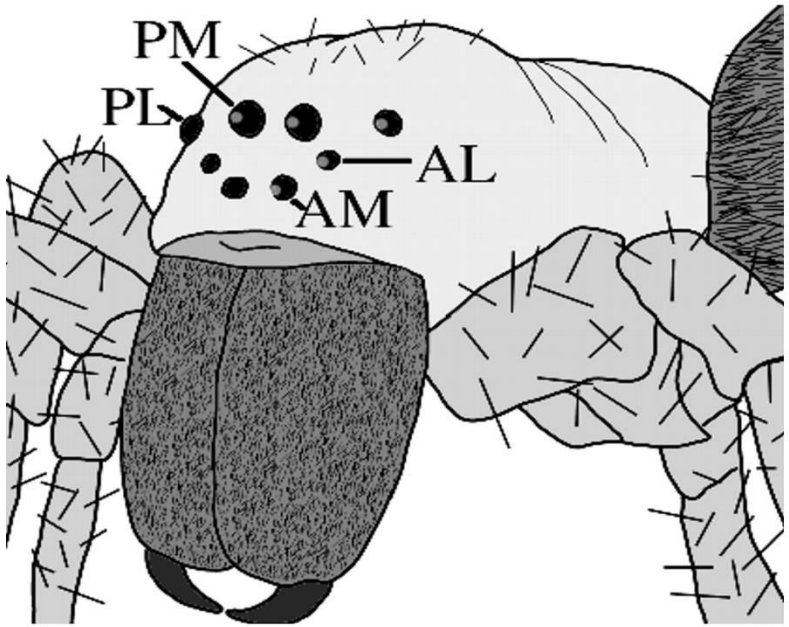


https://www.google.com/search?q=eyes+of+spiders&hl=en&sxsrf=ALeKk03MPwHBA8hK0MNcLPNSAuHBGFOVoQ:1606355925079&source=lnms&tbm=isch&sa=X&ved=2ahUKEwj89K-Sjp_tAhU1rHEKHfiCALgQ_AUoAXoECBEQAw&biw=1360&bih=657

Eyes in spiders

- Important Taxonomic character
- Eye arrangement in different families is unique

Morphological Identifications of Model Arthropods



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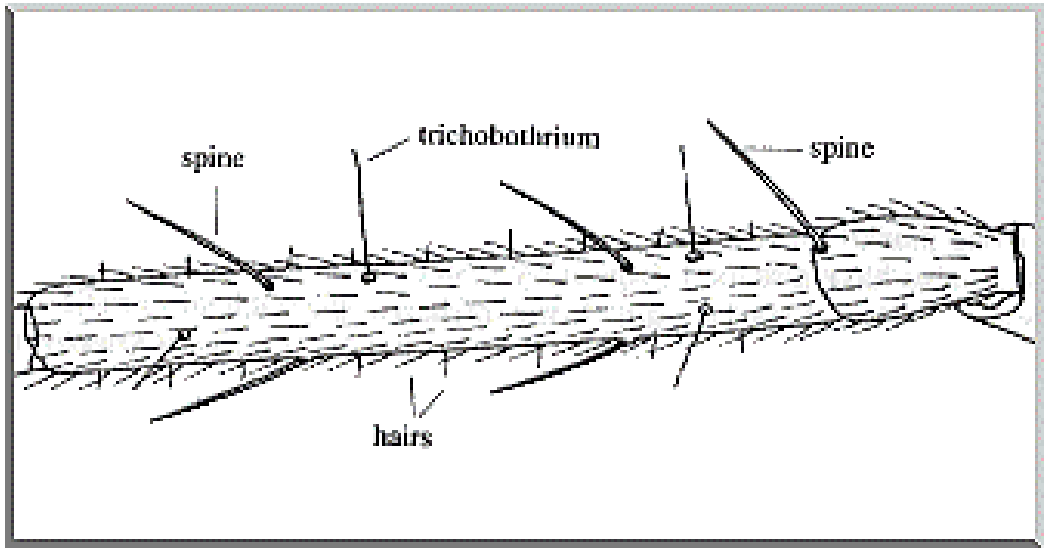
Eyes in spiders

- 99% of all spiders have 8 eyes
- 1% nearly all have 6, but there are a few exceptions.
- Varying number of eyes in the same spider family.
- For example, there are spiders in Cybaeidae that have eight eyes, six eyes, and two eyes

Morphological Identifications of Model Arthropods



https://www.google.com/search?q=sensory+structures+in+spiders&hl=en&sxsrf=ALeKk00SfvWnMast8XUR1tzxfi9s0_SzWA:1606356825967&source=lnms&tbm=isch&sa=X&ved=2ahUKEwiw2fm_kZ_tAhWdQxUIHQyYBj4Q_AUoAXoECBEQAw&biw=1360&bih=657#imgrc=ZPrWQHreEBXZcM&imgdii=9tD2Mquz-ppQyM

Sensory structures

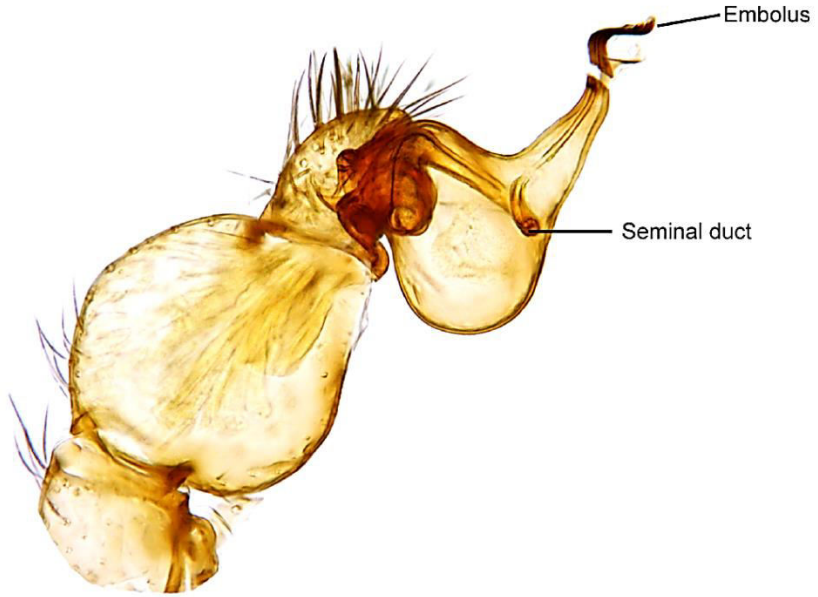
- **Spiders**, in fact, do taste, and also smell, through special **sensory organs** on their legs, as well as on their pedipalps.
- And they hear – or, more specifically, they sense vibrations – through hairs and tiny slits distributed over much of their body.
- A **spider's** sensitivity to vibrations is finely tuned.

Principles of Systematics

Module# 94

Morphological Identifications of Model Arthropod

Morphological Identifications of Model Arthropods



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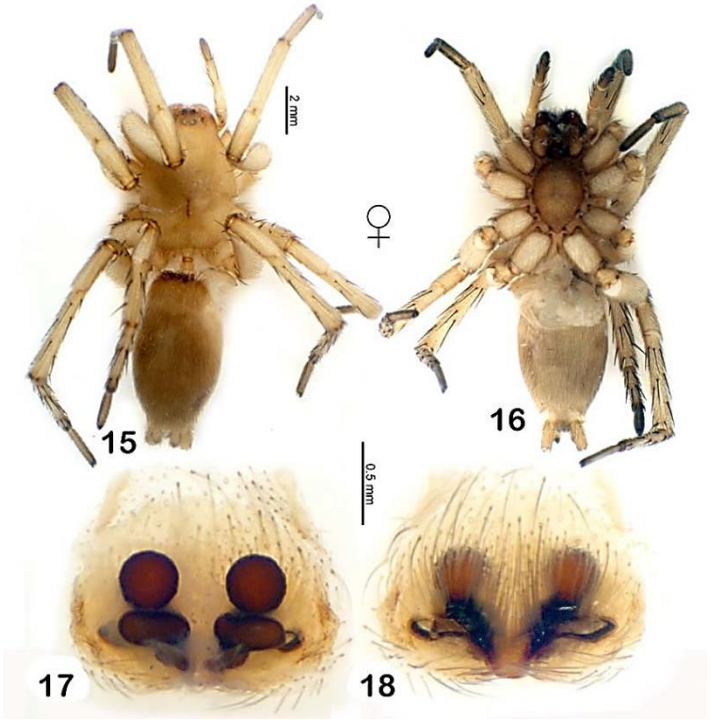
Padipalp/Epigyne

- **Pedipalps** (commonly shortened to palps or palpi) are the second pair of appendages of chelicerates

- **Epigyne** or epigynum is the external genital structure of female spiders.

Morphological Identifications of Model Arthropods



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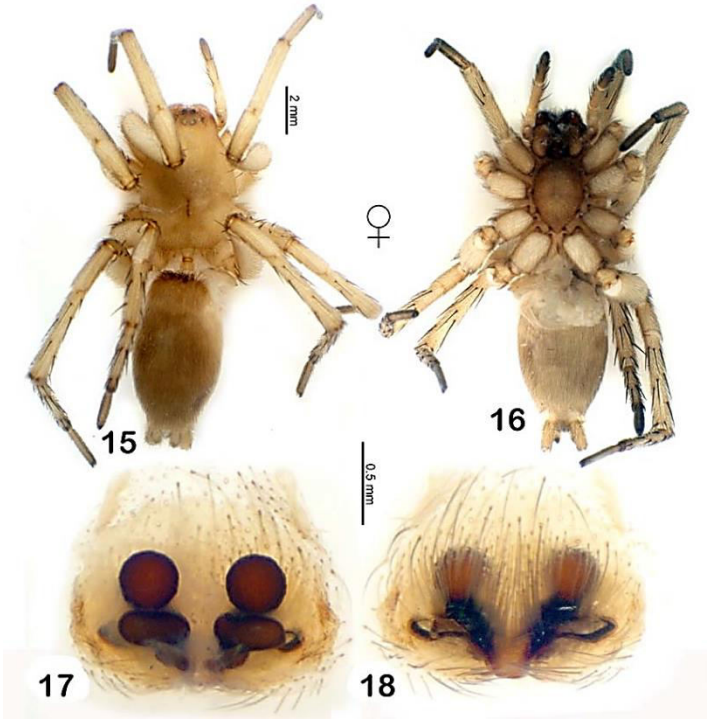
Padipalp/Epigyne

- As the **epigyne** varies greatly in form in different species, even in closely related ones.
- Distinctive characteristic for recognizing species.

Morphological Identifications of Model Arthropods



https://www.google.com/search?q=eyes+of+spiders&hl=en&sxsrf=ALeKk03MPwHBA8hK0MNcLPNSAuHBGFOVoQ:1606355925079&source=lnms&tbm=isch&sa=X&ved=2ahUKEwj89K-Sjp_tAhU1rHEKHfiCALgQ_AUoAXoECBEQAw&biw=1360&bih=657

Reproduction in Spiders

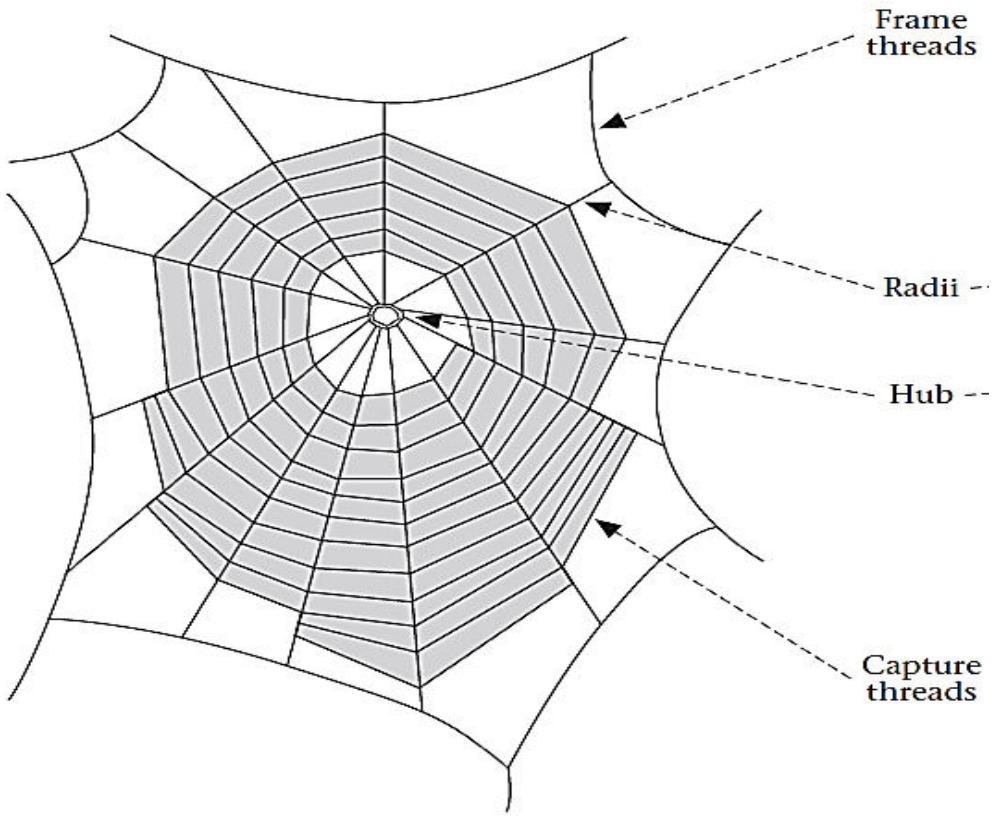
- Male find a female
- A male spider deposits his sperm into a sperm web, holding it in his palps.
- Female spiders produce either one egg sac containing several to a thousand eggs or several egg sacs each with successively fewer eggs. Females of many species die after producing the last egg sac.

Principles of Systematics

Module# 95

Morphological Identifications of Model Arthropod

Morphological Identifications of Model Arthropods



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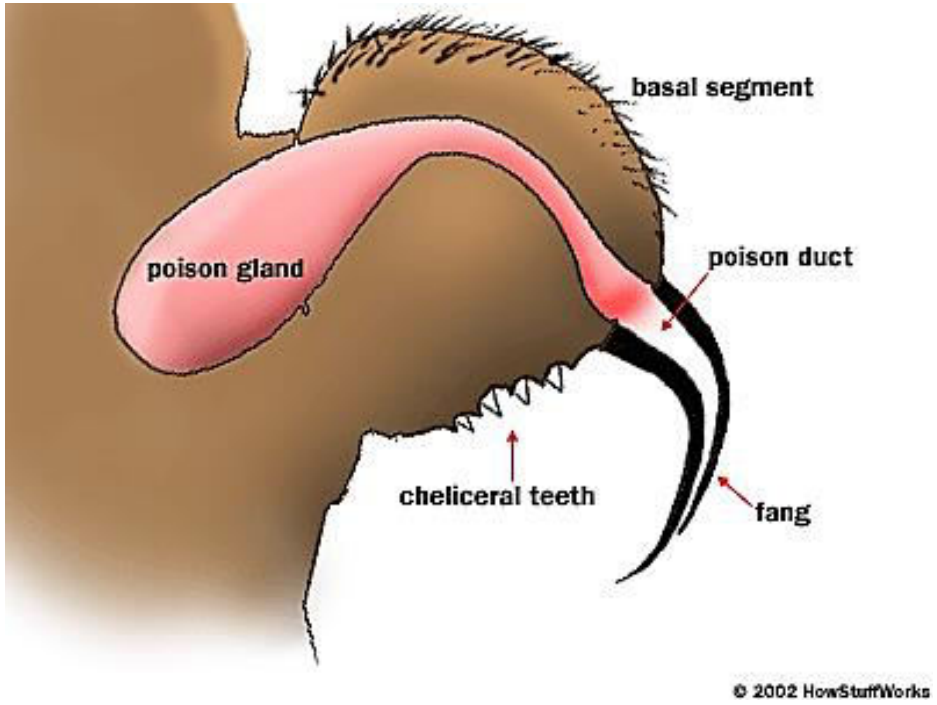
Spider web

- Orb-web spiders
- Hunters
- Design of web
- Web/prey size
- Wind director/prey density
- Repair/vibrations/self-eating/sticky nature/quality of silk

Morphological Identifications of Model Arthropods



https://www.google.com/search?q=venom+gland+of+spiders&sxsrf=ALeKk03L0hhd-Hg2f_BL_MYyBTdPVy7X5w:1606355341110&source=lnms&tbm=isch&sa=X&ved=2ahUKEwjWm_X7i5_tAhWCThUIHVHSDmEQ_AUoAXoECBEQAw&biw=1360&bih=657#imgrc=JnmQ_pY9IUUA_M

Morphological Identifications of Model Arthropods

- 2nd part is abdomen
- No skeleton inside their bodies
- Hard outer shell called exoskeleton
- Molting
- Males are smaller than females

END

Principles of Systematics

Module# 96

Principles of Systematics

Identifications of Family Salticidae

Identification of Family Salticidae

https://www.google.com/search?q=salticidae+family+eyes&tbm=isch&ved=2ahUKEwjbw9fQnK3tAhVE3hoKHZygBl4Q2-cCegQIABAA&oq=salticidae+family+eyes&gs_lcp=CgNpbWcQAzoGCAAQCBAeUKuKAVi9lQFg9JgBaABwAHgBgAG5BogB8g-SAQkzLTIuMS4wLjGYAQCgAQGqAQtnd3Mtd2l6LWltZ8ABAQ&sclient=img&ei=EnLGX9vyCsS8a5zBmvAF&bih=600&biw=1360

- As of 2019, it contained over 600 described genera
- Over 6000 described species.
- Largest family of spiders
- 13% of all species

Identification of Family Salticidae

https://www.google.com/search?q=salticidae+family+eyes&tbm=isch&ved=2ahUKEwjbw9fQnK3tAhVE3hoKHZygBl4Q2-cCegQIABAA&oq=salticidae+family+eyes&gs_lcp=CgNpbWcQAzoGCAAQCBAeUKuKAVi9lQFg9JgBaABwAHgBgAG5BogB8g-SAQkzLTIuMS4wLjGYAQCgAQGqAQtnd3Mtd2l6LWltZ8ABAQ&sclient=img&ei=EnLGX9vyCsS8a5zBmvAF&bih=600&biw=1360

- Jumping spiders are generally recognized by their eye pattern.
- All jumping spiders have four pairs of eyes, with the anterior median pair being particularly large.
- Best vision among arthropods

Identification of Family Salticidae

- The legs are not covered with any very prominent spines.
- Their front four legs generally are larger than the hind four
- Not held in the outstretched-arms attitude characteristic of the Thomisidae.
- The body length from 1 to 25 mm (0.04–0.98 in).

Identification of Family Salticidae

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- Prior to jumping, the spider attaches a silk thread to the surface beneath it, so it can climb quickly back to its perch if needed.
- Jumping spiders hunt and feed on small insects.
- All are carnivorous, but a few species also eat pollen and nectar.

Identification of Family Salticidae

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- Female jumping spiders build a silk case around their eggs and often stand guard over them until they hatch.
- Young jumping spiders emerge from the egg sac looking like miniature versions of their parents.
- They molt and grow into adulthood.

Principles of Systematics

Module# 97

Principles of Systematics

Identifications of Family Salticidae

Identification of Family Salticidae

- Jumping spider can jump quite far, achieving distances more than 50 times its body length.
- Legs not strong or muscular in appearance.
- Salticids are able to quickly increase the blood pressure in their

Identification of Family Salticidae

- Some jumping spiders mimic other insects such as ants.
- Others are able to camouflage themselves to blend into their surroundings, helping them sneak up on prey.
- Salticids hunt diurnally as a rule, which is consistent with their

Identification of Family Salticidae

https://www.google.com/search?q=salticidae+family+eyes&tbm=isch&ved=2ahUKEwjbw9fQnK3tAhVE3hoKHZygBl4Q2-cCegQIABAA&oq=salticidae+family+eyes&gs_lcp=CgNpbWcQAzoGCAAQCBAeUKuKAVi9lQFg9JgBaABwAHgBgAG5BogB8g-SAQkzLTIuMS4wLjGYAQCgAQGqAQtnd3Mtd2l6LWltZ8ABAQ&sclient=img&ei=EnLGX9vyCsS8a5zBmvAF&bih=600&biw=1360

- Although jumping spiders are generally carnivorous.
- Many species have been known to include nectars in their diets and one species, *Bagheera kiplingi* feeds primarily on plant matter

Identification of Family Salticidae

https://www.google.com/search?q=salticidae+family+eyes&tbm=isch&ved=2ahUKEwjbw9fQnK3tAhVE3hoKHZygBl4Q2-cCegQIABAA&oq=salticidae+family+eyes&gs_lcp=CgNpbWcQAzoGCAAQCBAeUKuKAVi9lQFg9JgBaABwAHgBgAG5BogB8g-SAQkzLTIuMS4wLjGYAQCgAQGqAQtnd3Mtd2l6LWltZ8ABAQ&sclient=img&ei=EnLGX9vyCsS8a5zBmvAF&bih=600&biw=1360

- During courtship "dance“ colored or iridescent parts of the body are displayed.
- In addition to the display of colors, jumping spiders perform complex sliding, vibrational, or zigzag movements to attract females.
- Many males have

Identification of Family Salticidae

- These amplified sounds presented to the females resemble buzzes or drum rolls.
- Species vary greatly in visual and vibratory components of courtship.
- If receptive to the male, the female assumes a passive, crouching position.
- In some species, the

Identification of Family Salticidae

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- The male then extends his front legs towards the female to touch her.
- If the female remains receptive, the male climbs on her back and inseminates her with his palps.

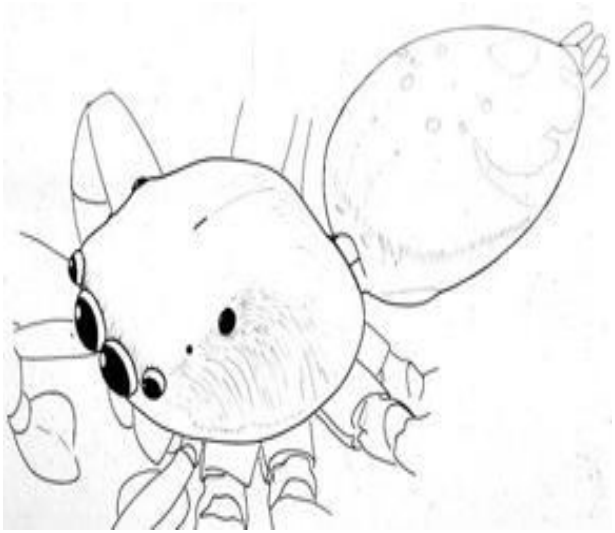
Principles of Systematics

Module# 98

Principles of Systematics

Identifications of Family Salticidae

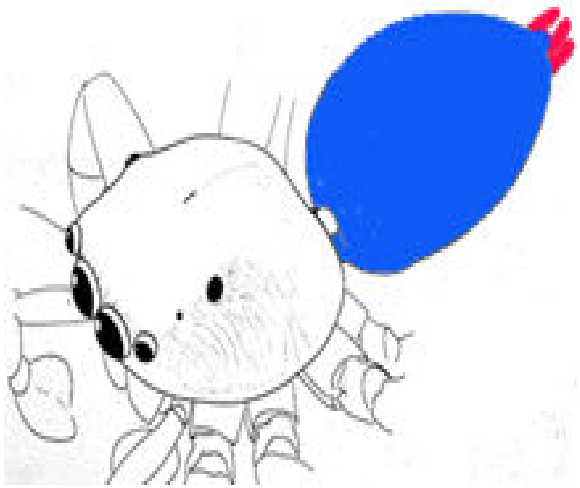
Identification of Family Salticidae



- Note that there are eight eyes, four in front and four on top.

http://tolweb.org/accessory/Jumping_Spider_External_Anatomy?acc_id=1952

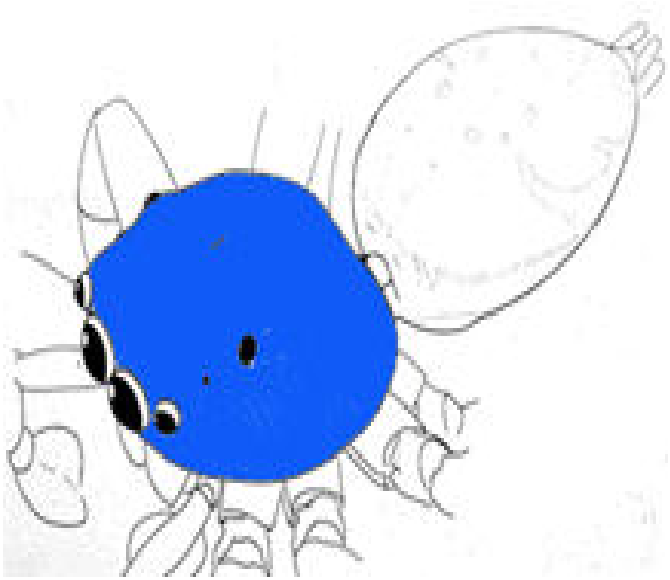
Identification of Family Salticidae



http://tolweb.org/accessory/Jumping_Spider_External_Anatomy?acc_id=1952

- The body is divided into two main parts, the posterior of which is the opisthosoma or abdomen, shown in blue:
- The spinnerets are shown in red. These are the silk-producing appendages.
- Jumping spiders, although they do not usually use silk to catch

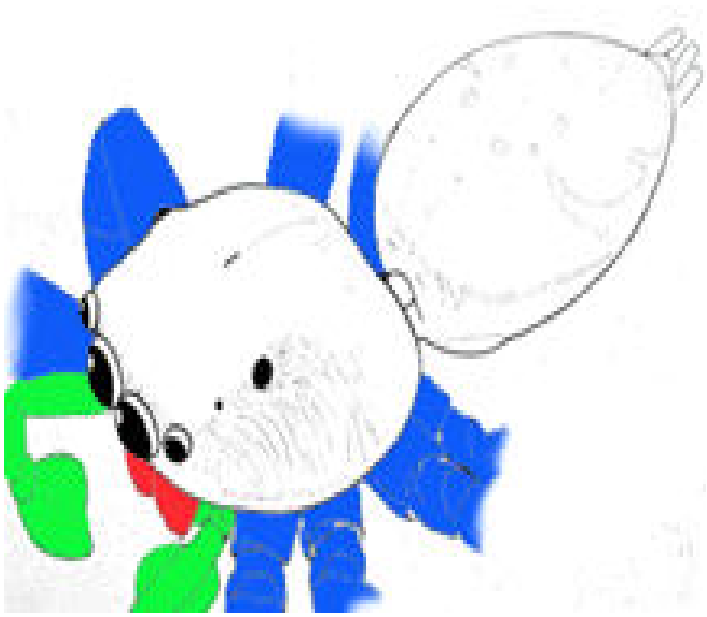
Identification of Family Salticidae



- The anterior body part is the prosoma or cephalothorax.
- It is covered above by a shell or carapace:

http://tolweb.org/accessory/Jumping_Spider_External_Anatomy?acc_id=1952

Identification of Family Salticidae



http://tolweb.org/accessory/Jumping_Spider_External_Anatomy?acc_id=1952

- Belonging to the prosoma are the main appendages:
- In this diagram the chelicerae (jaws with fangs) are shown in red, the palpi in green, and the legs in blue.

Principles of Systematics

Module# 99

Principles of Systematics

Identifications of Family Oxyopidae

Identification of Family Oxyopidae

https://www.google.com/search?q=lynx+spider&tbm=isch&ved=2ahUKEwiMyJWp07ftAhWO0YUKHcLSCQoQ2-cCegQIABAA&oq=lynx+spider&gs_lcp=CgNpbWcQA1DijCdY4ownYJCPJ2gAcAB4AIABAIgBAJIBAJgBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=mOnLX4ycJ46jlwTCpadQ&bih=657&biw=1360#imgrc=Zf71CWvaO3PBXM&imgdii=gpTVaNJ4dzXzsM

- Lynx spiders
- The common name refers to their quickness and agility; lynx-like in nature.
- Family Oxyopidae Thorell, 1870
- 9 genera worldwide, 419 species (WSC)

Identification of Family Oxyopidae

https://www.google.com/search?q=lynx+spider&tbm=isch&ved=2ahUKEwiMyJWp07ftAhWO0YUKHcLSCQoQ2-cCegQIABAA&oq=lynx+spider&gs_lcp=CgNpbWcQA1DijCdY4ownYJCPJ2gAcAB4AIABAIgBAJIBAJgBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=mOnLX4ycJ46jlwTCpadQ&bih=657&biw=1360#imgrc=Zf71CWvaO3PBXM&imgdii=gpTVaNJ4dzXzsM

- Oxyopid body length ranges from 3-25mm.
- Primarily tropical and subtropical in distribution, but also in temperate zones
- Primarily vegetation dwellers found on grasses, shrubs, and trees.

Identification of Family Oxyopidae

https://www.google.com/search?q=lynx+spider&tbm=isch&ved=2ahUKEwiMyJWp07ftAhWO0YUKHcLSCQoQ2-cCegQIABAA&oq=lynx+spider&gs_lcp=CgNpbWcQA1DijCdY4ownYJCPJ2gAcAB4AIABAIgBAJIBAJgBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=mOnLX4ycJ46jlwTCpadQ&bih=657&biw=1360#imgrc=Zf71CWvaO3PBXM&imgdii=gpTVaNJ4dzXzsM

- Feed on Insects and spiders
- Oxyopids are active hunters during both day and night.
- They use their vision to detect and legs to catch prey, sometimes jumping up to 2cm into the air to catch insects in flight

Identification of Family Oxyopidae

https://www.google.com/search?q=lynx+spider+eyes&sa=X&bih=657&biw=1360&hl=en&sxsrf=ALeKk003VLVpvTPYJB8licBIuXuhwHQEtA:1607200368560&tbm=isch&source=iu&ictx=1&fir=lC6Pj2HMuL3D0M%252CluWdmL9rJcOQ1M%252C_&vet=1&usg=AI4_-kSrgqaPxx4TYZ8ivqKJD-_e84Bo2Q&ved=2ahUKEwjf37P417ftAhWHWxUIHUiPBRcQ9QF6BAgLEAE#imgrc=lC6Pj2HMuL3D0M

- Some lynx's also capture prey by jumping from a stationary location much like a Salticid.
- An interesting member of *Tapinillus* is a web building, social lynx spider.
- Eight eyes

Identification of Family Oxyopidae

https://www.google.com/search?q=lynx+spider+eyes&sa=X&bih=657&biw=1360&hl=en&sxsrf=ALeKk003VLVpvTPYJB8licBIuXuhwHQEtA:1607200368560&tbm=isch&source=iu&ictx=1&fir=lC6Pj2HMuL3D0M%252CluWdmL9rJcOQ1M%252C_&vet=1&usg=AI4_-kSrgqaPxx4TYZ8ivqKJD-_e84Bo2Q&ved=2ahUKEwjf37P417ftAhWHWxUIHUiPBRcQ9QF6BAgLEAE#imgrc=lC6Pj2HMuL3D0M

- Most species make little use of webs
- Instead spending their lives as hunting spiders on plants.
- Many species frequently visit flowers in particular, ambushing pollinators, much as crab spiders do
- Most of them have

Identification of Family Oxyopidae

https://www.google.com/search?q=lynx+spider+eyes&sa=X&bih=657&biw=1360&hl=en&sxsrf=ALeKk003VLVpvTPYJB8licBIuXuhwHQEtA:1607200368560&tbm=isch&source=iu&ictx=1&fir=lC6Pj2HMuL3D0M%252CluWdmL9rJcOQ1M%252C_&vet=1&usg=AI4_-kSrgqaPxx4TYZ8ivqKJD-_e84Bo2Q&ved=2ahUKEwjf37P417ftAhWHWxUIHUiPBRcQ9QF6BAgLEAE#imgrc=lC6Pj2HMuL3D0M

- Oxyopidae in general rely on keen eyesight in stalking, chasing, or ambushing prey, and also in avoiding enemies.
- Active hunters/Chase prey
- Some members of the genus *Oxyopes* are abundant enough to be

Principles of Systematics

Module# 100

Principles of Systematics

Identifications of Family Oxyopidae

Identification of Family Oxyopidae

- High carapaces
- Distinctive eyes
- Numerous spines on their legs
- Bright colors in some species
- Some have cryptic coloration.

Identification of Family Oxyopidae

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- The abdomen is usually oval shaped and tapers to a point.
- The long legged hunting spiders are capable of running very fast and jump on their prey like a cat. This is why their common name is lynx spider.
- Although their eyesight is not as good as that of the jumping spiders

Identification of Family Oxyopidae

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- The male performs a dance before the female using it legs, abdomen and especially his palps.
- The female spider spins a cocoon that she attaches to a plant.
- She will guard the eggs until they hatch.
- Identified by the very

Identification of Family Oxyopidae

END

- Their abdomen is like a pencil ending in a point.
- Some oxyopids may produce several egg sacs, while others only one.
- Lynx's are very protective of their egg sacs, guarding them avidly.

- Many will not eat while

Principles of Systematics

Module# 101

Principles of Systematics

Identifications of Family Oxyopidae

Identification of Family Oxyopidae

- Generalist predator that feeds on a wide variety of insect prey, including several economically important insect pests.

- This family is one of the most abundant beneficial spider species in gardens, yards, and agricultural fields throughout world

Identification of Family Oxyopidae

- To capture prey, the striped lynx spider will sit and wait in grassy areas and herbaceous vegetation for small insects to land close enough for them to stalk and capture.

- This spider can detect odors given off by insect prey and will spend time in areas

Identification of Family Oxyopidae

- There is evidence that spiderlings can learn specific odors during foraging and retain specific prey preferences from these experiences later in life.

- Newly hatched spiderlings can feed on nectar, which can help them live longer and reduce cannibalism

Identification of Family Oxyopidae

- Other sources of sugars, including honeydew, may be potentially important to immature spiders

- The striped lynx spider is considered a major predator of bollworm and tobacco budworm larvae in cotton fields and the southern green stink bug in soybean

Identification of Family Oxyopidae

END

- The striped lynx spider is significantly affected by insecticide use that causes decreased survival and other sublethal effects, such as causing males to decrease the amount of time they spend courting and mating with females

- Predatory potential and

Principles of Systematics

Module# 102

Principles of Systematics

Identifications of Family Araneidae

Identification of Family Araneidae

- The family Araneidae is a member of the order Araneae (Foelix, 1982).
- The Araneidae include the spiders who spin the elaborate webs most often seen.
- Araneids are one of the most successful of the spider families; they have over 3,500 species

Identification of Family Araneidae

- The spiders vary greatly in size as well, some species of *Singa* are about 2 mm in length while some of the *Nephila* can grow to over 50 mm, even killing small birds.
- Because of this great variety of species a great variety of orb web designs exists.

Identification of Family Araneidae

- Eight eyes
- Eyes are arranged in two horizontal rows
- Males are usually smaller than females
- Spiders in the family Araneidae will demonstrate several different strategies of predation.

Identification of Family Araneidae

- Some will immediately wrap their prey in the web then bite
- Some wrap, move, then bite
- Some just seize their prey with their chelicerae, move them to the hub of the web, and then wrap up their prey.

Identification of Family Araneidae

- Spiders are found in woods, caves, grasses and almost any other habitat imaginable
- Being bitten by an orb weaver is very uncommon
- Only bite if they feel threatened and trapped without a chance for escape
- A bite is often compared to a bee sting, and for most people, is nothing serious. Should be observe in their environments

END

Principles of Systematics

Module# 103

Principles of Systematics

Identifications of Family Araneidae

Identification of Family Araneidae

- Present anywhere with abundant prey and structures to support the web
- Most noticeable in late summer in fall, when webs and adults reach their largest size.
- Food sources vary, but typically any small insects they catch in

Identification of Family Araneidae

- Some of the bigger orb weavers (Argiope genus) have been observed eating small frogs and humming birds (only if ensnared within the web).

- The adult males will wander in search of a mate, and as such, typically do not stay on a web for long, if they

Identification of Family Araneidae

- The most commonly noticed orb weaver is female, since she sits on her web, feeding and eventually waiting for the males to find her.
- Toward the end of fall, the females will lay their last clutch of eggs, and then die at the first frost.
- The eggs can survive

Identification of Family Araneidae

- The warmth of the spring will cause the eggs to mature, releasing a new generation of orb weavers.
- A single egg sac can contain upwards of several hundred eggs
- Juvenile orb weavers will spin "perfect" orb

Identification of Family Araneidae

- Orb weavers are very docile, non-aggressive spiders that will flee at the first sign of a threat (typically they will run or drop off the web).

- They **are not dangerous** to people & pets, and are actually quite beneficial because they will catch and eat a lot of pest-

Identification of Family Araneidae

- Feed on medium-sized insect in the web
- Spider bite the prey, wrap it in silk, wait for it to die, then begin to eat.
- As a first step in eating, the spider will literally vomit digestive fluid over the prey.
- The prey is chewed with the "jaws" (chelicerae), and the

Identification of Family Araneidae

- The spider repeats this process as often as necessary to digest, and ingest, all but the inedible hard parts.
- What is discarded afterwards is a small ball of residue.

END

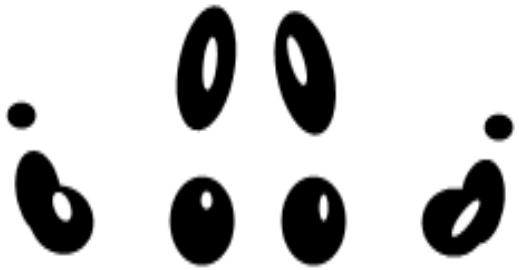
Principles of Systematics

Module# 104

Principles of Systematics

Identifications of Family Araneidae

Identification of Family Araneidae



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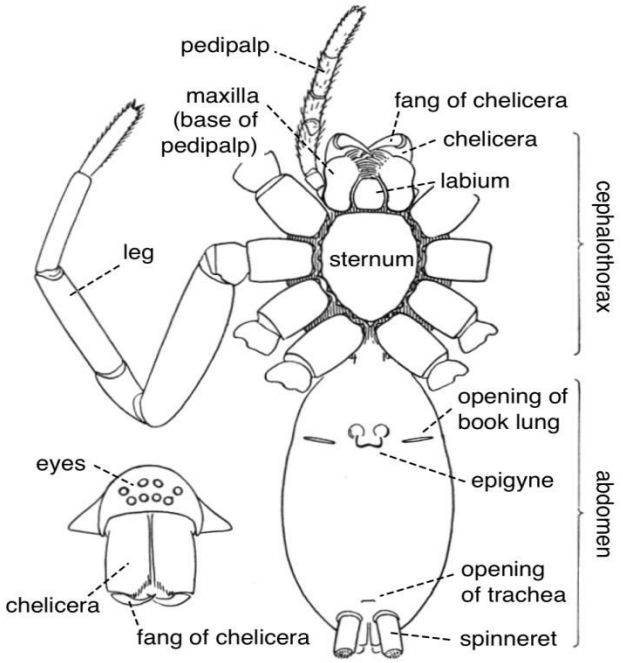
- Eight eyes
- Three tarsal claws
- The eyes in 2 rows of 4 with the lateral eyes widely separated from the medial eyes.
- The carapace is often flat with a distinct head region.

Identification of Family Araneidae



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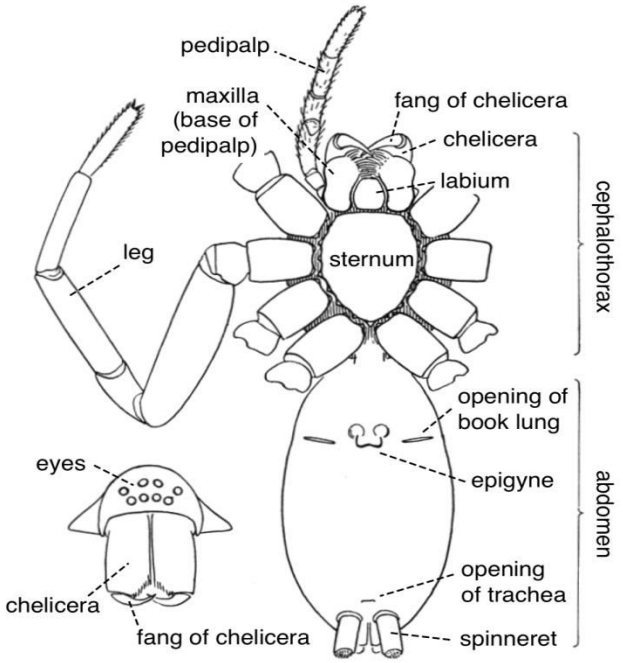
- Labium is wider than long and swollen at anterior edge
- Maxilla are widest anteriorly
- Legs with 3 claws and often furnished with strong spines and trichobothria on all segments except tarsi.

Identification of Family Araneidae



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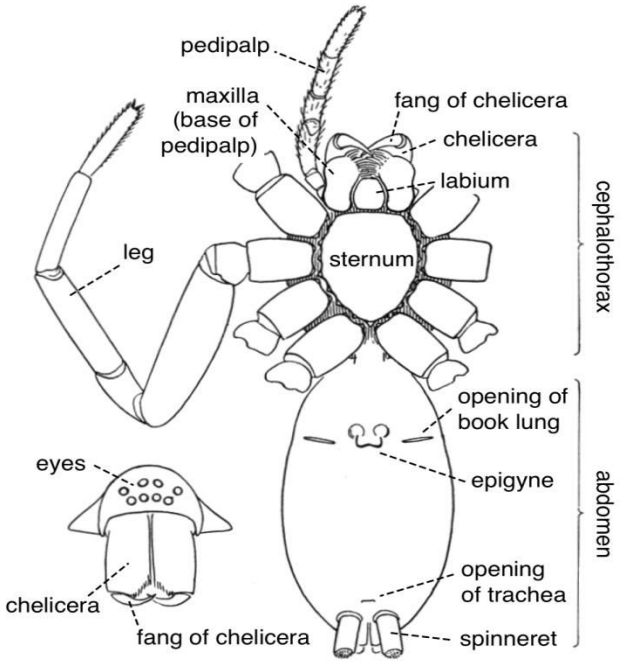
- Often legs are clearly annulated.
- The abdomen is usually globose and nearly always with species-specific often bright colour patterns.
- The tracheal spiracle is situated close to spinnerets.

Identification of Family Araneidae



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- Neoscona
- Tetragnathidae
- Regular webs/Time of web making
- Repair/new web
- Common in agro-ecosystems
- Biocontrol agent

Principles of Systematics

Module# 105

Principles of Systematics

Identifications of Family Lycosidae

Identification of Family Lycosidae

- Wolf Spiders
- Active hunters
- Ground spiders
- Some species clime plant to chase prey
- Widespread from the Arctic to the Subtropics.

Identification of Family Lycosidae

- Preferred habitat varies between species but includes open grasslands, suburban lawns, deciduous forests, deserts, coastal dunes, sandy soil and wet terrain such as marshes and swamps

Identification of Family Lycosidae

- Preferred habitat varies between species but includes open grasslands, suburban lawns, deciduous forests, deserts, coastal dunes, sandy soil and wet terrain such as marshes and swamps

- Mainly feed on insects

Identification of Family Lycosidae

https://bugguide.net/node/view/89936

- Eight dark eyes of unequal size arranged in three rows, the first having four eyes.
- The abdomen and the cephalothorax are usually as long as wide.
- The long legs have three microscopic claws at each tip.

Identification of Family Lycosidae

https://bugguide.net/node/view/89936

- Nocturnal
- Their dark mottled colors help camouflage them among the leaves.
- Do not spin webs
- Some dig burrows in the ground, others make holes under rocks, and many have

Identification of Family Lycosidae

https://bugguide.net/node/view/89936

- They are robust and agile hunters with excellent eyesight.
- Do not spin webs
- Some dig burrows in the ground, others make holes under rocks, and many have no retreat at all.
- Live and hunt alone

Identification of Family Lycosidae

https://bugguide.net/node/view/89936

- The abdomen must be held in a raised position to keep the egg case from dragging on the ground.
- However, despite this handicap, they are still capable of hunting
- Another aspect unique to wolf spiders is their

Identification of Family Lycosidae

https://bugguide.net/node/view/89936

- Spiderlings emerge from their protective silken case
- Climbed up their mother's legs and crowd onto the dorsal side of her abdomen.
- The mother carries the spiderlings for several weeks before they are

Identification of Family Lycosidae

https://bugguide.net/node/view/89936

- Wolf spiders inject venom if continually provoked.
- Symptoms of their bites include swelling, mild pain, and itching.
- American Wolf spider
- Australian wolf spiders have also been associated with

Principles of Systematics

Module# 106

Principles of Systematics

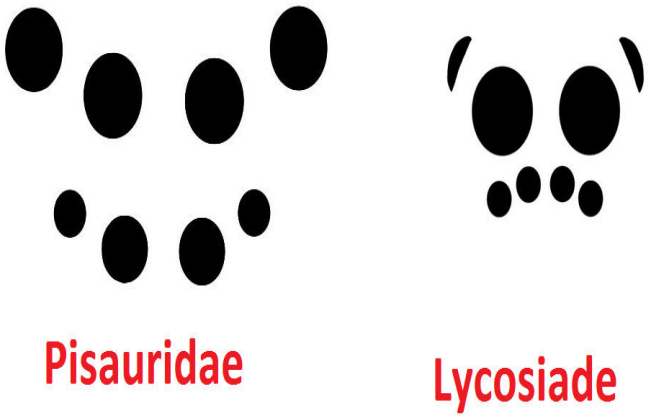
Identifications of Family Lycosidae

Identification of Family Lycosidae

https://www.biodiversityexplorer.info/arachnids/spiders/lycosidae/index.htm

- They are robust and agile hunters with excellent eyesight
- They live mostly in solitude and hunt alone
- Wolf spiders resemble Nursery Web Spiders (family Pisauridae), but wolf spiders carry their egg sacs by attaching them to their spinnerts, while the Pisauridae carry

Identification of Family Lycosidae



https://bugguide.net/node/view/89936

- Two of the wolf spider's eight eyes are large and prominent, which distinguishes them from the nursery web spiders whose eyes are all of roughly equal size.
- 10-35 mm size
- Unlike most other arachnids, which are generally blind or have poor vision, wolf spiders have excellent eyesight.

Identification of Family Lycosidae

- *Hogna* is the genus with the largest of the wolf spiders
- Found everywhere
- Body colour is brown and dull so inconspicuous on ground
- Collector notice them when spiders are disturbed by ground vibration

Identification of Family Lycosidae

- Do not spin web
- *Hippasa* and *Sosippus* are seen build funnel retreat, outside of which expended into sheet web
- These spiders wait for prey
- Prey into sheet/spiders come out/bite/kill/suck prey

Identification of Family Lycosidae

- No particular location of these webs on ground
- Danger/Extreme end of tube
- Morning travelling on vehicle/Train
- *Hippasa* webs look very conspicuous due to deposition of dew drops on them.

END

Principles of Systematics

Module# 107

Principles of Systematics

Identifications of Family Lycosidae

Identification of Family Lycosidae

- Members of genus *Lycosa* inhabit varied habitat
- Found on dry ground
- Under stones/Logs/ in crevices
- Seen in large numbers near heaps of cow dung, kept for drying

Identification of Family Lycosidae

Why near cow dung:

- For preying on insects like Diptera which come to lay their eggs in cow dung
- Other insects are attracted by smell.
- Seen at night chasing their prey and occasionally coming in human habitations for food search.

Identification of Family Lycosidae

- Some members of Lycosa prefer
- Cool, moist cavities under stone, where they lie in day time.
- *Paradosa* and *Aractosa* prefer humid atmosphere
- Frequently found at the edes of ponds, lakes, rivers etc.

Identification of Family Lycosidae

- *Paradosa birmanica. Paradosa samatrana* common near water source.
- When disturb, they hide under stones and soil crevices
- Run on water surface
-
- Jump in water to chase prey

END

Principles of Systematics

Module# 108

Principles of Systematics

Identifications of Family Lycosidae

Identification of Family Lycosidae

- *Lycosids* mainly feed on insects
- Diptera
- Lepidoptera
- Hymeniptera
- Coleoptera
- Collembola

Identification of Family Lycosidae

- May consume young spiders of their on species
- As well as different species
- Cannibalistic/importanc e
- Feed at night
- Feeding mechanism

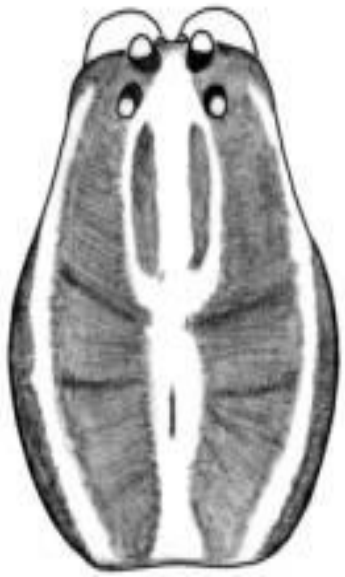
Identification of Family Lycosidae

- The family name and common name are derived from the Greek word "lycosa" meaning "wolf" due to the spiders' hunting method of ambushing and running down its prey.
- Research has shown that the Lycosidae are important in agriculture, as they are efficient controlling agents of insect pests. They are harmless to man.

Identification of Family Lycosidae

- Lycosids are often parasitized by wasps probably because they are free roaming and do not enjoy the protection of a web.

Identification of Family Lycosidae



https://www.google.com/search?q=fovea+in+lycosidae&sxsrf=ALeKk020bjj-vHKg9xn2OOMtsk0pchcuDg:1607331271899&source=lnms&tbm=isch&sa=X&ved=2ahUKEwjEwf3Lv7vtAhXMN8AKHe2ADgMQ_AUoAXoECA8QAw&biw=1360&bih=600#imgrc=--c7FlH2ulkqXM&imgdii=AIqodCv-Qo7AjM

- Centre of thoracic region is provided with fovea.
- Male has enlarged swollen palp
- Female genetalia is covered externally by heavey sclerotized plate called epigynal plate

Identification of Family Lycosidae

- Centre of thoracic region is provided with fovea.
- Male has enlarged swollen papl
- Female genetalia is covered externally by heavey sclerotized plate called epigynal plate
- Structure of genetalia is species specific.

Principles of Systematics

Module# 109

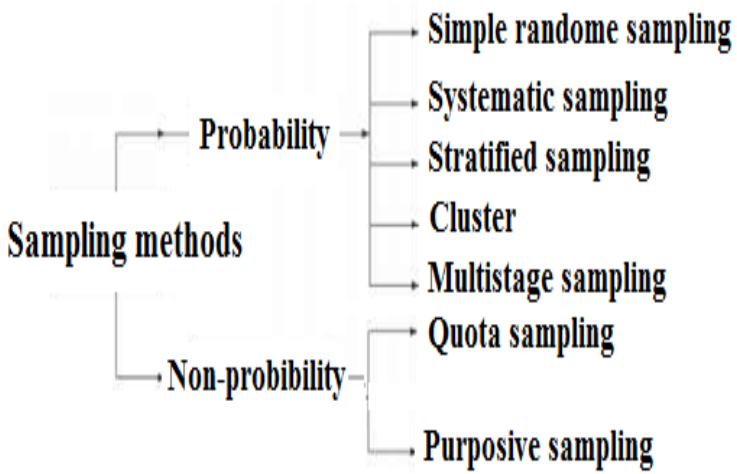
Principles of Systematics

Sampling Methods

Sampling Methods

- Probability Sampling
- Non-Probability Sampling

Sampling Methods



Probability Sampling

- Random Sampling
- Systematic Sampling
- Stratified sampling
- Cluster Sampling
- Multistage Sampling

Sampling Methods

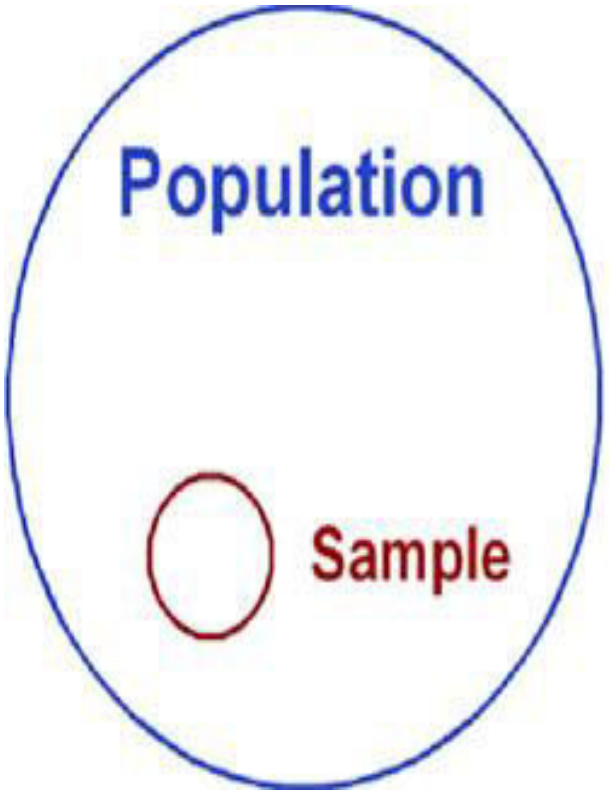
Non-Probability Sampling

- Purposive sampling
- Quota Sampling

Sampling Methods

- **Sampling:** It is the process of selecting a representative group from the population. The target population may be as board as all humans but it may be a smaller group as people who are diabetics or only the teenagers.

Sampling Methods



- **Sample:** A sample is a group of units selected from a population by a defined procedure or simply a part of a statistical population.
- The sample should be representative of the general population. 5-10% portion of the sample size of total

Sampling Methods

END

- **Population:**
  Population is the total group of individuals from which the sample might be drawn.
- All members of population possess the same basic and defined characteristics, but varying in amount or quality from one member to another. For

Principles of Systematics

Module# 110

Principles of Systematics

Random Sampling

Random Sampling

- If every individual of target population has an equal chance of selection then it will be called random sampling. Simple random sampling is the easiest form of probability sampling.

- No biasness and expert opinion is required in this method.

END

Random Sampling

- All the researcher needs to do is assure that all the members of the population are included in the list and then randomly select the desired number of subjects.

Examples:

- Selection of plants from rice fields
- Selection of students from class
- When carrying out a survey, it would be impractical to study a whole population.

Principles of Systematics

Module# 111

Principles of Systematics

Systematic Sampling

Systematic Sampling

- In this sampling method first sample from the population is taken randomly but after that samples are selected systematically at a regular interval.

Examples:

- Shopping centre
- Selection of plants
- Selection of soil sample

END

Principles of Systematics

Module# 112

Principles of Systematics

Stratified Sampling

Principles of Systematics

Module# 113

Principles of Systematics

Cluster Sampling

Cluster Sampling

- It is a sampling technique where the entire population is divided into groups, or clusters and a random sample of these clusters are selected.
- Members of one cluster have similarity with each other and also with other clusters; minor differences can be there between different clusters.
- Individual selection from each cluster is random.

Cluster Sampling

- Cluster of plants for study
- Cluster of animals

END

Cluster Sampling

- It is a sampling technique where the entire population is divided into groups, or clusters and a random sample of these clusters are selected.
- Members of one cluster have similarity with each other and also with other clusters; minor differences can be there between

END

Principles of Systematics

Module# 114

Principles of Systematics

Multistage Sampling

Multistage Sampling

- It is complicated form of cluster sampling. In this type of sampling we divide our population into groups (cluster) and each group is further divided into sub-groups in such a way that each lower group is adjusted into higher group.

Multistage Sampling

- For example in a survey of tuberculosis prevalence in world, if we divide whole world into continents, then select 10countries from each continent randomly, then 10districts from each country, 10 cities from each district, 10 villages from each city and finally 10 families from each villages randomly. This type of sampling is multistage sampling.

END

Principles of Systematics

Module# 115

Principles of Systematics

Non-Probability sampling

Non-Probability Sampling

- **Quota sampling:** In this type of sampling we divide the population intro different groups or strata based on some specific criterion but selection of individual from each group is not random.

Non-Probability Sampling

- **Purposive sampling:** In this technique researcher selects sample based on their knowledge and professional judgment. Purposive sampling can also be known as judgmental, selective or subjective sampling. For example purchase of car from market based on expert opinion or selection of staff for biology teaching in a college by a panel of subject experts.

END

Module# 116 Construction of Phenogram

- A tree diagram depicting taxonomic relationships among organisms based on overall similarity of many characteristics without regard to evolutionary history

Construction of Phenogram

Characters	A	B	C	D	E
No. of eyes	8	8	6	6	8
No. of spinnerts pairs	2	2	2	3	3
Shape of sternum	Heart shape	Heart shape	Heart shape	Oval	Oval
Size of Median eyes	Large	Small	Small	small	Large
size of chelicera	Small	small	Large	Large	Large

Construction of Phenogram

Characters	A	B	C	D	E
tail	+	+	-	+	-
Nail	-	+	+	-	-
fins	-	+	-	-	+
Wings	+	-	-	-	+
hairs	+	+	+	-	-

Construction of Phenogram

	A	B	C	D	E
A	x				
B	9	x			
C	5	2	x		
D	4	6	8	x	
E	3	5	7	6	x

Construction of Phenogram

Rules

- Start from lowest value
- Try to make maximum separate clusters
- Do not repeat
- Incase of same number multiple time, you can chose any combination
- Value of previous similarity should be

Construction of Phenogram

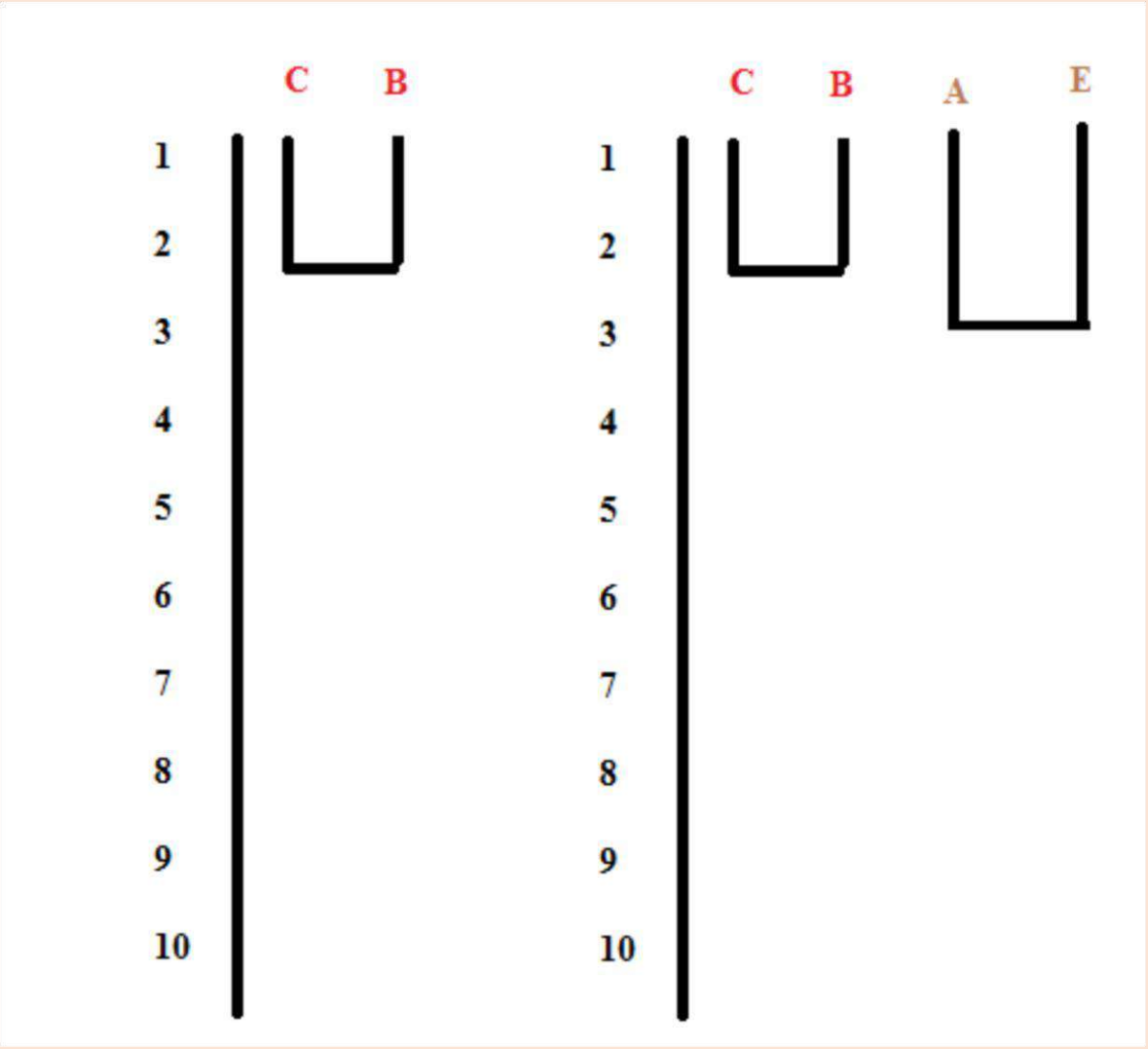
END

higher than afterwards

Construction of Phenogram

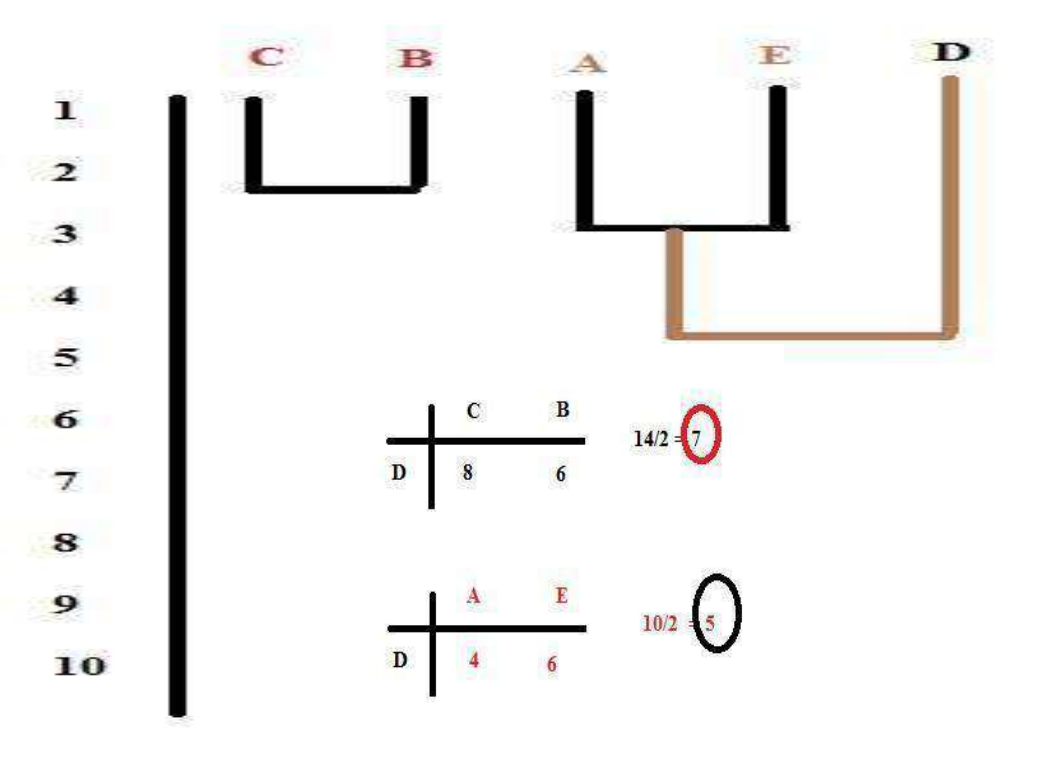
Module# 117 Construction of Phenogram

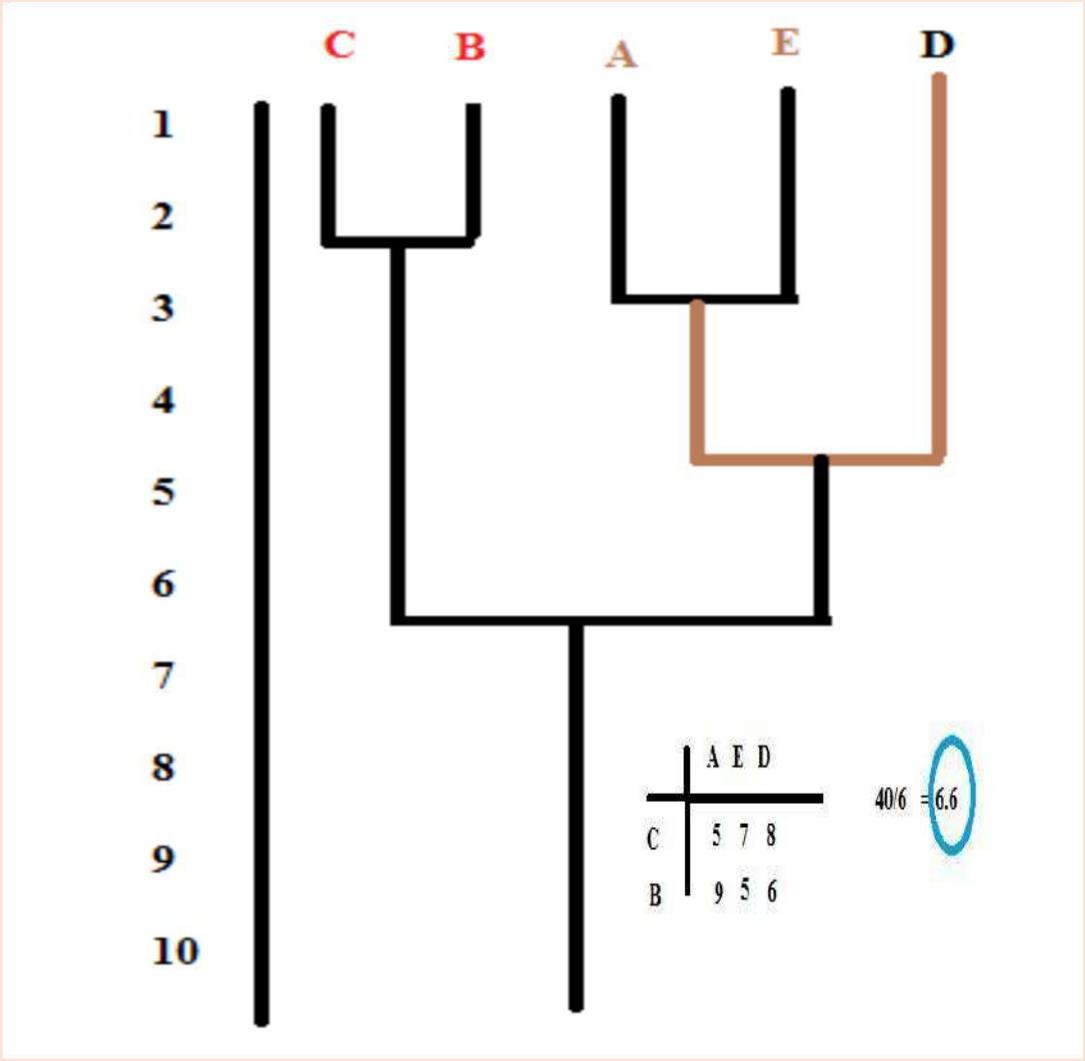
	A	B	C	D	E
A	x				
B	9	x			
C	5	2	x		
D	4	6	8	x	
E	3	5	7	6	x


C
B
1
2
3
4
5
6
7
8
9
10
C
B
A
E
1
2
3
4
5
6
7
8
9
10

Construction of Phenogram

Construction of Phenogram

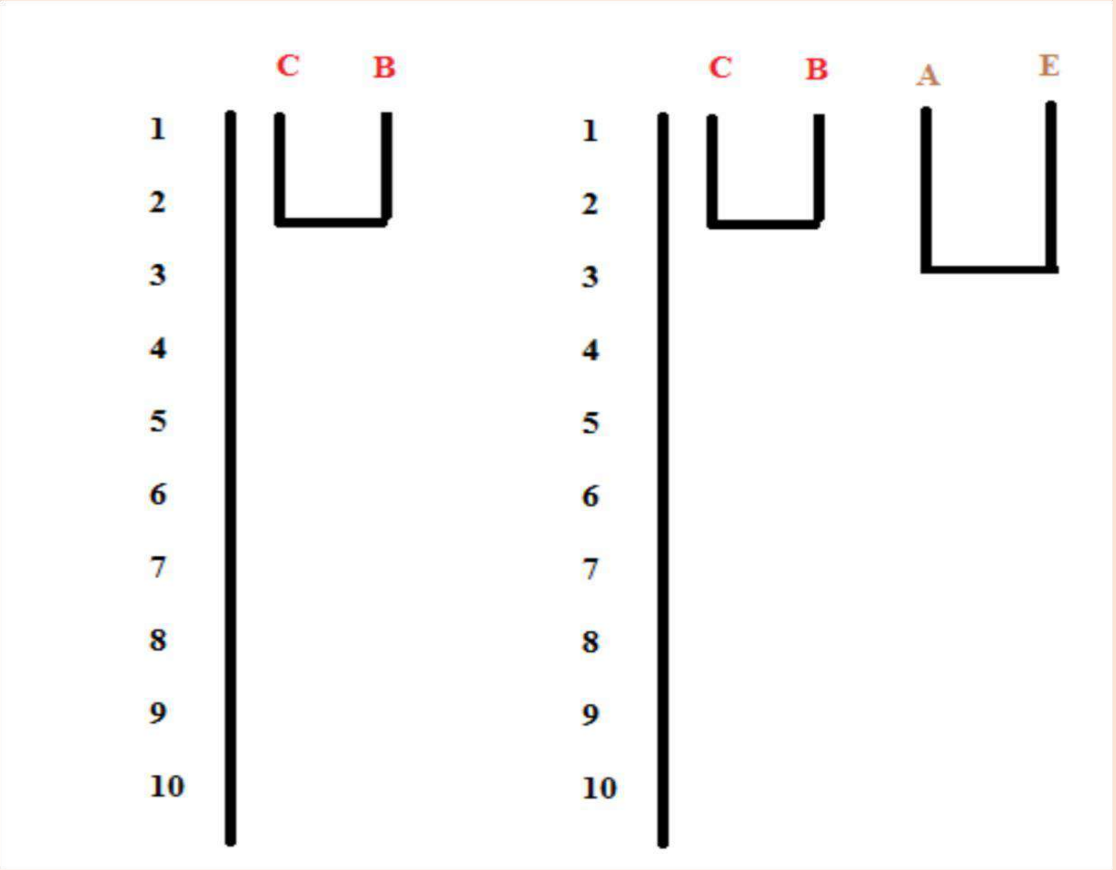



C
B
A
E
D
1
2
3
4
5
6
7
8
9
10
A E D
C 5 7 8
B 9 5 6
40/6 = 6.6

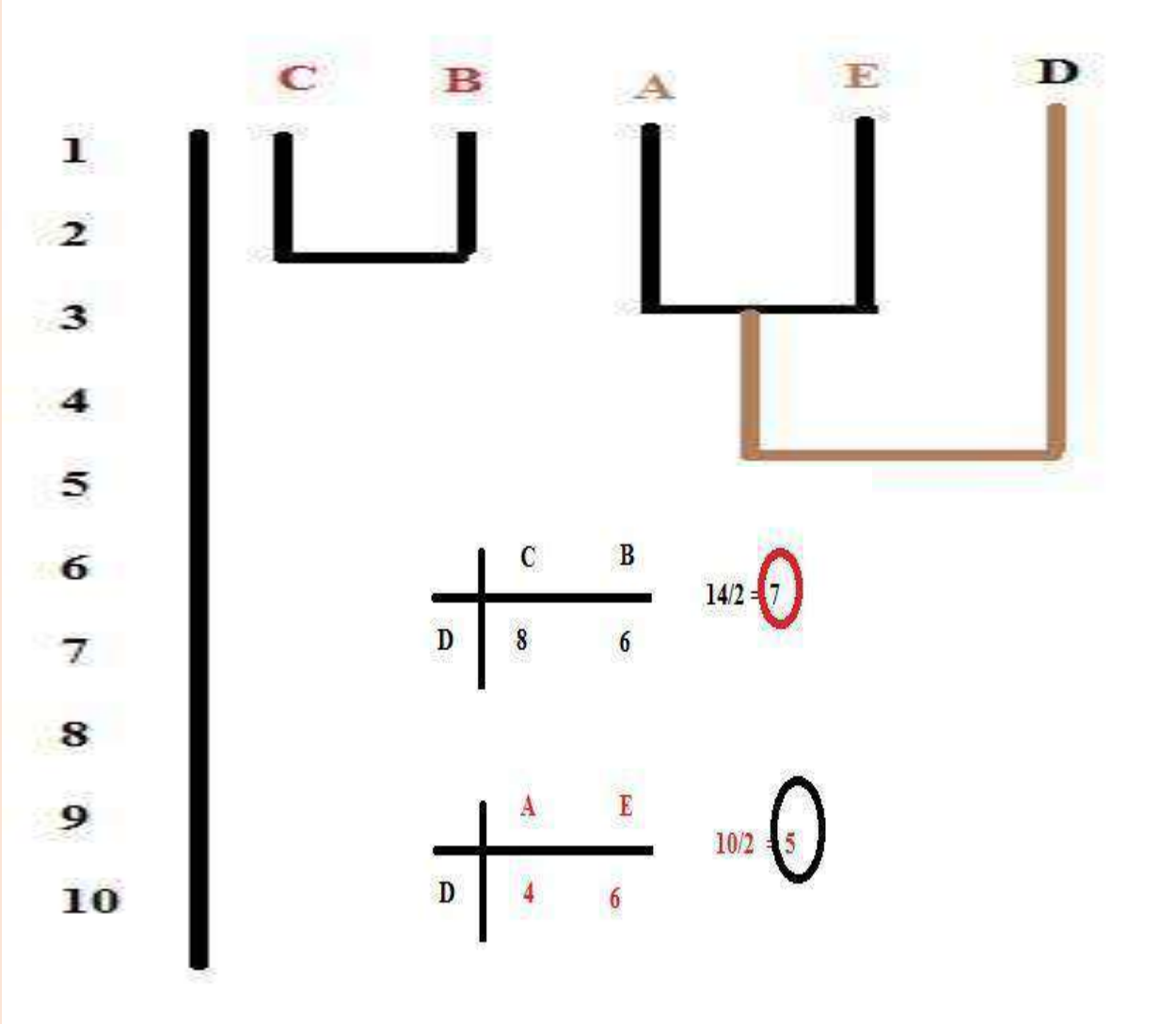
Construction of Phenogram

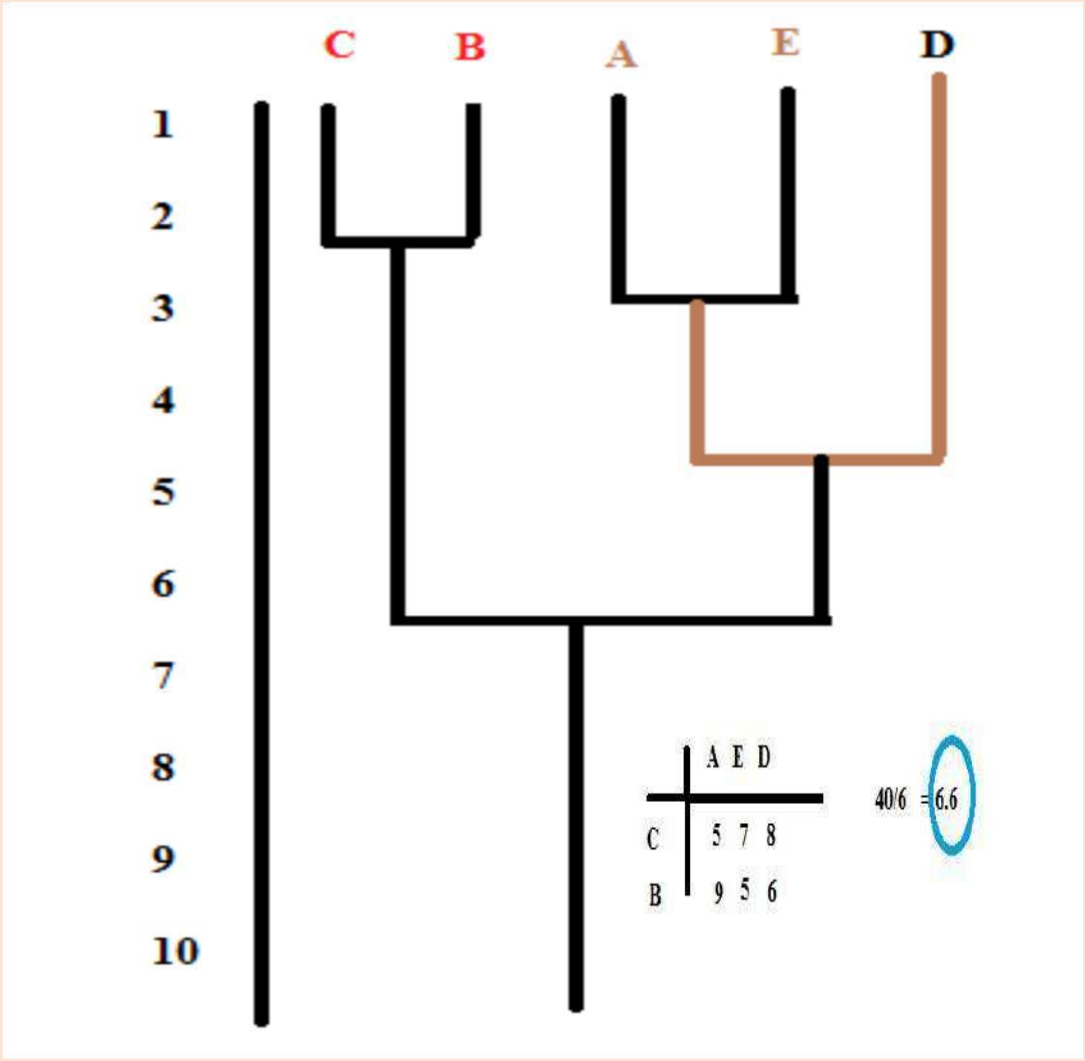
Module# 118 Construction of Phenogram

	A	B	C	D	E
A	x				
B	9	x			
C	5	2	x		
D	4	6	8	x	
E	3	5	7	6	x


C
B
1
2
3
4
5
6
7
8
9
10
C
B
A
E
1
2
3
4
5
6
7
8
9
10

Construction of Phenogram




C
B
A
E
D
1
2
3
4
5
6
7
8
9
10
A E D
C 5 7 8
B 9 5 6
40/6 = 6.6

Construction of Phenogram
Module# 119 Describing a New Species

- Describing new species is not something that is done by non-taxonomists.

- To describe a new species, you need to have the appropriate

credentials, experience, and knowledge to do so.

- Assuming you have all

Describing a New Species

- Carrying out the necessary measurements (scales, size, morphological characteristics, etc.) needed in a description.

- Record location, habitat, and other info such as DNA

Describing a New Species

- Evidence of comparative studies demonstrating that your species is indeed new or undescribed (this can

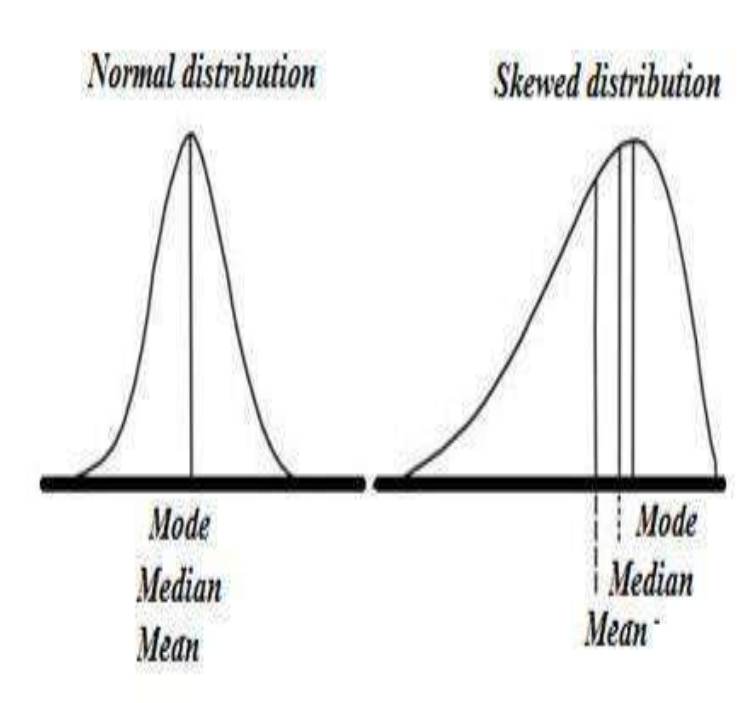
take a long time to do properly.

- Finding the right journal for publishing your description.

- Zootaxa, ZooKeys

END

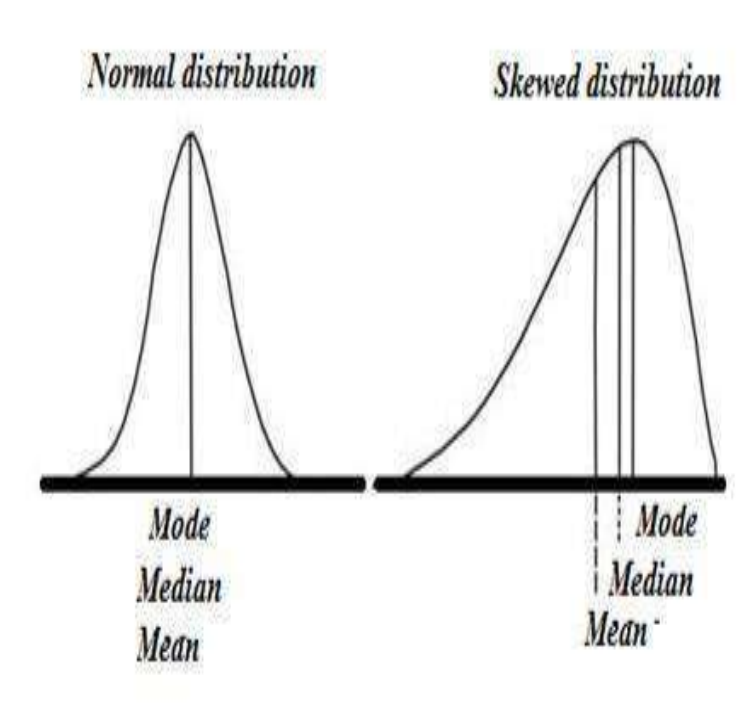
Module# 120 Normal and Skewed Data



- In normal data values of means, median mode are equal but this not the case for asymmetric or skewed data.
- Normal data follow a bell shape curve. In skewed

distribution one tail is larger than other as in above figure.

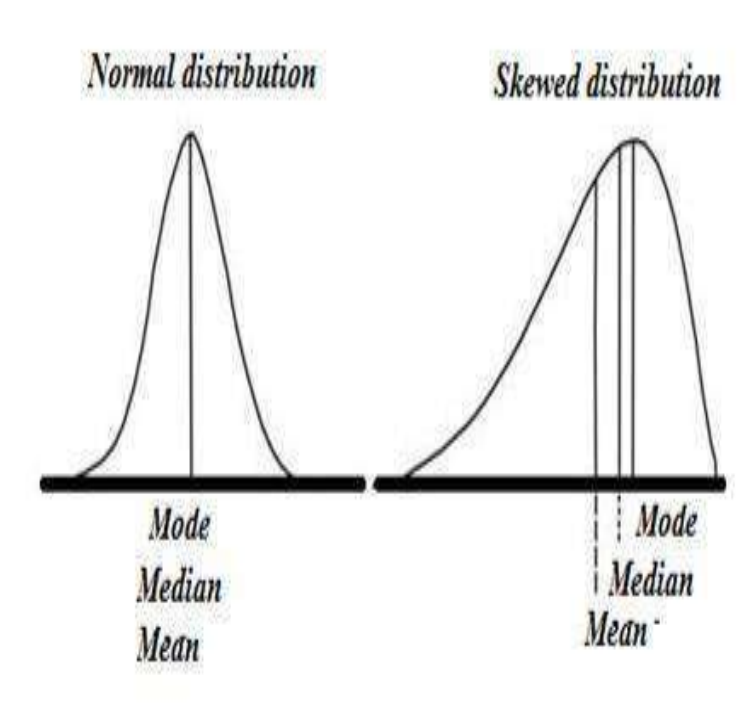
Normal and Skewed Data



- In normal data most of the individual values are close to the mean of the data. However, in skewed that the difference between individual values and mean values is higher.

- Extreme values are almost negligible in normal data but higher

Normal and Skewed Data



- Values of standard deviation are less for normal data compared to skewed data.
- Measure of central tendency for normal data

is mean but it is median
for skewed data.

Module# 121 Parametric Vs Non-Parametric Tests

- Parametric tests are used for Normal data
  - T-test
  - Z-test
  - F-test

- ANOVA
- Regression
- Correlation

Parametric Vs Non-Parametric Tests

- Non-parametric tests for skewed data.
- Wilcoxon Signed Rank Test

- Mann Whitney U Test
- Kruskal-Wallis test
- Principal Component Test
- Friedman Tests

Parametric Vs Non-Parametric Tests

Parametric Test	Non-Parametric test
Make assumptions about the population	Does not make assumptions
Uses normal probabilities	Distribution is arbitrary
Uses mean value for central tendency	Uses median value for central tendency
Require previous knowledge about the population	Does not require previous knowledge about the population

- For measuring the degree of association between two quantitative variables, **Pearson's coefficient of correlation** is used in the parametric test, while **Spearman's rank correlation** is used in the nonparametric test.

Module# 122 Types of data Distribution

- Normal Distribution
- Binomial Distribution
- Poison Distribution

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Types of data Distribution

- **Normal Distribution**

- Data which is normally distributed and follow the normal curve is called normal data. It can be continuous and discrete in value.
- **Binomial Distribution**

- Data in which there are only two outcomes from an event and numbers of events are

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Types of data Distribution

predecided is called binomial distribution. For example in five test match series, outcomes of each match are two i.e. to win or lose. Outcome of the event is a discrete number.

- **Poison Distribution**
- The data distribution in which total numbers are not predetermined and some time there number can reach to infinite. These events

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Types of data Distribution

occur in nature rarely. Outcome of such events is complete discrete value not in de cimal points. For example number of deaths in

END

Lahore per hour. Number of deaths can be infinite if some unusual event occurs.

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Module# 123 Measure of Central Tendency and Measure of Variability

Measure of central tendency

- It provide information about average value of a data set.

- Mean, Median and Mode are commonly used measures of central tendency.

- Data set has only one and only one Mean.
- Value of mean is influenced by all observations in the data set.

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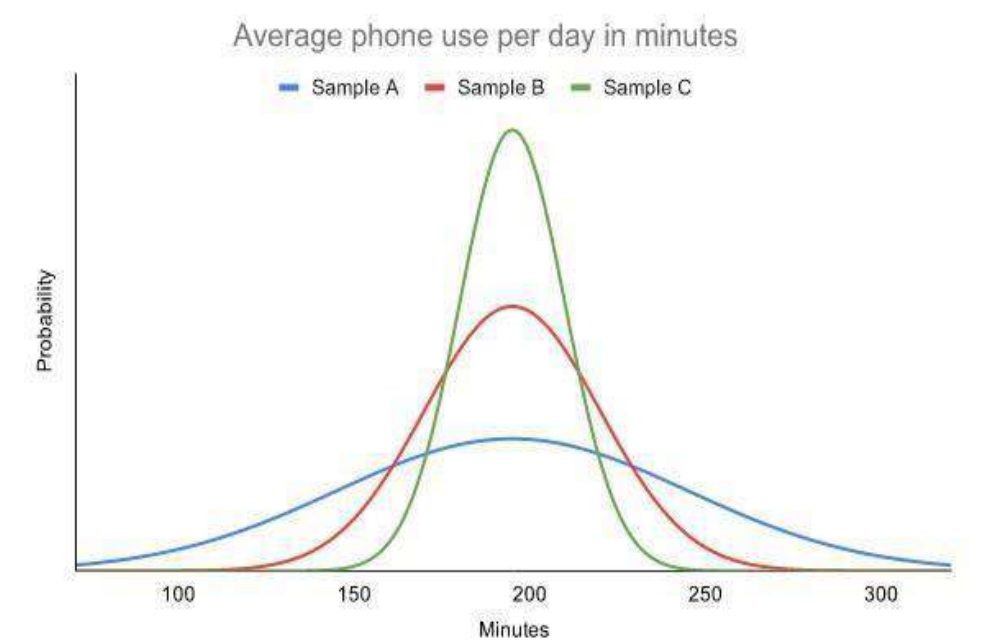
Measure of Central Tendency and Measure of Variability



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Measure of variability

- Mean deviation
- Variance
- Standard deviation
    - Standard error of mean

- Range

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- Less vs high value

Module# 124 Z-test

- Z-test is used to compare the means of two samples when sample size is greater than 30 (n >30).

$$Z = \frac{\overline{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

- This test like t -test assume a normal distribution of the data.

$$Z = \frac{\bar{X} - \mu}{SE}$$

Z-test

- This test is most useful when the standard deviation is known.

$$z = \frac{\overline{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

- Test the hypothesis that there is no difference in the calculated sample mean with given population means.

- Population mean: μ = 100, n= 55, Sample mean: and = 12
- By putting the values in formula:
- Z-calculated= -2.7

Z-test

$$z = \frac{\overline{x} - \mu}{\frac{\sigma}{\sqrt{n}}}$$

- Critical value at 95% confidence interval for Z -test is ±1.96. Zcalculated is not within the acceptable range so our null hypothesis is rejected.

Module# 125 ANOVA

- Analysis of variance (ANOVA) is a technique used to compare means of more than two samples.

- This technique can be used only for numerical data as it is a parametric test.”

ANOVA

Assumptions

- Number of samples must be more than two.
- Data should be normally distributed.
- Samples must be independent.

- Homogeneity of variances.

ANOVA

Examples

- A group of psychiatric patients are trying three different therapies: counseling, medication and biofeedback. You want to see if one therapy is better than the others.

- Students from different colleges take the same exam. You want to see if one college outperforms

END

the other.

Module# 126 ANOVA

Source of variation	SS	DF	MS	F-value
Between group variation				
Within group variation				
Total				

ANOVA

Source of variation	SS	DF	MS	F-value
Between group variation				
Within group variation				
Total				

- SS (total) [Sum of square] = (Each observation – Grand mean) ^{2}
- SS (within) = ((Each observation – Group mean) ^{2}
- SS (between) = (Group meanGrand mean)^2 × Number of observation in each group
- DF (between) = Total number of Groups – 1

END

- DF (within group) = Total number of Observations – Total number of Groups

Module# 127 One-Way ANOVA

- A one-way ANOVA uses one independent variable, while a Two-way ANOVA uses two independent variables.

- As a crop researcher, you want to test the effe ct of three different fertilizer mixtures on crop yield. You can use a one -way ANOVA to find out if there is a difference in crop yields between the three groups.

- You are interested to compare the effectiveness of three medicines in treating cancer patients

- Comparison of four feeds on weight of fish

One-Way ANOVA
One-Way ANOVA

Replicates	Lahore	Sialkot	Sargodha
1	8	10	8
2	14	14	6
3	3	11	5
4	7	9	1
Mean	**8**	**11**	**5**

- **Grand Mean** = 8+11+5/3 = 8
- SS (total) [Sum of square] = (Each observation – Grand mean) ^{2}
  - **SS(Total)** = $(8-8)^2+(14-8)^2+(38)^2+(7-8)^2+(10-8)^2+(14-8)^2+(118)^2+(9-8)^2+(8-8)^2+(6-8)^2+(6-8)^2+(5-8)^2+(1-8)^2$
- **SS (Total) = 174**

One-Way ANOVA

Replicates	Lahore	Sialkot	Sargodha
1	8	10	8
2	14	14	6
3	3	11	5
4	7	9	1
Mean	**8**	**11**	**5**

- **SS (within)** = ((Each observation – Group mean) ^{2}
- SS(within) = $(8-8)^2+(14-8)^2+(38)^2+(7-8)^2+(10-11)^2+(14-11)^2+(11-11)^2+(9-11)^2+(8-5)^2+(6-5)^2+(5-5)^2+(1-5)^2$
- **SS (with in)** = 102

Module# 128 One-Way ANOVA

Replicates	Lahore	Sialkot	Sargodha
1	8	10	8
2	14	14	6
3	3	11	5
4	7	9	1
Mean	**8**	**11**	**5**

- **SS (between)** = (Group meanGrand mean)^2 × Number of observation in each group
- SS (between) = [(8 -8)^2 × 4] + [(118)^2 × 4] + [(5-8)^2 × 4] = 72

One-Way ANOVA

Source of variation	SS	DF	MS	F-value
Between group variation	72	2	36	3.17
Within group variation	102	9	11.33	
Total	174	9+2 = 11	-	

- **DF (between)** = Total number of groups – 1
- DF (between) = 3-1 = 2
- **DF (within group)** = Total number of observation – Total number of groups
- DF (within group) = 12-3 = 9

One-Way ANOVA

Source of variation	SS	DF	MS	F-value
Between group variation	72	2	36	3.17
Within group variation	102	9	11.33	
Total	174	9+2 = 11	-	

MS (between)

- It can be calculated by dividing the SS (between group) with DF (between group)
- 72/2 = 36

MS (within)

It can be calculated by dividing the SS (within group) with DF (within group)

- 102/9 =11.33

One-Way ANOVA

Source of variation	SS	DF	MS	F-value
Between group variation	72	2	36	3.17
Within group variation	102	9	11.33	
Total	174	9+2 = 11	-	

F-calculated

- 3.11

F- Tabulated

- Comparison of both
- Decision about hypothesis

Module# 129 Two-Way ANOVA

- In two-way ANOVA there are two independent variables and one dependent variable.

- In one-way ANOVA there was only one independent variable

Two-Way ANOVA

Assumptions

- The population must be close to a normal distribution.
- Samples must be independent.
- Population variances must be equal.

Two-Way ANOVA

- Groups must have equal sample sizes.
- One could use a **two-way ANOVA** to understand whether there is an interaction between gender and educational level on test anxiety amongst university students, where gender (males/females) and education level

Two-Way ANOVA

(undergraduate/postgraduate) are your independent variables, and test anxiety is your dependent

- For example, a two -way ANOVA allows a company to compare worker productivity based on two independent variables, such as department and gender.

Two-Way ANOVA

- It is utilized to observe the interaction between the two factors. It tests the effect of two factors at the same time.
- The effective life (in hours) of batteries is compared by material type (1, 2 or 3) and operating temperature: Low (-10˚C), Medium (20˚C) or High (45˚C).

Two-Way ANOVA

- Twelve batteries are randomly selected from each material type and are then randomly allocated to each temperature level.

END

Module# 130 T-test

- T-test is appropriate whenever you want to compare the means of two samples.

- This test is used to find out whether the means of two samples are statistically different from each other or not.

T-test

- Data should be normally distributed.
- Samples should be taken randomly from the population
- Sample size should be 30 or less than 30.
- All randomly collected samples should have the same variances

(within the limits of sampling error).

- Standard deviation may or may not be known.

T-test

- Independent t -test/two sample ttest
- Paired t-test

Module# 131 Independent t-test

- For two sample t -test, the two populations must be

  independent

- It is not necessary that sample size should be same for both samples.

- **Example**s: Test the hypothesis that mea ns heights (cm) of plants randomly selected from two wheat fields are same.

Independent t-test

Data of field I (X_1)= 10, 15, 20. 25 and 30.

Data of field II (X_2)= 4, 8, 10. 15 and 10.

Null hypothesis: $\bar{X}_1 = \bar{X}_2$

Alternative hypothesis is $\bar{X}_1 \neq \bar{X}_2$.

Formula:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

- For two sample t-test, the two populations must be independent
- It is not necessary that sample size should be same for both samples.

Independent t-test

- **Example**s: Test the hypothesis that means heights (cm) of plants randomly selected from two wheat fields are same.

Independent t-test

X_1	X_2	X_1^2	X_2^2
10	4	100	16
15	8	225	64
20	10	400	100
25	15	625	225
30	10	900	100
$\sum X_1 = 100$	$\sum X_2 = 47$	$\sum X_1^2 = 2250$	$\sum X_2^2 = 505$

Independent t-test

$$\sigma_1^2 = \frac{\Sigma(X_1 - \bar{X}_1)^2}{n-1}$$

$$= \frac{\Sigma X_1^2 - \frac{(\Sigma X_1)^2}{n}}{4} = \frac{2250 - 2000}{4} = 62.5$$

$$\sigma_2^2 = \frac{\Sigma(X_2 - \bar{X}_2)^2}{n-1}$$

$$= \frac{505 - \frac{(47)^2}{5}}{4} = \frac{505 - 441.8}{4} = \frac{63.2}{4} = 15.8$$

$$t = \frac{20 - 9.4}{\sqrt{\frac{62.5}{5} + \frac{15.8}{5}}}$$

$$= \frac{10.6}{\sqrt{12.5 + 3.6}} = \frac{10.6}{\sqrt{15.66}} = \frac{10.6}{3.95} = 2.68$$

Independent t-test

Null hypothesis: $\bar{X}_1 = \bar{X}_2$

t = 2.68 (calculated value)

As both samples are independent so degree of freedom will be n-1 (from sample X_1) and n-1 (from sample X_2).

5-1 = 4 (X_1) and 5-1 = 4 (X_2)

Degree of freedom: 4+4 =8

Tabulated value/standard value/critical value = 2.3

Result: As our calculated value is lower than tabulated value so our null hypothesis is accepted.

Module# 132 Paired t-test

X	Y	d	d^2
9	12	-3	9
12	14	-2	4
13	19	-6	36
20	25	-5	25
5	10	-5	25
		$\sum d = -21$	$\sum d^2 = 99$

Paired t-test

$$t = \frac{\bar{d}}{\sqrt{\frac{\sigma^2 d}{n}}}$$

$$\sigma^2 d = \frac{\sum (d - \bar{d})^2}{n - 1}$$

$$= \frac{\sum d^2 - (\bar{d})^2}{n-1}$$

$$= \frac{99 - 17.64}{4} = \frac{81.36}{4} = 20.34$$

Paired t-test

$$\bar{d} = \frac{-21}{5} = -4.2$$

$$t = \frac{-4.2}{\sqrt{\frac{20.34}{5}}} = \frac{-4.2}{\sqrt{4.068}} = \frac{-4.2}{2.016} = -2.08$$

T calculated: -2.08

T tabulated $=2.77$

As observation were paired so degree of freedom (n-1)= 4.

Our null hypothesis is rejected as T- calculated is smaller than T-tabulated.

Module# 133 Standard Deviation

Data of field I (X_1)= 10, 15, 20. 25 and 30.

Data of field II (X_2)= 4, 8, 10. 15 and 10.

Null hypothesis: $\bar{X}_1 = \bar{X}_2$

Alternative hypothesis is $\bar{X}_1 \neq \bar{X}_2$.

Formula:

$$t = \frac{\bar{X}_1 - \bar{X}_2}{\sqrt{\frac{\sigma_1^2}{n_1} + \frac{\sigma_2^2}{n_2}}}$$

- It is positive square root of variance.
- Standard deviation "provides a sort of average of the differences of all scores from the mean.
- It is a measure of the dispersion of scores around the mean.
- Generally, a low standard deviation means that a set of scores is not very widely dispersed around the mean.

Standard Deviation

- While a high standard deviation indicates that the scores are more widely dispersed

$$SD = \sqrt{\frac{\sum(x - \bar{x})^2}{n - 1}}$$

- Its advantage over variance is that its physical unit are same as that of individual observations

Standard Deviation

- The standard error is also a measure of spread.

$$SE = \frac{\sigma}{\sqrt{n}} \quad \text{or} \quad SE = \frac{SD}{\sqrt{n}}$$

- It is represented by SE.
- Higher value of SE indicates higher spread of the data.
- The difference in the standard error and standard deviation is that standard error uses sample data but standard deviations use population data.
- SE tells us that how far our sample mean deviate from

Standard Deviation

$$SE = \frac{\sigma}{\sqrt{n}} \quad \text{or} \quad SE = \frac{SD}{\sqrt{n}}$$

population means.

Module# 134 Standard Deviation

The **sample standard deviation formula** is:

$$s = \sqrt{\frac{\sum(X - \bar{X})^2}{n - 1}}$$

where,

s = sample standard deviation
$\sum$ = sum of...
$\bar{X}$ = sample mean
n = number of scores in sample.

- The standard deviation is a measure of the spread of scores within a set of data.
- Usually, we are interested in the standard deviation of a population.
- However, as we are often presented with data from a sample only, we can estimate the population

Standard Deviation

standard deviation from a sample standard deviation.

The **population standard deviation** formula is:

$$\sigma = \sqrt{\frac{\sum(X-\mu)^2}{n}}$$

where,

σ = population standard deviation
$\sum$ = sum of...
μ = population mean
n = number of scores in sample.

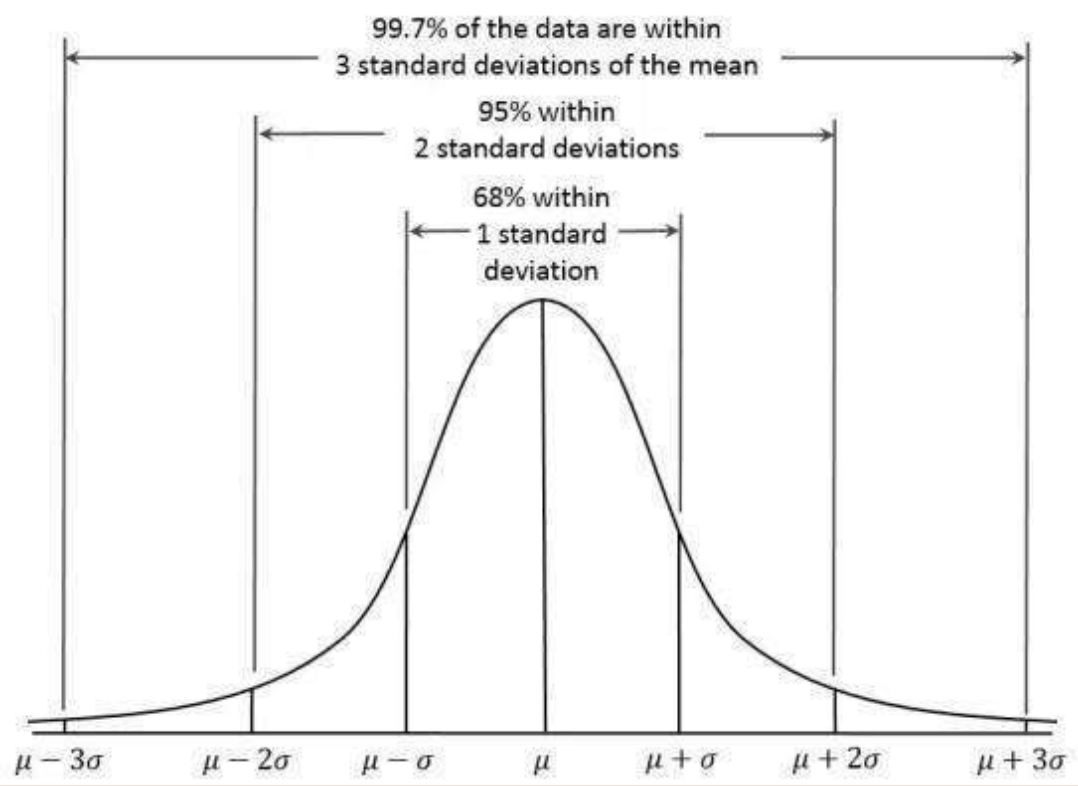
- The standard deviation is a measure of the spread of scores within a set of data.

- A **sample standard deviation** is a statistic. This means that it is calculated from only some of the individuals in a **population**.
- The **population standard deviation** is a parameter, which is a fixed value calculated from every individual in

Standard Deviation

the **population**.

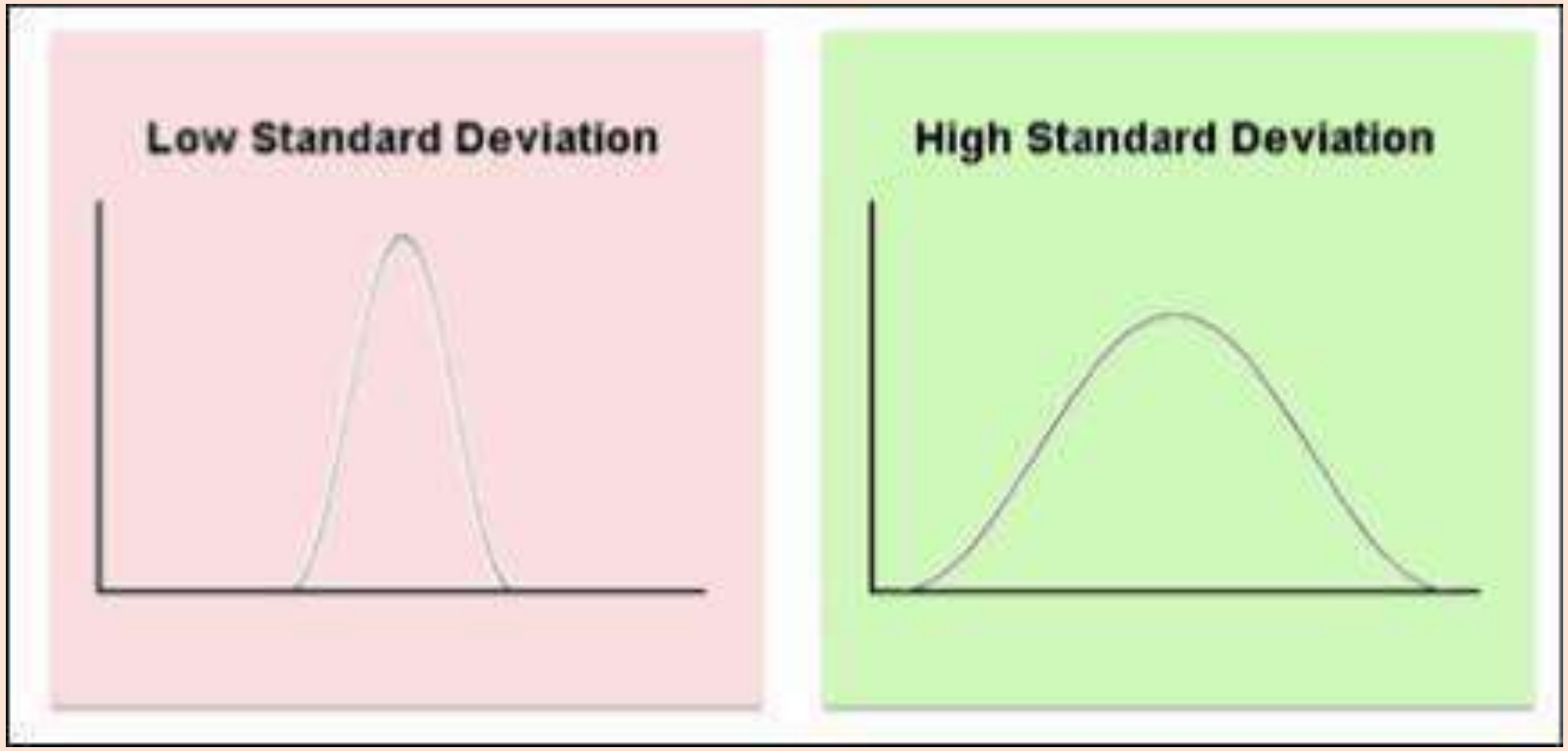
Standard Deviation



Relationship between Variance and SD

- The **standard deviation** is the square root of the **variance**. The **standard deviation** is expressed in the same units as the mean is, whereas the **variance** is expressed in squared units,

Standard Deviation



Module# 135 Introduction to Spiders

- Most abundant natural predators
  (Maloney et al., 2003)
- Ranked 7th in animal diversity
  (WSC, 2018)
- More than 46000 species
- Strictly carnivorous

Standard Deviation

- Cause wasteful killing
- Show functional and numerical response

Introduction to Spiders

https://www.google.com/search?q=spiders+eating&tbm=isch&ved=2ahUKEwi0ybiUp9vtAhUBRBoKHWkMA0IQ2-cCegQIABAA&oq=spiders+eating&gs_lcp=CgNpbWcQAzICCAAyAggAMgIIADICCAAyAggAMgIIADICCAA6BAgjECc6BAgAEENQ6SNYvTJg6jVoAHAAeACAAa4CiAGZDJIBBzih=657&biw=1360#imgrc=WaGecw9pWlST7M

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b

- Best survivors under food shortage
- Prey switching
- Cannibalistic
- Top down effect
- Indirect killing

Module# 136 Introduction to Spiders

- Ground runners
- Stalkers

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cCegQIABAA&oq=spiders+eating&gs_lcp=CgNpbWcQAzICCAAyAggAMgIIADICCAAyAggAMgIIADICCAA6BAgjECc6BAgAEENQ6SNYvTJg6jVoAHAAeACAAa4CiAGZDJIBBzAuMS41LjGYAQCgAQGqAQtnd3Mtd2l6LWltZ8ABAQ&sclient=img&ei=IJveX7S5JIGIaemYjJAE&b

- Web builders

Introduction to Spiders

- Bio-control agents
- Model organism for research
- Spiders silk

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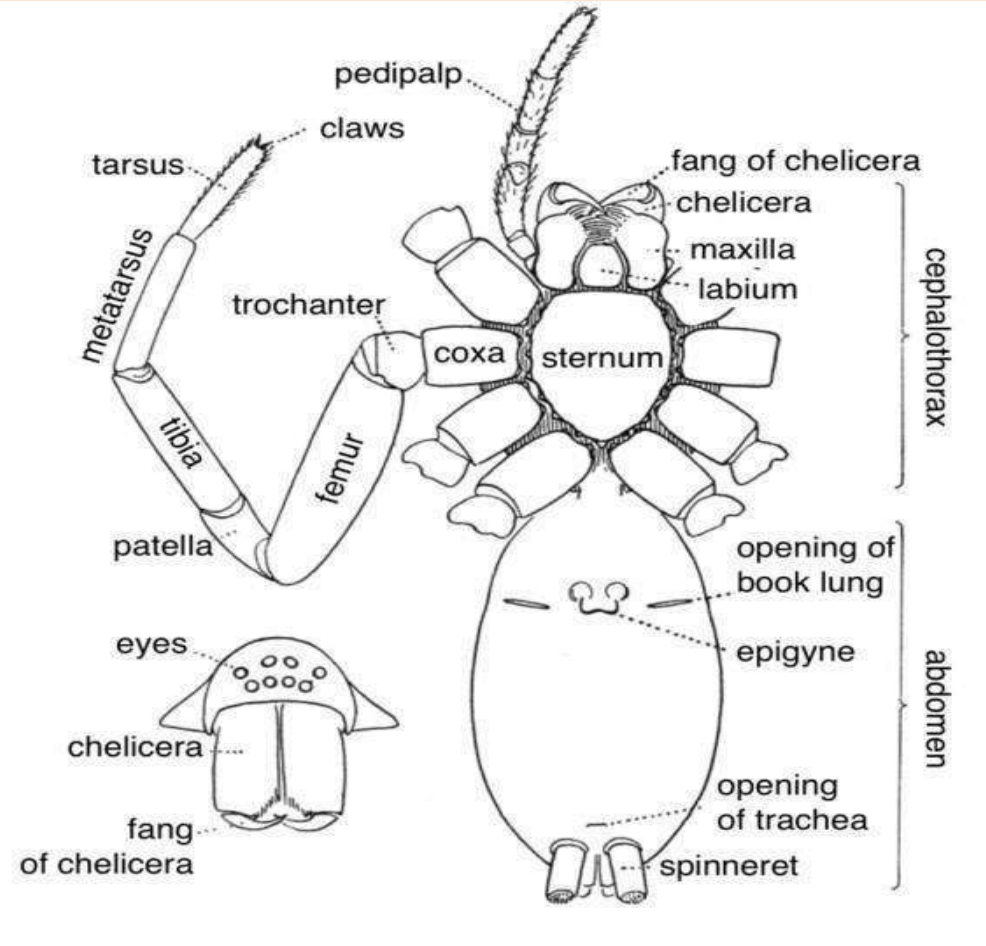
cCegQIABAA&oq=spiders+eating&gs_lcp=CgNpbWcQAzICCAAyAggAMgIIADICCAAyAggAMgIIADICCAAyAggAMgIIADICCAA6BAgjECc6BAgAEENQ6SNYvTJg6jVoAHAAeACAAa4CiAGZDJIBBzAuMS41Lj

GYAQCgAQGqAQtnd3Mtd2l6LWltZ8ABAQ&sclient=img&ei=IJveX7S5JIGIaemYjJAE&b

- Spider venom peptides

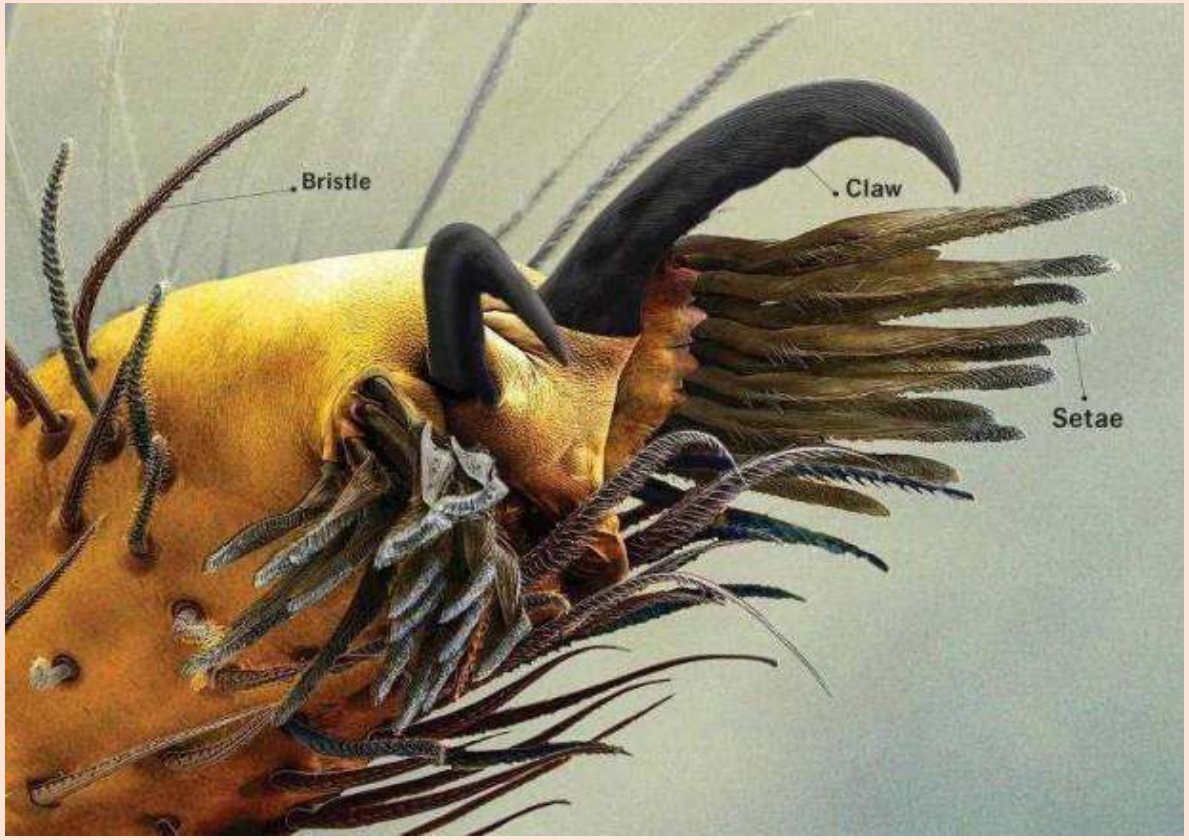
Module# 137 Morphology of spider Leg



- Spiders typically have eight walking legs (insects have six)
- They do not have antennae
- the pair of app endages in front of the legs are the pedipals(or just palps)
- Spiders' legs are made up of

Morphology of spider Leg

https://en.wikipedia.org/wiki/Spider_anatomy#/media/File:Spider_external_anatomy_en.png



seven segments

- The tip of the tarsus bears claws, which vary in number and size
- Spiders that spin webs typically have three claws, the middle one being small
- Hunting spiders typically have

Morphology of spider Leg

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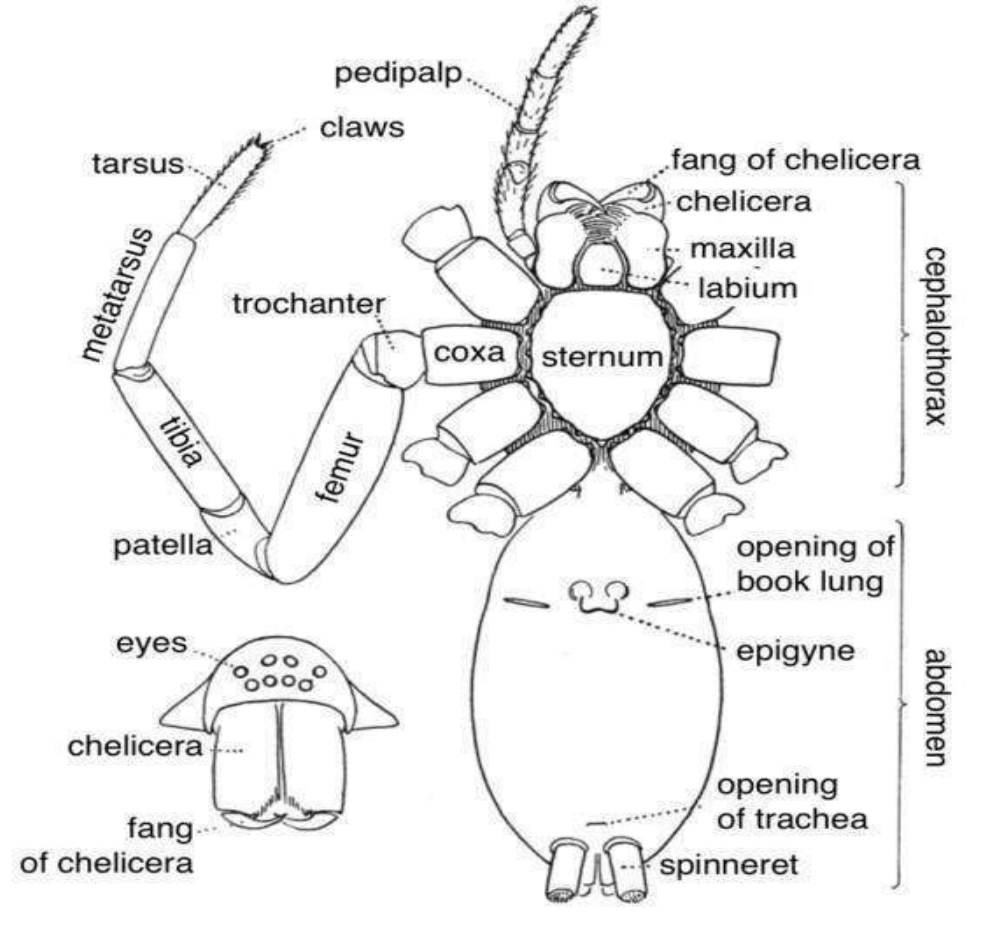
only two claws

Morphology of spider Leg



- Starting from the body end, these are the:
- Coxa
- Trochanter
- Femur
- Patella • Tibia

Morphology of spider Leg

https://en.wikipedia.org/wiki/Spider_anatomy#/media/File:Spider_external_anato

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Tarsus.

Morphology of spider Leg

- Since they do not have antennae, spiders use specialized and **sensitive setae** on their legs to pick up scent, sounds, vibrations and air currents.
- Some spiders, such as

Morphology of spider Leg

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the **Australia crab spiders**, do not have claws.

Morphology of spider Leg

- The pedipalps have only six segments: the metatarsus is missing.
- In adult males, the tarsus of each palp is modified for mating (variously called a palpal pulb, palpal organ or copulatory bulb)

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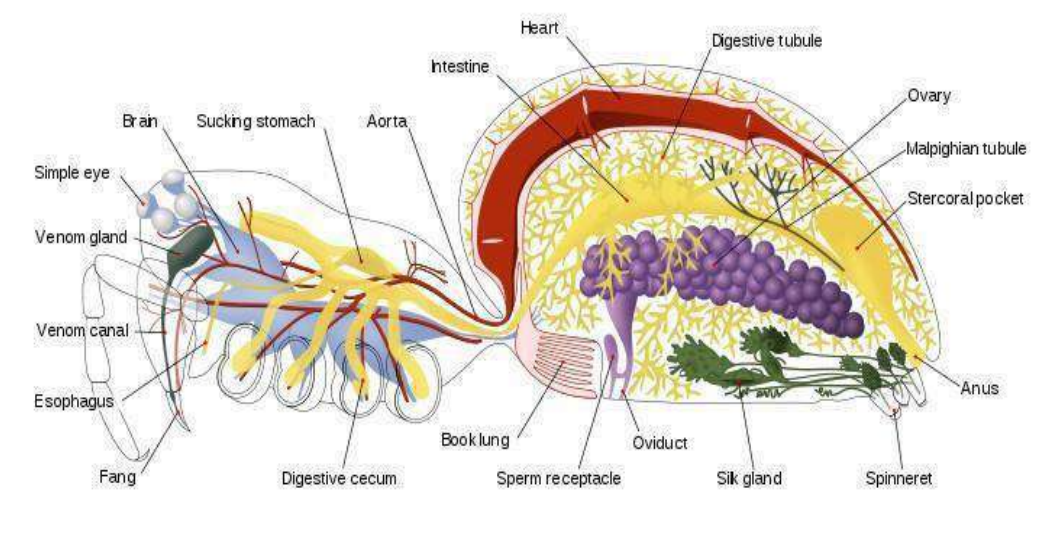
Module# 138 Morphology of spider's Abdomen



- The abdomen is also known as the **opisthosoma.**
- On the ventral side of the abdomen are two hardened plates covering the book lungs.

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QE&sclient=img&ei=tLHeX_L8Acu4aLe6vNgD&bih=657&biw=1360#imgrc=HeEJxStweWNH

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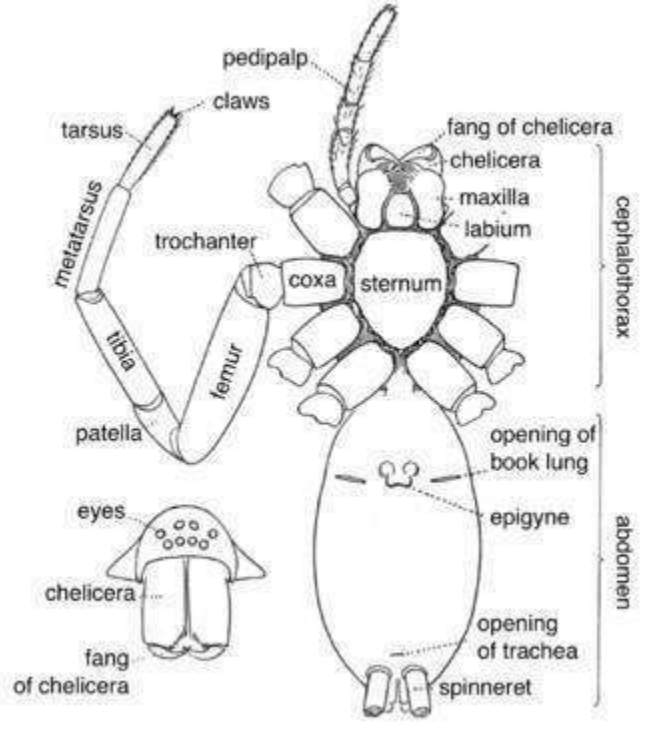
Morphology of spider's Abdomen



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- These are called the **epigastric plates**.
- Females genetal organ is also present on ventral side of the abdomen.
- It is called epigyne.

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cCegQIABAA&oq=palpal+bub+in+spiders&gs_lcp=CgNpbWcQAzoECCMQJzoICAAQsQMQgw

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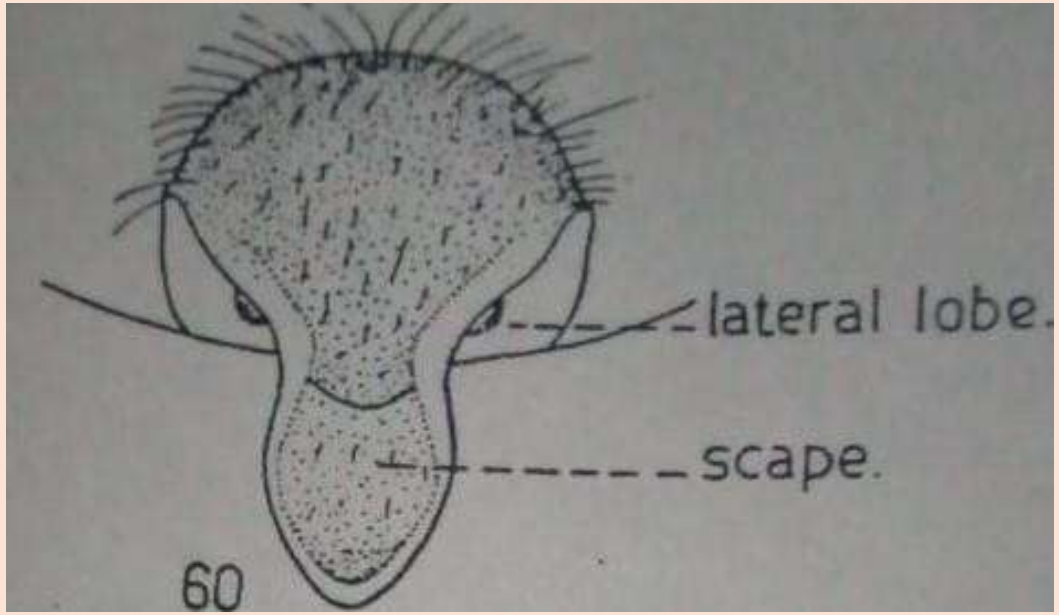
QE&sclient=img&ei=tLHeX_L8Acu4aLe6vNgD&bih=657&biw=1360#imgrc=HeEJxStweWNH

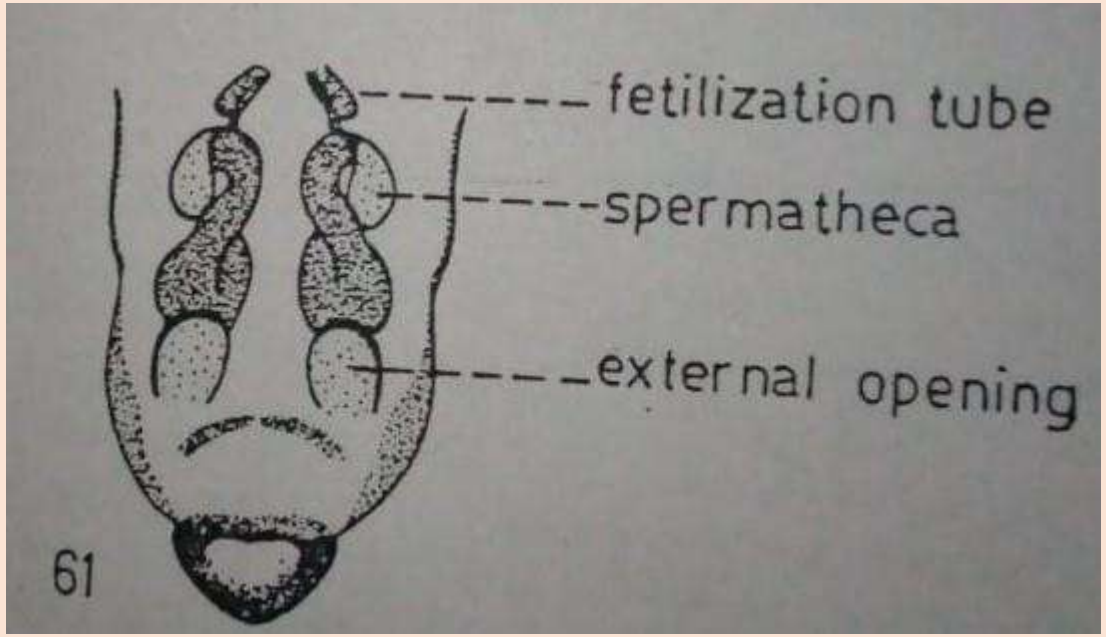
Morphology of spider's Abdomen



- In many cases epigyne is provided with a tongue like seclerotic projection called scape.
- Scape often have paired lateral lobes as in Neoscon

Tikadar, 1982. The Funa of India

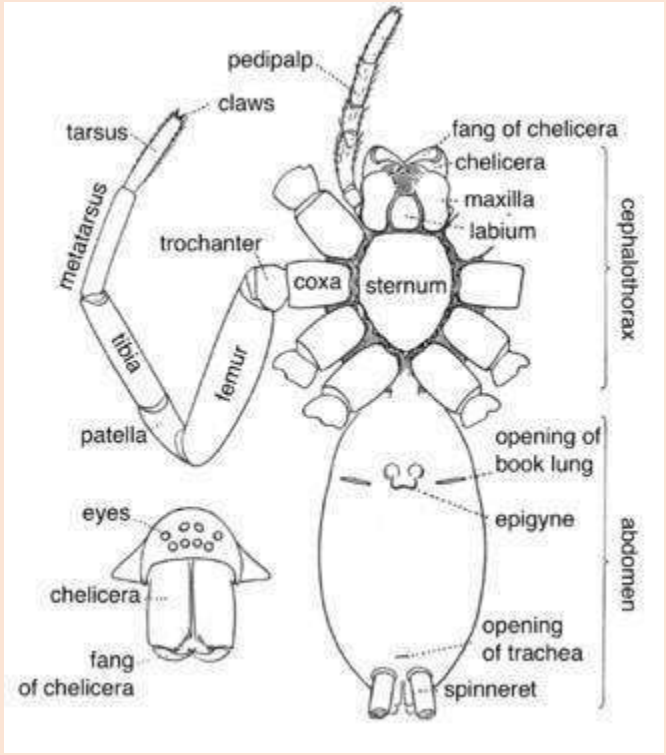
Morphology of spider's Abdomen



Tikadar, 1982. The Funa of India

- Epigine has two external openings which lead to spermathecae inside and paired of narrow curved tubes called fertilization tube
- Scape often have paired lateral lobes as in Neoscon

Morphology of spider's Abdomen



- A fold, known as the **epigastric furrow**, separates the region of the **book lungs** and **epigyne**.
- In the middle of this furrow is the opening of the oviduct (in females) and at either end are the lung slits.

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Morphology of spider’s Abdomen

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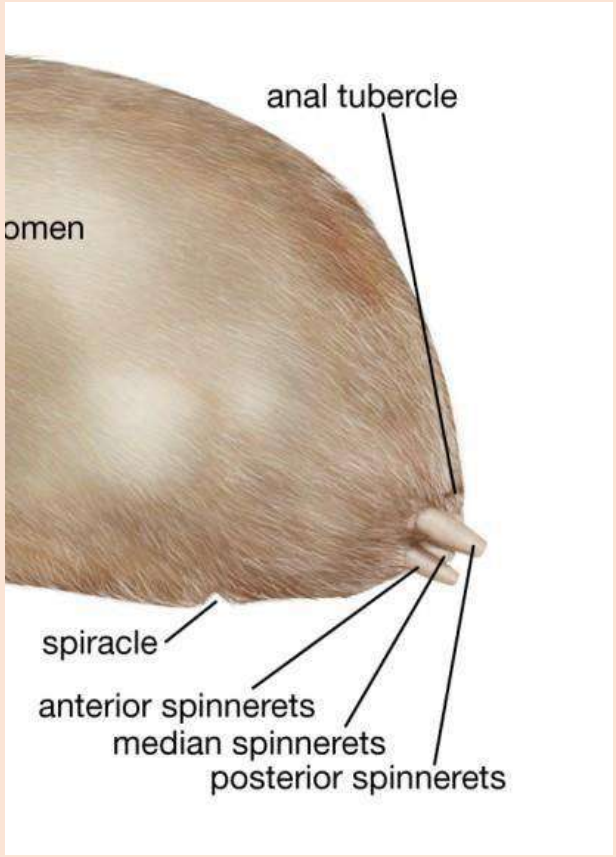
Morphology of spider's Abdomen



- On posterior side of the abdomen there is conical anal tubercle that bear three pairs of spinnerets ventrally.
- On ventral side there are two pairs of book lungs.

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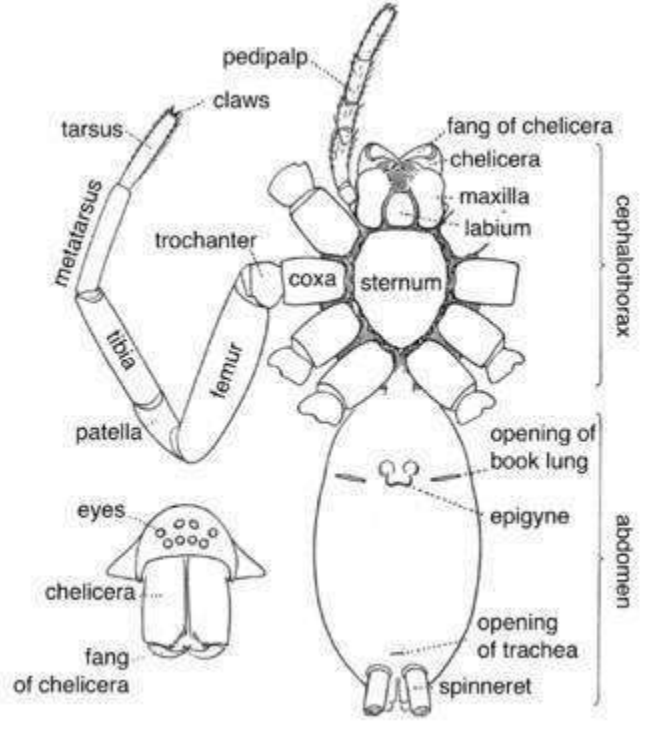
Morphology of spider’s Abdomen



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Morphology of spider’s Abdomen

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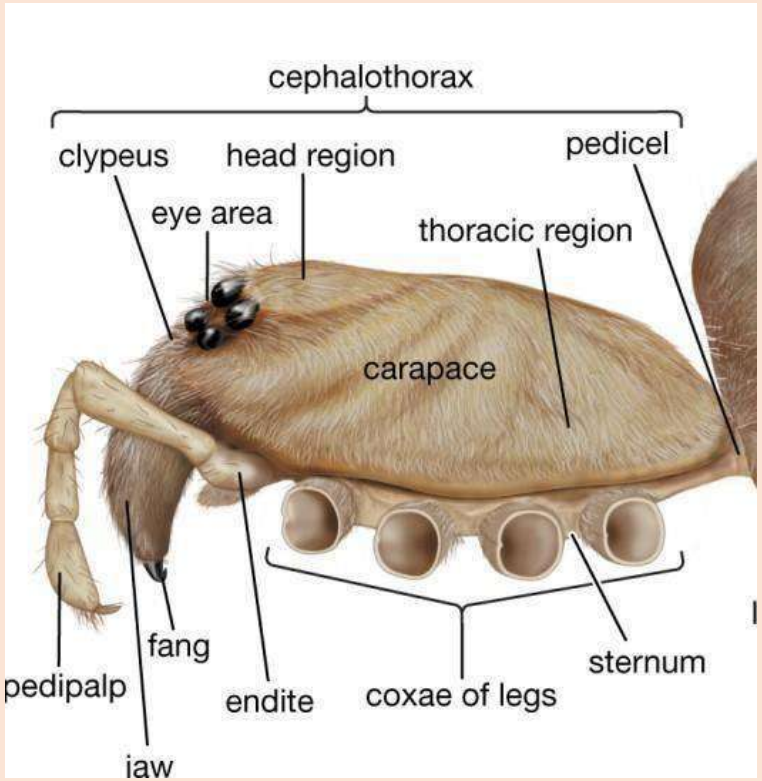
NM&imgdii=tDDYazdFPMxm0M

Module# 139 Morphology of spider's Cephalothorax



- Cephalic region has 6-8 eyes
- Eyes or black or diurnal
- White or nocturnal
- If only one type of eyes (homogenous)
- If both type of eyes or present

Morphology of spider's Cephalothorax

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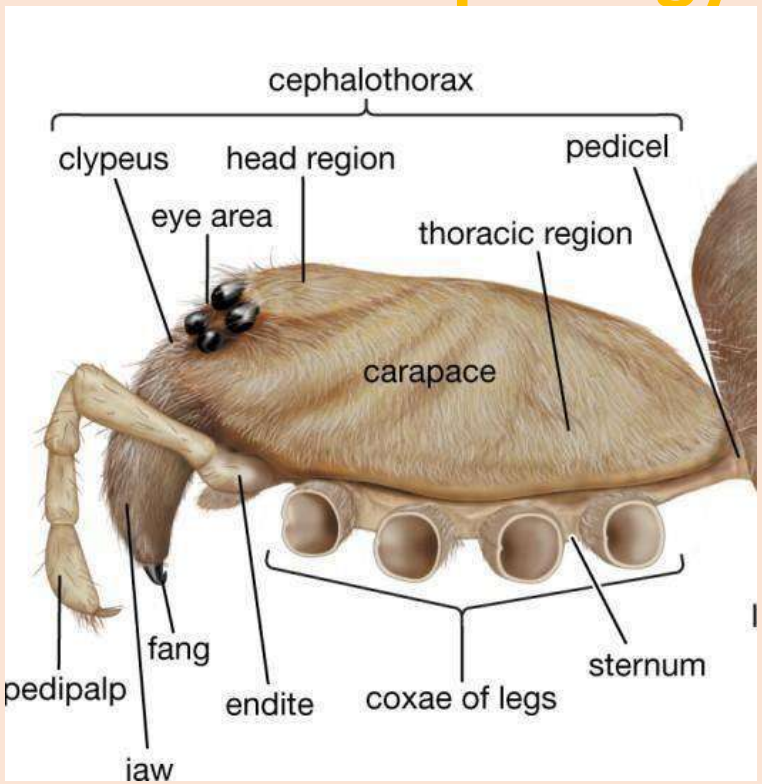
(heterogeneous)

Usually arranged in

- two rows

Morphology of spider's Cephalothorax



- Each row usually contain 4 eyes.
- Rows of eyes are frequently curved.
- Anterior median/Anterior lateral
- Posterior median/Posterior lateral

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- Cephalic area of eyes is called **ocular area.**

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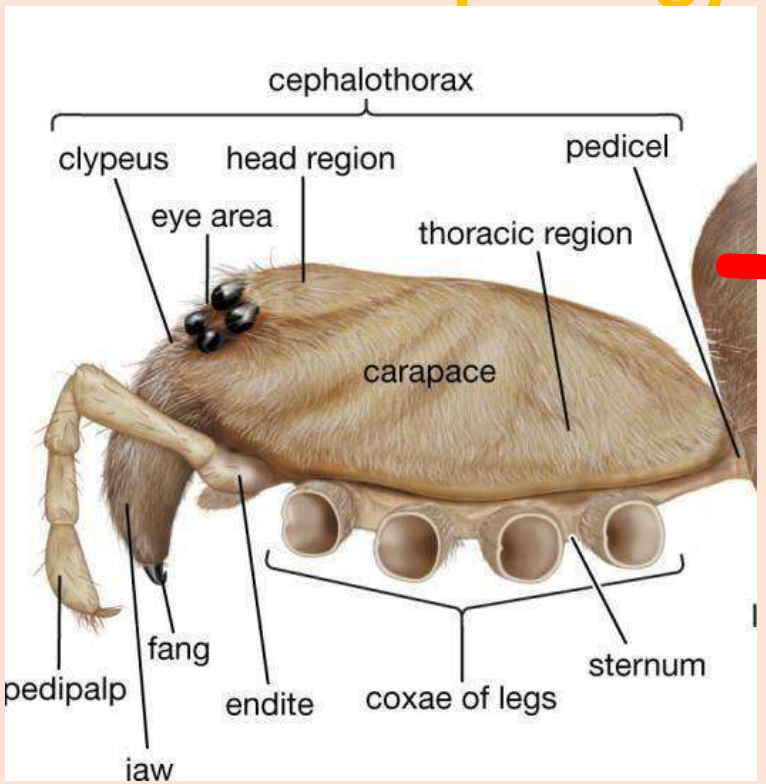
Morphology of spider's Cephalothorax



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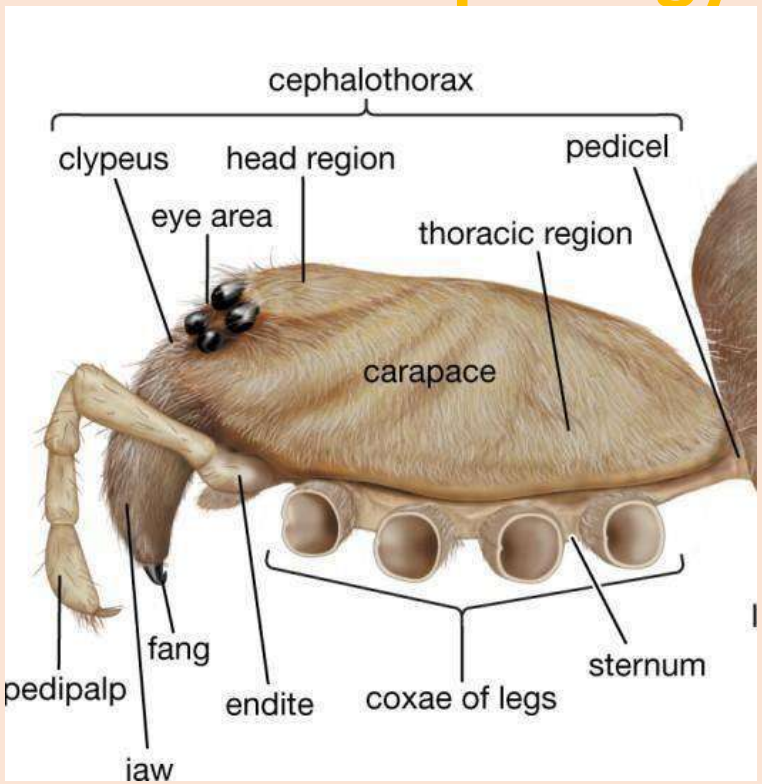
- Area of four median eyes is called ocular quad.
- The area between anterior row of eyes and anterior margin of carapace is **clypeus.**
- The depression in the middle of thorax is called **thoracic groove or fovea.**

Morphology of spider's Cephalothorax



- Appendages
- Chelicerae/Fangs
- Padipalp (2nd pair of appendages)
- Papl has six parts/metatarsus missing.
- In males tarsus/modified copulatory organ

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Morphology of spider's Cephalothorax

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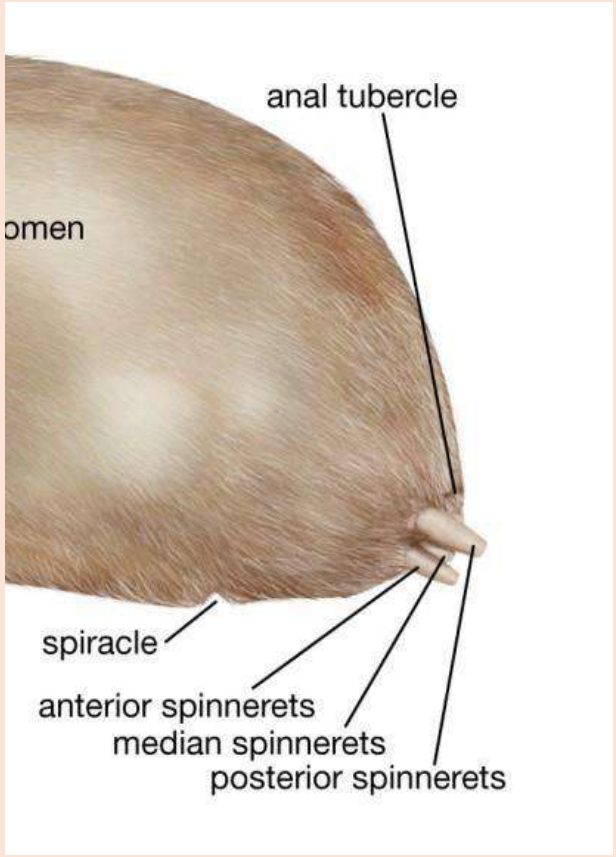
- Ventral side is sternum

- Oval/heart shape

Module# 140 Morphology of spider's Spinnerets



- The abdomen of spiders has modified pairs of movable telescoping organs called **spinnerets**, which produce silk.
- The suborder **Mesothelae** is unique in having only two types

Morphology of spider’s Spinnerets

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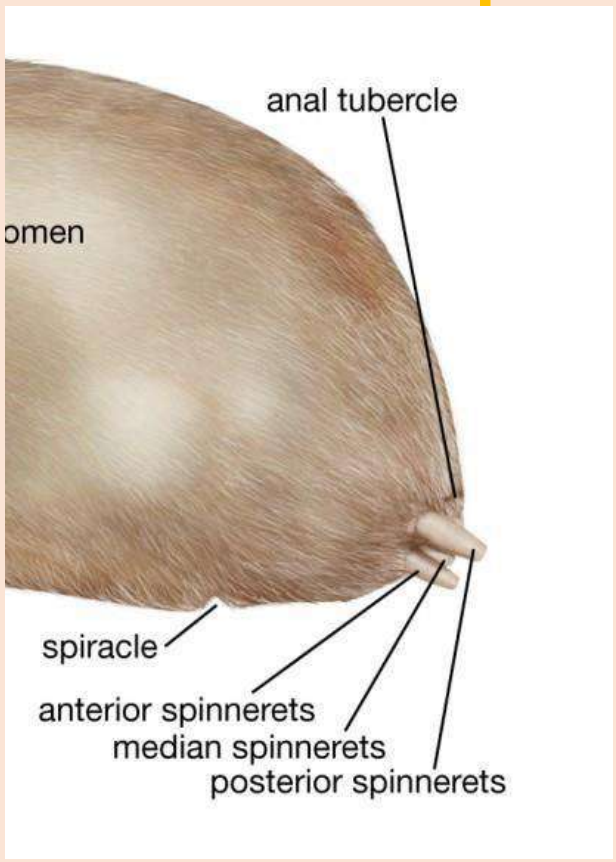
of silk glands – thought to be the
ancestral condition.

Morphology of spider's Spinnerets



- Depending on the complexity of the species, spiders will have two to eight spinnerets, usually in pairs.
- Behind each spinneret visible on the surface of the spider lies a gland.

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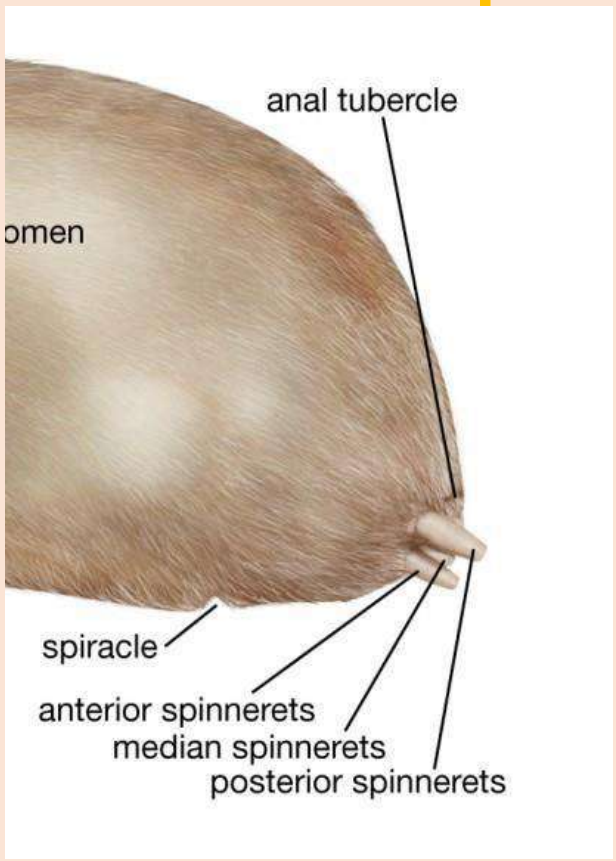
Morphology of spider's Spinnerets



- Each spigot has a valve that controls the thickness and speed of the extruded material. As the spigots pull the **fibroin protein molecules** out of the ducts and extrude them into the air, the

Morphology of spider’s Spinnerets

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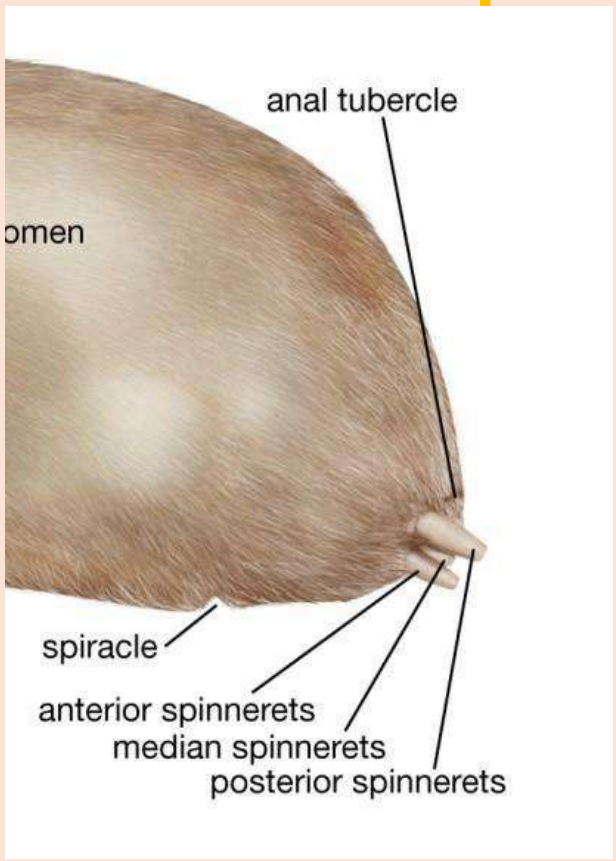
molecules are stretched out and
linked together to form long strands.

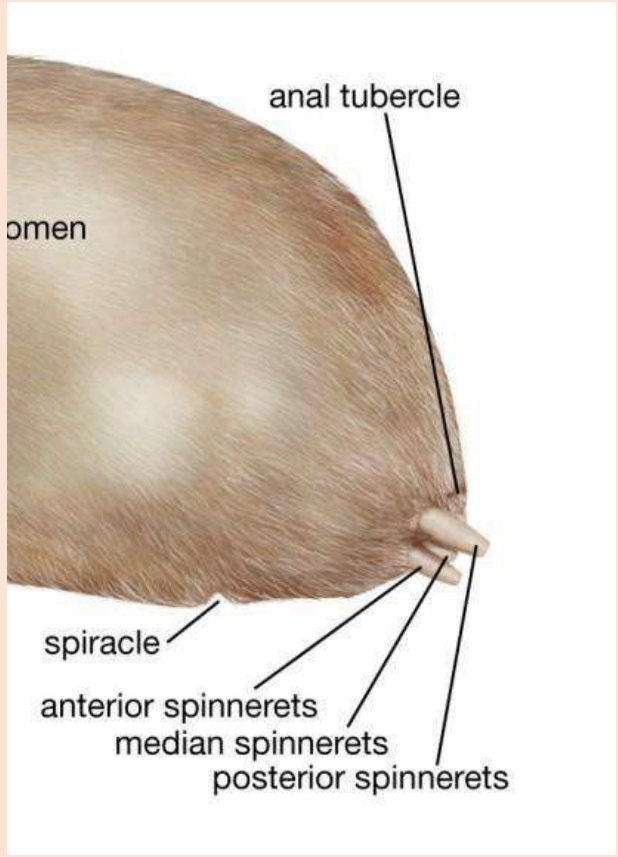
Morphology of spider's Spinnerets



- The spinnerets wind these strands together to form the sturdy silk fiber.
- Most spiders have multiple silk glands, which secrete different types of silk material optimized for different purposes.

https://www.google.com/search?q=cephalolthorax+of+spiders&tbm=isch&ved=2ahUKEwiy29yN5eftAhXI0oUKHUvcAnoQ2-cCegQIABAA&oq=cephalolthorax+of+spiders&gs_lcp=CgNpbWcQAzoECCMQJzoICAAQsQMQgwE6BQgAELEDOgQIABBDOgcIABCxAxBDOgIIADoECAAQHjoGCAAQChAYOgQIABAYUKuwT

Morphology of spider’s Spinnerets



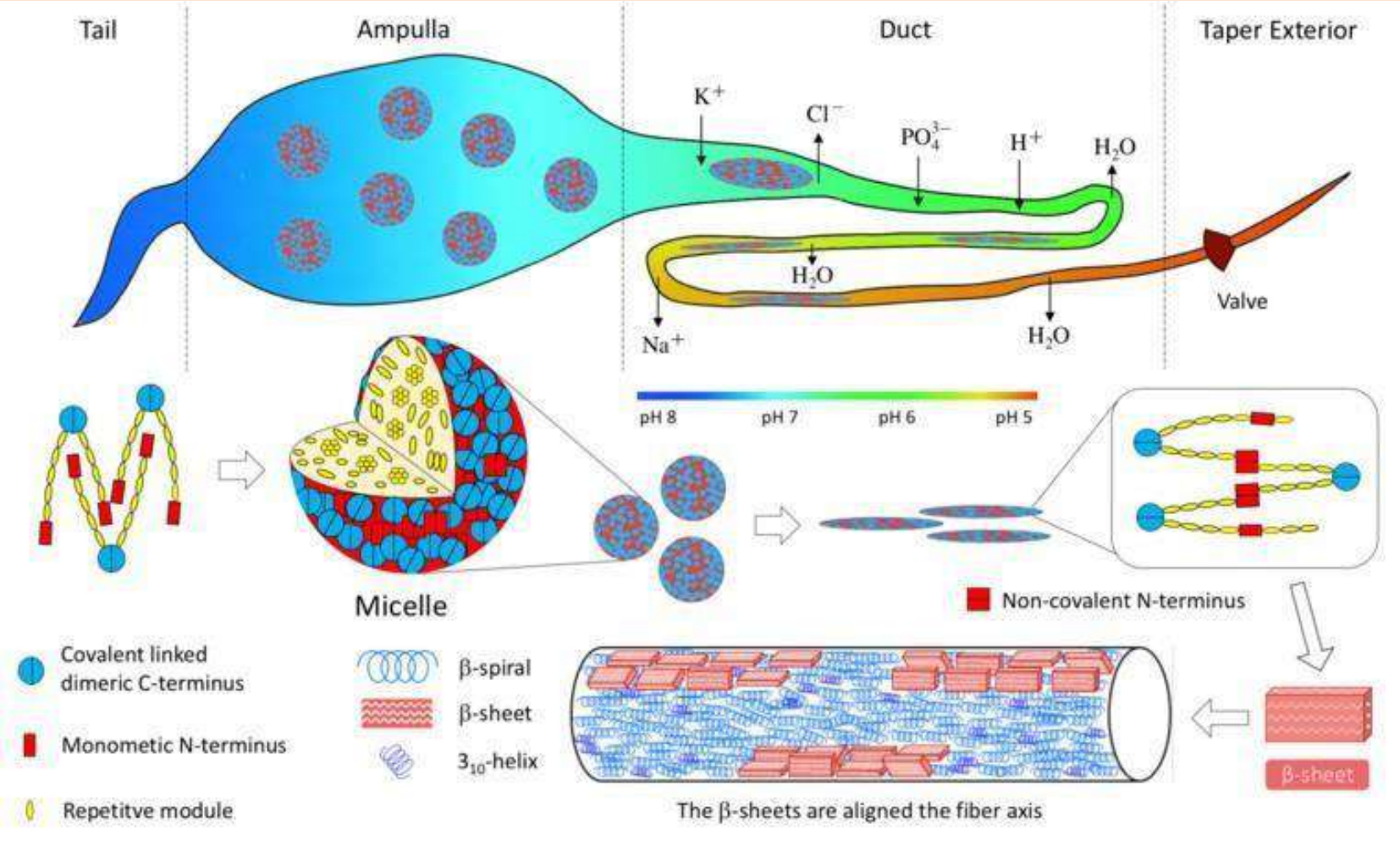
- Spider silk is incredibly strong and flexible. Five times as strong as an equal mass of **steel** and twice as strong as an equal mass of **Kevlar.**
- Elasticity

Morphology of spider's Spinnerets

https://www.google.com/search?q=cephalolthorax+of+spiders&tbm=isch&ved=2ahUKEwiy29yN5eftAhXI0oUKHUvcAnoQ2cCegQIABAA&oq=cephalolthorax+of+spiders&gs_lcp=CgNpbWcQAzoECCMQJzoICAAQsQMQgwE6BQgAELEDOgQIABBDOgcIABCxAxBDOgIIADoECAAQHjoGCAAQChAYOgQIABAYUKuwTFiihU1gzYhNaAFwAHgAgAGHBIgBqjqSAQoyLTE4LjYuMS4xmAEAoAEBqgELZ3dzLXdpei1pbWfAAQE&sclient=img&ei=siblX7K6FsillwTLuIvQBw#imgrc=MENLiSCsscXroM

- Size
- Temperature/PH/uses

Morphology of spider's Spinnerets



Morphology of spider’s Spinnerets

https://en.wikipedia.org/wiki/Spider_silk

Module# 141 Introduction to Scorpions

https://en.wikipedia.org/wiki/Scorpion

- **Scorpions** are predatory arachnids of the order **Scorpiones**.
- hey have eight legs, and are easily recognized by a pair of grasping pincers and a narrow, segmented tail, often carried in a characteristic forward curve over the back and ending with

Introduction to Scorpions

- Scorpions primarily prey on insects and other invertebrates, but some species take vertebrates.
- They use their pincers to restrain and kill prey.

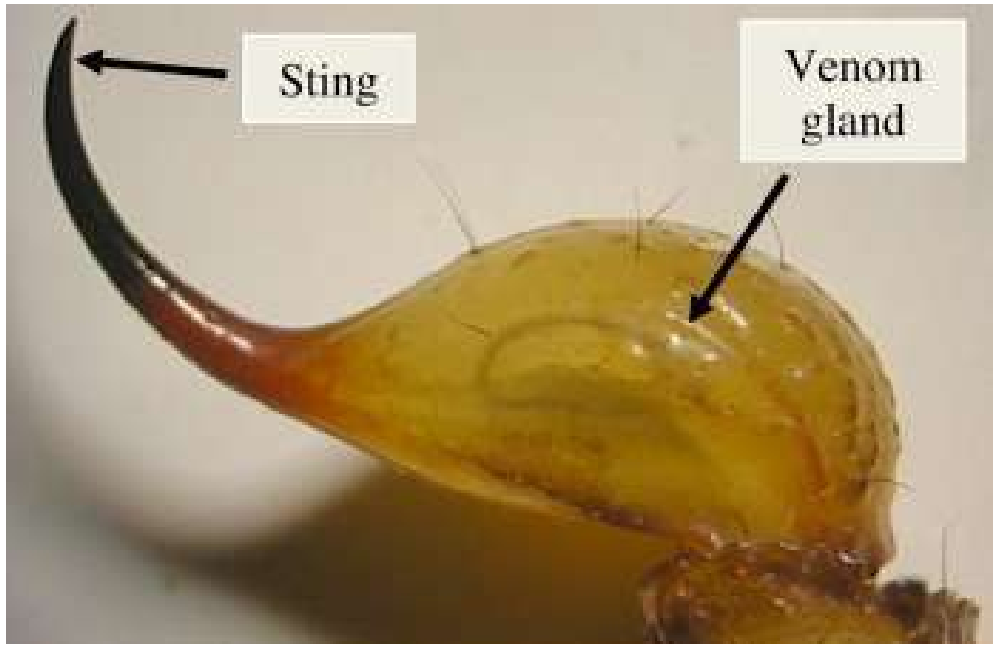
https://en.wikipedia.org/wiki/Scorpion

Introduction to Scorpions

https://en.wikipedia.org/wiki/Scorpion

- The vast majority of species do not represent a serious threat to humans, and healthy adults usually do not need medical treatment after being stung.
- Poisonous scorpions
- Habitat/desert/Mountains
- Collection/UV/Turning stone/pitfalls

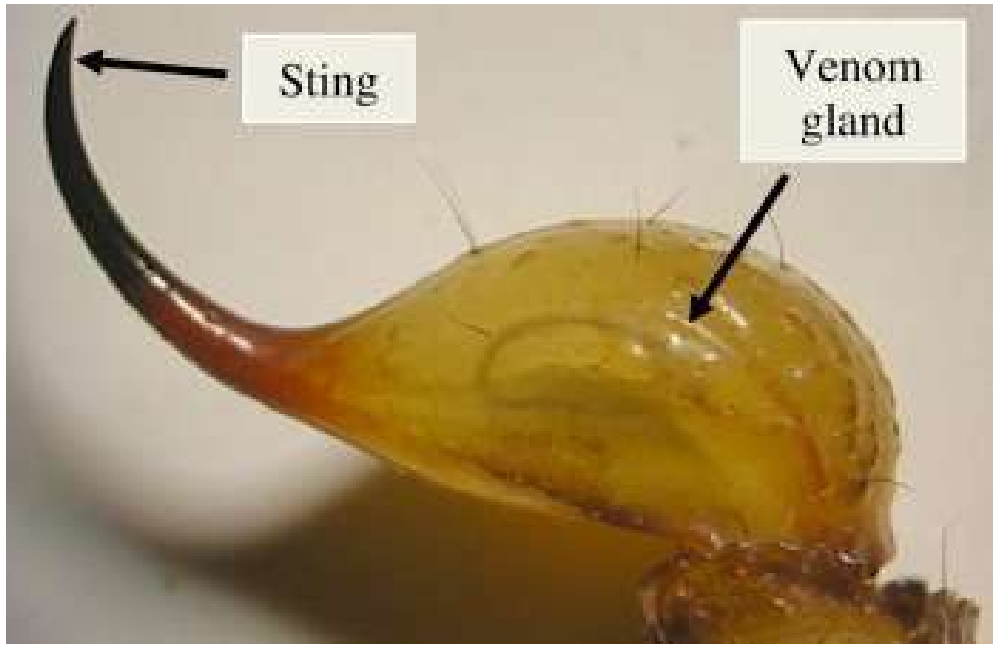
Module# 142 Introduction to Scorpions



https://en.wikipedia.org/wiki/Scorpion

- Scorpion bites are highly painful
- Venom peptides
- Type/Long chain/Short chain
- Medical importance/Buthidae
- Treatment of diseases
- Cancer treatment
- Venom price
- Venom collection

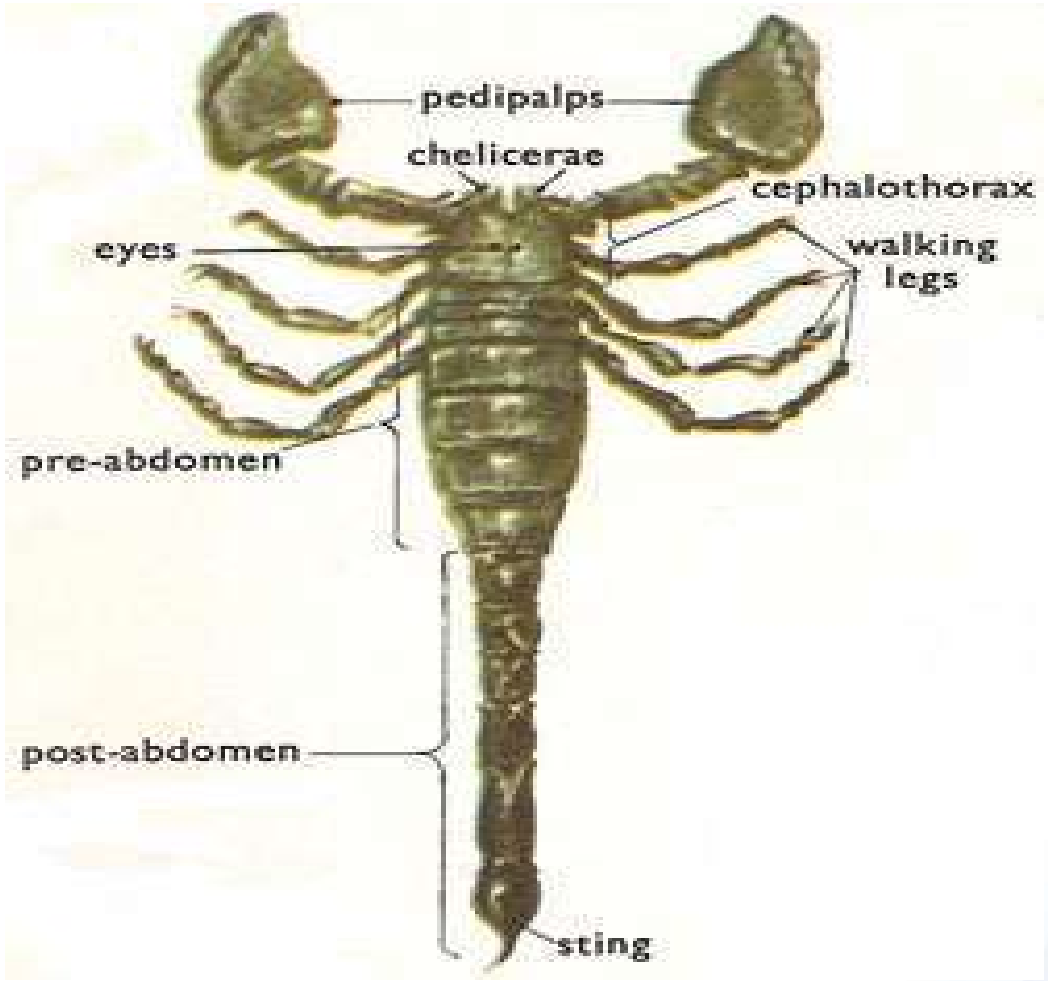
Introduction to Scorpions



- How to collect venom
- Effect of temperature on venom production
- Amount of venom
- Method of venom collection

https://en.wikipedia.org/wiki/Scorpion

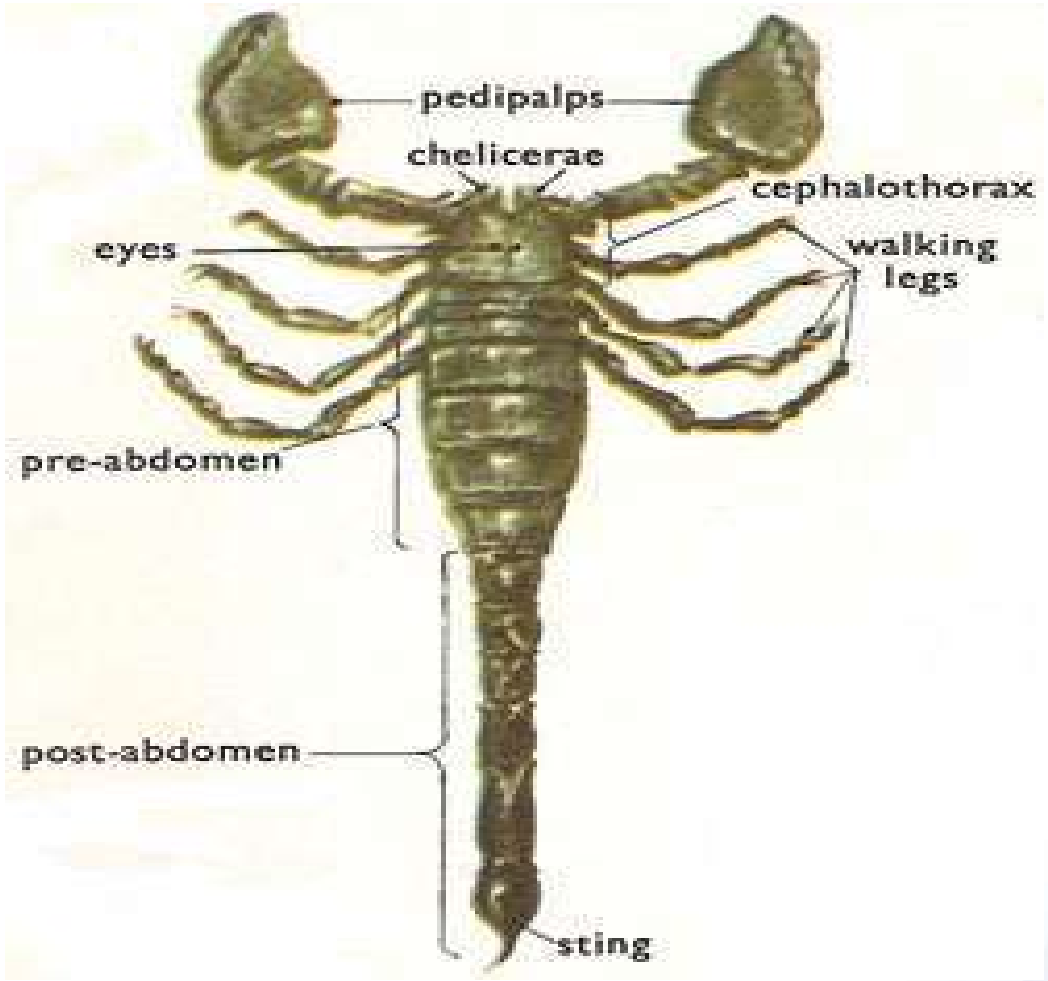
Module# 143 Morphology of Scorpions



https://en.wikipedia.org/wiki/Scorpion

- Cephalothorax
- Padipal
- Chelicera
- Abdomen
- Walking legs
- Sting
- Body is covered with a cuticle consisting of chitin and tanned proteins

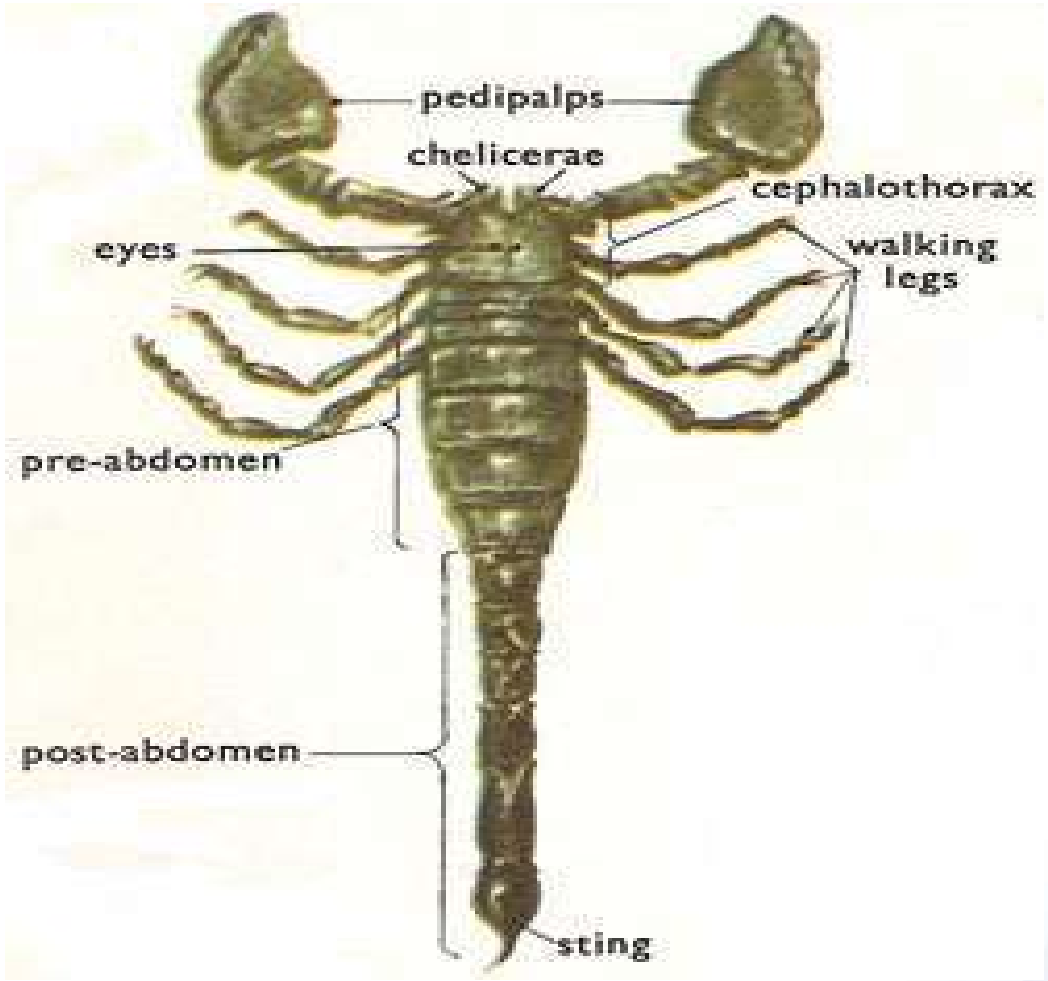
Morphology of Scorpions



https://en.wikipedia.org/wiki/Scorpion

- 8.5-23mm
- The opisthosoma is subdivided into a broad anterior portion, the mesosoma or pre-abdomen, and a narrow tail-like posterior, the metasoma or post-abdomen.
- Most species are nocturnal.
- Feeding habit/cannabalism

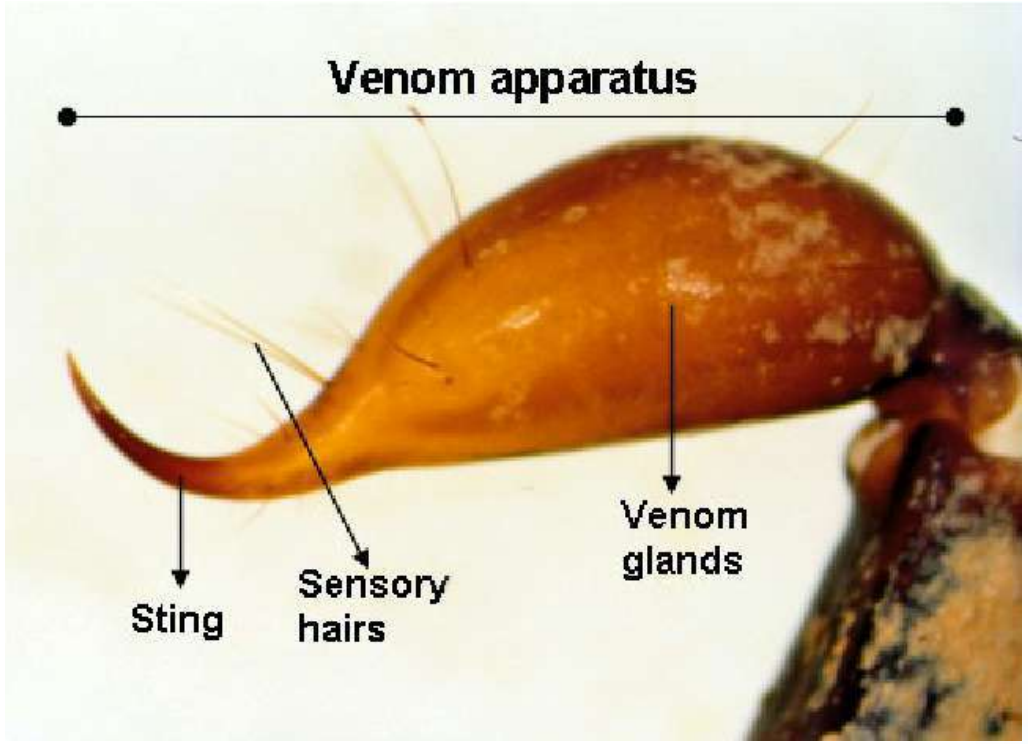
Morphology of Scorpions



https://en.wikipedia.org/wiki/Scorpion

- The telson includes the vessicle, which contains a symmetrical pair of venom glands.
- Externally it bears the curved stinger, the hypodermic aculeus, equipped with sensory hairs.
- Abdominal segments These segments have keels, setae and bristles which may be used for

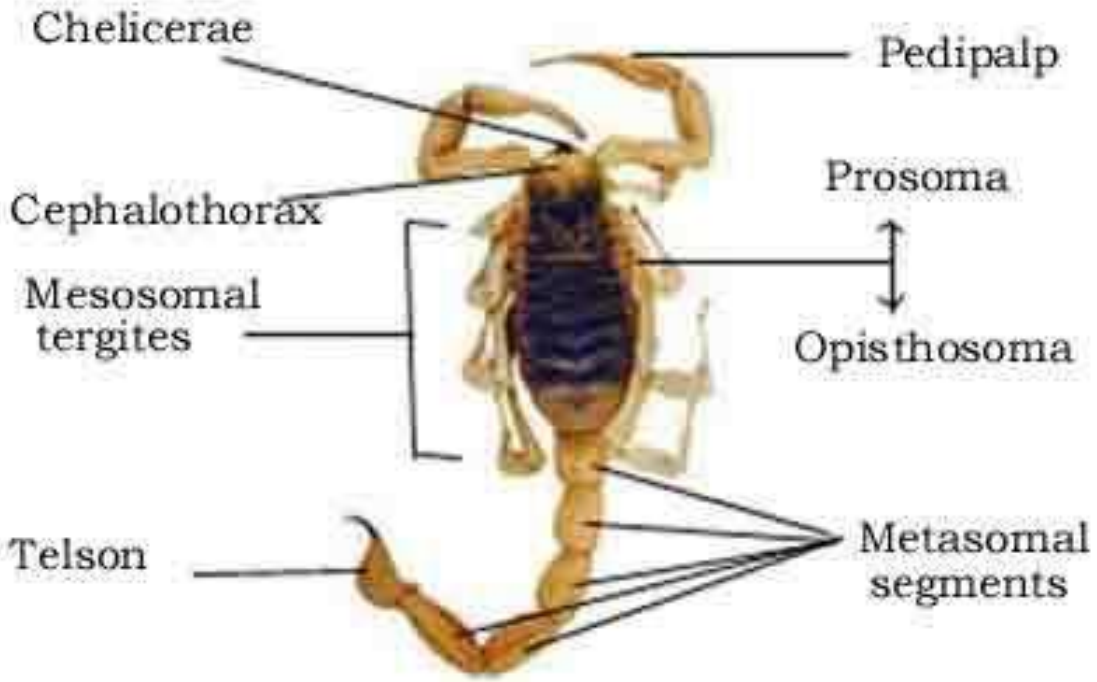
Morphology of Scorpions



https://en.wikipedia.org/wiki/Scorpion

- Each of the venom glands has its own duct to convey its secretion along the aculeus from the bulb of the gland to immediately subterminal of the point of the aculeus.

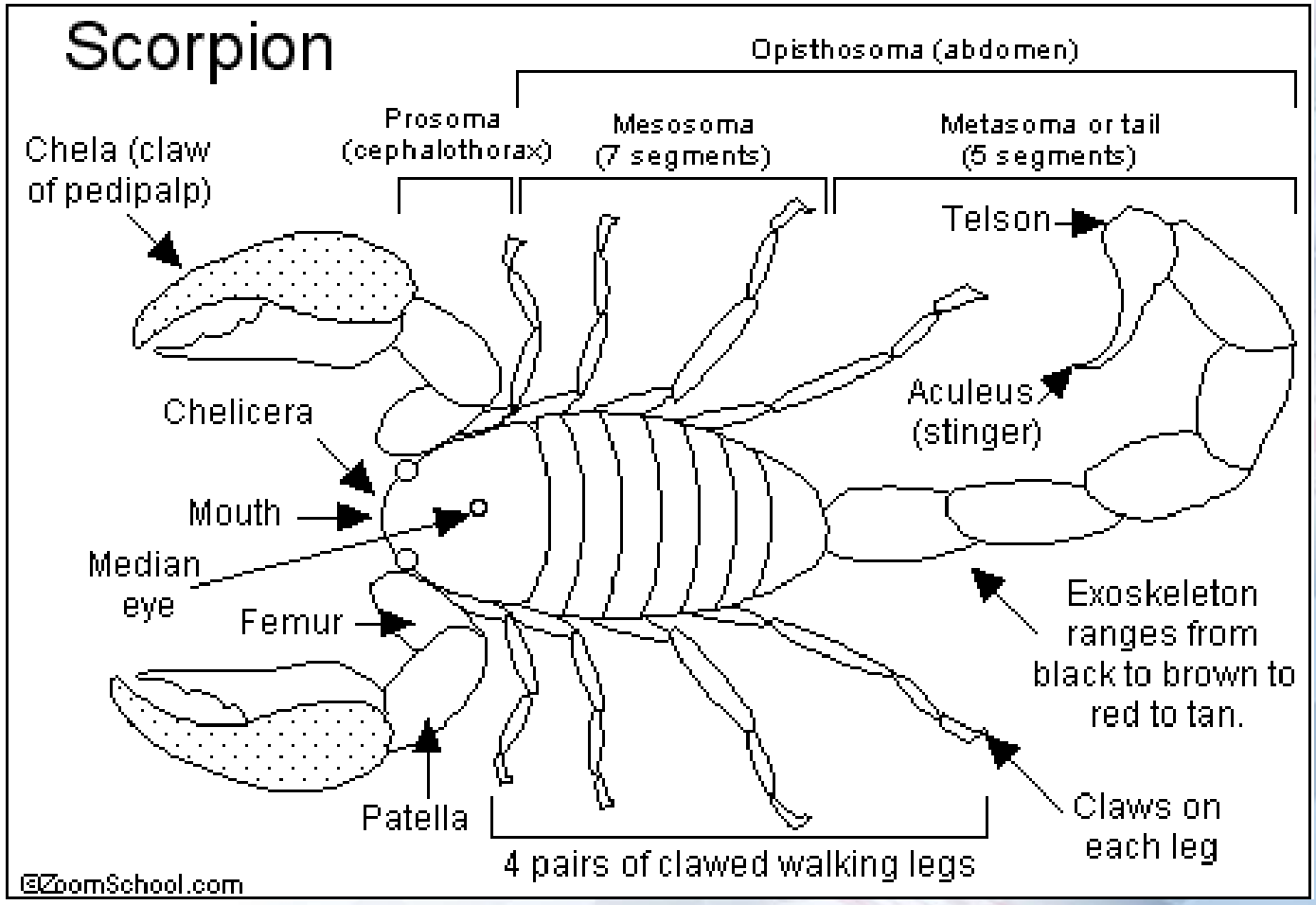
Morphology of Scorpions



https://en.wikipedia.org/wiki/Scorpion

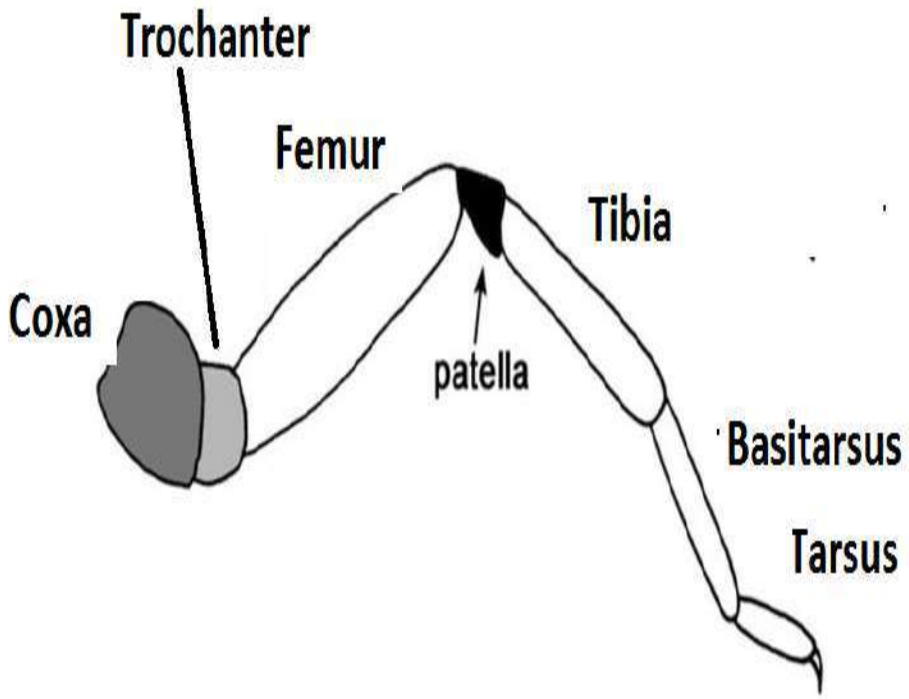
- Parts of scorpion body

Morphology of Scorpions



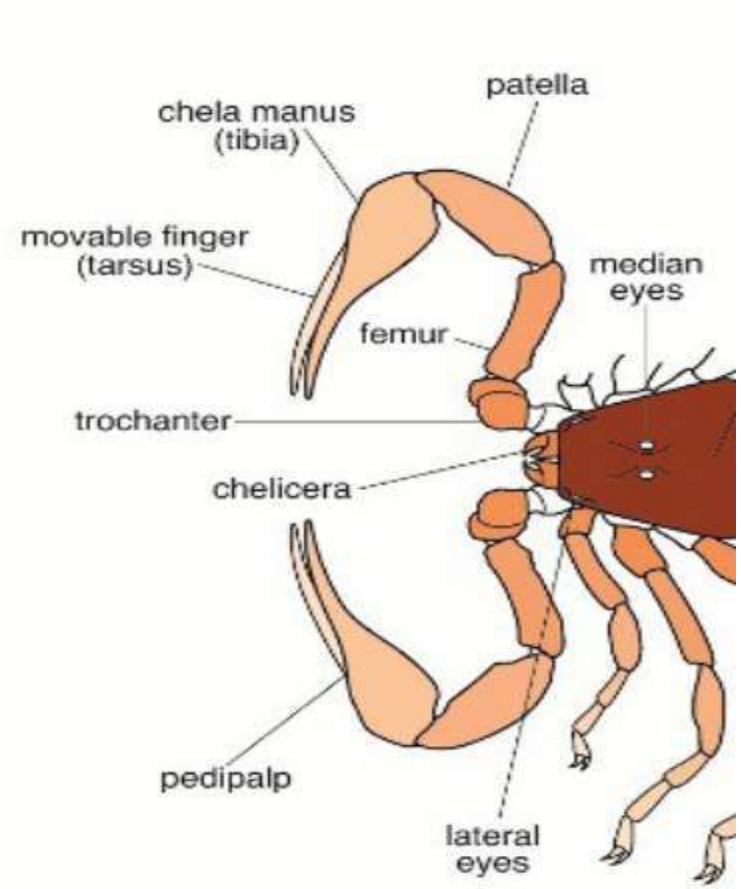
https://www.enchantedlearning.com/subjects/arachnids/scorpion/Scorpionprintout.shtml

Module# 144 Morphology of Scorpion Leg



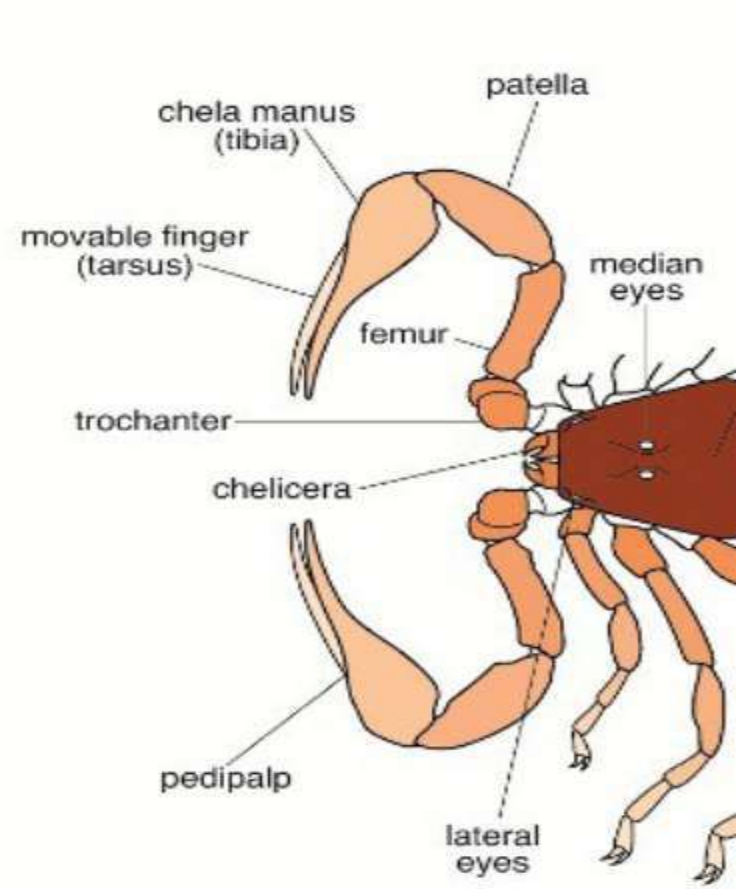
- Four pairs of legs
- Each leg is terminating in a pair of claws.
- A series of trichobothria, the number and positions of which are taxonomically informative, is present on the pedipalp femur, patella, and tibia.

Morphology of Scorpion Leg



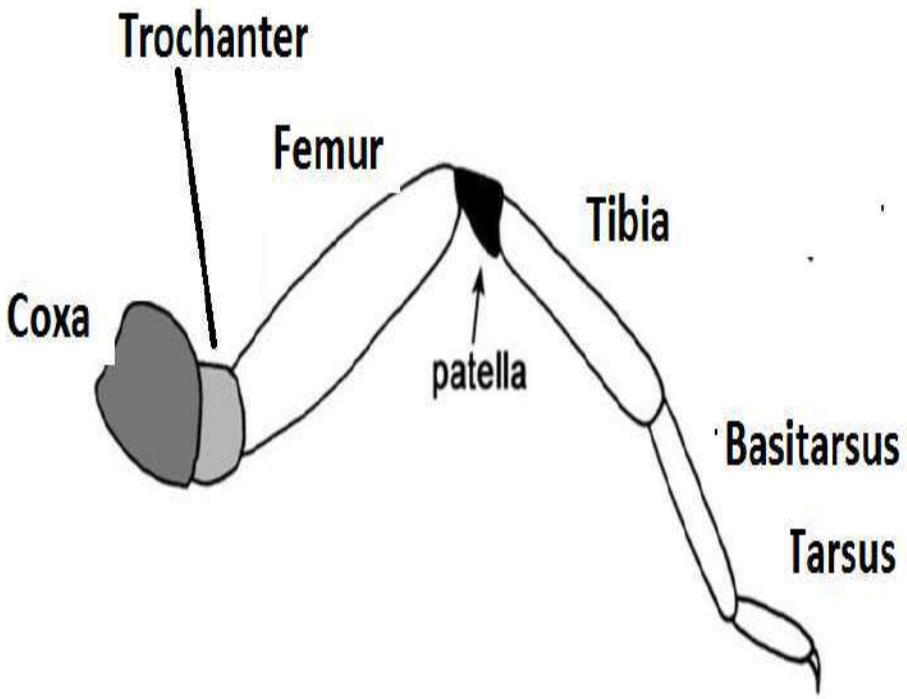
- Each chelicera comprises three segments, including a chela manus with fixed finger and a movable finger.
- Each pedipalp comprises a coxa, trochanter, femur, patella, chela manus with fixed finger (tibia), and movable finger (tarsus).

Morphology of Scorpion Leg



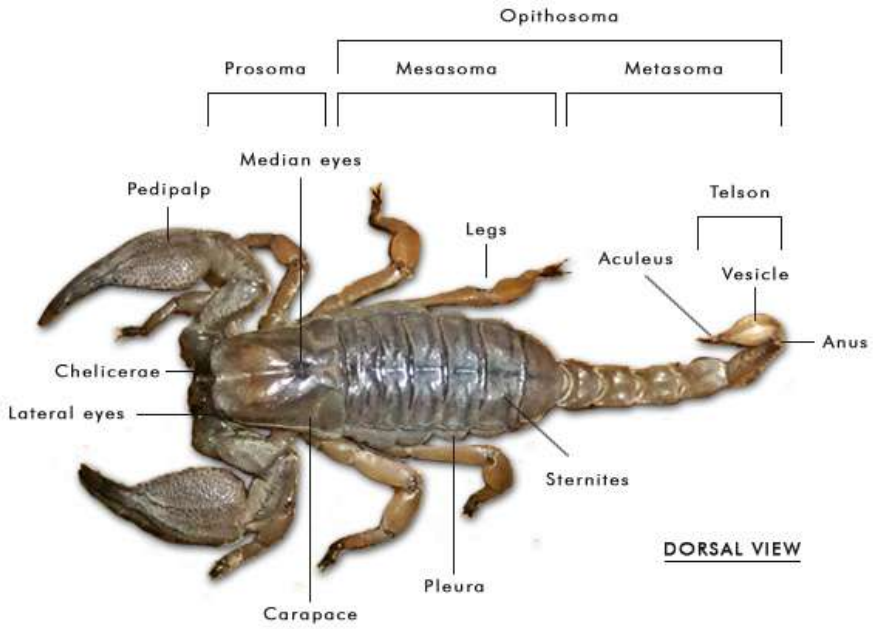
- **Setae** are thick hairs which cover the legs, body, and pedipalps (leg-like mouthparts).
- They are sensitive to the touch.
- Also located on the pedipalps are smaller hairs known as **trichobothria,** which sense air vibrations.

Morphology of Scorpion Leg



- Ground vibrations are sensed by other organs located on the tip of the legs.

Module# 145 Morphology of Scorpion Telson



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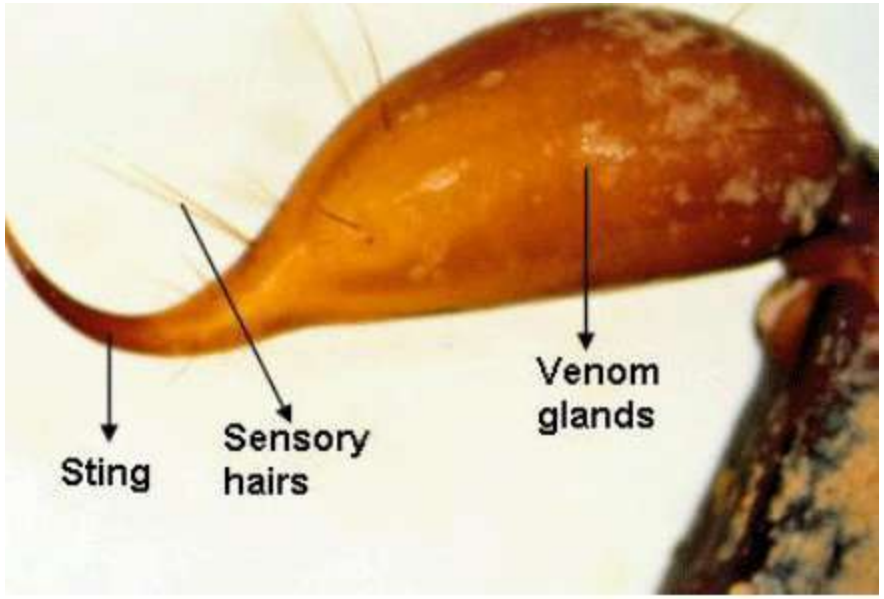
- The telson is present in Metasoma.
- Metasoma is also called as post-abdomen.
- The postabdomen includes five metasomal segments, the last of which contains the anus and bears the telson.
-

Morphology of Scorpion Telson



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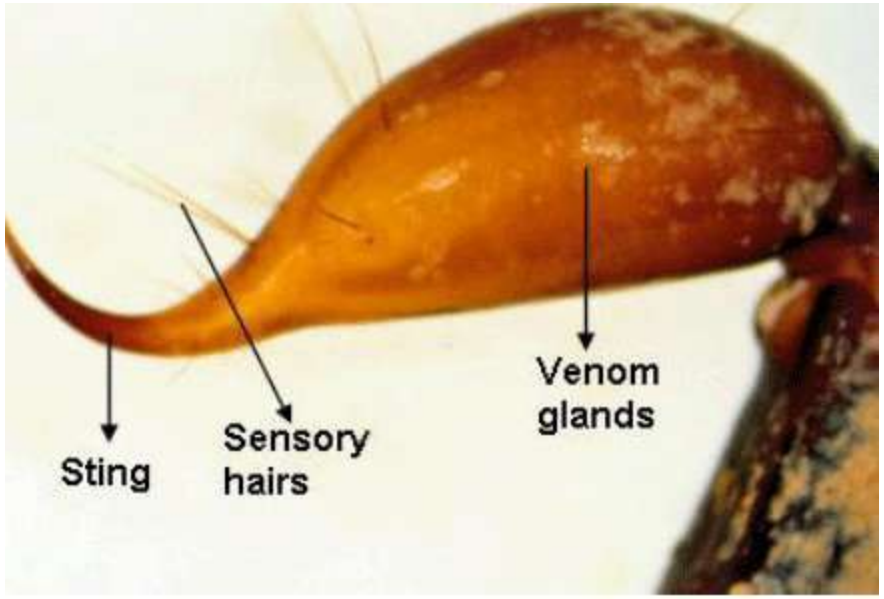
- The telson consists of the vesicle, which holds a pair of venom glands, and the hypodermic **aculeus,** the venom-injecting sting.

Morphology of Scorpion Telson



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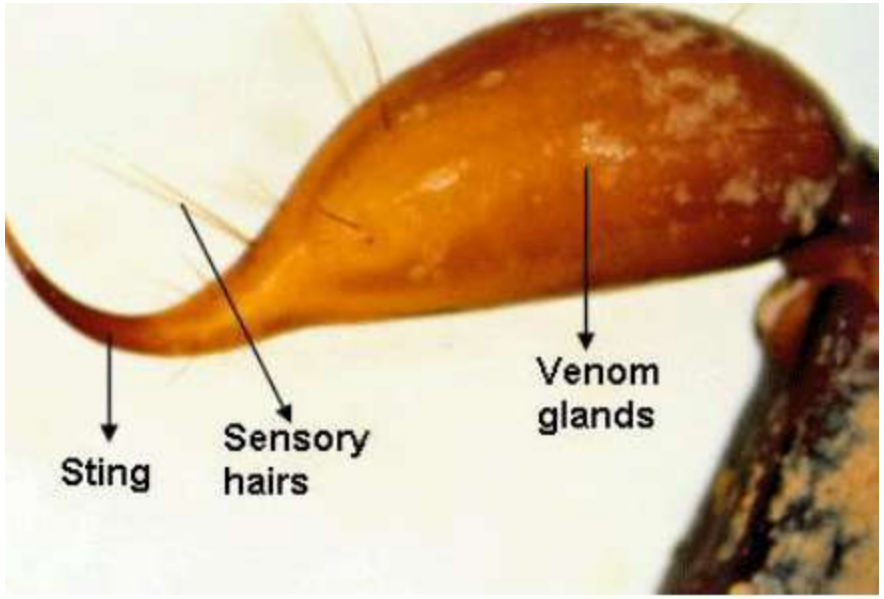
- Each of the venom glands has its own duct to convey its secretion along the aculeus from the bulb of the gland to immediately subterminal of the point of the aculeus, where each of the paired ducts has its own venom pore

Morphology of Scorpion Telson



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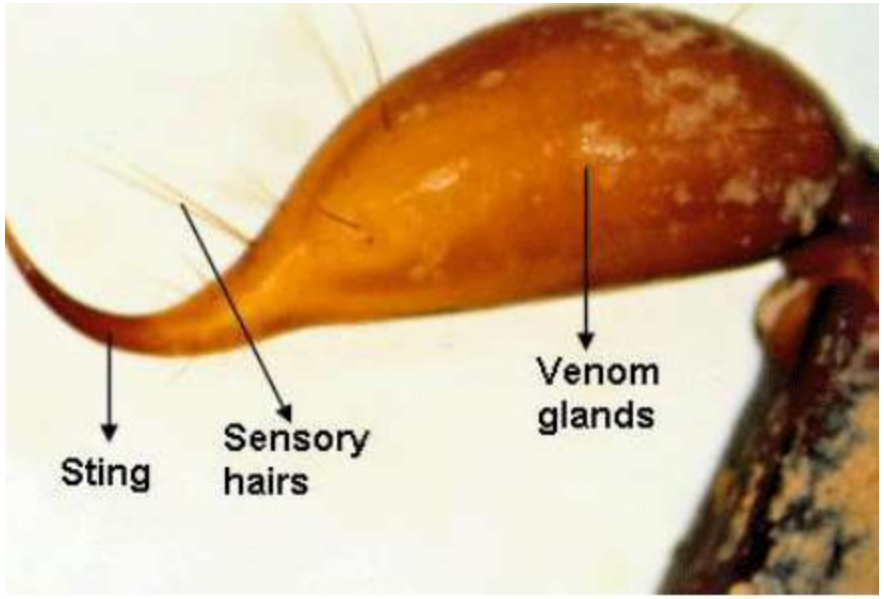
- An extrinsic muscle system in the tail moves it forward and propels and penetrates with the aculeus, while an intrinsic muscle system attached to the glands pumps venom

Morphology of Scorpion Telson



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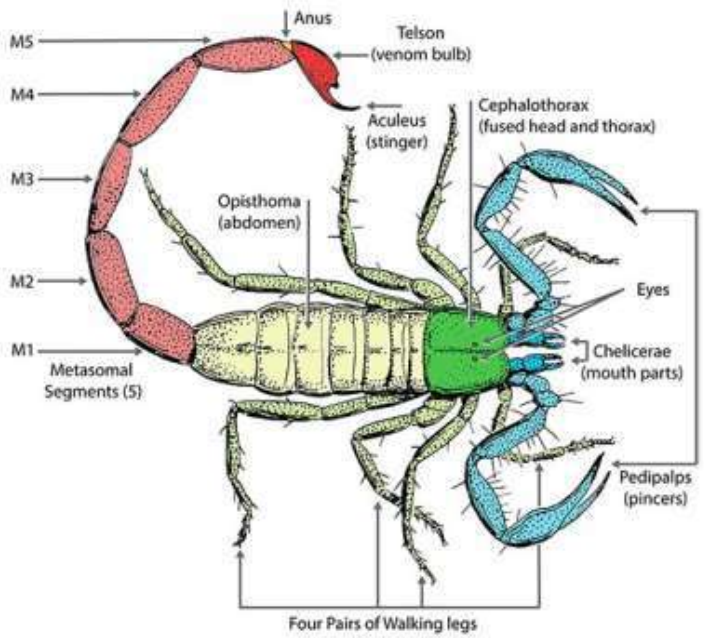
- All scorpion species possess venom, which is used to kill or paralyze prey and as a defense against predators.

Module# 146 Morphology of Chelicerae and Abdomen



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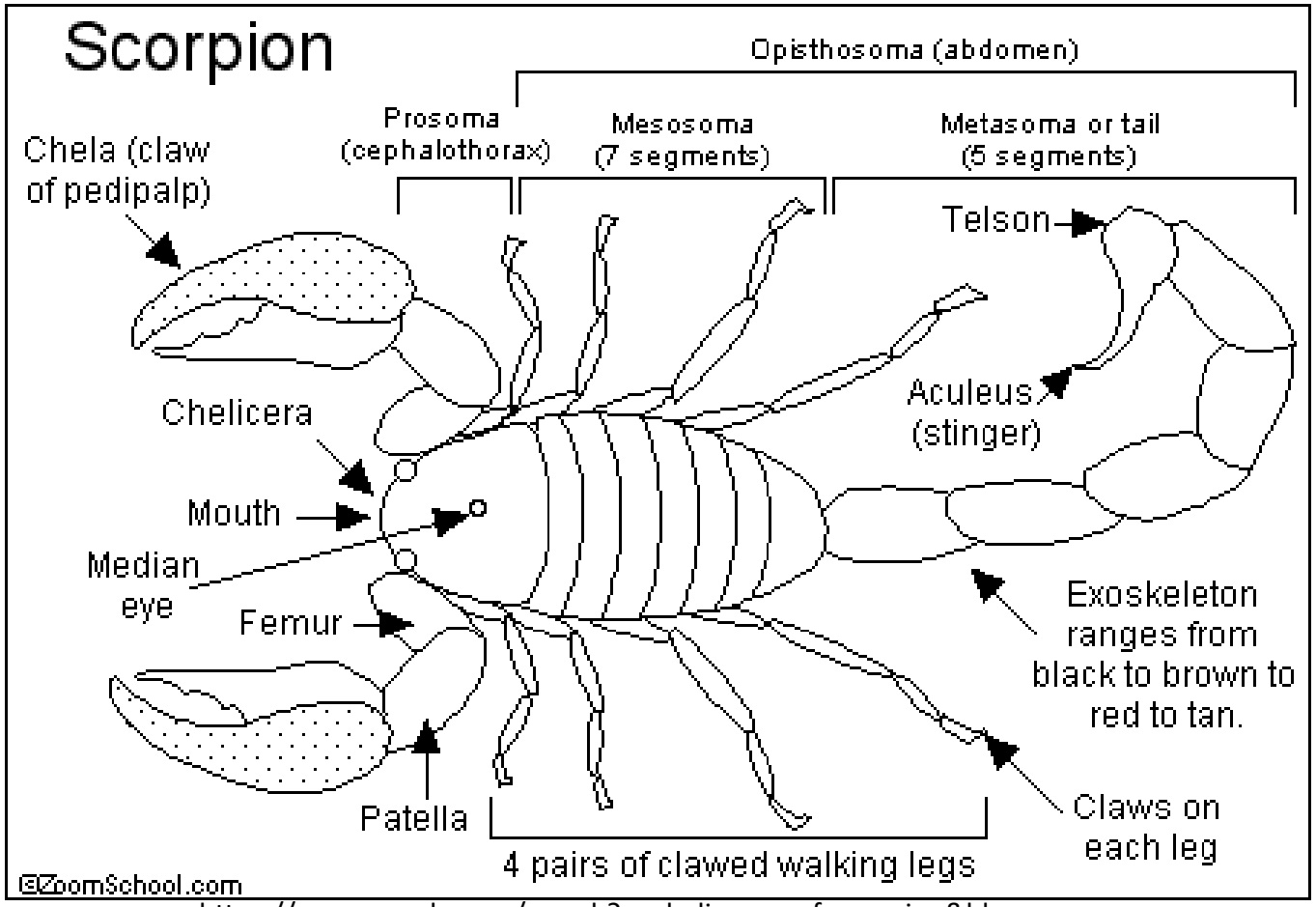
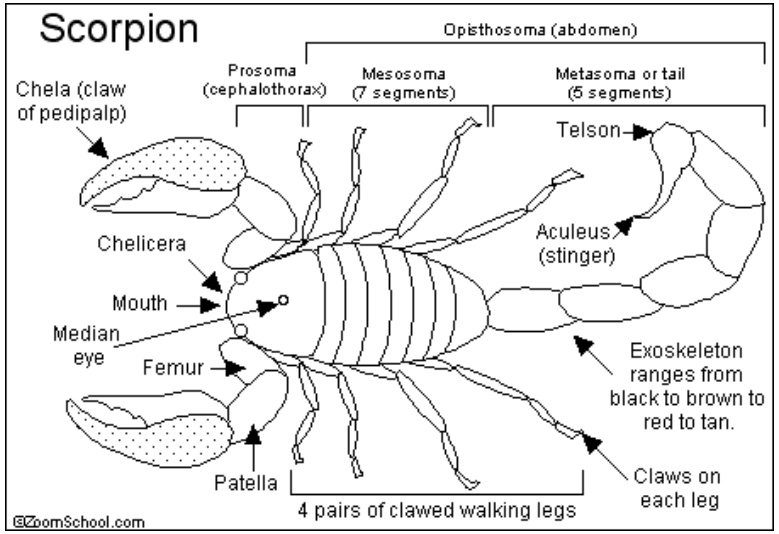
- The chelicerae of scorpion is present in cephalothorax region.
- They are scorpions jaws.
- Each chelicera comprises three parts a **coxa, tibia** which has a fixed finger, and the **tarsus** forming a moveable finger.

Morphology of Chelicerae and Abdomen



https://www.google.com/search?q=chelicerae+of+scorpion&hl=en&sxsrf=ALeKk038xnnZQit8v9nSosB_7rPe5ABxcQ:1609049933235&source=lnms&tbm=isch&sa=X&ved=2ahUKEwiJxciNwu3tAhXNWxUIHeYLDlUQ_AUoAXoECBoQAw&biw=1360&bih=657#imgrc=EzHwRyQb55Ah_M

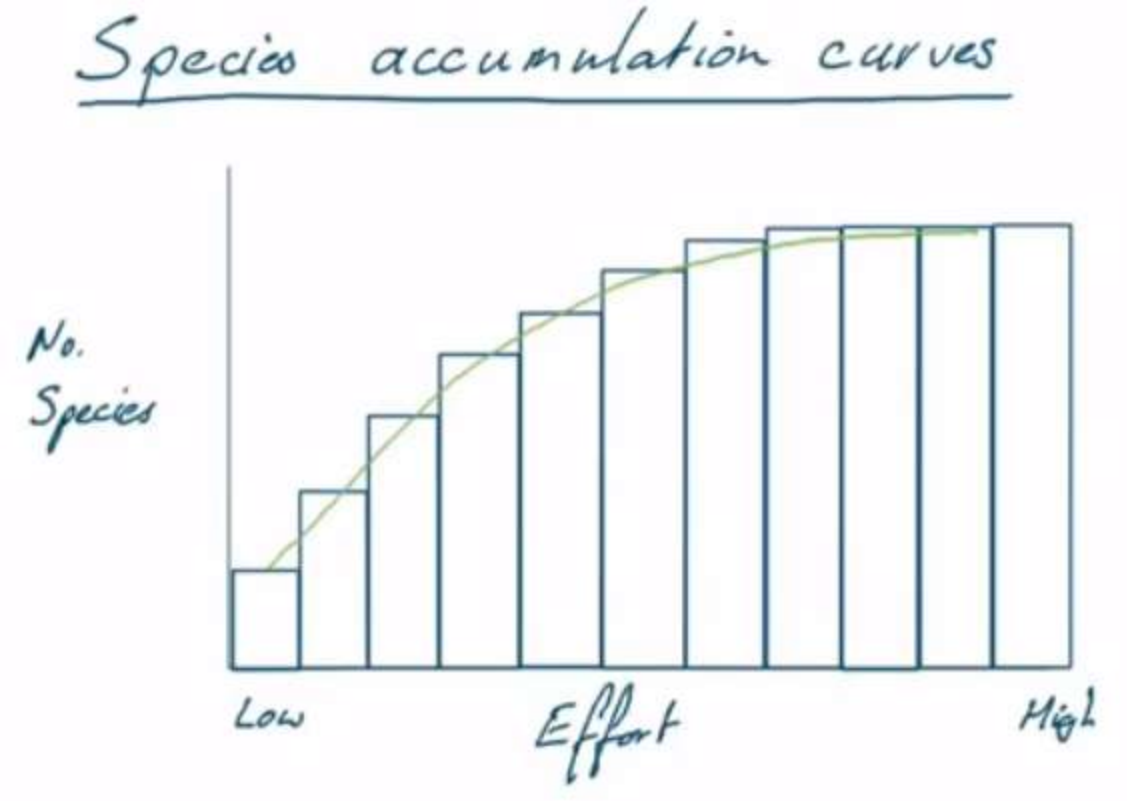
Morphology of Chelicerae and Abdomen



https://www.google.com/search?q=chelicerae+of+scorpion&hl=en&sxsrf=ALeKk038xnnZQit8v9nSosB_7rPe5ABxcQ:1609049933235&source=lnms&tbm=isch&sa=X&ved=2ahUKEwiJxciNwu3tAhXNWxUIHeYLDlUQ_AUoAXoECBoQAw&biw=1360&bih=657#imgrc=EzHwRyQb55Ah_M

- Within the abdomen are the scorpion's respiratory system, which is a set of thin plates called **book lungs.**
- A spiracle leads to each pair of book lungs on each segment.
- The opisthosoma has eight legs arranged into four pairs.

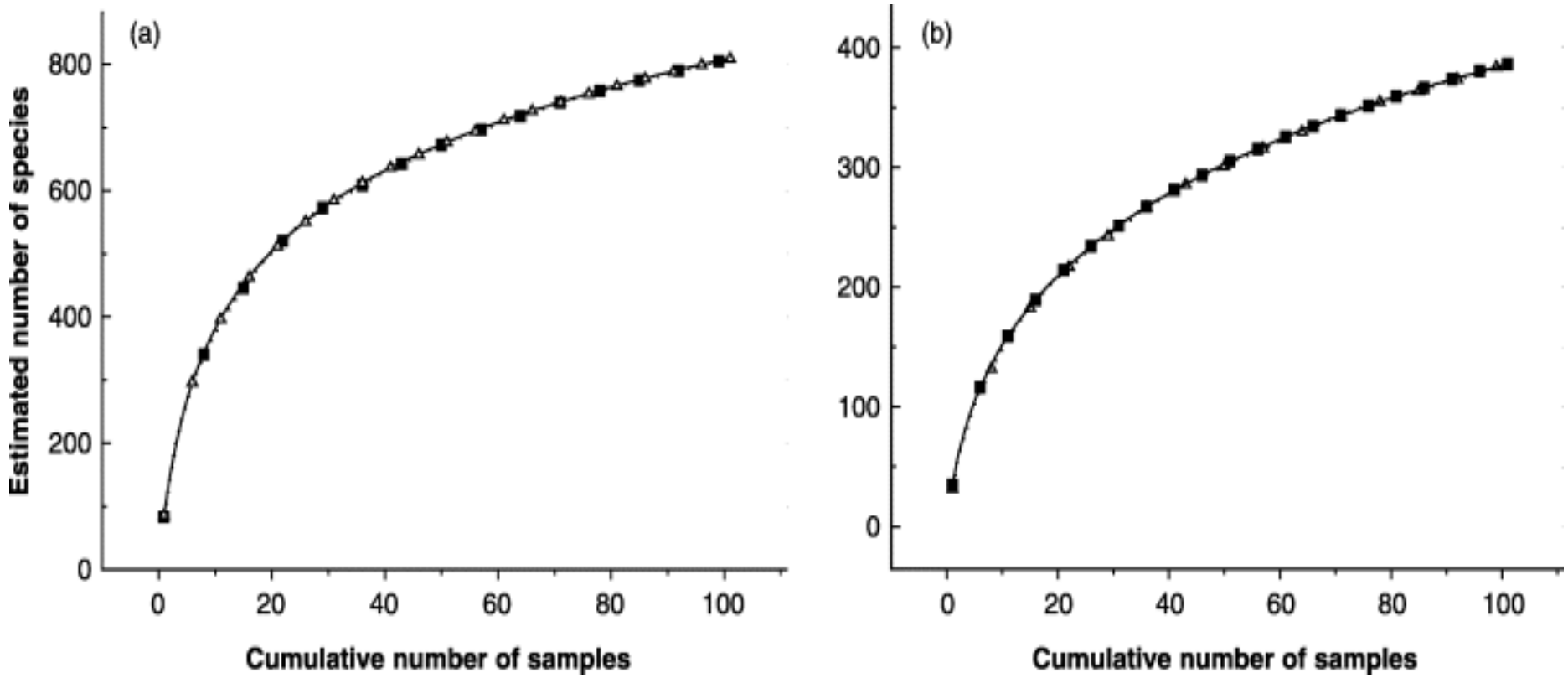
Module# 147 Accumulation Curve



https://www.youtube.com/watch?v=Jj7LYrU_6RA

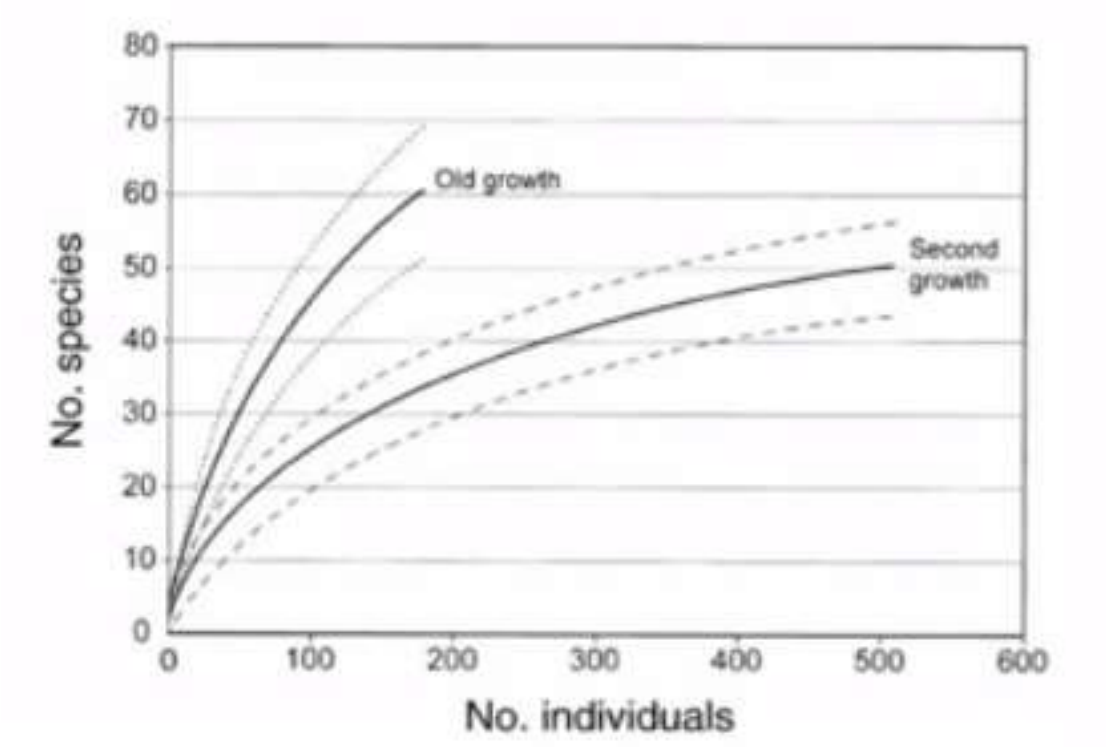
- The species discovery curve or species accumulation curve is a graph recording the cumulative number of species of living things recorded in a particular environment as a function of the cumulative effort expended searching for them.

Accumulation Curve



https://besjournals.onlinelibrary.wiley.com/doi/full/10.1046/j.1365-2656.2003.00748.x

Accumulation Curve



https://www.youtube.com/watch?v=Jj7LYrU_6RA

- Types of accumulation curve
- Importance of accumulation curve
- Accumulation curve vs trapping effort
- Accumulation curve vs trapping method
- Accumulation curve vs biasness

Module# 148 Calculation of Allelic and Genotypic Frequency

$$p + q = 1$$

$$(p + q)^2 = 1$$

$$p^2 + 2pq + q^2 = 1$$

Hardy-Weinberg Equilibrium Model

- P = frequency of Allele A
- q = Frequency of Allele a
- P2 = Frequency of AA genotype
- 2pq= Frequency of Aa genotype
- q2 = Frequency of aa genotype

Calculation of Allelic and Genotypic Frequency

	AA	Aa	aa	
Individuals	70	80	50	(200)
Genotypic Array	70/200 = 0.35	80/200 = .40	50/200 = 0.25	= 1
"A' alleles	140	80	0	220/400 = 0.55
'a' alleles	0	80	100	180/400 = 0.45

http://eweb.furman.edu/~wworthen/bio102/evolution219b.htm

Module# 149 Calculation of Allelic and Genotypic Frequency

Hardy-Weinberg Equilibrium Model

- In a population there are 320AA individuals, 160 heterozygous (Aa) and 20aa individuals.
- Calculate the genotypic and allelic frequencies.

Calculation of Allelic and Genotypic Frequency

Alleles = A, a

Genotypes = AA, Aa, aa

Frequency of allele A: $f(A) = f(AA) + 1/2\, f(Aa)$

Frequency of allele a: $f(a) = f(aa) + 1/2\, f(Aa)$

Hardy-Weinberg Equilibrium Model

- In a population there are 320AA individuals, 160 heterozygous (Aa) and 20aa individuals.
- Calculate the genotypic and allelic frequencies.

Calculation of Allelic and Genotypic Frequency

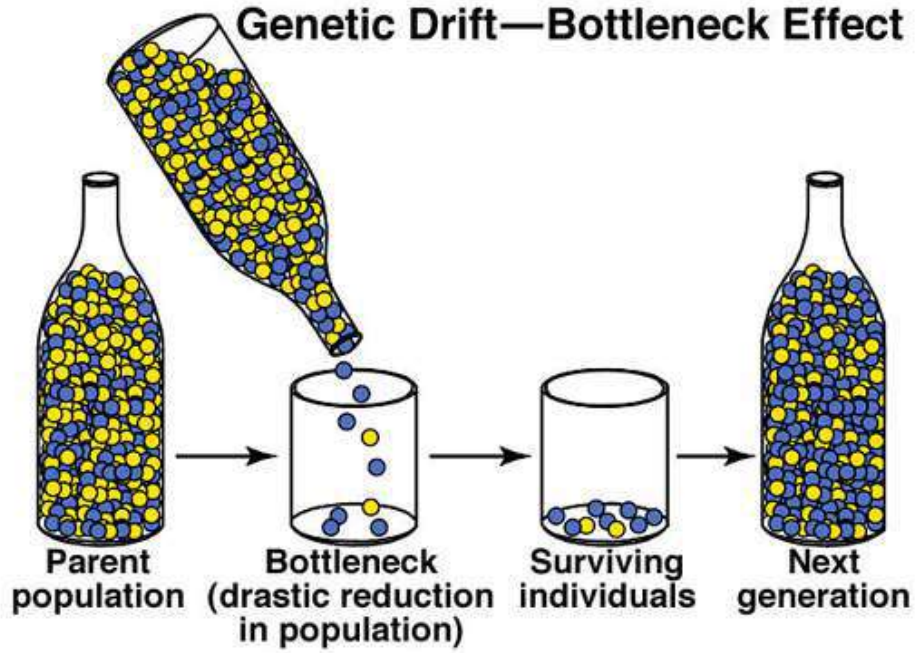
Genotypes	AA	Aa	aa
Number of individuals	320	160	20
Genotypic frequencies	320/500=0.64	16/500=0.32	20/500=0.04

Allelic Frequencies	
A	320+320+160/1000 =0.8
a	160+40/1000 =0.2

Hardy-Weinberg Equilibrium Model

- In a population there are 320AA individuals, 160 heterozygous (Aa) and 20aa individuals.
- Calculate the genotypic and allelic frequencies.

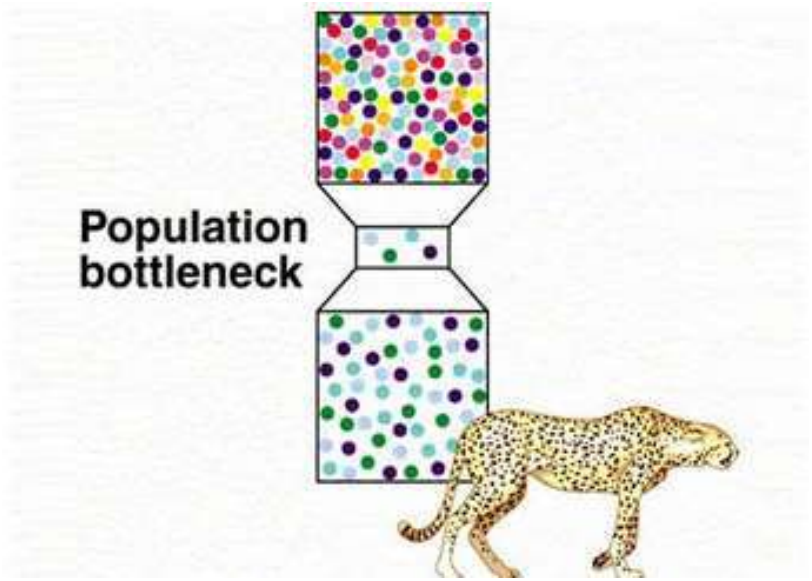
Module# 150 Genetic Drift and its impact on Allele Frequency



https://www.google.com/search?q=genetic+drift&tbm=isch&ved=2ahUKEwjVurHI-evtAhVPw4UKHYbCBvoQ2-cCegQIABAA&oq=genetic+&gs_lcp=CgNpbWcQARgAMgQIABBDMgQIABBDMgQIABBDMgIIADICCAAyAggAMgIIADICCAAyAggAMgIIADoFCAAQsQM6BAgjECc6BwgAELEDEENQ3KsjWNe6I2D6zCNoAHAAeACAAfUDiAGCE5IBCTItNi4wLjEuMZgBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=BVXnX5WkA8-GlwSGhZvQDw&bih=657&biw=1360#imgrc=bg-AkcfJaUxBGM

- Change in allege frequency which occur by chance.
- Loss in alleles
- Loss in genetic diversity
- Lost genes for ever
- Bottle neck effect
- Unique genes
- Examples

Genetic Drift and its impact on Allele Frequency



https://www.google.com/search?q=genetic+drift&tbm=isch&ved=2ahUKEwjVurHI-evtAhVPw4UKHYbCBvoQ2-cCegQIABAA&oq=genetic+&gs_lcp=CgNpbWcQARgAMgQIABBDMgQIABBDMgQIABBDMgIIADICCAAyAggAMgIIADICCAAyAggAMgIIADoFCAAQsQM6BAgjECc6BwgAELEDEENQ3KsjWNe6I2D6zCNoAHAAeACAAfUDiAGCE5IBCTItNi4wLjEuMZgBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=BVXnX5WkA8-GlwSGhZvQDw&bih=657&biw=1360#imgrc=bg-AkcfJaUxBGM

- Change in allege frequency which occur by chance.
- Loss in alleles
- Loss in genetic diversity
- Lost genes for ever
- Bottle neck effect
- Unique genes
- Examples

Module# 151 Principles of Systematics

Concept of Fitness

Concept of fitness

Genetic Load = Max. fit individual - Ave. fitness of population

Genetic Load = 9-6 =3

- Survival+ reproductive success
- Relative fitness
- Ideal population
- Maximum value of fitness
- Genetic load/Genetic burden
- Population with high load
- Population with low load
- Variability in fitness

Module# 152 Taxonomic publications

Kinds of publications

- Short research paper
- Synopsis & Reviews
- Revisions
- Monographs
- Faunal/Floral Work
- Field guide/Manuals
- Catalogues & Check lists

- No taxonomic study is complete until its results are published.
- Advances in any area of science is made trough publications.
- Taxonomic publications range from short description of species (may be one page) to monograph or handbook (several volumes)

Taxonomic publications

Kinds of publications

- Short research paper
- Synopsis & Reviews
- Revisions
- Monographs
- Faunal/Floral Work
- Field guide/Manuals
- Catalogues & Check lists

- They are ordinary descriptive papers of new species, subspecies or combined all of these or some.
- May or may not contain key for identification.
- May contain a brief check list.

Taxonomic publications

Kinds of publications

- Short research paper
- Synopsis & Reviews
- Revisions
- Monographs
- Faunal/Floral Work
- Field guide/Manuals
- Catalogues & Check lists

- They are brief summaries of current knowledge of a group.
- Not necessary to include new material or new interpretations.
- Such work bring scattered work together
- Helpful for Revisionary or monographic work.

Module# 153 Taxonomic publications

Kinds of publications

- Short research paper
- Synopsis & Reviews
- Revisions
- Monographs
- Faunal/Floral Work
- Field guide/Manuals
- Catalogues & Check lists

- Revisions are representations of new material or new interpretations that have been integrated with previous knowledge with brief summaries i.e.
- Restudy of complete unit of particular group of Taxa.

Taxonomic publication

Kinds of publications

- Short research paper
- Synopsis & Reviews
- Revisions
- Monographs
- Faunal/Floral Work
- Field guide/Manuals
- Catalogues & Check lists

- Revisions usually include super families, families or part of families.
- It can also deal with genus or species.
- Generic revision
- Species revision

Taxonomic publications

Kinds of publications

- Short research paper
- Synopsis & Reviews
- Revisions
- Monographs
- Faunal/Floral Work
- Field guide/Manuals
- Catalogues & Check lists

- It include re-description or diagnosis of little known taxa, list of synonymy, new names along with keys.

Taxonomic publications

Kinds of publications

- Short research paper
- Synopsis & Reviews
- Revisions
- Monographs
- Faunal/Floral Work
- Field guide/Manuals
- Catalogues & Check lists

- A comprehensive systematic study of a particular taxonomic group.
- Always taxon-based, not geography-based.
- Its preparations require more complete material than other types of publications.

Taxonomic publications

Kinds of publications

- Short research paper
- Synopsis & Reviews
- Revisions
- Monographs
- Faunal/Floral Work
- Field guide/Manuals
- Catalogues & Check lists

- 3-5 years for completion.
- Monographs try to review all the facts about a taxonomic group.
- A **Monograph** is a complete global account of a **taxon** of any rank – family, genus or species at a given time.
- Book format

Taxonomic publications

Kinds of publications

- Short research paper
- Synopsis & Reviews
- Revisions
- Monographs
- Atlases
- Faunal/Floral Work
- Field guide/Manuals
- Catalogues & Check lists

- They show comprehensive characters of animals in picture form.
- Purpose of atlases are purely taxonomic
- Very helpful taxonomist for identification
- Comparison of species

Module# 154 Taxonomic publications

Kinds of publications

- Short research paper
- Synopsis & Reviews
- Revisions
- Monographs
- Atlases
- Faunal/Floral Work
- Field guide/Manuals
- Catalogues & Check lists

- It deals with detail description faunal/Floral work of particular region.
- It may include information on geographical distribution.
- Also include quantitative data and ecological comments.
- Only experience specialists.

Taxonomic publications

Kinds of publications

- Short research paper
- Synopsis & Reviews
- Revisions
- Monographs
- Atlases
- Faunal/Floral Work
- Field guide/Manuals
- Catalogues & Check lists

- Example: Altas of scale insects of North America
- Some works are done by taxonomists for non-taxonomists to help them identify common animals.
- Include easy and understandable simple key characters.

Taxonomic publications

Kinds of publications

- Short research paper
- Synopsis & Reviews
- Revisions
- Monographs
- Atlases
- Faunal/Floral Work
- Field guide/Manuals
- Catalogues & Check lists

- Manuals are also written in simple language.
- For students and layman

Taxonomic publications

Kinds of publications

- Short research paper
- Synopsis & Reviews
- Revisions
- Monographs
- Atlases
- Faunal/Floral Work
- Field guide/Manuals
- Catalogues & Check lists

- A catalogue is an index to taxa arranged in such a way to give a vivid picture of references for both zoological and nomenclature purpose:
- Catalogue contain following information:

Taxonomic publications

Kinds of publications

- Short research paper
- Synopsis & Reviews
- Revisions
- Monographs
- Atlases
- Faunal/Floral Work
- Field guide/Manuals
- Catalogues & Check lists

- Catalogue contain following information:
- The original description reference
- Later references
- Type locality
- Depository detail
- Type of genus

Taxonomic publications

Kinds of publications

- Short research paper
- Synopsis & Reviews
- Revisions
- Monographs
- Atlases
- Faunal/Floral Work
- Field guide/Manuals
- Catalogues & Check lists

- Taxa are written alphabetically
- Merits depends on completeness
- Its preparation is tedious jobe
- Useful to know many species/genera are reported from particular area.

Module# 155 Parts of thesis & Research paper

Thesis

- Abstract/Summary
- Introduction
- Review of literature
- Materials and Methods
- Results
- Discussion
- References

Parts of thesis & Research paper

END

Research paper

- Abstract
- Introduction
- Materials and Methods
- Results
- Discussion
- Acknowledgements
- References

Module# 156 Parts of thesis & Research paper

Parts of references

- Name of authors
- Year of publication
- Title/topic of article
- Name of Journal
- Volume number
- Issue number
- Page number

Parts of thesis & Research paper

Pakistan J. Zool., vol. 41(2), pp. 85-93, 2009.

Effects of Different Management Practices and Field Margins on the Abundance of Ground Spiders in Rice Ecosystems

Hafiz Muhammad Tahir
Department of Zoology University of the Punjab Lahore, Pakistan

Pakistan J. Zool., vol. 41(2), pp. 85-93, 2009.

Effects of Different Management Practices and Field Margins on the Abundance of Ground Spiders in Rice Ecosystems

Hafiz Muhammad Tahir* and Abida Butt
Department of Zoology University of the Punjab Lahore, Pakistan

Reference in Text

- Spider is carnivorous animal (Tahir, 2009)

- Spider is carnivorous animal (Tahir and Butt, 2009)

Parts of thesis & Research paper

Effect of Pesticides on agrobiont spiders in Laboratory and Field

Sher Muhammad Sherawat, Abida Butt and Hafiz Muhammad Tahir

Pakistan Journal of Zoology Vol 47, issue 4

pages: 120-129.

Reference in Text

- Spider is carnivorous animal (Sherawat et al., 2009)

- Spider is carnivorous animal (Tahir, 2008; Tahir, 2012; Sherawat et al., 2009)

Parts of thesis & Research paper

Effect of Pesticides on agrobiont spiders in Laboratory and Field

Sher Muhammad Sherawat, Abida Butt and Hafiz Muhammad Tahir

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pages: 120-129.

Reference in Text

- Spider is carnivorous animal (Sherawat et al., 2009a)

- Spider is carnivorous animal (Sherawat et al., 2009b)

Module# 157 How to write References

Effect of Pesticides on agrobiont spiders in Laboratory and Field

Sher Muhammad Sherawat, Abida Butt and Hafiz Muhammad Tahir

Pakistan Journal of Zoology Vol 47, issue 4

pages: 120-129.

Full reference

- Serawat, S.M., Butt, A. and Tahir, H.M., 2009. Effect of pesticides on agrobiont spiders in laboratory and field. Pakistan Journal of Zoology, 42 (4): 120-129.

How to write references

Effect of Pesticides on agrobiont spiders in Laboratory and Field

Abida Butt and Hafiz Muhammad Tahir

Pakistan Journal of Zoology Vol 47, issue 4

pages: 120-129.

Full reference

- Butt, A. and Tahir, H.M., 2009. Effect of pesticides on agrobiont spiders in laboratory and field. Pakistan Journal of Zoology, 42 (4): 120-129.

How to write References

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Lahore

Reference of Thesis

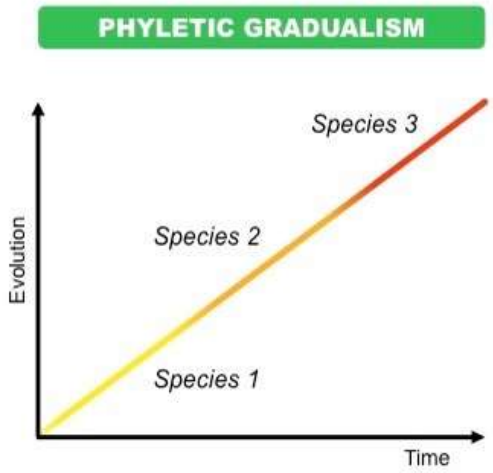
- Tahir, H.M., 2009. Effect of pesticides on agrobiont spiders in laboratory and field. Thesis submitted to University of the Punjab, Lahore, Pakistan.

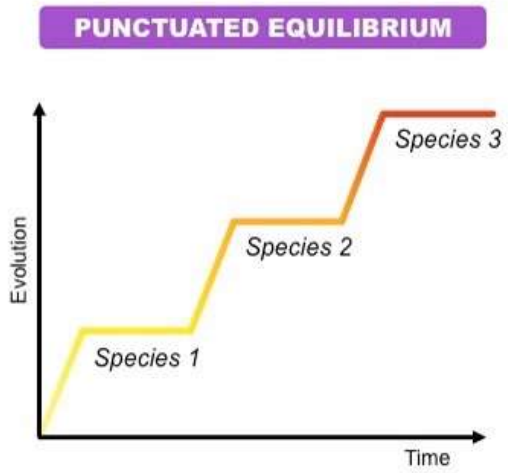
How to write References

Reference of Book

- Tahir, H.M., 2009. Effect of pesticides on agrobiont spiders in laboratory and field. Oxford University press, England. Pages. 120-129.

Module# 158 Phyletic Gradualism Vs Punctuated Equilibrium

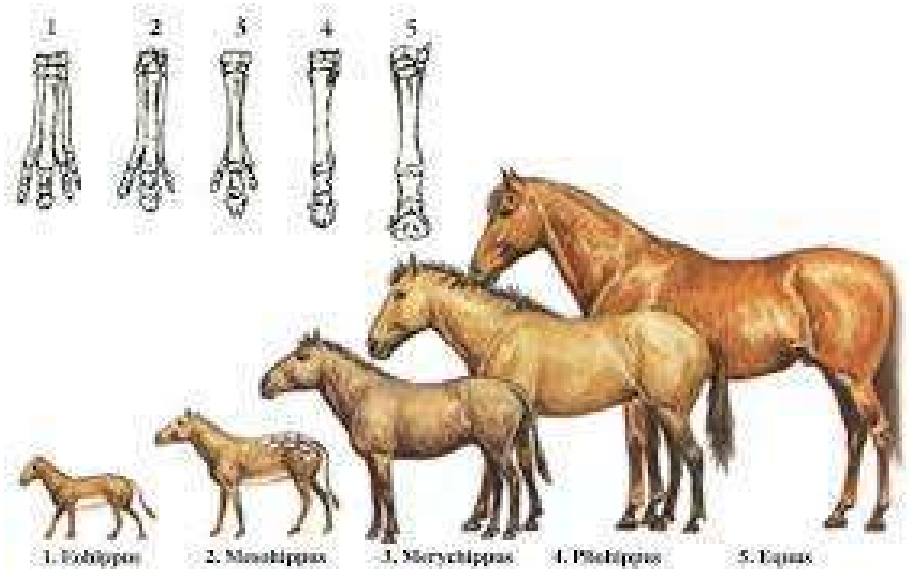




Phyletic gradualism

- Evolution is a continuous process
- Bring small variation to particular species
- Occurred over a long period of time
- Describe as slow process
- Not proven by fossil record

Phyletic Gradualism Vs Punctuated Equilibrium

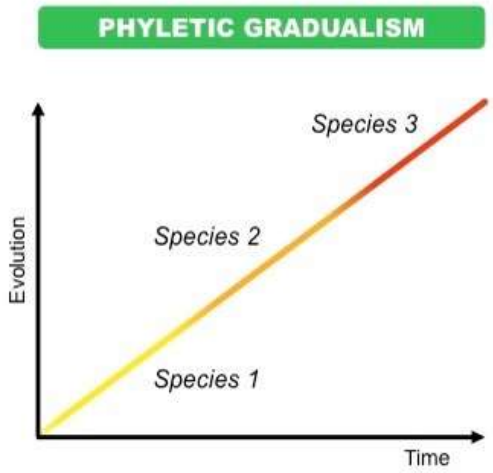


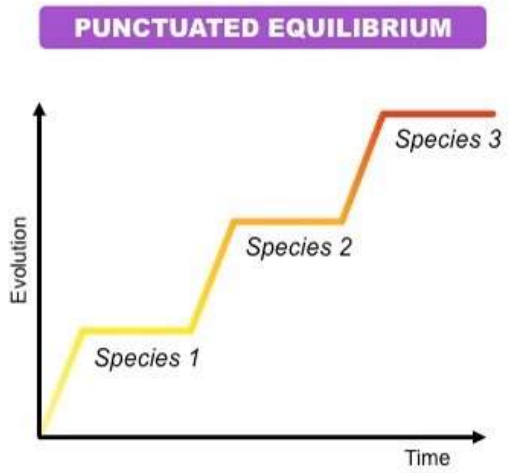
https://www.google.com/search?q=evolution+of+horses&sxsrf=ALeKk02aH8tlJqhmp23ipjOZarZGrJkUhA:1609052837923&source=lnms&tbm=isch&sa=X&ved=2ahUKEwig1tD2zO3tAhV-UxUIHTvEDe4Q_AUoAXoECB4QAw&biw=1360&bih=657

Phyletic gradualism

- It is natural theory of natural selection.
- Small changes are pased from one generation to other generation.
- Horses 50 million years (from small to model day horse)

Phyletic Gradualism Vs Punctuated Equilibrium

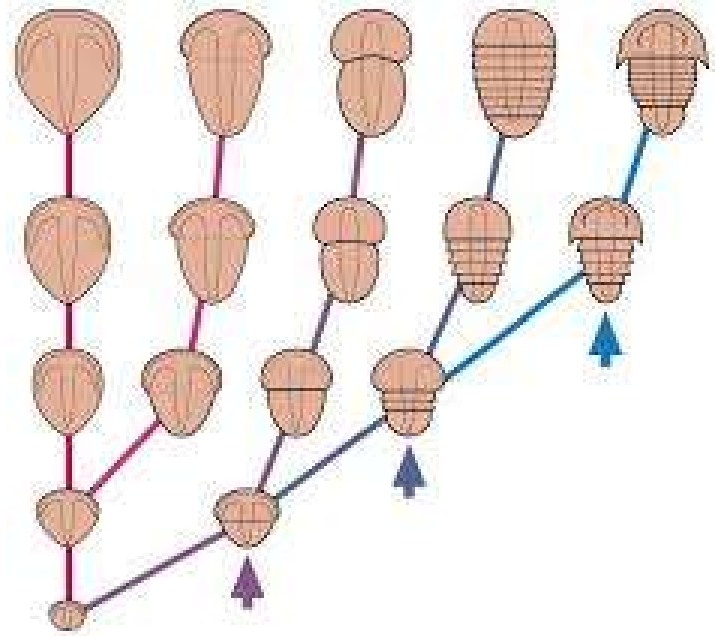




Punctuated Equilibrium

- Evolution is not a continuous process/fast at one time and slow at other time
- Caused consistent and accumulative changes
- Occurred within short period of time

Phyletic Gradualism Vs Punctuated Equilibrium



https://www.google.com/search?q=evolution+of+trilobites&tbm=isch&ved=2ahUKEwjY7rH5zO3tAhUQ4oUKHUqKCnIQ2-cCegQIABAA&oq=evolution+of+trilobites&gs_lcp=CgNpbWcQAzICCAAyBAgAEBg6BAgjECc6BAgAEENQ_bkFWIzzBWCH9QVoAHAAeACAAcQCiAHwK5IBBjItMTkuM5gBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=qzLoX5jtKpDElwTKlKqQBw&bih=657&biw=1360

Punctuated Equilibrium

- Better model to describe species
- Evolution of Trilobites

Module# 159 Adaptation

- Adaptation refers to all the structural, functional and behavioral characteristics that enhance the organism's reproductive success in its natural environment.

Adaptation

Examples

- **Frogs:** Have adapted to freezing temperatures by evolving a glycol like substance in their bloodstreams - like antifreeze.
- **Bats:** Have adapted their hearing to hunt at night via sonic response.

Adaptation

- **Butterflies:** non toxic ones have adapted to look like their more poisonous cousins.
- **A behavioral adaptation** in the oceans is the use of loud, low-frequency calls by fin whale to communicate with other whales over great distances.
-

Adaptation

- **One physical adaptation** used in the intertidal zones is a crab's hard shell, which protects it from predators, drying out and being crushed by waves.
-

Module# 160 Taxonomy and Ecology of spiders

Phylum	Arthropoda
Class	Arachnida
Sub-class	Chelicerata
Class	Arachnida
Order	Araneae

- Spiders are poisonous, predacious and fluid feeding carnivorous animals.
- They are found all over the world.

- Body is divided into two main divisions (Prosoma and Opisthosoma)
- Connected by pedicel

Taxonomy and Ecology of spiders

Phylum **Arthropoda**

Sub-Phylum **Chelicerata**

Class **Arachnida**

- Prosoma is dorsally covered by carapace
- Ventrally covered by two chitinous plates (small labium and large sternum)
- Prosoma bears eyes and six pair of appendages
- Chelicerae are provided with poisonous gland
- Four walking legs having 7 segments

Taxonomy and Ecology of spiders

Phylum	**Arthropoda**
Sub-Phylum	**Chelicerata**
Class	**Arachnida**

- Opisthosoma contains spinnerets, Female genital organ, book lungs
- May be nocturnal or diurnal
- Wolf spiders chase their prey
- Sit and wait predators
- Jumping spiders has largest eyes and keenest vision
- Crab spiders lurk among flower petals and fallen flowers, in wait of prey

Taxonomy and Ecology of spiders

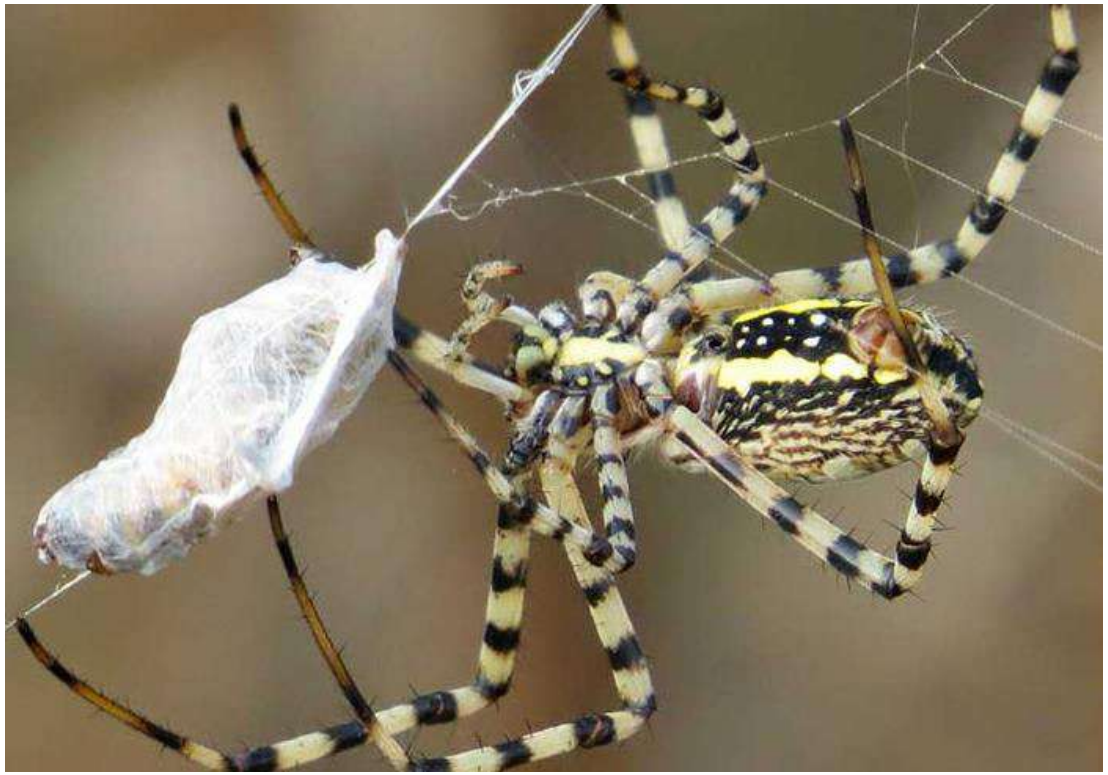
Phylum **Arthropoda**

Sub-Phylum **Chelicerata**

Class **Arachnida**

- Mostly feed on insects
- Survive without food for many months
- Only few species are dangerous to man (i.e. Latrodectus)
- Power of regeneration (leg)
- Sexual dimorphism (male smaller)

Taxonomy and Ecology of spiders



- Female suck body sap of male after mating
- Some species carry cocoon on the back of abdomen

Module# 161 Taxonomy and Ecology of Lady beetle

Phylum	Arthropoda
Class	Insecta
Order	Coleoptera
Family	Coccinellidae

- Ladybugs, also called lady beetles or ladybird beetles, are a very beneficial group.
- They are natural enemies of many insects, especially aphids and other sap feeders.
- A single lady beetle may eat as many as 5,000 aphids in its lifetime.

Taxonomy and Ecology of Lady beetle

https://entomology.ca.uky.edu/ef105

- Adult lady beetles have very characteristic convex, hemispherical to oval shaped bodies that can be yellow, pink, orange, red, or black, and usually are marked with distinct spots.
- This is a type of warning coloration to discourage other animals that may try to eat them.

Taxonomy and Ecology of Lady beetle

https://entomology.ca.uky.edu/ef105

- Like many other brightly-colored insects, they are protected by an odorous, noxious fluid that seeps out of their joints when the insects are disturbed.

Taxonomy and Ecology of Lady beetle

https://entomology.ca.uky.edu/ef105

- Adult females usually lay clusters of eggs on plants close to aphid, scale, or mealybug colonies.
- The alligator-like larvae are also predators.

Taxonomy and Ecology of Lady beetle

https://entomology.ca.uky.edu/ef105

- The alligator-like larvae are also predators.
- They are spiny and black with bright spots.
- Although they look dangerous, lady beetle larvae are quite harmless to humans.
- After feeding on insect prey for several weeks, the larva pupates on a leaf.

Taxonomy and Ecology of Lady beetle

https://entomology.ca.uky.edu/ef105

- Some lady beetle species have several generations each year while others have only one.
- During the summer months, all stages often can be found at the same time.
- Adults of some species spend the winter clustered together in large groups under leaf litter, rocks, or other debris.

Taxonomy and Ecology of Lady beetle

https://entomology.ca.uky.edu/ef105

- Important biological control agents
- Should be conserved
- Learn to recognize the different stages of these beneficial insects.
- Make insecticide applications only when necessary and use selective insecticides or limited treatments to avoid killing lady

Taxonomy and Ecology of Lady beetle

https://entomology.ca.uky.edu/ef105

- Important biological control agents
- Should be conserved
- Learn to recognize the different stages of these beneficial insects.
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Taxonomy and Ecology of Lady beetle

https://entomology.ca.uky.edu/ef105

- Birds are ladybugs' main predators, but they also fall victim to frogs, wasps, spiders, and dragonflies.
- Ladybugs are happy in many different habitats, including grassland, forests, cities, suburbs, and along rivers.

Taxonomy and Ecology of Lady beetle

https://entomology.ca.uky.edu/ef105

- Highly susceptible to pesticides
- Ladybugs are most active from spring until fall. When the weather turns cold, they look for a warm, secluded place to hibernate, such as in rotting logs, under rocks, or even inside houses. These hibernating colonies can contain thousands of ladybugs.
- 5000 species/different consumption rate

Module# 162 Taxonomy and Ecology of Green lacewing

Phylum	Arthropoda
Class	Insecta
Family	Chrysopidae
Genus	Chrysoperla
Species	C. carnea

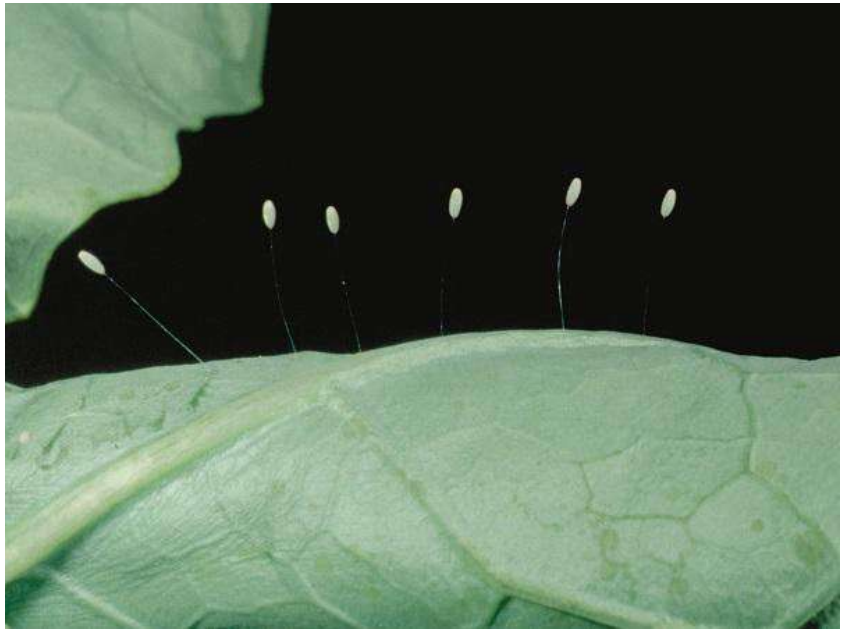
- Green lacewings are an often under-appreciated group of beneficial insects.
- As with lady beetles, these natural enemies are important predators of many types of soft bodied insects and insect eggs.

Taxonomy and Ecology of Green lacewing

https://entomology.ca.uky.edu/ef105

- The adult green lacewing is about 3/4 inch long, light green and has a delicate appearance with lacy wings.
- One unusual characteristic are its eyes, they look like two golden hemispheres.
- They are weak fliers and are commonly found near aphid colonies.

Taxonomy and Ecology of Green lacewing



https://entomology.ca.uky.edu/ef105

- The adults feed mostly on nectar, pollen, and honeydew but with some species the adults will feed on insects.
- The eggs are either laid singly or in small groups. Each is always found perched on the tip of a hairlike stalk that is about 1/2 inch long.
- This helps to reduce cannibalism of the eggs by sibling larvae.

Taxonomy and Ecology of Green lacewing

https://entomology.ca.uky.edu/ef105

- Females will usually deposit the egg close a food source for the larvae.
- The larvae are brown and white and may grow up to about l/2 inch in length.
- Larvae are called aphidlions, because they feed on other soft-bodied insects as well as aphids.
- They are voracious feeders, attacking with large, curved, hollow mandibles.

Module# 163 Taxonomy and Ecology of Stink bug

Phylum	Arthropoda
Class	Insecta
Order	Hemiptera
Family	Pentatomidae
Sub-Family	Asopinae

- **Pentatomidae** is a family of insects belonging to the order Hemiptera, generally called **shield bugs** or **stink bugs**.
- Pentatomidae has over 4700 species
- Have piercing sucking mouthparts
- Most are phytophagous, severe pests on agricultural crops.

Taxonomy and Ecology of Stink bug

Phylum	Arthropoda
Class	Insecta
Order	Hemiptera
Family	Pentatomidae
Sub-Family	Asopinae

- However, some species, particularly in the subfamily Asopinae, are predatory and may be considered beneficial.
- They have ability to release a pungent defensive spray when threatened, disturbed, or crushed.
- The composition of this spray may vary between species, and even by sex or age

Taxonomy and Ecology of Stink bug

https://www.uky.edu/Ag/CritterFiles/casefile/insects/bugs/stinkbugs/stinkbugs.htm

- but generally includes aldehyde and alkanes.
- Moderate to large size, ranging in length from 7 to 25 mm, and are broadly elliptical in shape.
- The subfamily Asopinae belongs to the family Pentatomidae of the suborder Heteroptera (true bugs).
- About 300 species in 69 genera have been described worldwide.

Taxonomy and Ecology of Green lacewing

https://www.uky.edu/Ag/CritterFiles/casefile/insects/bugs/stinkbugs/stinkbugs.htm

- Adult stink bugs are strong fliers and typically migrate into cotton from alternate hosts outside the field, including adjacent fields of maturing corn and soybean.
- Stink bugs are difficult to scout.
- Adults tend to aggregate, and the distribution of stink bugs within a field may be highly concentrated, particularly along borders.

Taxonomy and Ecology of Stink bug

https://www.uky.edu/Ag/CritterFiles/casefile/insects/bugs/stinkbugs/stinkbugs.htm

- Despite being relatively large insects, they can be difficult to see.
- Stink bugs can be scouted for visually, but most thresholds are based on drop cloth samples or the occurrence of internal symptoms of feeding damage to bolls.

Taxonomy and Ecology of Stink bug

https://www.uky.edu/Ag/CritterFiles/casefile/insects/bugs/stinkbugs/stinkbugs.htm

- Predatory stink bugs are a gardener's friend, feeding on more than 100 species of insect pests.
- Both nymphs and adults attack insects larger than themselves, and suck the body fluids from their prey with a needle-like beak.

Taxonomy and Ecology of Green lacewing

https://www.uky.edu/Ag/CritterFiles/casefile/insects/bugs/stinkbugs/stinkbugs.htm

- Predatory stink bugs tend to have a shorter, stouter beak than the long, thin beak of plant-feeding stink bugs.
- Adults overwinter in sheltered locations and leaf litter, emerging in early spring to stalk and devour their prey.

Taxonomy and Ecology of Stink bug

https://www.uky.edu/Ag/CritterFiles/casefile/insects/bugs/stinkbugs/stinkbugs.htm

- **Life stage(s) that feed on pests**: Nymphs and adults.
- **Insect(s) fed on**: Spined soldier bugs prefer larger insects such as grubs and the larvae of various beetles and moths, and are especially fond of caterpillars. Two-spotted stink bugs often eat the eggs and larvae of Colorado potato beetle, and smaller caterpillars and larvae.

Taxonomy and Ecology of Stink bug

https://www.uky.edu/Ag/CritterFiles/casefile/insects/bugs/stinkbugs/stinkbugs.htm

- 1st instar nymphs of two-spotted stink bugs feed on plant sap, as will adults in early spring before prey are available, but do no significant damage to plants.
- **Where to find:** Generally throughout the garden, particularly on apples, asparagus, beans, cole crops, cucurbits, eggplant, potatoes, onions, sweet corn and tomatoes.

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Taxonomy and Ecology of Stink bug

https://www.uky.edu/Ag/CritterFiles/casefile/insects/bugs/stinkbugs/stinkbugs.htm

- **How to attract and conserve:** Pheromones have been used to attract spined soldier bugs to the garden. Year-round ground cover in garden will provide overwintering sites and shelter, as well as plant sap to feed on early in the season. Avoid or reduce use of broad-spectrum insecticides.

Module# 164 Taxonomy and Ecology of parasitiod wasp

Phylum	Arthropoda
Class	Insecta
Order	Hymenoptera
Family	Eulophidae
Genus	Tamarixia
Species	*T. radiata*

- Wasps, a very large evolutionary group of hymenopteran insects,
- *Tamarixia radiata* is a species specific ectoparasitoid of the Asian citrus psyllid (ACP) *Diaphorina citri*.
- *Diaphorina citri* is an economically important pest of citrus in many citrus growing regions of the world (Halbert and Manjunath, 2004).

Taxonomy and Ecology of parasitiod wasp

https://biocontrol.entomology.cornell.edu/parasitoids/Tamarixia.php

- The psyllid vectors the phloem limited gram-negative bacterium *Candidatus* Liberibacter asiaticus, one of the causal organisms of the devastating citrus disease “huanglongbing” (HLB) or citrus greening disease. HLB is responsible for the decline of most trees seen in disease affected regions of the world

Taxonomy and Ecology of parasitiod wasp

https://biocontrol.entomology.cornell.edu/parasitoids/Tamarixia.php

- *Tamarixia radiata* was first imported to Florida from Taiwan and south Vietnam and limited numbers were released soon after psyllid was discovered in the state (Hoy et al., 1999).
- Successfully managed the pest
- Saved citrus industry of Florida
- Tamarixia radiata was first described from northwestern India (Punjab) which is now part of Pakistan.

Taxonomy and Ecology of parasitiod wasp

https://biocontrol.entomology.cornell.edu/parasitoids/Tamarixia.php

- *Tamarixia radiata* is not yet available commercially.
- The female usually deposits one and very rarely two eggs on the ventral side of the *D. citri* nymph, mostly in the groove between thorax and abdomen and close to the point of attachment of hind coxae.

Taxonomy and Ecology of parasitiod wasp

https://biocontrol.entomology.cornell.edu/parasitoids/Tamarixia.php

- The neonate larva sucks fluid from the site of attachment
- The parasitized nymph remain indistinguishable from unparasitized nymph for couple of days until almost all body contents of the parasitized nymph are consumed and only wing buds and dorsal covering are left.

Taxonomy and Ecology of parasitiod wasp

https://biocontrol.entomology.cornell.edu/parasitoids/Tamarixia.php

- At that point, the larva is mature and attaches itself to the leaf or shoot surface with a network of fine anchoring silken threads
- Pupation takes place underneath the mummified nymph and the adult emerges by chewing a circular exit hole in the integument of the thoracic region.

Taxonomy and Ecology of parasitiod wasp

https://biocontrol.entomology.cornell.edu/parasitoids/Tamarixia.php

- Development rate varies with temperature. It takes about 12 days to go from egg to adult at 25 ºC and 14L:10D.
- A single female can lay up to 300 eggs at 25-30 ºC (Chu and Chien, 1991; Étienne et al. 2001).

Taxonomy and Ecology of parasitiod wasp

PARASITIODS	PEST ATTACKED
***Ichneumonidae* wasps**	Larva of noctuids* (noctuids include helicoverpa, armyworms, cutworms and loopers)
Microplitis (WASP)	Helicoverpa larvae
Tachinid flies (Tachinidae)	Larva of noctuids
Trichogramma Wasp	Eggs of noctuids
Chalcid wasps	Aphids
Braconid wasps	beetles, caterpillars, flies and sawflies.

Module# 165 Taxonomy and Ecology of Whitefly

TAXONOMY

Phylum	Arthropoda
Class	Insecta
Order	Hemiptra
Family	Aleyrodidae

- There are1550 species worldwide under two subfamilies.
- 70 in Punjab Pakistan
- Cosmopolitan insects
- Most abundant in the tropical and subtropical countries of the world.

Taxonomy and Ecology of Whitefly

https://www.google.com/search?q=white+fly&sxsrf=ALeKk00g6UvbE7F9hqkatINSO3B_FjFJ5Q:1609560426799&source=lnms&tbm=isch&sa=X&ved=2ahUKEwifiO_rr_ztAhUQVRUIHeP6DAQQ_AUoAXoECBEQAw&biw=1360&bih=657#imgrc=VNa7b8IGlFu45M

Why called Whitefly?

- White appearance of adults due to deposition of wax on the on body and wings

Taxonomy and Ecology of Whitefly

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- Adult whiteflies are small, approximately 1-2 mm in length.
- Adults with wingspan less than 3mm.
- With a yellow body and two pairs of white wings.
- The compound eyes are red.

Taxonomy and Ecology of Whitefly

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- Sucking and piercing mouthparts.
- Both sexes have functional mouth parts and two pairs of functional membranous wings.
- Nymphs are initially semi-transparent, later becoming creamy yellow to pale-green.

Taxonomy and Ecology of Whitefly

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- The pupae or fourth nymphal stage bear 7 pairs of setae on their dorsum. It is called pseudopupa.

Taxonomy and Ecology of Whitefly

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- Produced large quantity of honeydew
- The honeydew emerges from the anus

Taxonomy and Ecology of Whitefly

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- Present under surface of the leaves.
- Both nymphs and adults feed.
- It has **160 host** plants.
- Polyphagous feed on a large number of host plants like Field crops, Fruit trees, Vegetables, Ornamentals , Weeds etc

Taxonomy and Ecology of Whitefly

https://www.google.com/search?q=how+dew+of+whitefly&tbm=isch&ved=2ahUKEwjl8PL_sPztAhUM4oUKHbGmA3sQ2-cCegQIABAA&oq=how+dew+of+whitefly&gs_lcp=CgNpbWcQAzoECCMQJzoICAAQsQMQgwE6BQgAELEDOgQIABBDOgIIADoECA

- One of the most noxious insect pest attaching fields crops.
- Loss is due to activities of sucking the sap from the leaves that weaken the plants, resulting in low fruiting with reduced yield.
- They excrete large amounts of honeydew and throw it on different parts of plants.
- Honeydew attract sooty mold which affect photosynthesis

Taxonomy and Ecology of Whitefly

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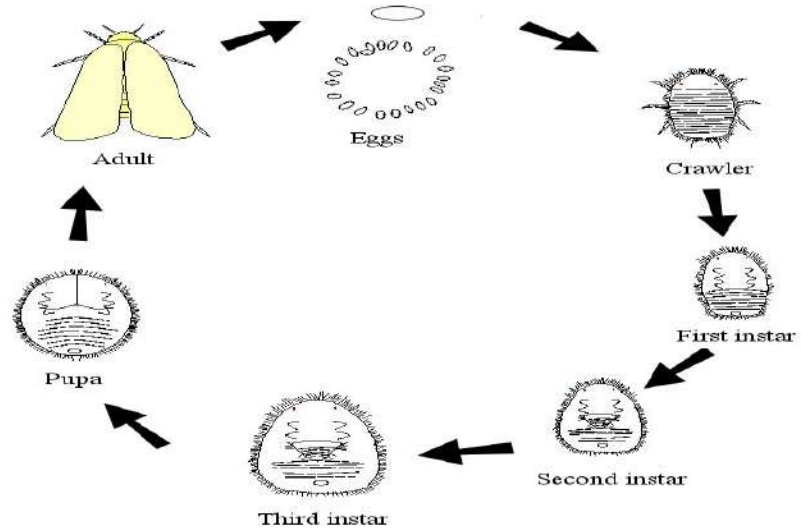
- They also act as vectors of many plant viruses, which adversely affect the growth and yield of crops.
- Cotton leaf curl virus (CLCuV) disease in Pakistan
- Sweet potato mild mottle virus
- Yellow dwarf disease
- They inject toxic saliva which results in plant disorders

Taxonomy and Ecology of Whitefly



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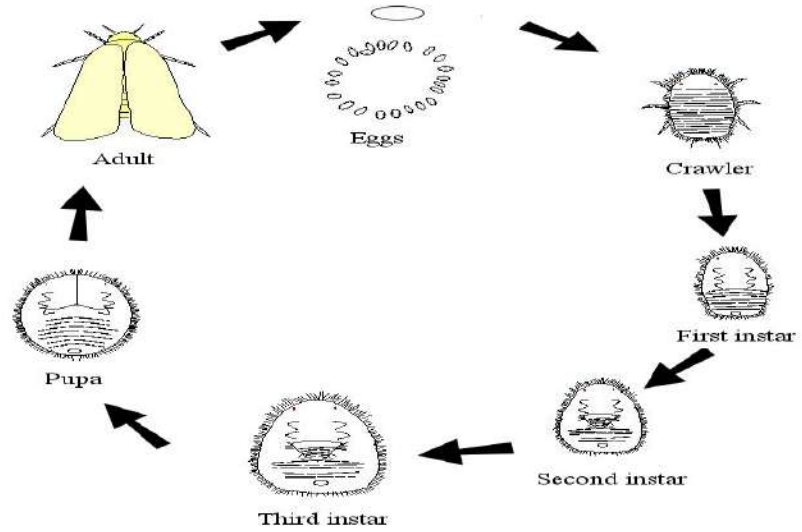
- Females can lay 10-150 eggs in her lifetime, often in fairy rings
- Eggs hatch into mobile crawler after 6-8 days.
- Crawlers settle at feeding site and lose the ability to walk. They insert their mouthparts into the leaf to feed on sap.
- 18-28 days in warm months

Taxonomy and Ecology of Whitefly



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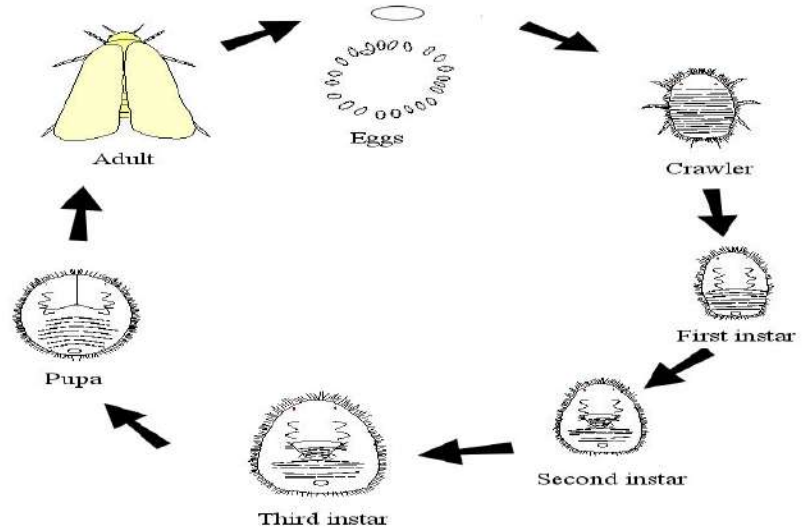
- Females can lay 10-150 eggs in her lifetime, often in fairy rings
- Eggs hatch into mobile crawler after 6-8 days.
- Adults can live three weeks and are often found in meeting pairs.
- Females are larger than males
- Resistance against pesticides.
- Rotation of insecticides

Taxonomy and Ecology of Whitefly



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Biological control

- Small birds
- Spiders
- Lacewings
- Hoverflies
- Ground beetles
- Mirid bugs
- Damsel bugs and pirate bug

Module# 166 Taxonomy and Ecology of Mealy bug

Phylum	Arthropoda
Class	Insecta
Order	Hemiptera
Family	Pseudococcidae
Genus	Phenacoccus
Species	*P. solenopsis*

- Found in all parts of worlds
- Native to USA (Tinsley 1898)
- Mostly found in moist, warm climates
- Many species are considered pests as they feed on plant juices
- Act as a vector for several plant diseases.

Taxonomy and Ecology of Mealy bug

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- Vehari in 2005
- Now almost in all cotton growing areas
- Affecting crop yields
- In 2006 (Multan, Bahawalpur, Kanewal, Vhari) as epidemic

Taxonomy and Ecology of Mealy bug

https://www.google.com/search?q=Cotton+mealy+bug,+classification&hl=en&sxsrf=ALeKk00sPhegtaV7Ff15-pZGu6cHFk9o3A:1609563987617&source=lnms&tbm=isch&sa=X&ved=2ahUKEwj9juaNvfztAhWmyIUKHannDcgQ_AUoAXoECBIQAw&biw=1360&bih=657#imgrc=zRh_k1UrcmqUyM

- Mealy bugs are sexually dimorphic
- Females are about 3 to 5 mm long.
- Their body is usually covered with a mealy waxy secretion, often extended into lateral and terminal filaments.
- They usually remain clustered around the terminal shoots, leaves or fruits.

Taxonomy and Ecology of Mealy bug

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- They are wingless and do not move unless disturbed.
- They often retain legs and can move.
- Females are provided with 9-segmented antennae.

Taxonomy and Ecology of Mealy bug

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- There are 18 pairs of very short, lateral wax filaments, and several slightly longer caudal filaments

Taxonomy and Ecology of Mealy bug

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- Female Mealybug eggs are very small and are laid under a white, loose woolly wax, which remains attached to the abdomen of the females.
- A female may lay between 50 to 600 eggs.
-

Taxonomy and Ecology of Mealy bug

https://www.google.com/search?q=Cotton+mealy+bug,+classification&hl=en&sxsrf=ALeKk00sPhegtaV7Ff15-pZGu6cHFk9o3A:1609563987617&source=lnms&tbm=isch&sa=X&ved=2ahUKEwj9juaNvfztAhWmyIUKHannDcgQ_AUoAXoECBIQAw&biw=1360&bih=657#imgrc=zRh_k1UrcmqUyM

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Taxonomy and Ecology of Mealy bug

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- Within 3-9 days crawlers come out and actively move and spread out and settle down for on plant parts and gradually develops waxy coating.

Taxonomy and Ecology of Mealy bug

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- Males are smaller, gnat-like and have wings.
- Male mealy bugs do exhibit a radical change during their life cycle, changing from wingless, ovoid nymphs to wasp-like flying adults.

Taxonomy and Ecology of Mealy bug

https://www.google.com/search?q=male%5D+mealy+bug%2C+classification&tbm=isch&ved=2ahUKEwiOgZLJvvztAhUC_RoKHT99BEkQ2-cCegQIABAA&oq=male%5D+mealy+bug%2C+classification&gs_lcp=CgNpbWcQA1Cc-DFY1oMyYPaJMmgAcAB4AIABgQOIAboKkgEFMi0zLjKYAQCgAQGqAQtnd3Mtd2l6LWltZ8ABAQ&sclient=img&ei=3ADwX46jHoL6a7_6kcgE&bih=657&biw=1360&hl=en#imgrc=lmRAc8BM0ZXGzM

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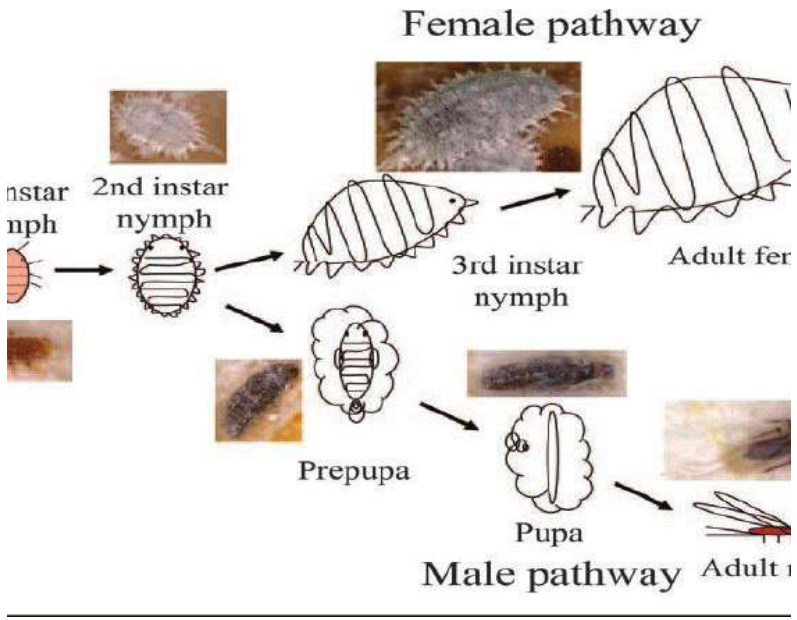
Taxonomy and Ecology of Mealy bug



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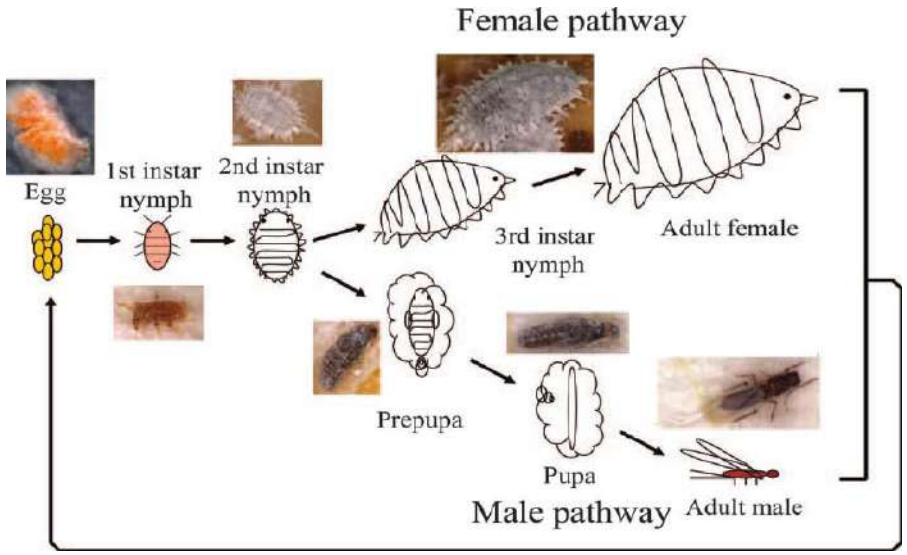
- Females feed on plant sap\Honey dew----Black sooty Moulds----Photosynthesis
- They attach themselves to the plant and secrete a powdery wax layer used for protection while they suck the plant juices.
- Honey dew---Ants—dispersal from plant to plant

Taxonomy and Ecology of Mealy bug



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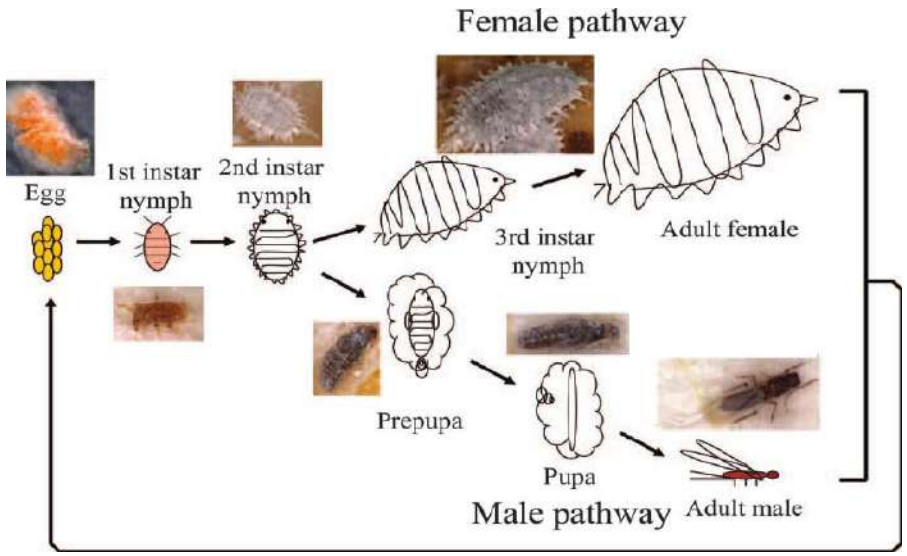
- The males on the other hand are short-lived as they do not feed at all as adults.
- They only live to fertilize the females.

Taxonomy and Ecology of Mealy bug



https://www.google.com/search?q=male%5D+mealy+bug%2C+classification&tbm=isch&ved=2ahUKEwiOgZLJvvztAhUC_RoKHT99BEkQ2-cCegQIABAA&oq=male%5D+mealy+bug%2C+classification&gs_lcp=CgNpbWcQA1Cc-DFY1oMyYPaJMmgAcAB4AIABgQOIAboKkgEFMi0zLjKYAQCgAQGqAQtnd3Mtd2l6LWltZ8ABAQ&sclient=img&ei=3ADwX46jHoL6a7_6kcgE&bih=657&biw=1360&hl=en#imgrc=8GCO-4rWcYOEzM

- The most venerable stage of the cotton mealy bug is larval stage.
- It can be control easily by applying different control measures.
- The control measures of cotton mealy bug are the traditional, biological, chemical.

Module# 167 Taxonomy and Ecology of Yellow stem borer

Phylum	Arthropoda
Class	Insecta
Order	Lepidoptera
Family	Pyralidae
Genus	Scirpophaga
Species	*S. incertula*

- Serious pests of rice
- Males are ochreous (natural soil colour, clay)
- The spot on the wings is inconspicuous
- Anal tuft is absent

Taxonomy and Ecology of Yellow stem borer

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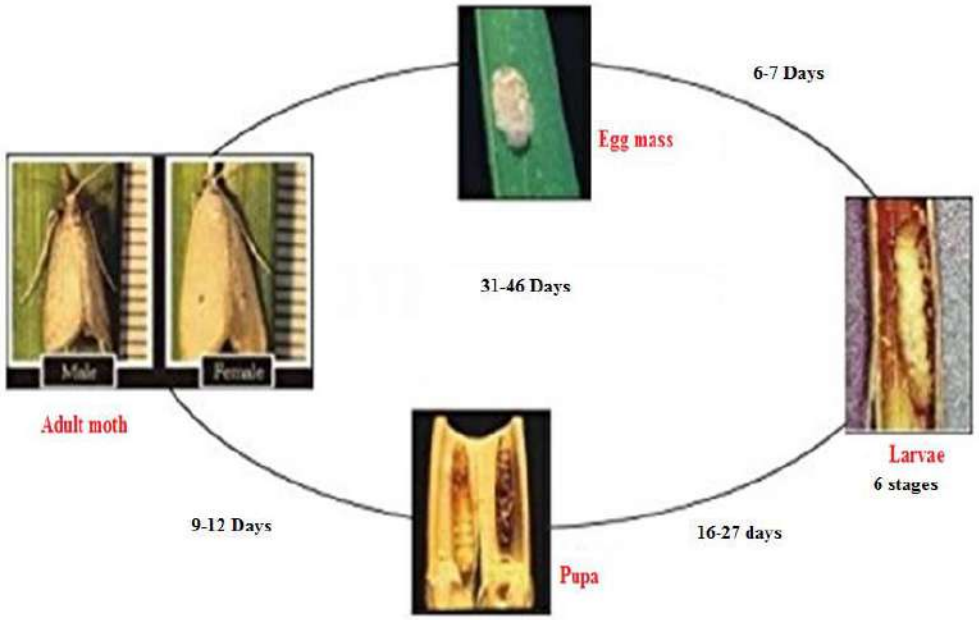
- Female are pale yellow with distinct dark spot on the front wings.
- Tip of the abdomen is equipped with tuft of white and brown colored hairs.
- Sexual dimorphism in this insect has often led to the conclusion that 2 different species are involved.

Taxonomy and Ecology of Yellow stem borer



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- Adults are active after dusk, mate
- Female lay eggs 120-150 underside leaves
- 2-5 clusters (60-100 eggs)
- Covered with yellowish brown hairs

Taxonomy and Ecology of Yellow stem borer

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- The larvae first feeds on the inside of leaf-sheath and bore into the plant stem as a result of which the central shoot of the young plant dries up causing “dead hearts”.
- In older plants the whole earheads become whitish and chaffy (white head).
- Loss caused to rice production by this very pest in different regions and seasons varies from 5-90%.

Taxonomy and Ecology of Yellow stem borer

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Control

- Destruction of weeds harbouring pests
- Pulling out infested rice plant
- Flooding the field after harvest
- Stubble (short cut stalks) left after harvest should be
- ploughed up and burn or submerged in water immediately after harvest
- Crop rotation

Taxonomy and Ecology of Yellow stem borer

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- Application of insecticides
- Collection and destruction of egg masses from nursery plants.
- Setting up of light traps to trap and destroy the moth.
- Destruction of seedlings with “dead hearts” while planting.

Module# 168 Taxonomy and Ecology of American Bollworm

Phylum	Arthropoda
Class	Insecta
Order	Lepidoptera
Family	Noctuoidea
Genus	Helicoverpa
Species	*H. armigera*

- American Bollworm is serious pest of cotton.
- It is distributed in Pakistan
- Central and southern Europe
- Temperate Asia and Africa

Taxonomy and Ecology of American Bollworm

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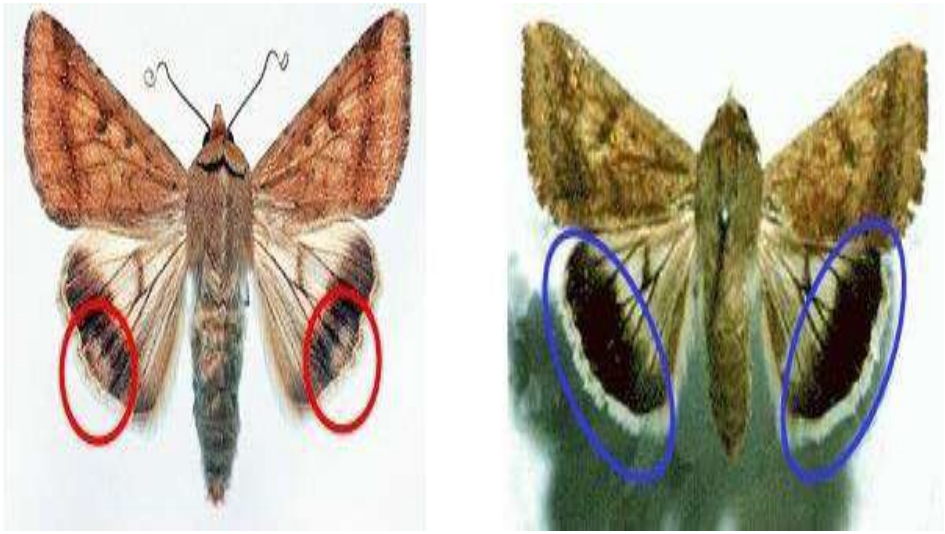
- The body length varies between 12 and 20 millimeters with a wingspan of 30–40 millimeters.
- The fore wings are yellowish to orange in females and greenish-gray in males
- The hind wings are a pale yellow with a narrow brown band at the external edge and a dark round spot in the middle.

Taxonomy and Ecology of American Bollworm



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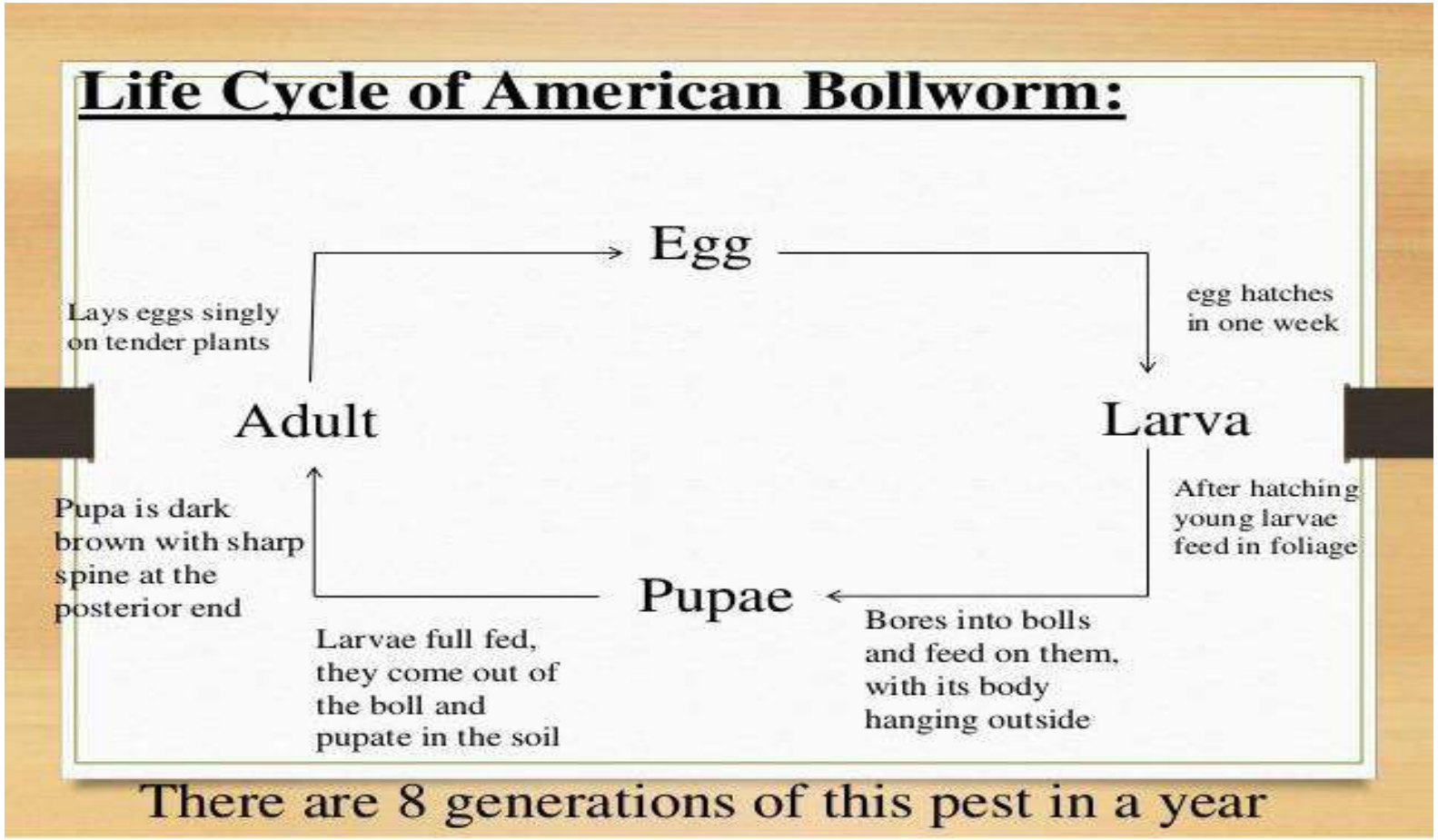
- The body length varies between 12 and 20 millimeters with a wingspan of 30–40 millimeters.
- The fore wings are yellowish to orange in females and greenish-gray in males
- The hind wings are a pale yellow with a narrow brown band at the external edge and a dark round spot in the middle.

Taxonomy and Ecology of American Bollworm



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Taxonomy and Ecology of American Bollworm

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- Caterpillar are polyhagous
- Feed on foliage and on the bolls of cotton
- Reduce yield
- Show prominent hole
- Dirty feces accumulate on the boll surface
- It shifts from one boll to another and damages dozens of bolls during life time.

Taxonomy and Ecology of American Bollworm

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- Avoid the cultivation of okra, moong and castor in and around the cotton crop
- Avoid growing of American cotton in the field
- Ichneumonid wasp, larval parasitoid of American bollworm should be promoted in the field
- Spray profenofos EC 800 ml/acre

Module# 169 Principles of Systematics

Taxonomy and Ecology of Pink Bollworm

Taxonomy and Ecology of Pink Bollworm

Phylum	Arthropoda
Class	Insecta
Order	Lepidoptera
Family	Noctuoidea
Genus	Pectinophora
Species	*P. gossypiella*

- The pink bollworms prefer areas where cotton and okra are being produced.
- They originated in the eastern Indian
- All over the worlds

Taxonomy and Ecology of Pink Bollworm

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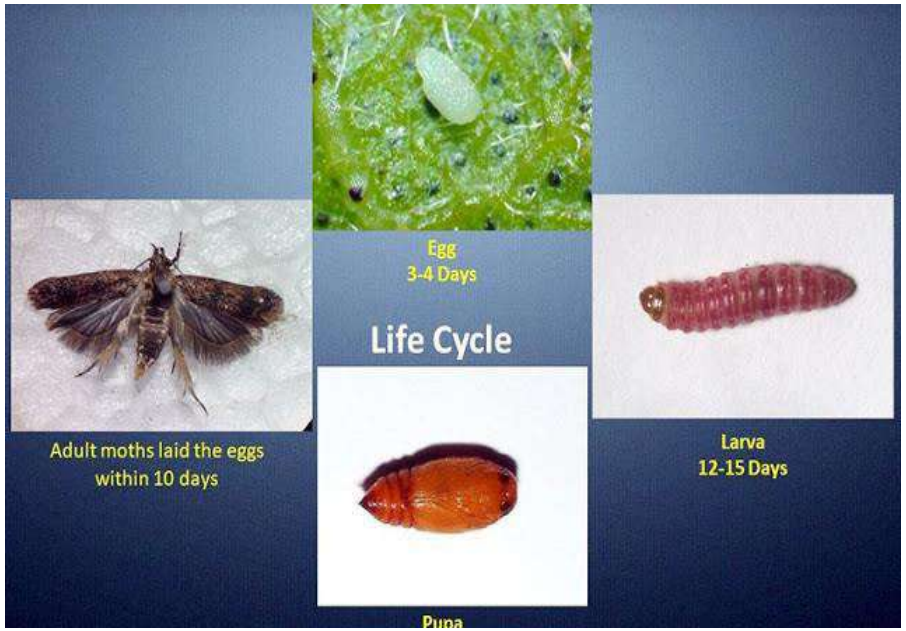
- Adult pink bollworms are small and grayish-brown in color with folded wings and a slender body.
- Young larvae are small caterpillars with dark brown heads.
- Once the larvae have matured, they are about 12 mm in length with pink bands on their backs.

Taxonomy and Ecology of Pink Bollworm



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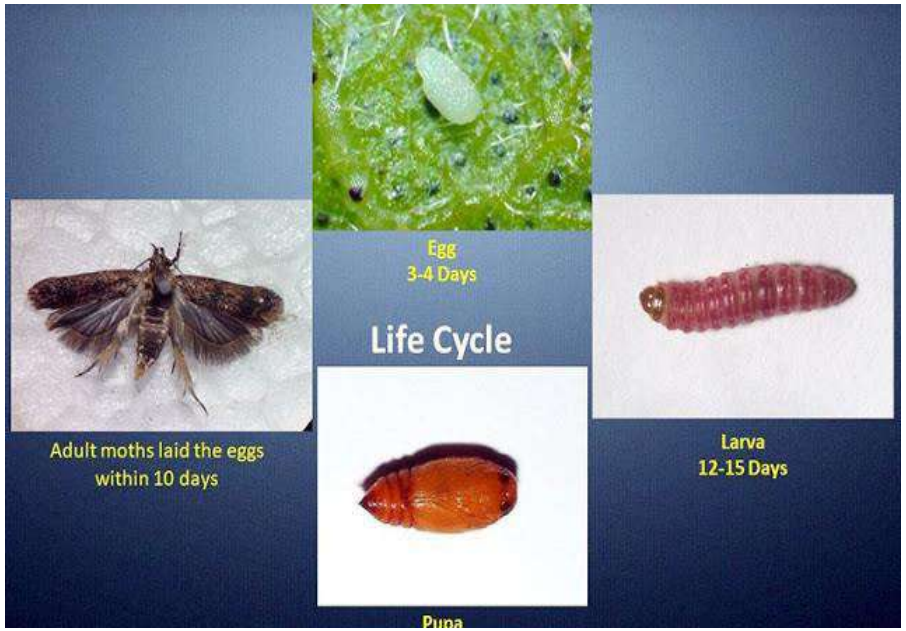
- Once laid, bollworm eggs take three to four days to hatch.
- Freshly hatched pink bollworm larvae are white with brown heads, turning pink in the fourth larval stage.
- Larvae bore into cotton plants to feed on their seeds.

Taxonomy and Ecology of Pink Bollworm



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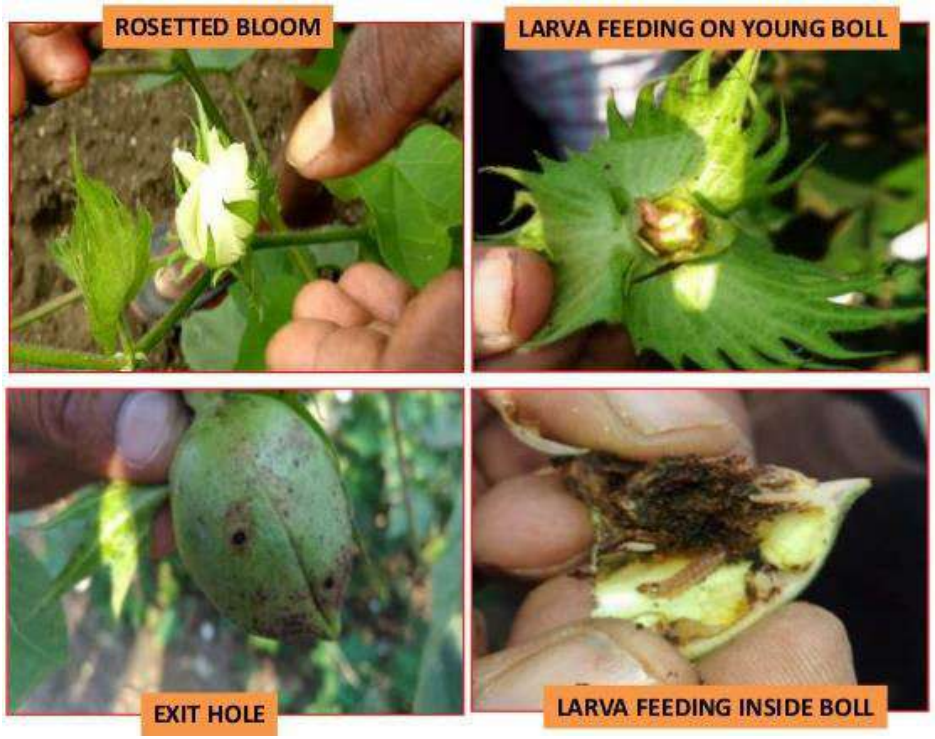
- The larvae take twelve to fifteen days to develop, then they move to the soil to pupate.
- The brown pupa is immobile for its first seven to eight days.

Taxonomy and Ecology of Pink Bollworm



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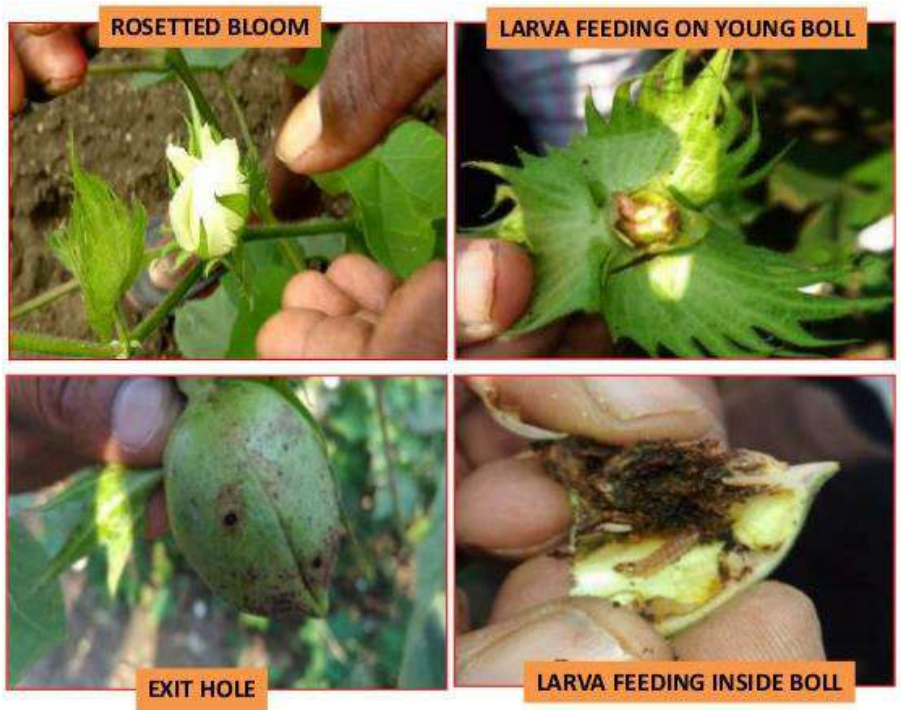
- Common signs of an infestation include shedding of flower buds, deformation of flowers, reduction of lint development and destruction of the seeds in lint.

Taxonomy and Ecology of Pink Bollworm



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- Bollworms' greatest harm is in the damage they do to cotton bolls.
- As the larva of the pink bollworm burrows into the boll, its stains the lint, rendering it commercially unusable.
- If the humidity is high, it only takes one or two larvae to completely destroy a boll.

Module# 170 Taxonomy and Ecology of Jassids

Phylum	Arthropoda
Class	Insecta
Order	Hemiptera
Family	Cicadellidae

- Major sucking pests of cotton.
- Eggs yellowish white
- Nymphs and adults are greenish yellow with a black spot on tip of each forewing.
- Jump on disturbing

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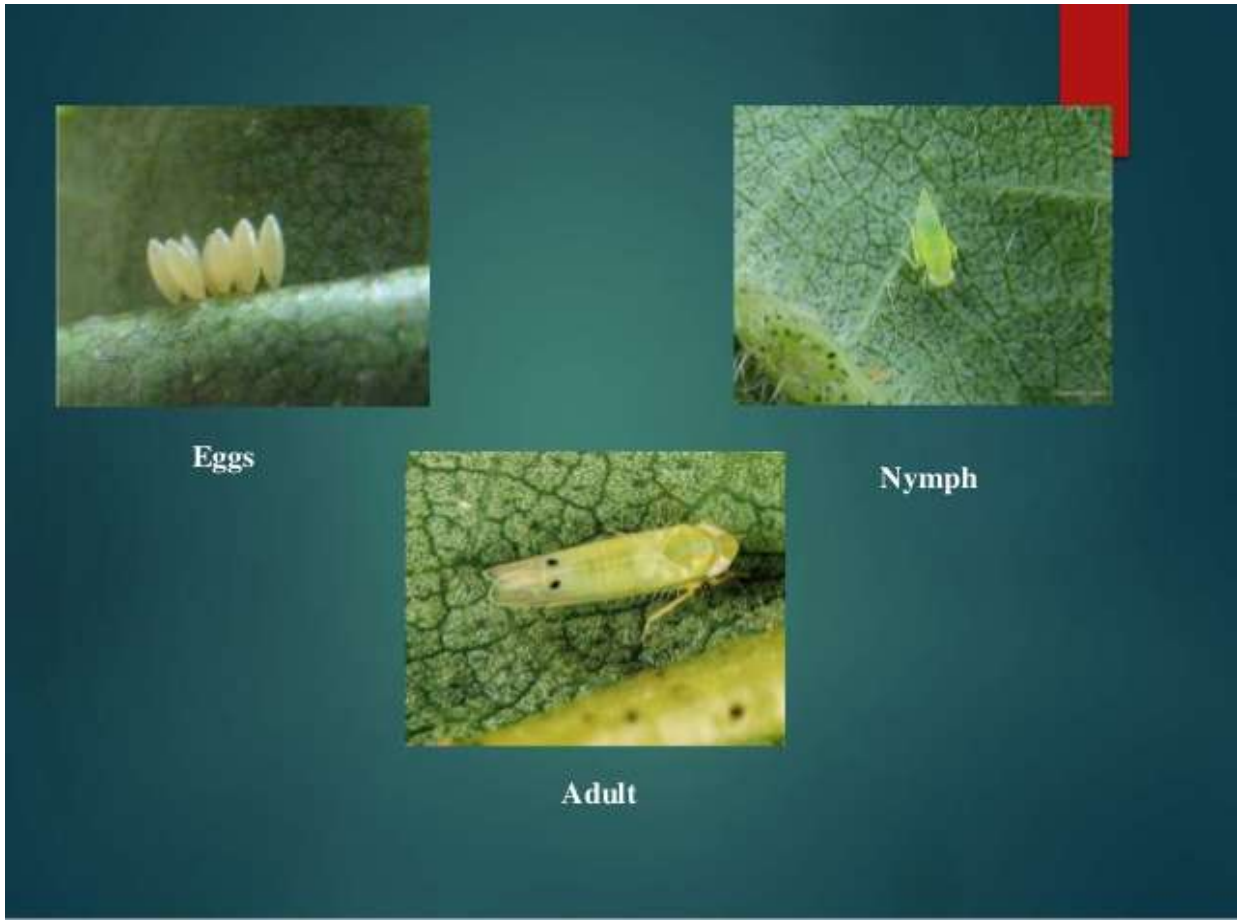
Taxonomy and Ecology of Jassids



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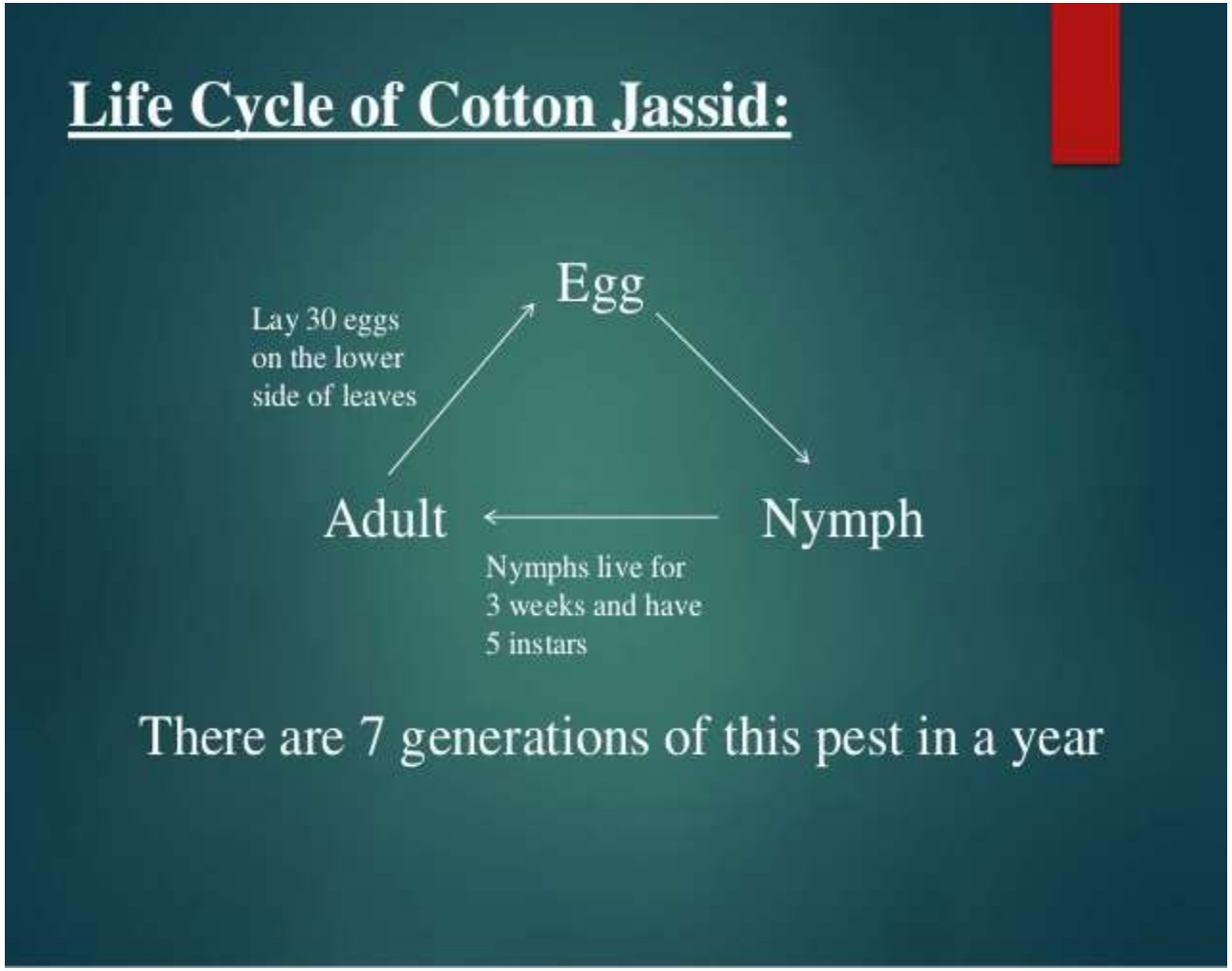
Taxonomy and Ecology of Jassids



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Taxonomy and Ecology of Jassids

https://www.slideshare.net/EhtishamAliHussain/cotton-jassid

- Nymphs and adults both suck sap.
- Leaves first show yellow and then reddish spots.
- Attacked leaves turn downwards, dry and fall
- Plants become weak and their fruit fall

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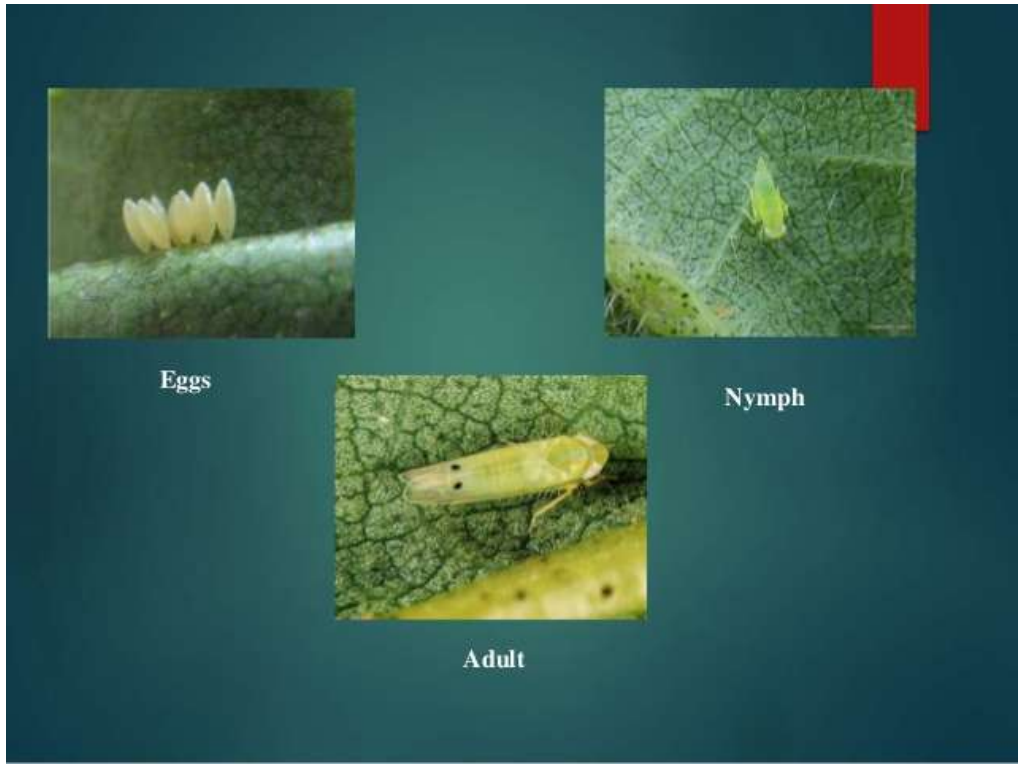
Taxonomy and Ecology of Jassids



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Control

- Clean cultivation and destruction of alternate host plant
- Selected varieties of cotton should be grown
- Green lace wing is a good predator of cotton jassid
- Spray of acetamiprid 125ml/acre
- Spiders

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Module# 171 Taxonomy and Ecology of Thrips

Phylum	Arthropoda
Class	Insecta
Order	Thysanoptera
Family	Thripidae

- Tiny sucking insects
- Their identification to species by standard morphological characters is often challenging.
- There is also considerable variability leading to many species being misidentified. Molecular sequence based approaches have increasingly been applied to their identification

Taxonomy and Ecology of Thrips

Phylum	Arthropoda
Class	Insecta
Order	Thysanoptera
Family	Thripidae

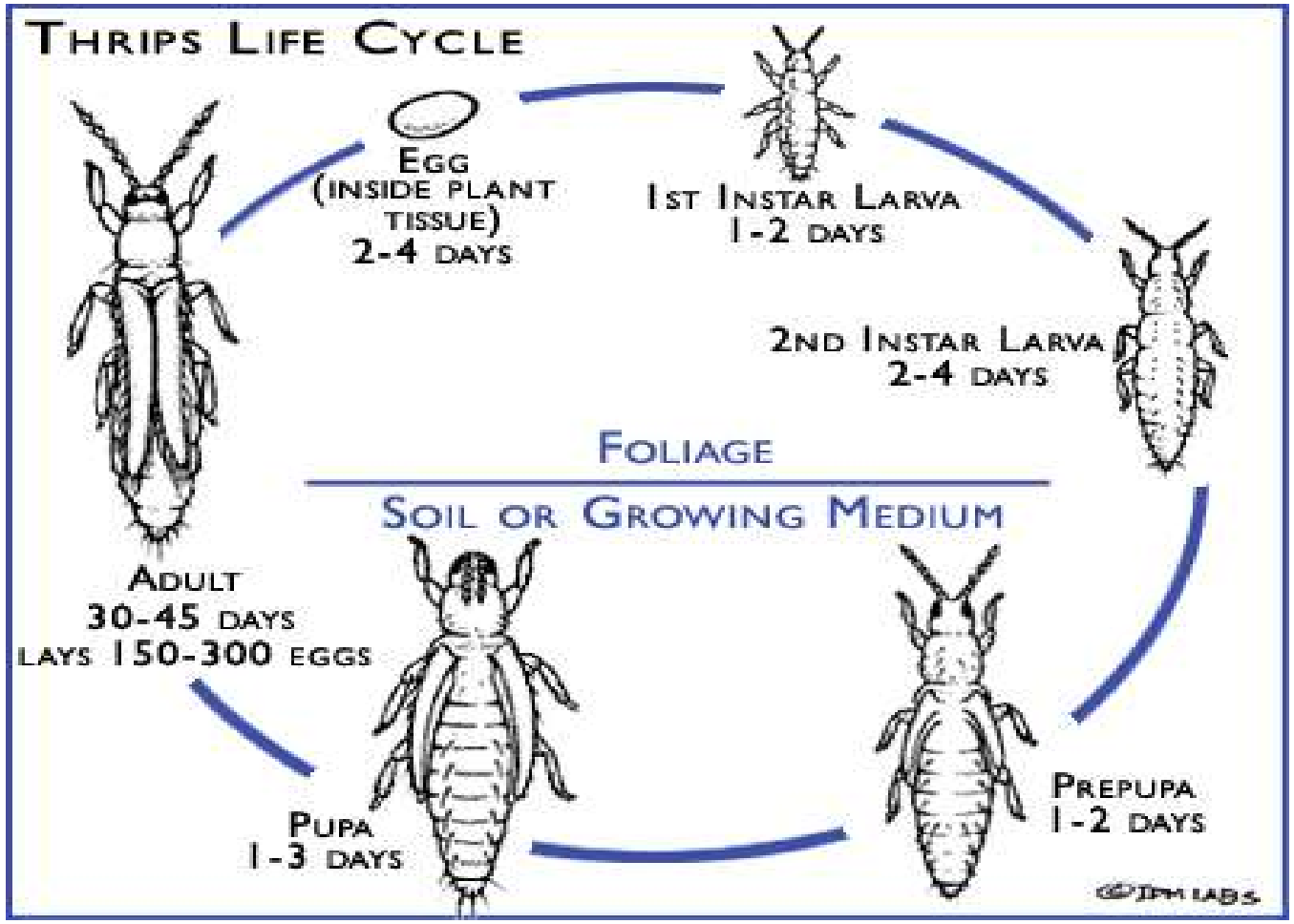
- Fringed wings
- Also transmit tomato spotted wilt virus to many plants
- Feed on wide range of plants world wide
- They prefer tender parts of the plant, such as buds, flowers and new leaves.
- Besides feeding on plant tissues, the common blossom thrips feeds on pollen grains and on the eggs of mites.

Taxonomy and Ecology of Thrips

Phylum	Arthropoda
Class	Insecta
Order	Thysanoptera
Family	Thripidae

- When the larva supplements its diet in this way, its development time and mortality is reduced,
- Adult females that consume mite eggs increase their fecundity and longevity.

Taxonomy and Ecology of Thrips



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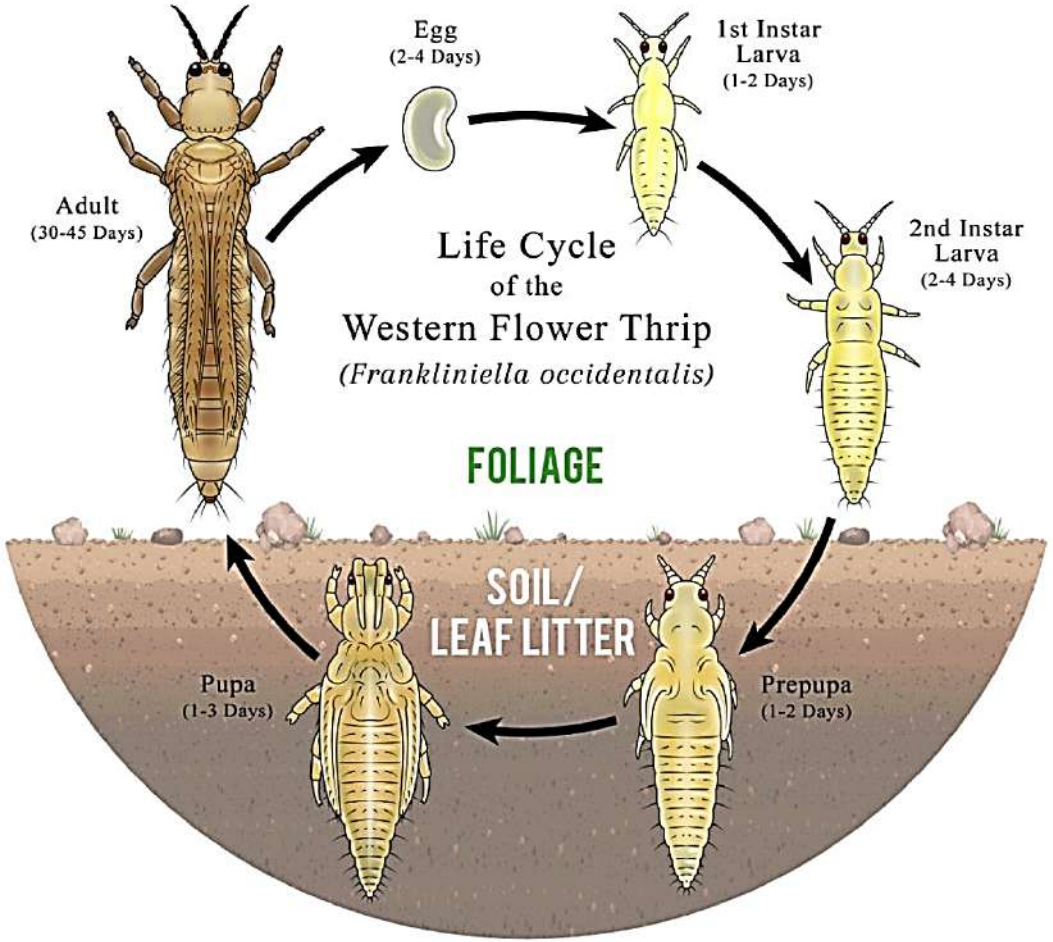
Taxonomy and Ecology of Thrips



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Taxonomy and Ecology of Thrips

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Damage

- In the right conditions, such as in greenhouses, many species can exponentially increase in population size and form large swarms because of a lack of natural predators coupled with their ability **to reproduce asexually**

Taxonomy and Ecology of Thrips

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Damage

- Sucks plant cells from many garden flowers, plants and fruits
- Damage includes streaks and small white patches
- Major infestation leads to stunted growth and damaged fruit and flowers

Taxonomy and Ecology of Thrips

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Control

- Try using yellow or blue sticky traps
- Spray dormant oil on fruit trees
- Shake branches of plants to remove the thrips
- Spray of insecticidal soap kills them
- Reflective mulches can be used

Module# 172 Taxonomy and Ecology of Rice Leaf Folder

- Very active
- Major pest of rice

Phylum	Arthropoda
Class	Insecta
Order	Lepidoptera
Family	Crambidae

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Taxonomy and Ecology of Rice Leaf Folder

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- Bright yellow or straw in color
- Two wavy lines in forewing and one in hindwing
- Eggs are laid singly or in groups in rows beneath the leaf surface
- Eggs are scaly white in color
- Incubation period is 4-8 days

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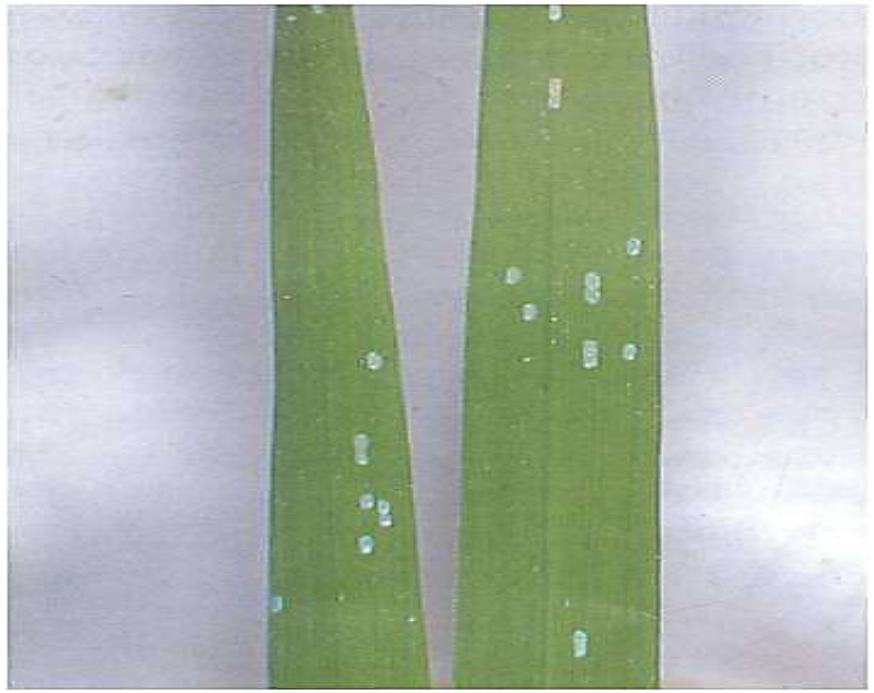
Taxonomy and Ecology of Rice Leaf Folder



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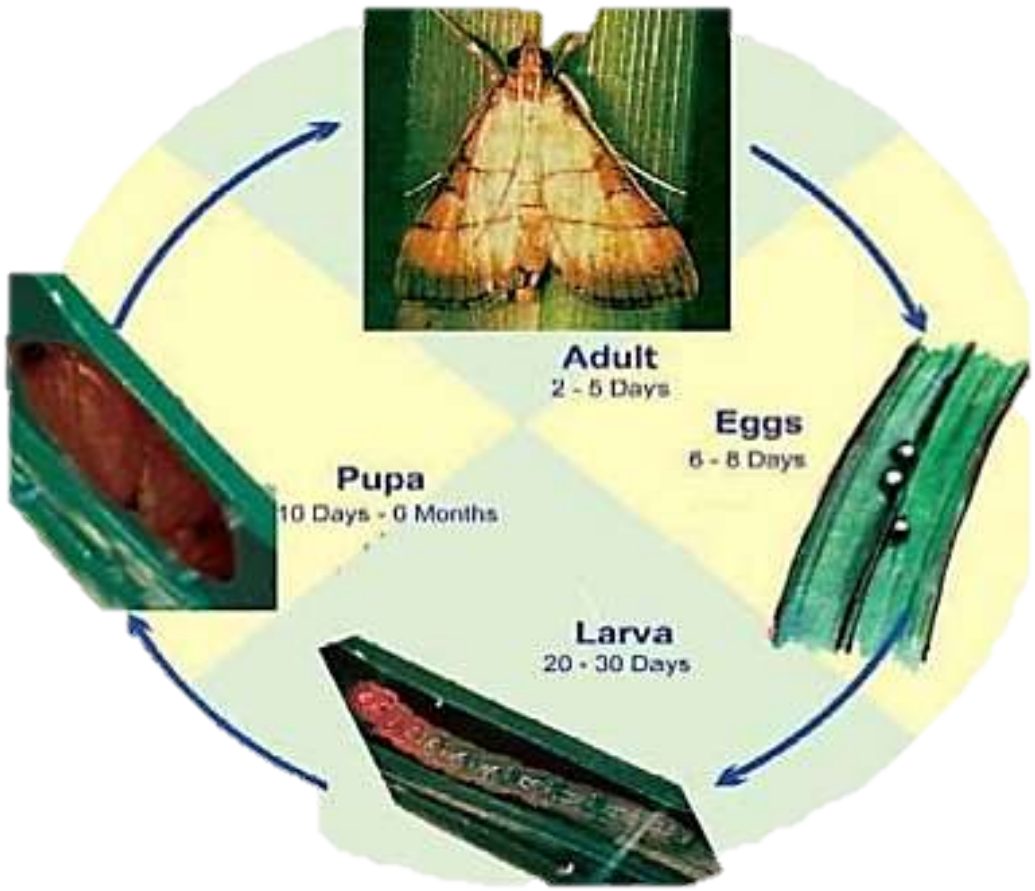
Taxonomy and Ecology of Rice Leaf Folder



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Taxonomy and Ecology of Rice Leaf Folder

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Damage

- Larval stage is damaging
- All stages of plants are damaged by this pest
- Newly hatched caterpillar cut leaf edges and folds the leaf
- Infested leaves appear white
- Attacked plant dries up
- Yield loss vary from 10-50%

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Taxonomy and Ecology of Rice Leaf Folder

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Control

- Reform tillage and cultivation system
- Light trap or lure can be used.
- Utilize natural enemies (beetles and *Trichogramma* spp.) as biological control
- Application of chlorpyrifos EC 1000ml/acre

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Module# 173 Taxonomy and Ecology of Citrus Leaf Miner

Phylum	Arthropoda
Class	Insecta
Order	Lepidoptera
Family	Gracillariidae

- Origin-Asia
- Distribution-worldwide
- Leaf miner: when the larvae being very small, live in between the two epidermal layers of leaves and feed on the food material inside.
- Examples: Citrus leaf miner, the cashew and mango leaf miner, and the buprestid leaf miner etc.

Taxonomy and Ecology of Citrus Leaf Miner

Life cycle stages

https://www.agric.wa.gov.au/citrus/citrus-leafminer

- Eggs whitish in color usually laid towards the petiole of leaf
- Larvae are translucent, light green in color and very difficult to detect.
- The pupa is yellowish-brown and darkens with age
- Adults females have a black and yellow color with a conspicuous yellow dorsal spot.

Taxonomy and Ecology of Citrus Leaf Miner

https://www.pestnet.org/fact_sheets/citrus_leafminer_240.htm

https://ucanr.edu/blogs/UCIPMurbanpests/blogfiles/58278_original.jpg

Damage

- Leaf mining results in
- Leaf deformation
- Leaf chlorosis
- Necrosis
- Leaf fall

Taxonomy and Ecology of Citrus Leaf Miner

http://ggc.gardencenternews.com/citrus-leafminer/

Control

- Monitoring:

  Pheromone traps can be used

- Biological control:

  General predators and parasitoids

- Chemicals:

❑Abamectin

❑Imidacloprid

❑Spinosad

Module# 174 Principles of Systematics

Taxonomy and Ecology of Citrus Psylla (*Diaphorina citri*)

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BS ZOOLOGY
O3336361244

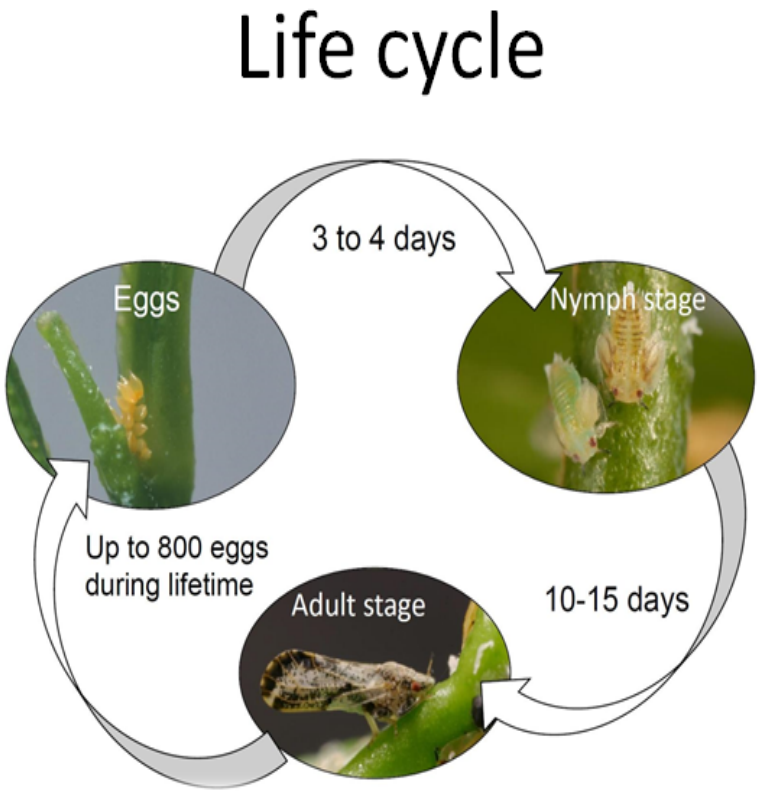
Taxonomy and Ecology of Citrus Psylla

Phylum	Arthropoda
Class	Insecta
Order	Hemiptera
Family	Liviidae

- Newly laid eggs are pale brown but the turn yellow and finally orange before hatching
- Nymph is generally yellowish-orange
- Adults are mottled brown
- Body is covered with whitish waxy secretion, making it appear dusty
- Greening disease or Huanglongbing

BAQIR HUSSAIN
BS ZOOLOGY
O3336361244

Taxonomy and Ecology of Citrus Psylla



https://www.agristudyinfo.com/citrus-psylla/

BAQIR HUSSAIN
BS ZOOLOGY
O3336361244

Taxonomy and Ecology of Citrus Psylla

http://entnemdept.ufl.edu/creatures/citrus/acpsyllid.htm

Damage

- Vectors of causative agent of Citrus greening disease or Huanglongbing
- Psyllid nymphs produce a toxin that causes the plant tips to die back or become contorted, preventing the leaves from expanding normally.

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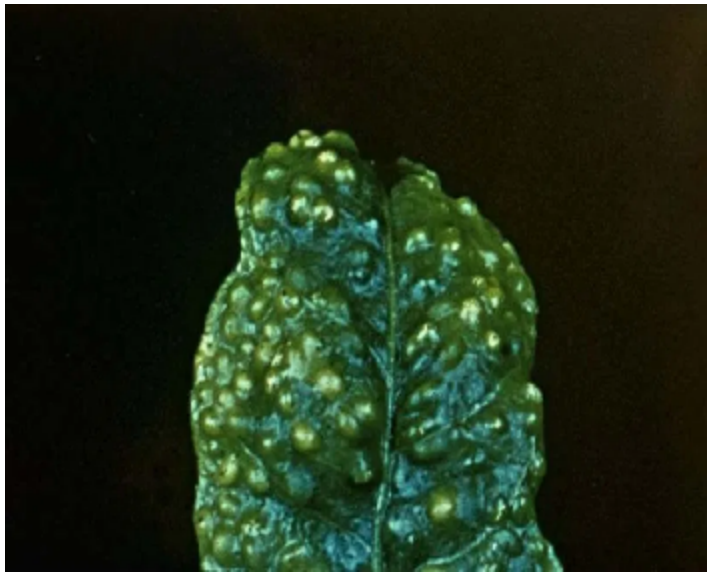
Taxonomy and Ecology of Citrus Psylla



https://agrihunt.com/articles/horti-industry/citrus-psylla-and-its-control/

- **Control:**

Traditional control:

Infected plants and trees should be eradicated before spread of disease

- **Chemical control:**

Insecticides

Injection of trees with tetracyclin and antibiotics

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Taxonomy and Ecology of Citrus Psylla

https://www.biocontrol.co.za/citrus-psyllid-control/

- Biological control:
  - Maggot Fly Larvae
  - Lady beetle
  - *Tamarixia radiata*
  - Lace wing
  - *Pterrastichus radiatus*

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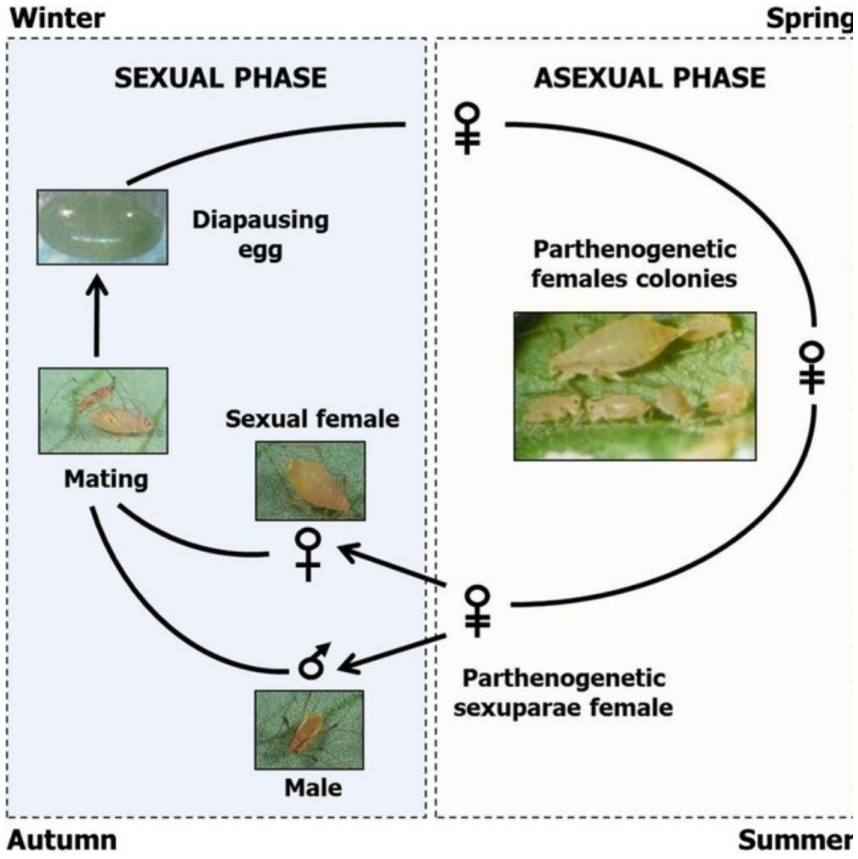
BS ZOOLOGY
O3336361244

Module# 175 Taxonomy and Ecology of Aphids

Phylum	Arthropoda
Class	Insecta
Order	Hemiptera
Superfamily	Aphidoidea

- Soft bodied-weak
- Compound and simple eyes
- Walking legs:skinny
- Head pointed downwards for feeding
- Most aphids have conicles (excrete pheromones)
- Also have cauda (flick honeydew)

Taxonomy and Ecology of Aphids



https://www.researchgate.net/figure/Pea-aphid-life-cycle-During-spring-and-summer-asexual-reproduction-phase-occurs-when_fig1_260211188

Taxonomy and Ecology of Aphids

https://iapps2010.me/2018/08/29/plant-viruses-manipulate-the-feeding-behavior-of-aphids/

Feeding habits

- Aphid feed from the phloem of plants which they trap into with the styletes of their probosocis. They gain access to the phloem vessels from 3 main parts of the plant, stems, leaves, and roots.

Taxonomy and Ecology of Aphids

https://www.hipchickdigs.com/2012/07/battling-aphids/

Damage

- Large populations of aphids can turn leaves yellow and stunt shoots; **aphids** can also produce large quantities of a sticky exudates known as honeydew, which often turns black with the growth of a sooty mold fungus.

Taxonomy and Ecology of Aphids

https://en.wikipedia.org/wiki/Aphid#/media/File:Wasp_&_aphid_May_2010-1.jpg

Control

Chemical control

- Using insecticides like carbamates, pyrethroids and organophosphates

Biological control

- Ladybug, Ladybug larvae, Lacewing, Parasitoids wasp, Mummies, Fungal Pathogen

Module# 176 Phenon, Taxon and Category

Phenon

A term used for the different forms or phenotypes that may occur within a single population.

It includes:

- Sexes
- Age stages
- Seasonal varieties
- Morphs

Phenon, Taxon and Category

Phenon

- The term *Penon* was first introduced by the Camp and Gilly (1943) to describe phenotypic ally homogeneous samples at the species level.
- It was later used by Sneath and Sokal (1973)

Phenon, Taxon and Category

Taxon

- Defined by simpson (1961) as

‘a group of real organisms recognized as a formal unit at any level of hierarchical classification’

- In other words
- A taxon is any rank

Phenon, Taxon and Category

Taxon

- In other words

A taxon is any rank that is considered sufficiently distinct by the taxonomists to be formally recognized and assigned to a definite category.

Two aspects:

1. A species is not a taxon but category
- Example: Robin (*Tundus migratorius*) is a taxon

Phenon, Taxon and Category

Taxon

Taxon must be formally recognized by the taxonomists Examples: Demes

Geographic isolates within a species become taxa only –subspecies. Wiley (1981) called Taxa as historical groups which are clearly individuals like and not classes.

1. Higher taxa: Thrushes, Birds, Vertebrates
2. Lower taxa: Bluebirds, Rbins

Phenon, Taxon and Category

Category:

‘Rank or level in a hierarchical classification’ a class whose members are all the taxa

- A class whose members are all the taxa that are assigned a given rank

Phenon, Taxon and Category

Examples:

The **species category** is a class whose members are species taxa.

20 or more categories fall into 3 categories

1. The species category
2. Infraspecific categories
3. Higher categories

END

Module# 177 Silkworm

Phyllum	Arthropoda
Class	Insecta
Order	Lepidoptera
Family	Bombycidae
Genus	Bombyx
Species	mori

- Silkworm, common name for the silk-producing larvae of any of several species of moths.
- Silkworms possess a pair of specially modified salivary glands called silk glands, or sericteries, which are used in the production of cocoons.

Silkworm

Phyllum	Arthropoda
Class	Insecta
Order	Lepidoptera
Family	Bombycidae
Genus	Bombyx
Species	mori

- The silk glands secrete a clear, viscous fluid that is forced through openings, called spinnerets, on the mouthparts of the larva; the fluid hardens as it comes into contact with air.
- The diameter of the spinneret determines the thickness of the silk thread produced.

Silkworm

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Species	mori

- The best-known silkworm is the larvae of the common, domesticated silkworm moth. This moth, native to China, was introduced into Europe and western Asia in the 6th century AD and into North America in the 18th century.

Silkworm

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Class	Insecta
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Species	mori

- The best-known silkworm is the larvae of the common, domesticated silkworm moth. This moth, native to China, was introduced into Europe and western Asia in the 6th century AD and into North America in the 18th century.
- Silk is know as queen of textile due to sheen and luster

Silkworm

Phyllum	Arthropoda
Class	Insecta
Order	Lepidoptera
Family	Bombycidae
Genus	Bombyx
Species	mori

- Silk is one of the most beautiful and precious fibers given to us by nature
- Its is considered bio-steel because of its strength
- China is the largest producer of silk followed by India

Silkworm

Phyllum	Arthropoda
Class	Insecta
Order	Lepidoptera
Family	Bombycidae
Genus	Bombyx
Species	mori

SERICULTURE IN PAKISTAN

- Despite the growing demand for natural silk, this industry could not flourish in Pakistan.
- This industry is facing numerous challenges such as lack of trained personnel, high mortality rate in local silk breed, and poor economic conditions of the farmers.

Silkworm

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Silkworm

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SERICULTURE IN PAKISTAN

- Unfortunately, this industry has not been on the list of priorities of our Government.
- Provide livelihood for rural poor families but silk proteins also have wide range of applications in textile, food, cosmetics and pharmaceutical industry as well as in traditional medicines.
- Currently, Pakistan is importing about 70% raw silk to meet the demand of local market.

Silkworm

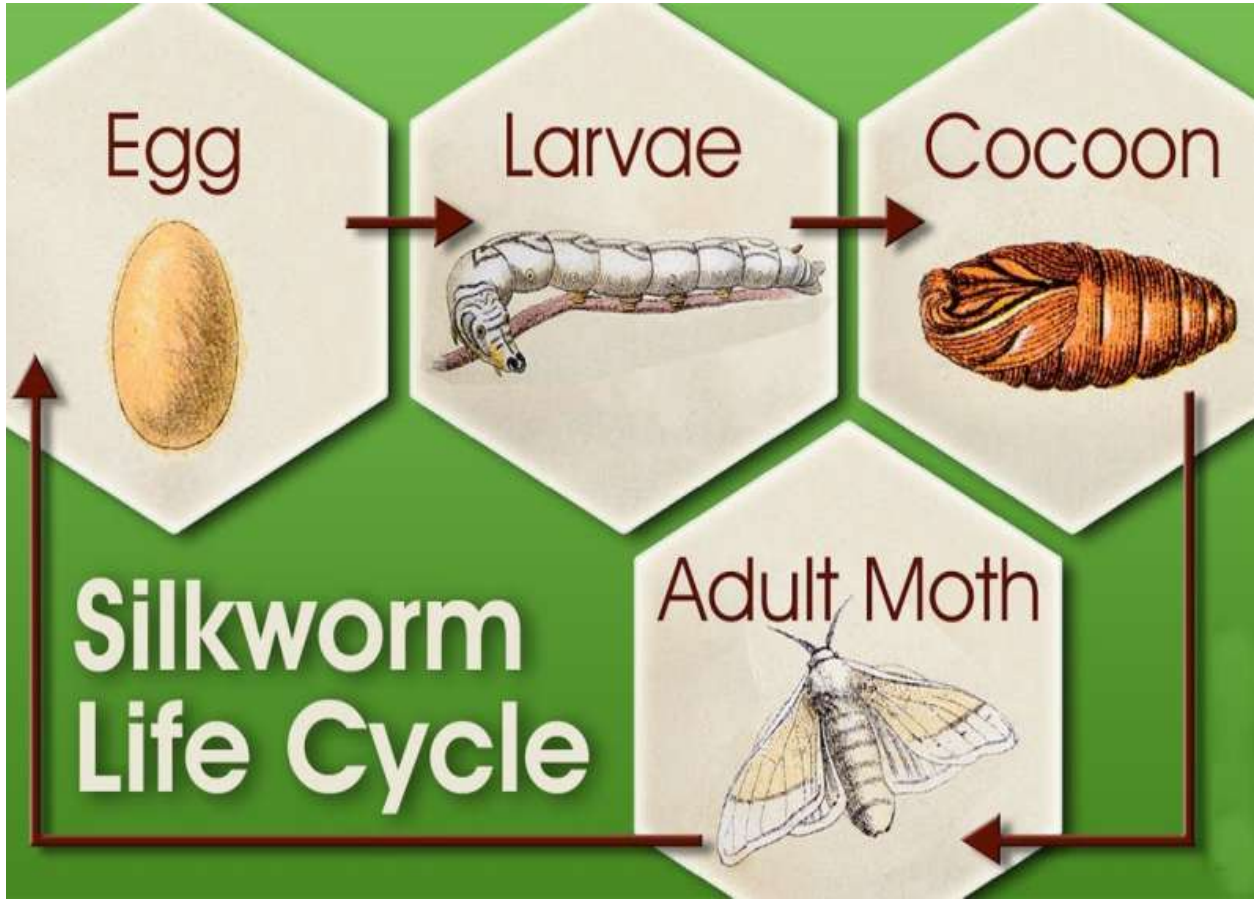
Feeding

- Mulberry silkworm feeds on mulberry plants of genus *Morus*
- *M. alba*
- *M. indica*
- *M. nigra*
- Deep rooted plants
- Grow well in loamy soil or red sandy loamy
- Plantation during early spring and late autumn

Silkworm

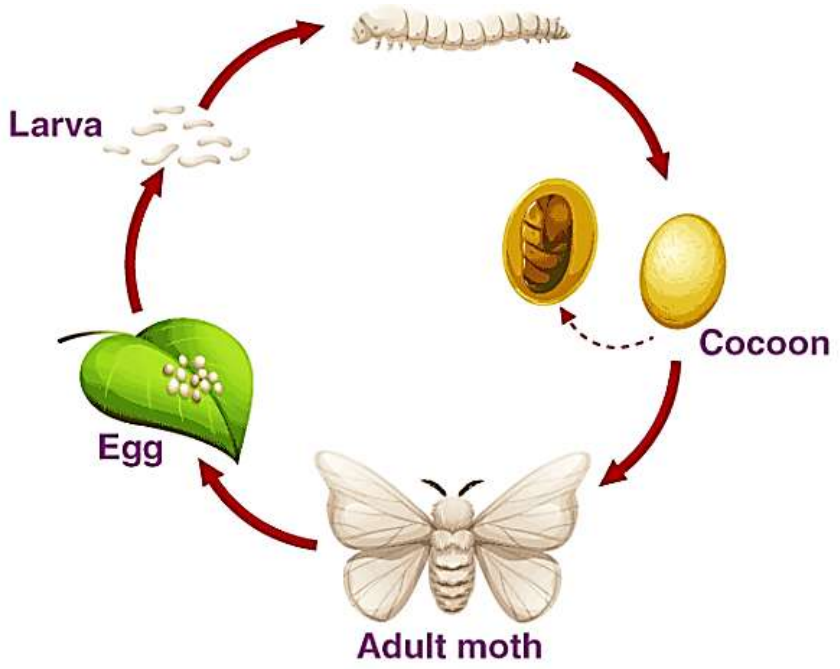
- Mulberry plants grow 15-37 degree temperature
- 600-2500mm rainfall
- 300-900m altitude above sea level

Module# 178 Life Cycle of Silkworm



https://www.google.com/search?q=life+cycle+of+bombyx+mori&tbm=isch&ved=2ahUKEwi97eq2wJDuAhVX04UKHX1UDLoQ2-cCegQIABAA&oq=life+cycle+of+bombyx&gs_lcp=CgNpbWcQARgAMgQIIxAnMgIIADICCAAyAggAMgYIABAFEB4yBggAEAUQHjIGCAAQCBAeMgQIABAYMgQIABAYMgQIABAYOgQIABBDOgUIABCxA1D5B1igLWDjOmgAcAB4AIABrASIAf0WkgEJMi03LjIuMC4xmAEAoAEBqgELZ3dzLXdpei1pbWfAAQE&sclient=img&ei=KX_6X_20LdemlwT9qLHQCw&bih=600&biw=1360#imgrc=T2GFif2nGeMvrM

Life Cycle of Silkworm

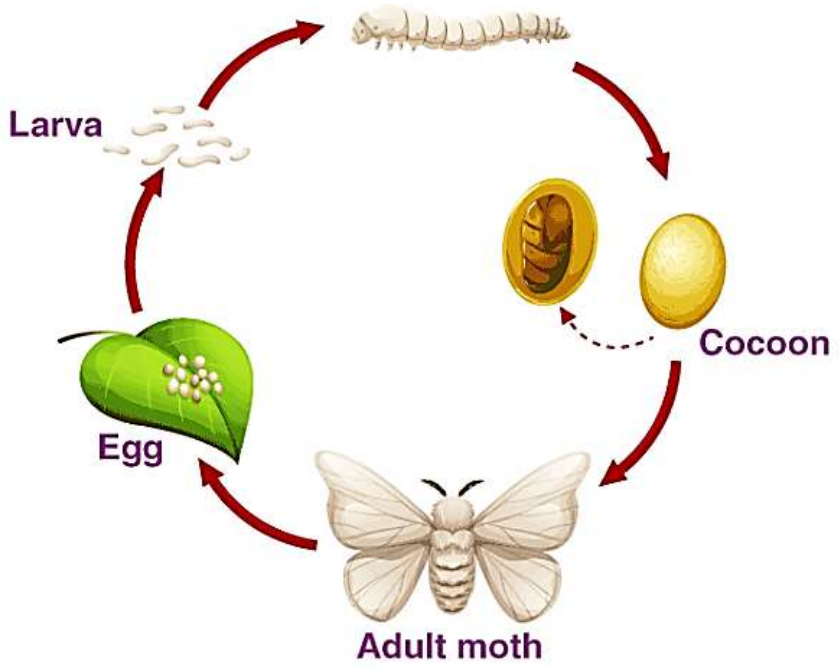


https://www.google.com/search?q=life+cycle+of+bombyx+mori&hl=en&sxsrf=ALeKk00DLnlNrorKjhrTMlGD6alt7iic-g:1610252730004&source=lnms&tbm=isch&sa=X&ved=2ahUKEwiNquHvwpDuAhVOZcAKHcNPDD8Q_AUoAXoECBAQAw&biw=1360&bih=600#imgrc=BluSQkTq1mUHlM

Copulation

- Following emergence from the cocoon, the females rest on the cocoon, spreading out her feeble wings and immediately begin to scent by exuding pheromone from her abdominal projection.
- The males being attracted by the pheromone find the female and start mating.

Life Cycle of Silkworm



https://www.google.com/search?q=life+cycle+of+bombyx+mori&hl=en&sxsrf=ALeKk00DLnlNrorKjhrTMlGD6alt7iic-g:1610252730004&source=lnms&tbm=isch&sa=X&ved=2ahUKEwiNquHvwpDuAhVOZcAKHcNPDD8Q_AUoAXoECBAQAw&biw=1360&bih=600#imgrc=BluSQkTq1mUHlM

Eggs lying

- Thus they remain over for 12-24 hours when the females get insemi-nated by the male.
- After mating for 12-24 hours, the pair gets separated and the female starts laying eggs while the male moths die.
- Most females lay 150-300 eggs over the course of 1-2 days, some can lay as many as 1000 also.

Life Cycle of Silkworm

https://www.google.com/search?q=life+cycle+of+bombyx+mori&tbm=isch&ved=2ahUKEwi97eq2wJDuAhVX04UKHX1UDLoQ2-cCegQIABAA&oq=life+cycle+of+bombyx&gs_lcp=CgNpbWcQARgAMgQIIxAnMgIIADICCAAyAggAMgYIABAFEB4yBggAEAUQHjIGCAAQCBAeMgQIABAYMgQIABAYMgQIABAYOgQIABBDOgUIABCxA1D5B1igLWDjOmgAcAB4AIABrASIAf0WkgEJMi03LjIuMC4xmAEAoAEBqgELZ3dzLXdpei1pbWfAAQE&sclient=img&ei=KX_6X_20LdemlwT9qLHQCw&bih=600&biw=1360#imgrc=xYs_vHjdFWKWiM

Life Cycle of Silkworm

Stages	Period (days)	Characters
Egg	9-12	Yellowish-white, semi round, 1 mg wt
1 instar	3-4	Last instar larvae is greenish, cylindrical caterpillar, 3-3.5 inches long, bear 3 pairs of thoracic legs, 5 pairs of pre-legs, last pair of leg modified to claspers, 1 thoracic and 8 abdominal spiracles, each larvae feed on 90 gm of leaves during larval period (voracious feeder)
II instar	2-3	
III instar	3-4	
IV instar	5-6	
V instar	7-8	
Prepupa	4-7	Non-feeding stage, cocoon spinning within 2-3 days
Pupa	10-12	
Adult	2-3	Robust creamy white, 2-3 day, bipectinate antennae, non-feeder, poor fliers, each female lays 400-600 eggs, secrete alkaline secretions to dissolve cocoon fibers.

Life Cycle of Silkworm

- Fifth instar larvae makes quick round movements of its head
- Rate of 65 movements per minutes while spinning its cocoon.

Life Cycle of Silkworm

Module# 179 Composition of Silk

https://www.google.com/search?q=sericin+and+fibroin&hl=en&sxsrf=ALeKk01htdAQq6Kj9SKjy0Hp29TwqyER6g:1610254161949&source=lnms&tbm=isch&sa=X&ved=2ahUKEwj2msiayJDuAhUt4YUKHWlCDGMQ_AUoAXoECBEQAw&biw=1360&bih=600#imgrc=HBldrPESaxbasM&imgdii=ohYUlslSWHcmpM

Silk Composition

- Silk consists of two main proteins, sericin and fibroin, fibroin being the structural center of the silk, and serecin being the sticky material surrounding it.

Fibroin	70-75%
Sericine (silk gum)	25-30%
Waxy substances	2-3%
Natural colours	1-1.5%
Mineral matter	0.5-1%

Fig- Silk Composition

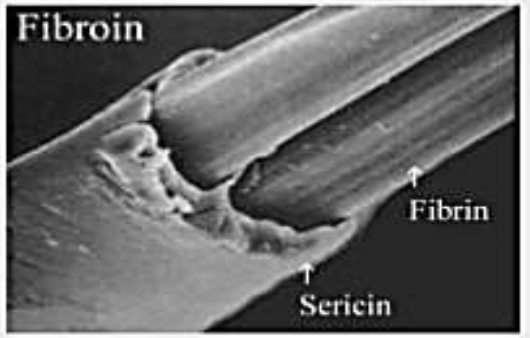


Fig -Raw Silk Microscope View

Composition of Silk

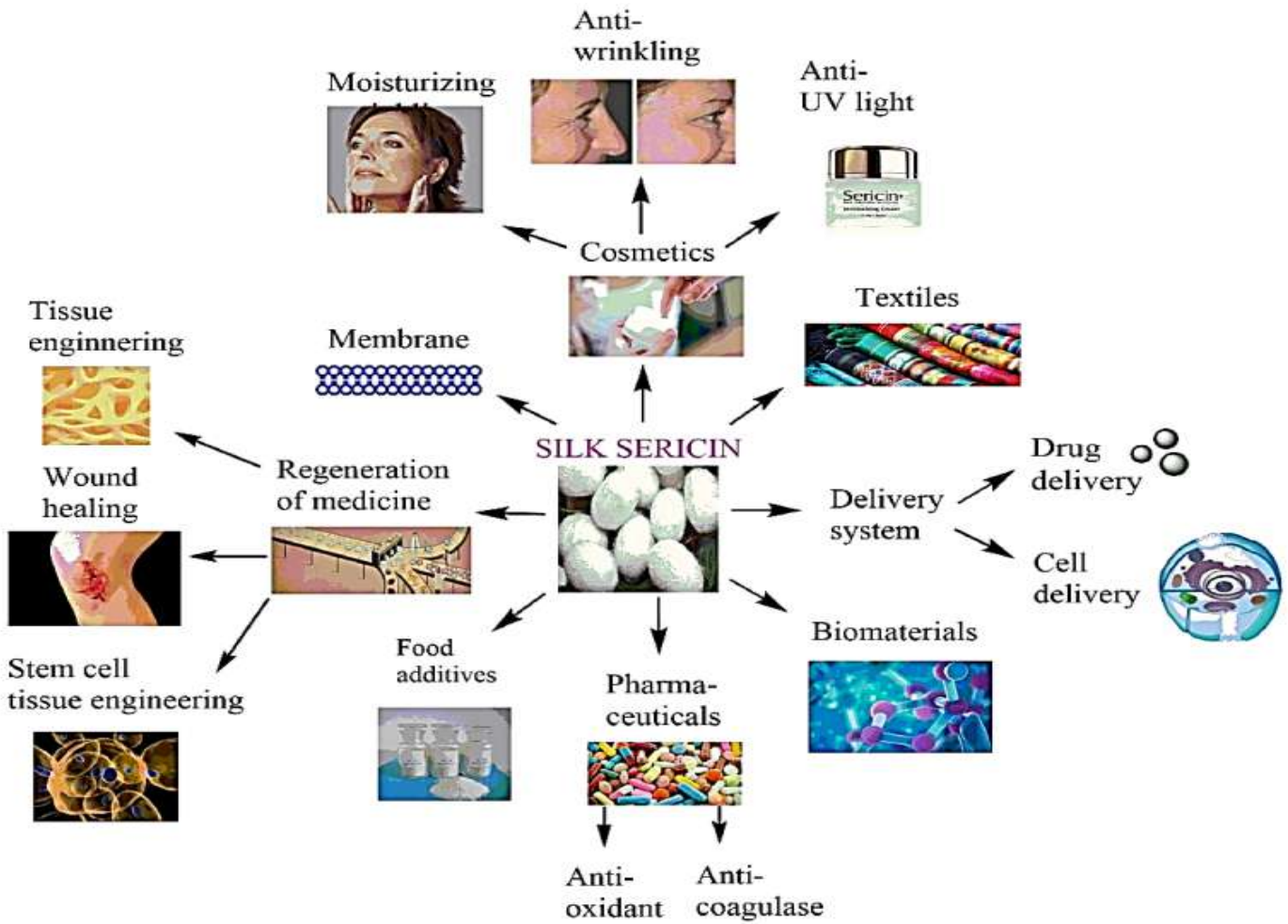


Figure 2. Applications of silk sericin.

Composition of Silk

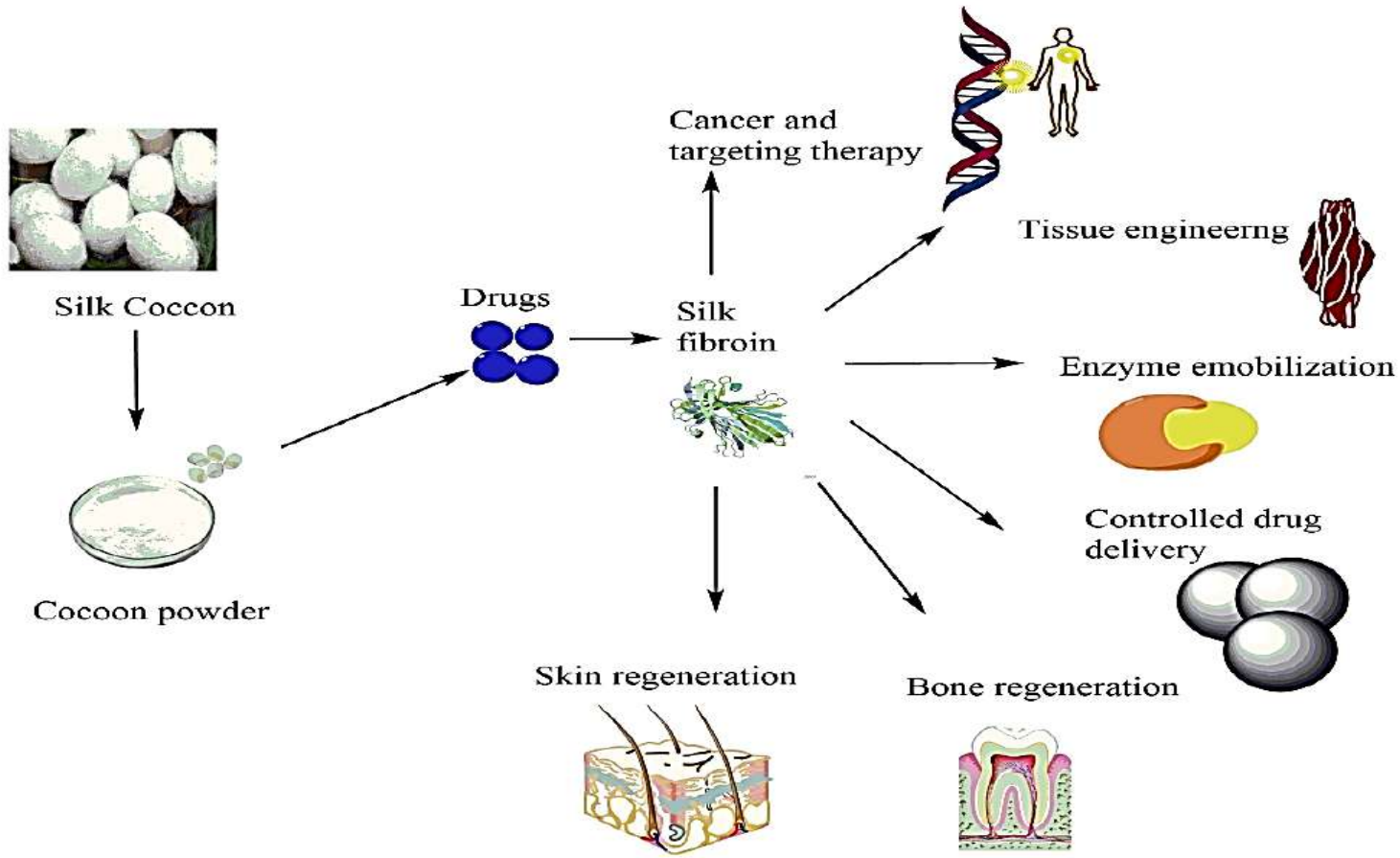


Figure 3. Applications of silk fibroin.

Module# 180 Lac Insect

Kingdom	Animalia
Phylum	Arthropoda
Class	Insecta
Order	Hemiptera
Family	Kerriidae
Genus	Kerria
Species	*Kerria lacca*

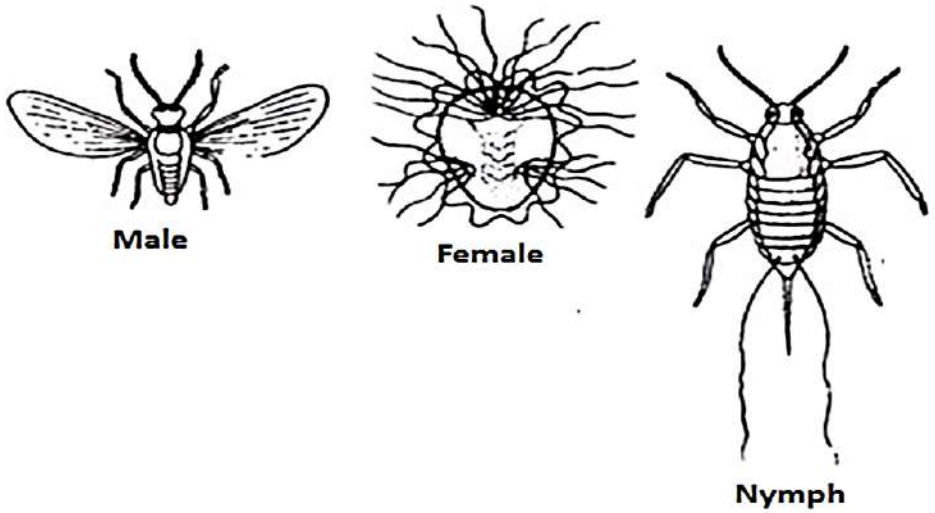
- Lac insect live as parasite on sap of certain tress and shrubs and shows sexual dimorphism
- *Babul (Acacia arabica)*
- *Khair (Acacia catcchu)*
- Its life is completed in 6 months
- Two crops can be obtained

Lac Insect

Sexual dimorphism

- The adult lac insect shows a marked phenomenon of sexual dimorphism. The male and female insect varies in:
- Shape
- Size and
- Presence or absence of certain body parts

Lac Insect



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- Shape
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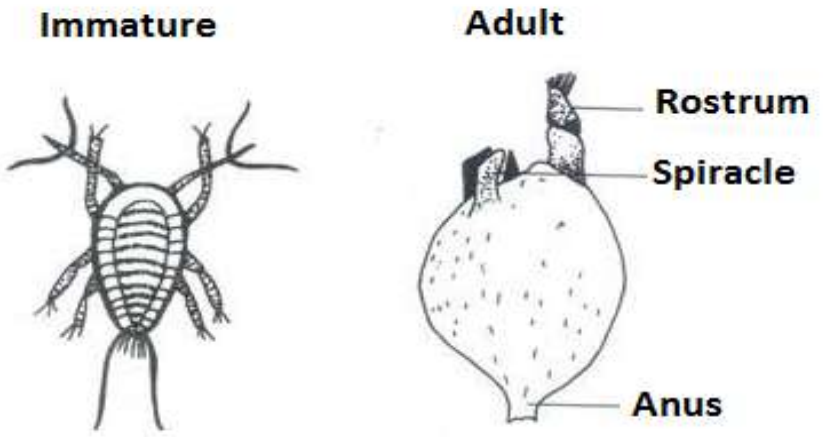
Lac Insect



Loss of some vegetative parts in an adult female

Female Lac Insect

- Small in size
- Head bear a single Antennae
- Single Proboscis
- Eyes are absent

- **Female produce bulk of Lac**

Lac Insect

Fertilization

- After attaining the maturity, males emerge out from their cells and walk over the lac incrustations.
- The male enters the female cell through anal tubular opening and inside female cell it fertilizes the female.

Lac Insect

Fertilization

- After copulation, the male dies. One male is capable of fertilizing several females.
- Females develop very rapidly after fertilization.
- They take more sap from plants and exude more resin and wax.

Lac Insect

https://www.google.com/search?q=life+cycle+of+lac+insect&tbm=isch&ved=2ahUKEwjExJ_b2JDuAhVBrhoKHbRgCZIQ2-cCegQIABAA&oq=life+cycle+of+lac+insect&gs_lcp=CgNpbWcQAzIECAAQQzICCAAyBggAEAUQHjIGCAAQBRAeMgQIABAYMgQIABAYMgQIABAYOgQIIxAnOgUIABCxAzoHCAAQsQMQQzoGCAAQCBAeOgQIABAeULnFQVik_0Fg2YJCaABwAHgAgAGDBogB516SAQsyLTIuNy4zLjYuNpgBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=oJj6X8T0IcHcarTBpZAJ&bih=600&biw=1360#imgrc=qaJmF6tx4lGHlM

LIFE CYCLE

Adult male emerges out of cell and enter female cell through anal tube and fertilizing it

↓

Female lay eggs into incubating chamber inside her body

↓ June-July/Ocotober-November

↓ After 6 weeks

Larvae (Feb-March)

These larvae then come come out of female body and fix themselves to the twing of plant

↓ Secrete resinous substance (LAC) around themselves and sit inside it and undergo moulting (3 times)

↓

Adult

THE CELL WILL BE ELONGATED, CIGAR SHAPED-

IF INSECT INSIDE IT IS MALE.

OVAL- FEMALE

Module# 181 Lac Insect

Swarming

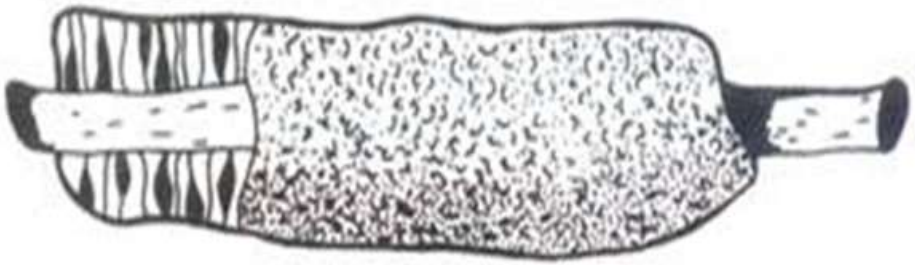
- These larvae begin to wander in search of suitable centre to fix them.
- This mass movement of larvae from female cell to the new off-shoots of host plant, is termed as “swarming”.

Lac Insect

Lac Secretion

1. Glands present in the abdomen of the lac insects.
2. Secrete immediately after the larval settlement on the new and tender shoots
3. This secretion appears first as a shining layer which soon gets hardened after coming in contact with air.
4. This makes a coating around the insect and the twig on which it is residing.

Lac Insect



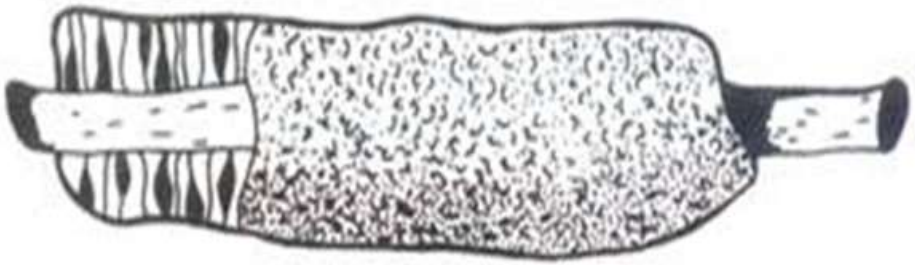
Lac incrustation

https://www.google.com/search?q=lac+incrustations&tbm=isch&ved=2ahUKEwiY7ITv3JDuAhVQ-YUKHVaIBOIQ2-cCegQIABAA&oq=lac+incrustations&gs_lcp=CgNpbWcQAzoECCMQJzoECAAQQzoCCABQqccXWLv0F2Dq9hdoAHAAeACAAe4EiAGmK5IBCTItOC4yLjQuMpgBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=-5z6X9iCMdDylwTWkJKQDg&bih=600&biw=1360#imgrc=5Ko97ssJN9oklM

Lac Secretion

5. As the secretion continues the coating around one insect meet and fuses completely with the coating of another insect.
6. *In this way a continuous or semi-continuous incrustation of lac is formed on the tender shoots.*

Lac Insect



Lac incrustation

https://www.google.com/search?q=lac+incrustations&tbm=isch&ved=2ahUKEwiY7ITv3JDuAhVQ-YUKHVaIBOIQ2-cCegQIABAA&oq=lac+incrustations&gs_lcp=CgNpbWcQAzoECCMQJzoECAAQQzoCCABQqccXWLv0F2Dq9hdoAHAAeACAAe4EiAGmK5IBCTItOC4yLjQuMpgBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=-5z6X9iCMdDylwTWkJKQDg&bih=600&biw=1360#imgrc=5Ko97ssJN9oklM

Lac Secretion

1. Proper care of host plant
2. Regular pruning of host plant
3. Propagation of insect (Inoculation or infection)
4. Control of insect pests
5. Collection and processing of lac

Lac Insect

Inoculation

- The method by which the lac insects are introduced to the new lac host plant is known as inoculation.
- This may be of two types, namely “Natural infection” and “Artificial infection”.
- When infection from one plant to other occurs by natural movements of insect, it is called natural infection.

Lac Insect

Inoculation

- This may be due to overcrowding of insect population and non-availability of tender shoots on a particular tree.

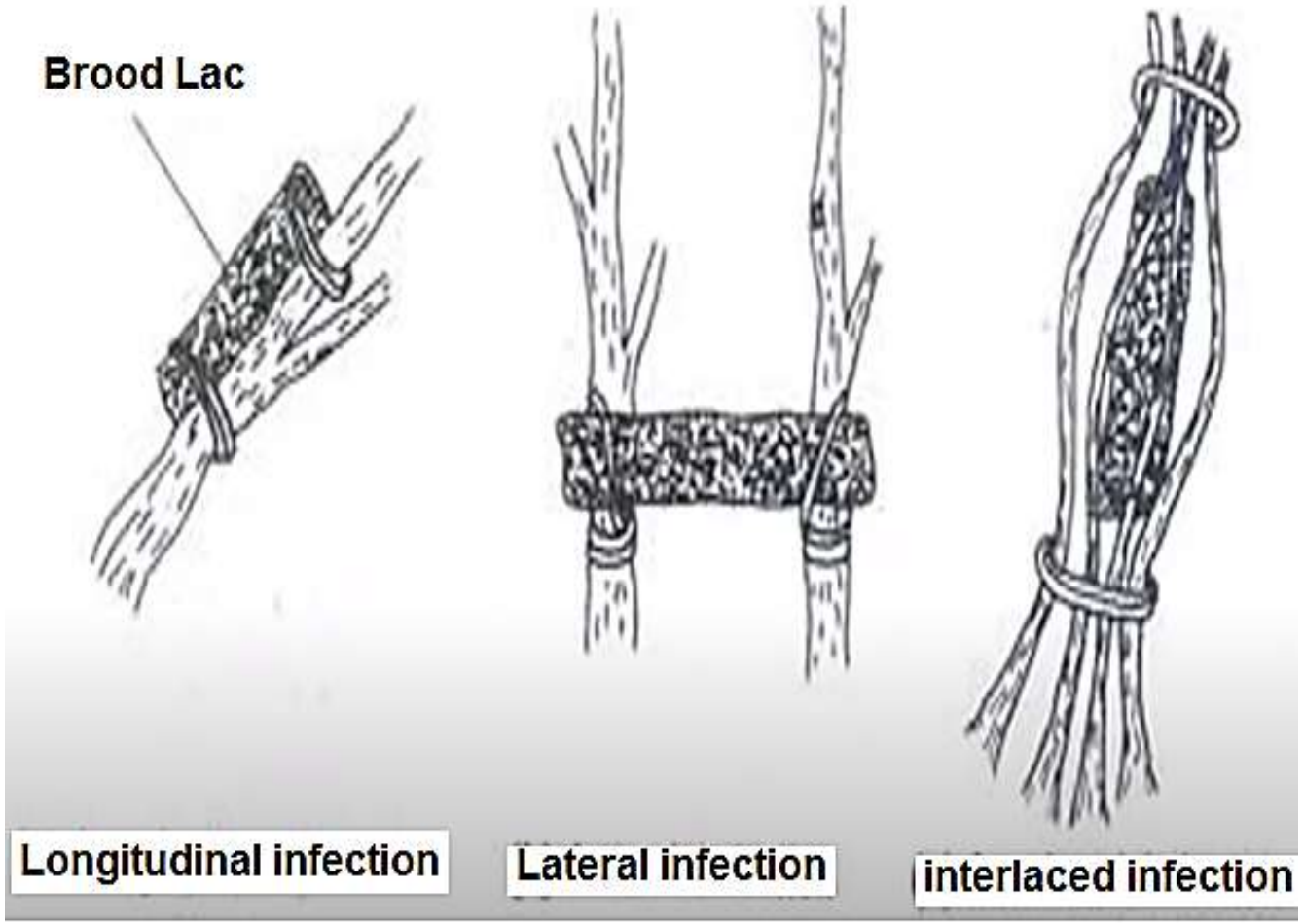
Module# 182 Lac Insect

Artificial Infection

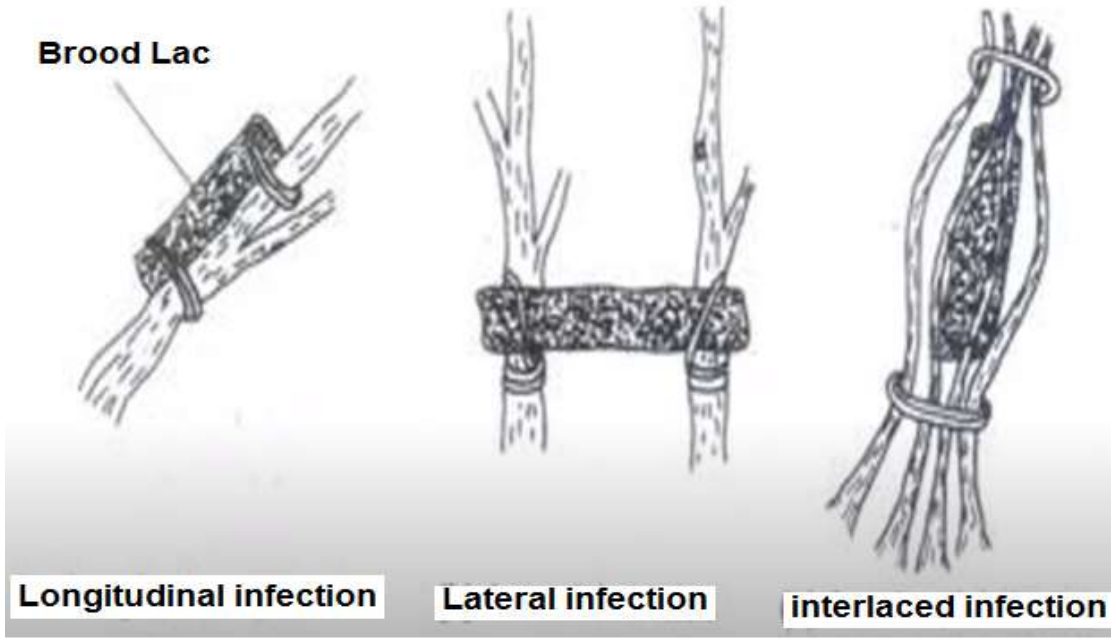
- Prior to about two weeks of hatching, lac bearing sticks are cut to the size of six inches. **They are called "Brood lac".**
- Brood lacs are then kept for about two weeks in some cool place.
- When the larvae start emerging from this brood lac, they are supposed to be ready for inoculation

Lac Insect

https://www.google.com/search?q=lac+artificial+infection&tbm=isch&ved=2ahUKEwiniZKr3pDuAhVKiRoKHY8wCAAQ2-cCegQIABAA&oq=lac+artificial+infection&gs_lcp=CgNpbWcQA1CZsy1YweAtYMniLWgAcAB4AIABAIgBAJIBAJgBAKABAaoBC2d3cy13aXotaW1nwAEB&sclient=img&ei=hp76X6euEcqSao_hIA&bih=600&biw=1360



Lac Insect



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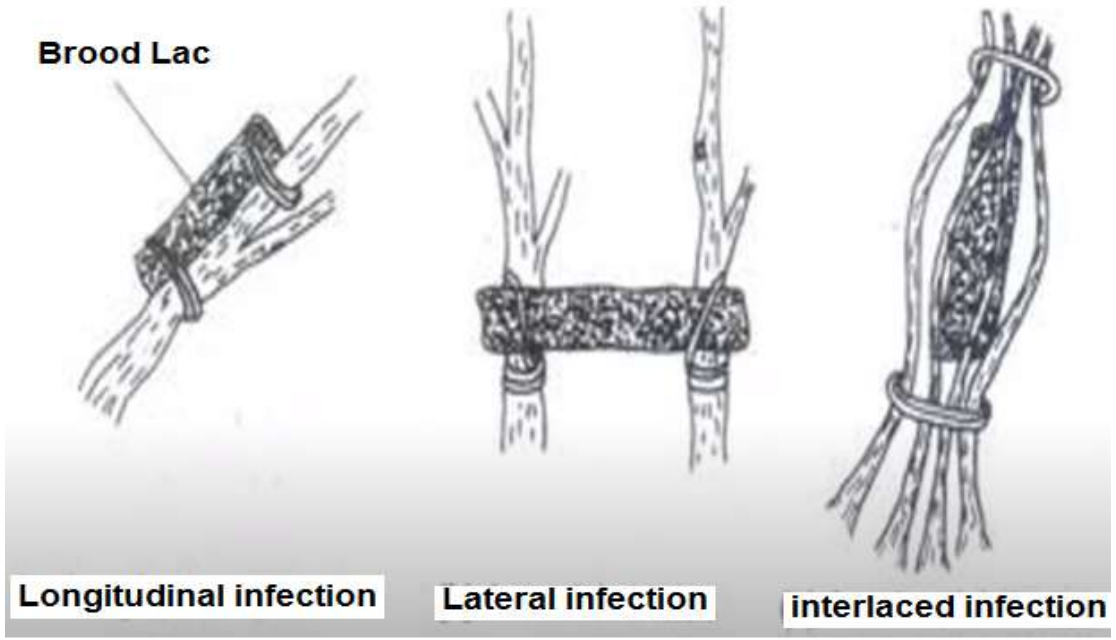
Extraction of Lac

- Lac cut from the host plant is called as "stick lac".
- Lac can be scraped from the twigs before or after the emergence of larvae.
- Ari lac (if used before emergence of larvae)
- Phunki lac (if used after swarming of larvae)
- Swarming

Lac Insect



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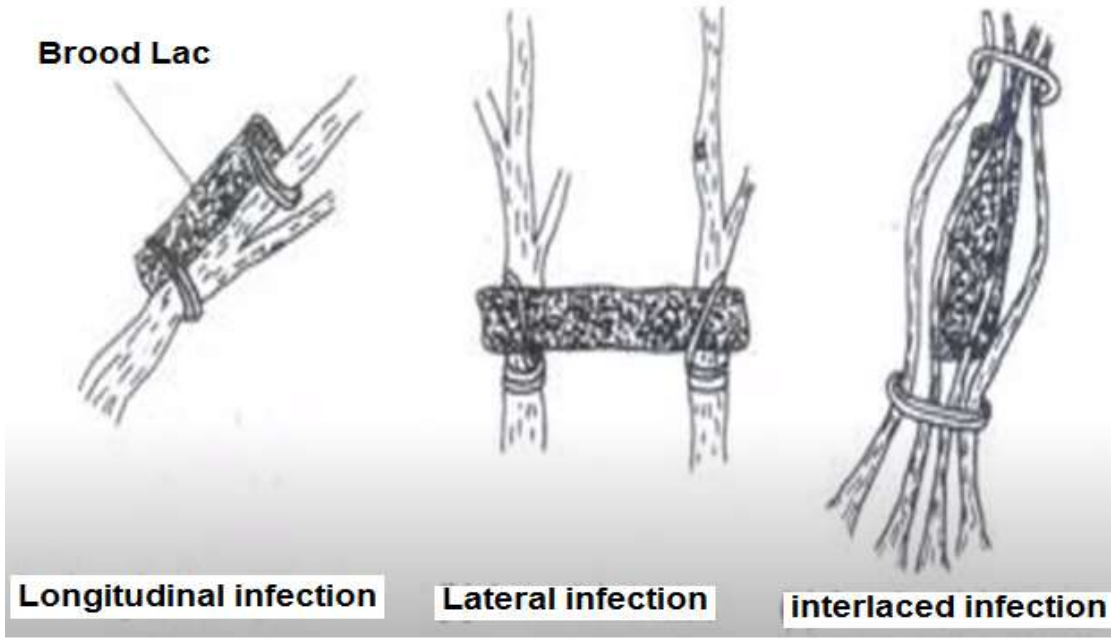
Scraping of Lac

- The scraping of lac from twig is done by knife, after which they should not be exposed to sun. The scraped lac is grinded in hard stone mills.
- The unnecessary materials are sorted out.

Lac Insect



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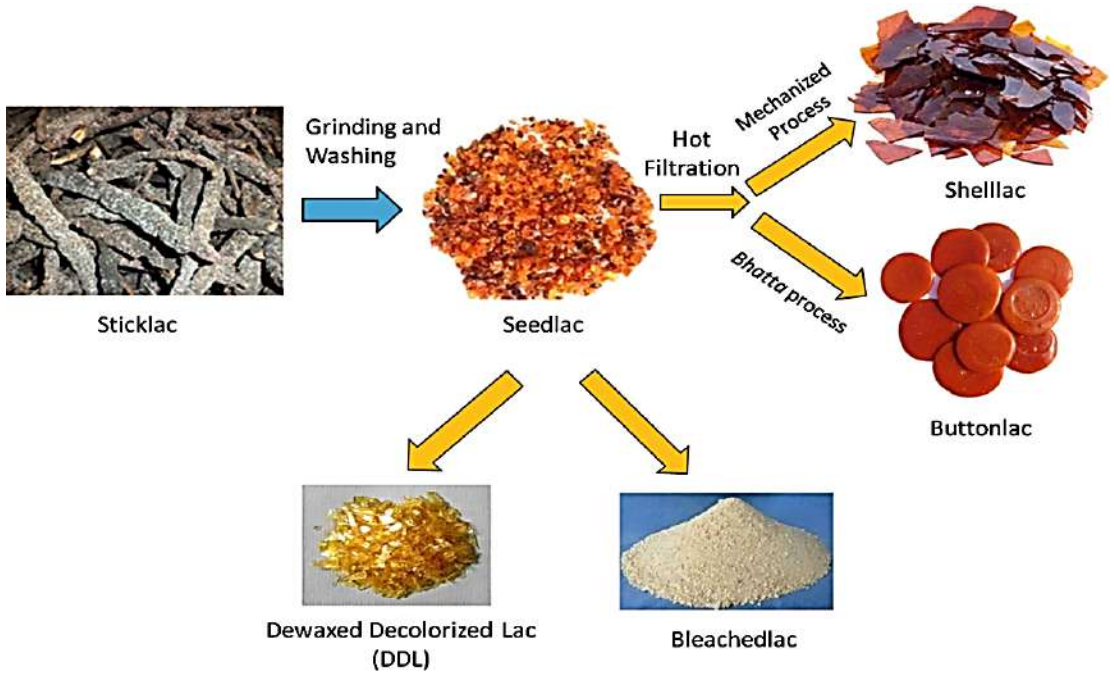
Scraping of Lac

- In order to remove the finer particles of dirt and colour, this lac is washed repeatedly with cold water.
- Now at this stage it is called as **“Seed lac”** and is exposed to sun for drying.

Lac Insect



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Scraping of Lac

- Bleach lac is a refined product obtained by chemical process.
- Shell lac is dissolved in Sodium carbonate solution and bleaching the solution with Sodium hypochlorite. precipitation of the resin with sulphuric acid.
- Deteriorate quickly and should be used within 2-3 months.

Module# 183 Principles of Systematic

Sub species

In 19th century, term subspecies replaced the term variety

Ant specialists use term not only for geographic races but also for sibling species and individual variants.

'An aggregate of phenotypic ally similar populations of a species inhabitation a geographic subdivision of the orange of that species and differing taxonomically from other populations of that species.'

Temporal sub species identified on the basis of paleontology

Principles of Systematic

Sub species

Difficulties in application of sub species

Wilson and brown (1953) and Inger (1961) pointed out four aspects:

1. The tendency of different characters to show independence trends
2. The independence occurrence of similar or phenotypic ally indistinguishable populations
3. The occurrence of micrographic races within subspecies
4. The arbitrariness of the degree of distinction-local population

Principles of Systematic

Sibling species

‘Reproductively isolated species without morphological difference are called sibling species’

- Sibling species are very similar in appearance, behaviour and other characteristics but they are reproductively isolated.
- In other words, sibling species are pairs or groups of genetically closely related species which are often morphologically indistinguishable, but are reproductively isolated, meaning that while they may interbreed, the offspring cannot reproduce.

Module# 184 Principles of Systematics

Polytypic species

Species that contain two or more subspecies are called polytypic species. Species that are not subdivided into subspecies are called monotypic species.

The term *formenkries* and *rassenkries* were formally applied to polytypic species

Principles of Systematics

Importance of recognition of Polytypic species

- Discovery of the fact that previously morphospecies were not reproductively isolate
- and did not qualify as a separate species hence…sub species
- Polytypic species provide simplicity of classification of

  birds snail, mammals and butterflies

  EXAMPLES: 1900 monotypic species of birds----------9040 polytypic species

Principles of Systematics

Difficulties of Polytypic species

1. Differentiation between allopatric and allochronic populations at subspecies level
2. Closely related species replace each other geographically
3. Species are on the boderline between sub species and species (Species under the process of evolution (Incipient species)

Principles of Systematics

Occurrence of Polytypic species in animal kingdom

- In animals 40-80% are polytypic
- *Formenkries and rassenkries* are the order terms for polytypic species

Module# 185 Phenetics Vs Cladistics

- In the field of taxonomy there are two methods of constructing phylogenetic threes i.e. Phenetics and Cladistics
- **Phnetics methods** construct threes (phenogram) by considering the current states of characters without considering the evolutionary history that brought the species to their current phenotype.

Phenetics Vs Cladistics

- However, **cladistics method** depend on the assumptions about ancestral relationships as well as current state.
- Which character states are deeply conserved characters
- Which character states are not deeply conserved characters
- Conserved character state are used for phylogenetics reconstruction

Phenetics Vs Cladistics

- In **cladistics approach** trees are drawn based on conserved characters
- **Phenetic approach** trees are drawn based on some measurements of morphological traits.
- **Molecular phylogenies** are inferred from molecular data (DNA sequence, Protein sequence)
- Phenetics or cladistics or mixture of both

Phenetics Vs Cladistics

- **Traditional taxonomy** is based on phenetics.
- Phenetics is based on overall similarity to group organisms into hierarchical levels called *Taxonomic groups.*
- **Faster and convenient but not accurate.**
- **Cladistics** use certain shared features to group the organisms by common ancestor **(better method of determining ancestry)**

Phenetics Vs Cladistics

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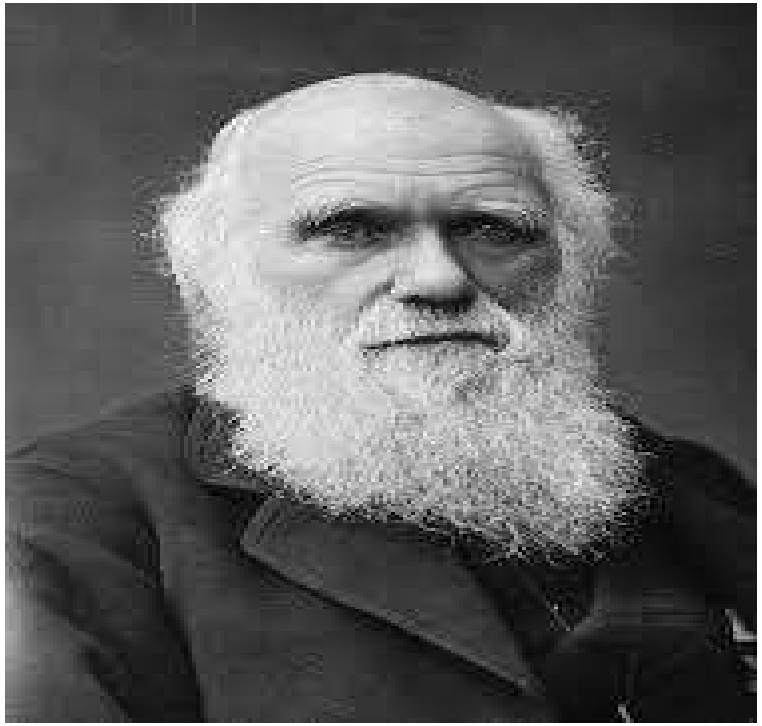
- **Linnaeus was a Phenetists**
- Although he did not believe in evolution by descent but his pattern of classification provide a framework for thinking about evolution from common ancestor.
- For him **“relatedness”** meant propinquity in creators' design.
- Phenetic clustering

Phenetics Vs Cladistics



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- **Darwin was a Cladist**
- According to him natural system based on descent with modification.
- The characters that show true affinity have been inherited from a common patent.
- All true classification is genealogical
- Phenetic classification is artificial
- Cladistic classification is natural

(Darwin, origin of species, 1859)

Phenetics Vs Cladistics

- **Phenetics**
- Computer algorithms based on the phenetic model rely on **distance method** to build tree from sequence data.
- Phenetic approach lead to faster algorithms and have nice statistical properties for molecular data.
- Error rate is less
- But final tree is not highly reliable

Phenetics Vs Cladistics

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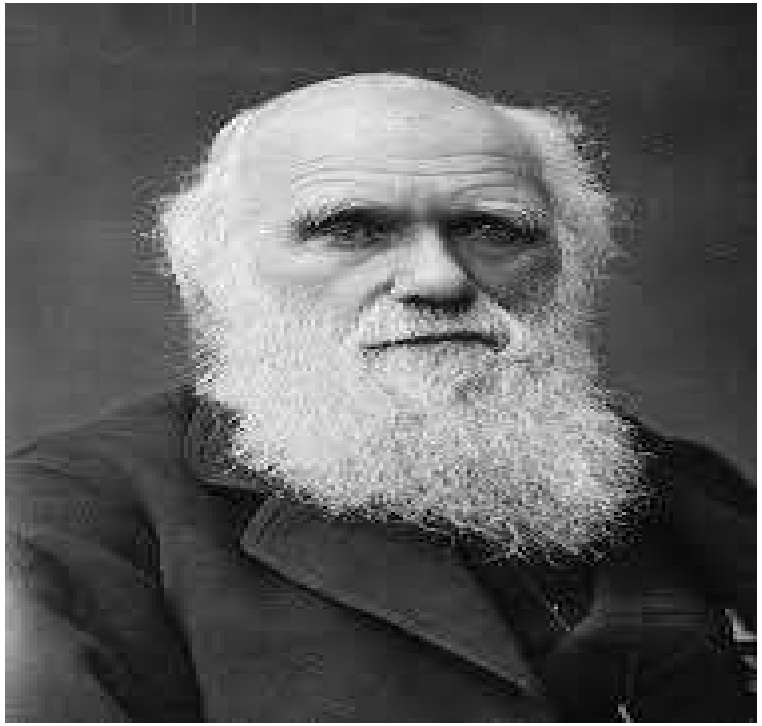
Phenetics Vs Cladistics



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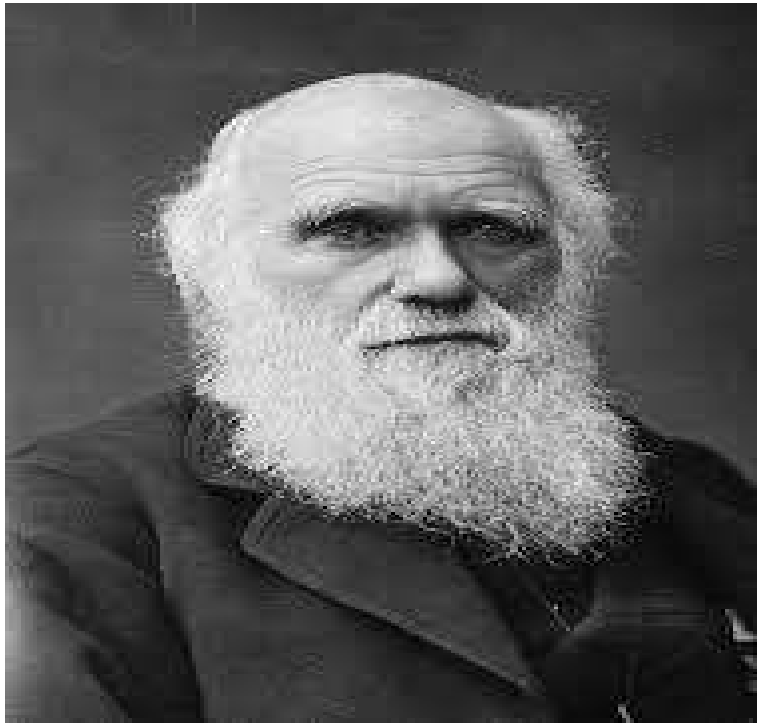
Cladistic

- Evolutionary relationships are documented by creating a branching structure **(phylogenetic tree)**
- It illustrates the relationships among sequences
- Cladistic method construct a tree (cladogram) by considering **various possible pathways** of evolution and chose from among the best.

Phenetics Vs Cladistics



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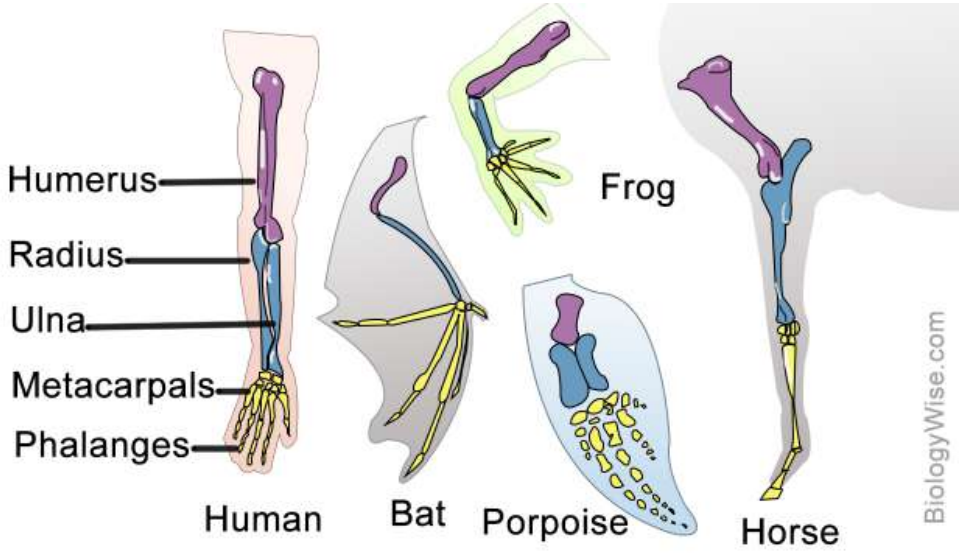
Cladistic

- Evolutionary relationships are documented by creating a branching structure (Cladogram)

- Phylogram is a tree with branching that are proportional to the evolutionary distances.
- It is more informative

Module# 186 Homology



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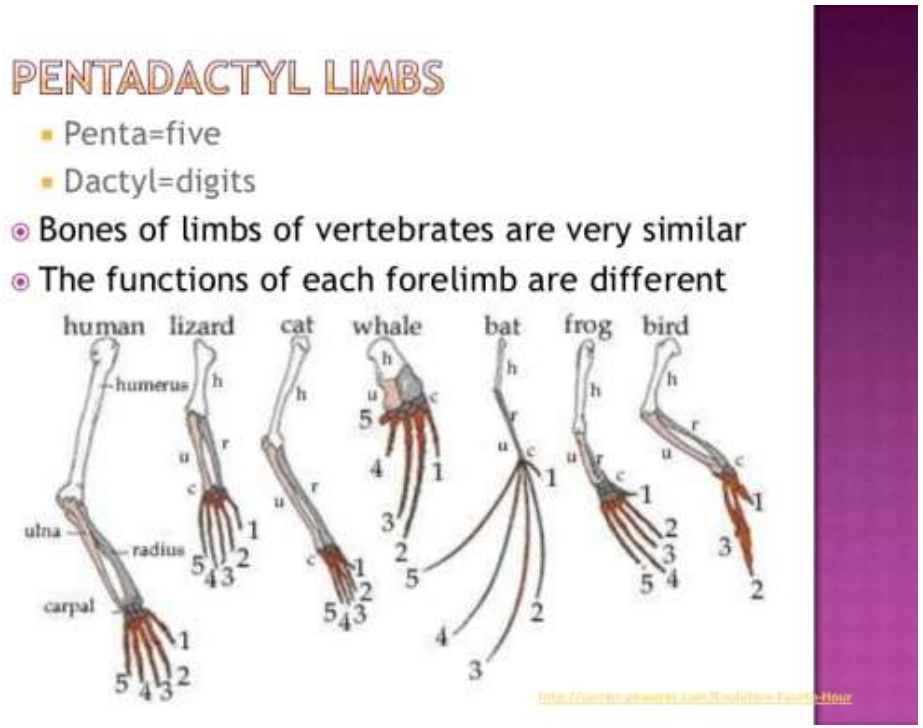
What are homologous structure

- A body part with a same structure between some groups due to inheritance from same ancestor.
- Not necessarily performing same function

Homology



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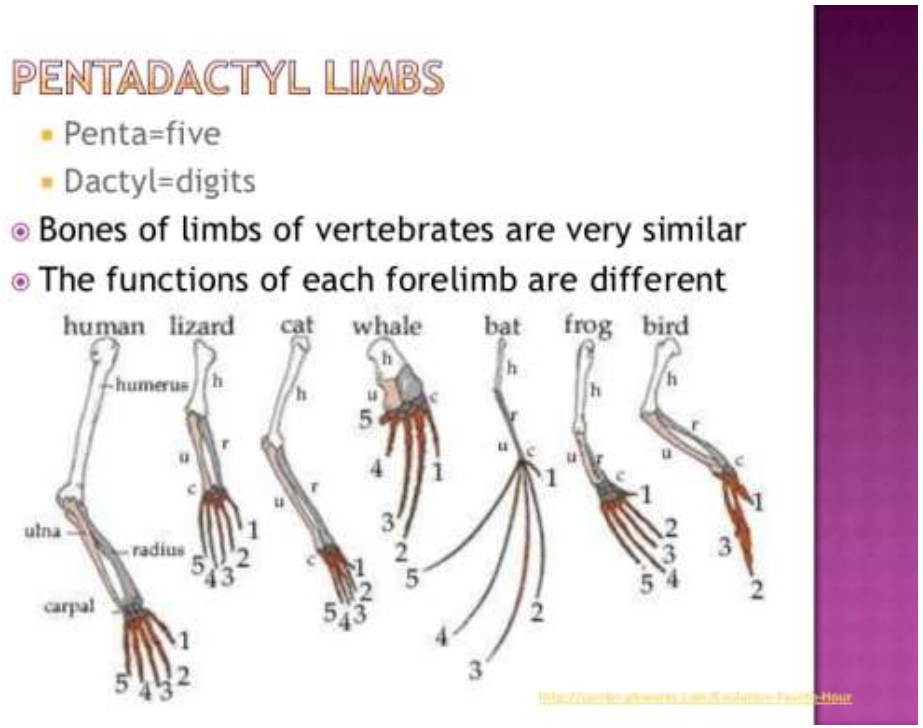
Importance of homologous structure

- To provide a path of evolutionary paths
- Diverges between species and sub-species

Homology



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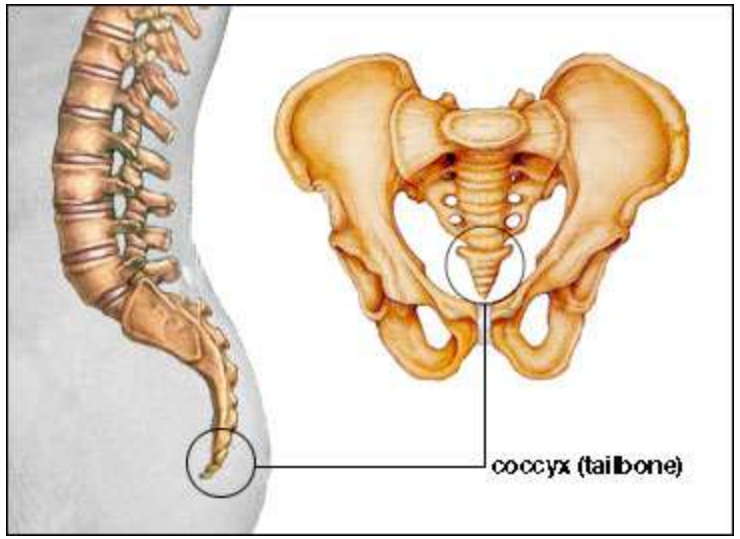
Common examples

- Common **example of homologous structures** is the forelimbs of vertebrates, where the wings of bats and birds, the arms of primates, the front flippers of whales and the forelegs of four-legged vertebrates like dogs and crocodiles are all derived from the same ancestral tetrapod **structure**.

Homology

Tail bone



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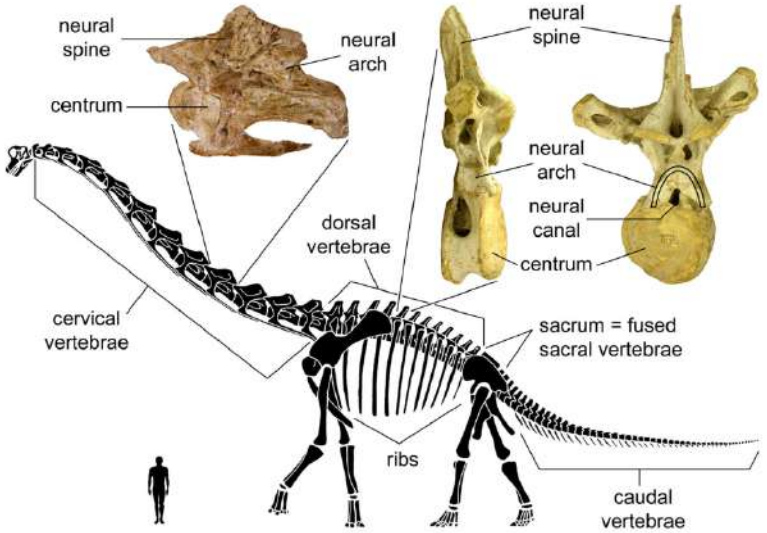
- The tail bone in human is a homologous structure to the beginning of many animals' tails, such as monkeys. It is known as a **“vestigial structure”** because it is the last vestige of what was once a tail. **This structure serves as evidence of having a common ancestor,** one that would have had a tail.

Homology

Neck bone



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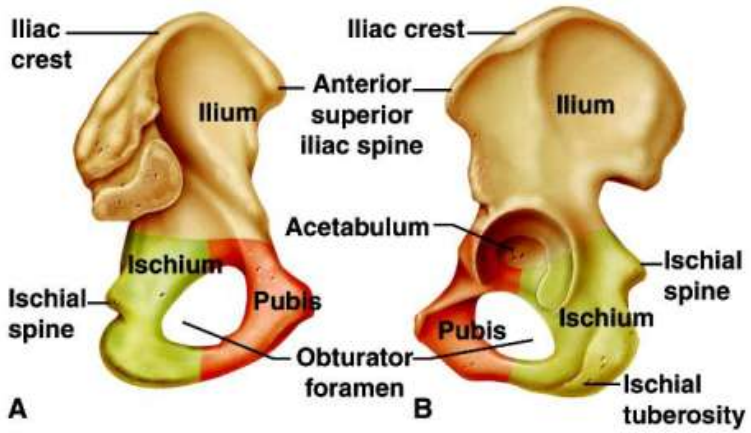
- In spite of its height, the giraffe has the very same number of neck bones (seven) as a giant whale and a tiny human being.

Homology

Pelvis bones



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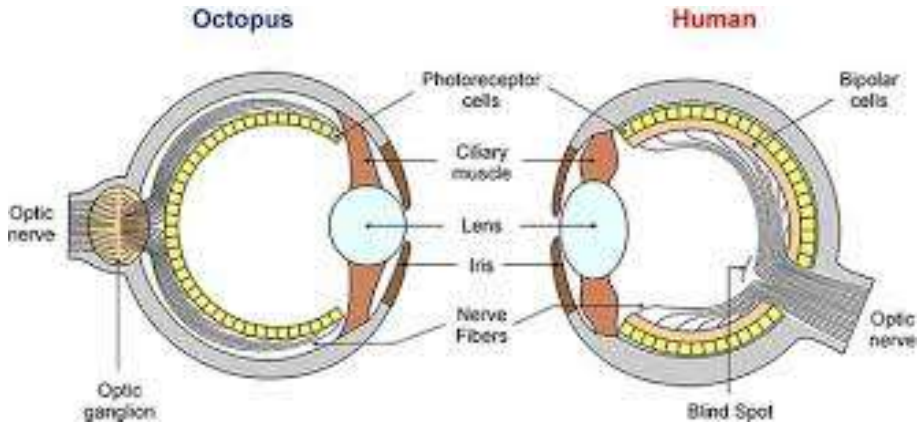
- Human beings, dogs, and cats all have similar pelvises, which are homologous structures to a vestigial pair of bones in snakes. These bones in snakes are the last remains of a pelvis, with no legs to attach.

Homology

Eyes structure



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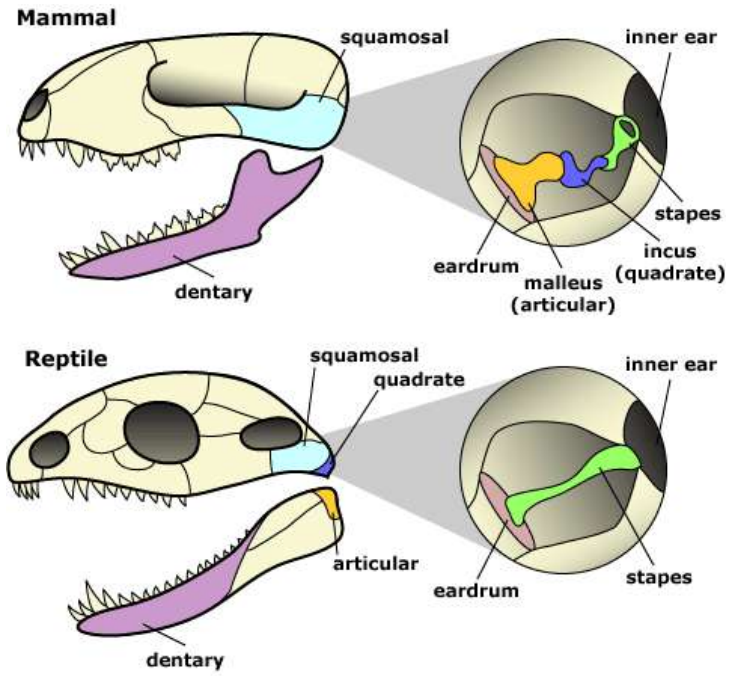
- Human eyes are homologous to the eye bulbs which blind creatures who live in caves – **like cave fish**-- have on their heads.

Homology

Plasma membrane

- All organisms contain homologous plasma membranes with what is called a phospholipid bilayer.

Homology



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Auditory bones

- An auditory bone that exists inside the ears of mammals is a homologous structure to a reptile's jawbone, including dinosaurs.

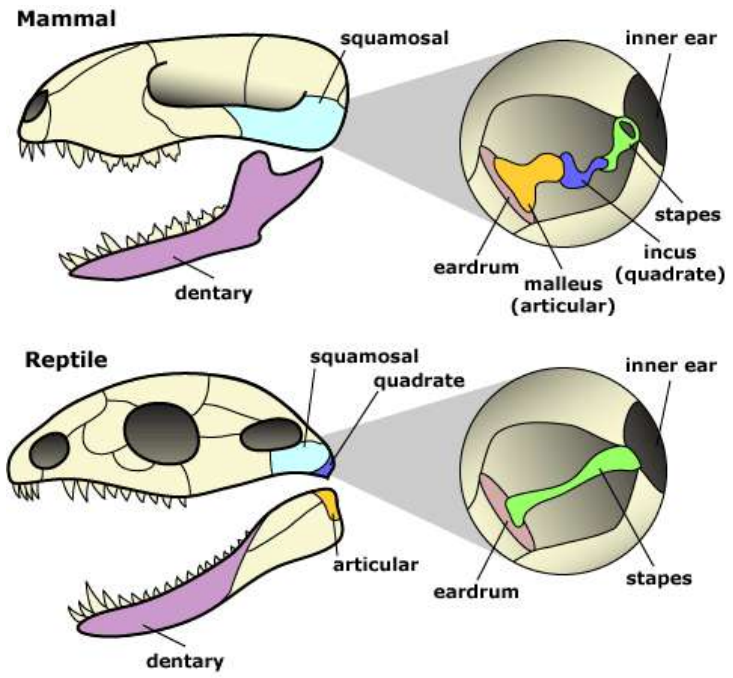
Homology

Stinger of wasps and bees and ovipositor of insects

- Wasps and bees have stingers that they can use when they feel they are in danger. This is a homologous structure to the ovipositor of other organisms like insects and certain fish; the feature allows these organisms to lay eggs.

END

Module# 187 ANALOGY



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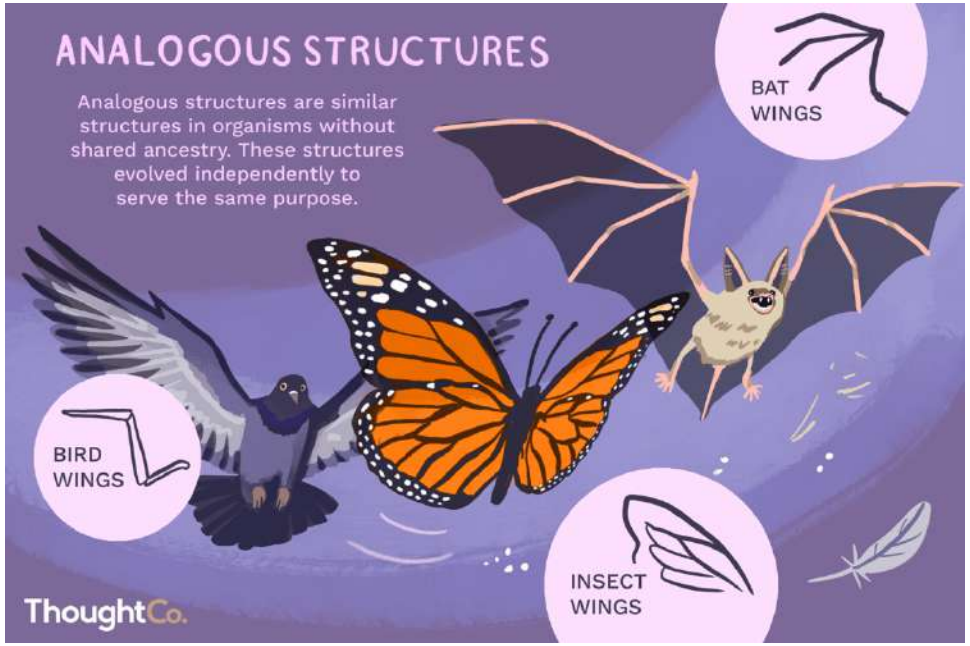
- The structure which are different in different species but perform similar function.
- Have similar appearance
- Not evolved together
- Do not share common ancestor
- Linnaeus started classification on the basis of morphological similarities.
- Wrong classification
- All similar looking organism do not have same origin

ANALOGY



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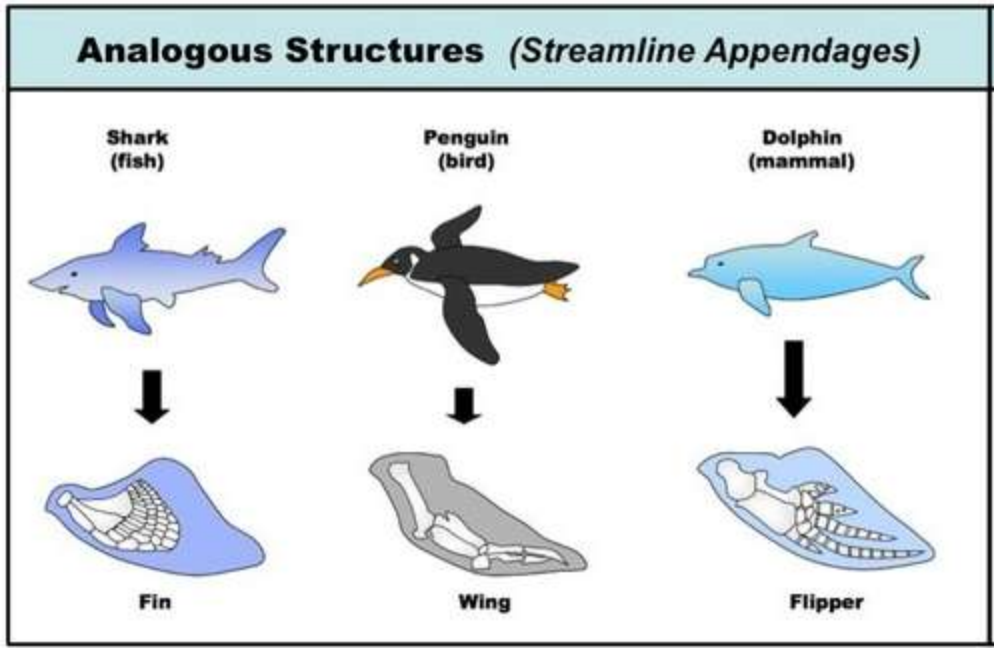
- Analogous structures evolves due to convergent evolution
- When different organism adapt to same environment
-
- Wings of insect and birds
- Fins of fish and penguin
- Limbs of human, horse and dolphin

ANALOGY



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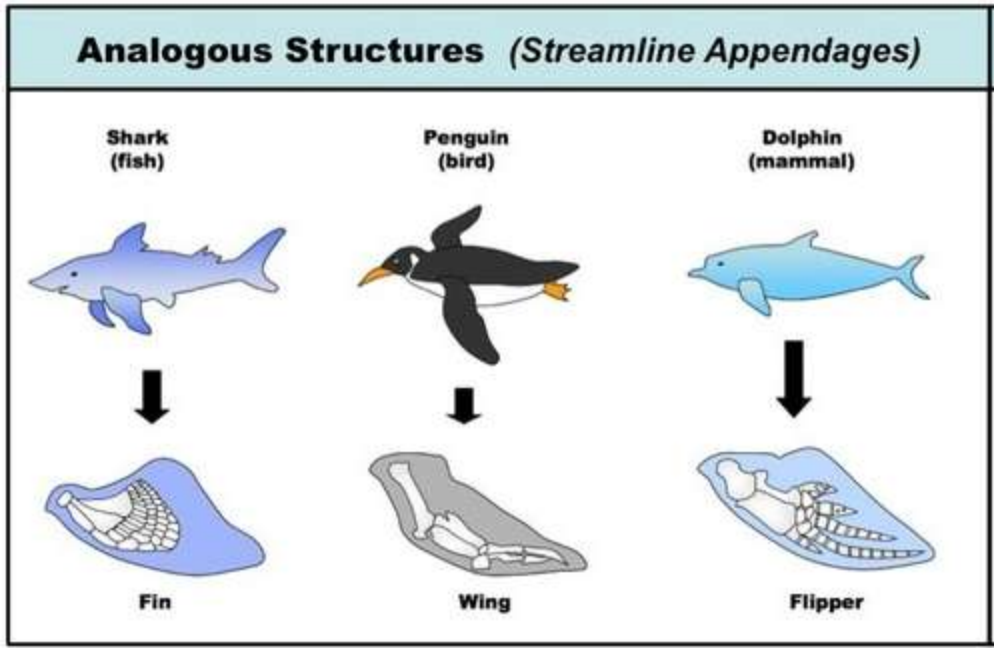
- For similarity we should not only depends on general appearance
-
- Analogous structure act as evidence that theory of natural selection persists and adaptations accumulates over time

ANALOGY



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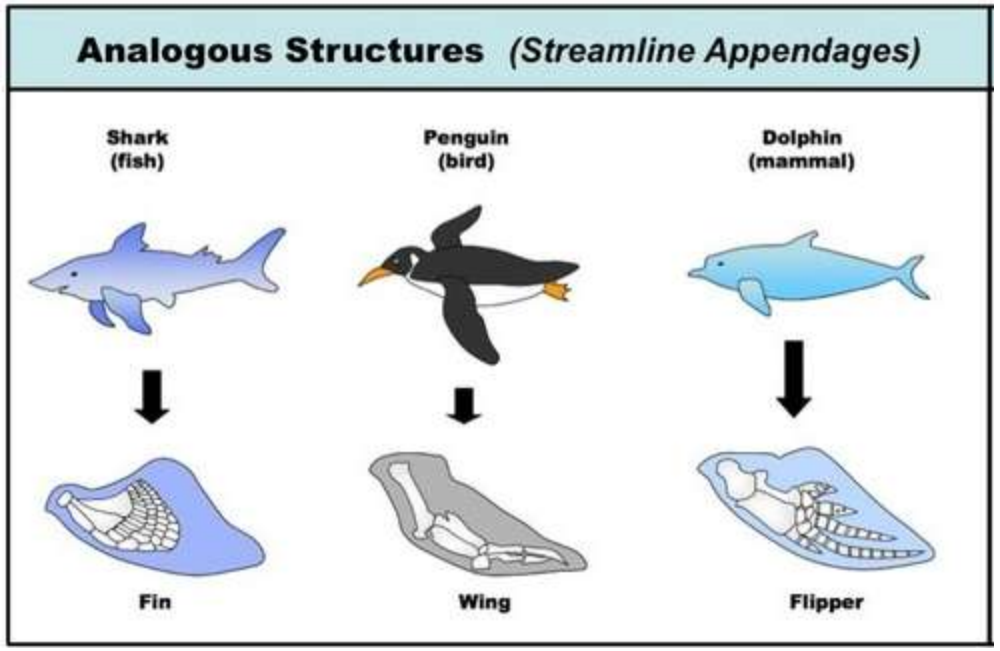
- An example is the convergence of the **streamlined form** in the bodies of squid, shark, seal, penguin, and ichthyosaur, animals of diverse ancestry.
- **Physiological processes and behaviour patterns** may also exhibit analogous convergence.

ANALOGY



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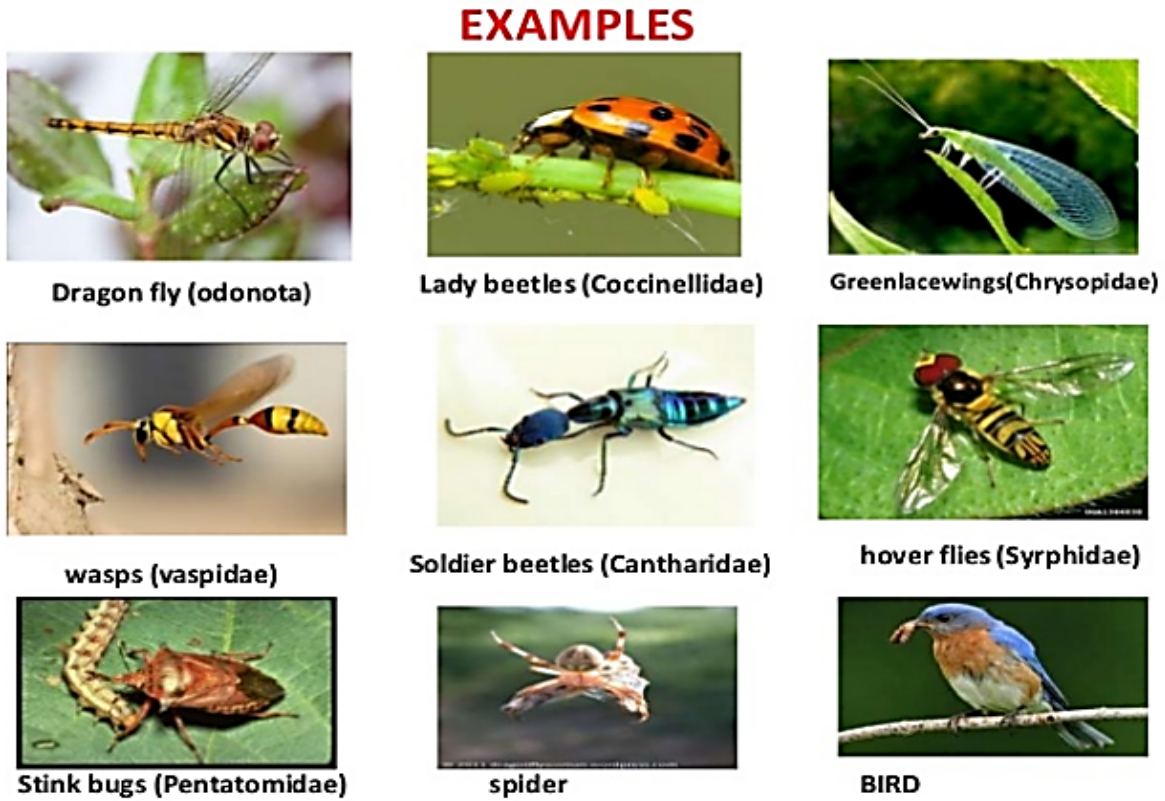
- Egg-guarding behaviour in the cobra, the stickleback, the octopus, and the spider is thought to have evolved independently among those animals, which are quite distant in their biological relationships.

Module# 188 Natural Enemies

- Predators
- Parasitoids
- Pathogens

Natural Enemies



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- Predators kills and consume other organisms (prey)
- Usually prey is smaller in size
- Weak
- Lion is a predators
- Spiders are predators
- Lady beetles are predators

Natural Enemies

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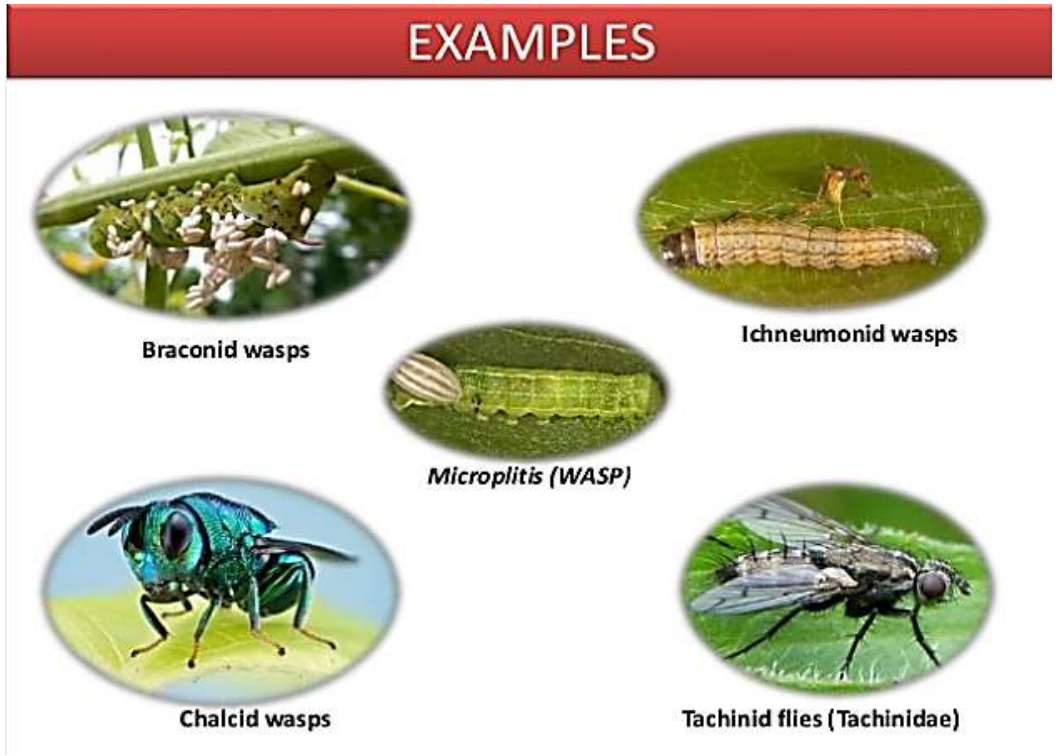
WHO EATS WHO??	
PREDATORS	**PESTS ATTACKED**
Lady beetles	Aphids, mites, Lepidoptera eggs
Green lacewing	Aphids, lepidoptera eggs and small larvae
Hoverflies	Aphids
Spiders	Generalist predator, including moths, larvae, eggs, bug nymphs and other spiders
Ants	Eggs and larvae of lepidoptera
Blue birds	caterpillars, beetles, grubs
Dragonfly	Termites, mosquito larvae, other aquatic insects

Natural Enemies



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- A **parasitoid** is an organism that has young that develop on or within another organism (the host), eventually killing it.
- **Parasitoids** have characteristics of both predators and parasites.
- **Parasitoids** are usually smaller than their selected host

Natural Enemies

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WHO EATS WHO???	
PARASITIODS	**PEST ATTACKED**
***Ichneumonidae* wasps**	Larva of noctuids* (noctuids include helicoverpa, armyworms, cutworms and loopers)
Microplitis (WASP)	Helicoverpa larvae
Tachinid flies (Tachinidae)	Larva of noctuids
Trichogramma Wasp	Eggs of noctuids
Chalcid wasps	Aphids
Braconid wasps	beetles, caterpillars, flies and sawflies.

Natural Enemies

- Pathogen is an infectious agent that feed on host body and cause damage to its host.

- Bacteria
- Viruses
- Protozoon

END

Module# 189 Mimicry

- Mimicry refer to similarities between species.
- Mimicry is a form of symbiosis where species may mimic colours, patterns or even behaviours for one of the two reasons:

A. Either to be able to get closer to unsuspecting prey

B. To gain protection by attaining similarity with predators or poisonous species.

Mimicry

Camouflage ➡
- To change color of body to shade into scenery

Mimicry ➡
- To take the form of other organism, such as plant or insects

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- Mostly in mimictic relationship advantage is one side.

Types of Mimicry

- Batesian mimicry
- Mullerian mimicry
- Wasmannian mimicry
- Peckhmian mimicry
- Self-mimicry

Mimicry

- Batesian mimicry is a type of mimicry in which platable animal mimics the bad-tasting or noxious prey so that predator may not distinguish b/e them.
- After several attempts of distasteful encounters predators learn to avoid such appearance.
- It is called Batesian mimicry in honor of English naturalist Henry W. Bates who published an explanation of this mimicry in 1862.

Mimicry

Example

- Viceroy butterfly (palatable) **(Mimic)**
- Monarch (unpalatable/bad-taste) **(Model)**

END

Module# 190 Mimicry

Batesian Mimicry

- Eastern coral snake (poisonous)
- Scarlet King snake (non-poisonous)
- Condition for successful Batesiam Mimicry
- Number of mimics should be lower than number of models
- Mimicry breakdown

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Mimicry

Mullerian Mimicry

- Mullerian mimicry is the form of protective mimicry in which two or more distasteful or harmful species especially insects, closely resemble each other and therefore avoided equally by all their natural predators (Muller (1821-1897)/German born Brazilian Zoologist
- **Larger the group greater will be the benefit.**

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Mimicry

Subspecies of *Heliconius erato* (left-hand column) and of *H. melpomene* on the right. Both species are toxic and form a local mimicry ring from a different area of Ecuador or northern Peru.

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Mimicry

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Module# 191 Mimicry

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Wasmannian Mimicry

- Wasmannian mimicry is a mimicry in which the mimics resembles its host in order to live within the same nest or structure.
- Several jumping spiders closely resemble ants.
- This works well in order to get "lost in the crowd" and thus avoid predators.

Mimicry

Wasmannian Mimicry

- Some beetles mimic ants in order to be provided with food, helter and protection.

Araeoschizus sp.

Reichenbachia spatulifer

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Mimicry

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Peckhamian Mimicry

- Peckhamian mimicry is also known as Aggressive mimicry.
- In this type of mimicry predator mimics its prey in order to catch it.
- Bolas spiders are type of aggressive mimics.
- They produce chemicals that mimics the sex pheromones of particular moth species.

Mimicry

Chemical Mimicry

Bola spiders and bids-dropping spiders emit pheromones that mimics the sex pheromones of certain moth species. Male moths come and get eaten

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- When male moths approach what they perceive to be a females in order to mate with her, they are caught by bolus spiders

Module# 192 Species Concepts

Nominnalistic Species Concept

- According to this concept " Nature produces individuals and nothing more”
- It is also known as Linneaus species concept.
- Proponents of this concept (Buffon, Bessey, Lamark etc.) belive that only individuals exits but they do not believe in existence of species.

Species Concepts

Nominnalistic Species Concept

- Species are man's own creations
- No actual existence in nature.
- Mental concept nothing more.
- Such mental concept of man has no value.
- This concept was popular in France in 18th Century.

Species Concepts

Criticism

- Simpson (1961), Rollins (1965) and Mayr (1969) stated that no biologist can agree with the idea that species do not exist as it is established fact.
- Species are product of evolution.

Species Concepts

Biological Species Concept

- **“A species is a group of interbreeding natural population that is reproductively isolated from other such species”**
- A species has three properties:
- Reproductive community
- Ecological Unit
- Genetic Unit

Species Concepts

Criticism

- Not application on asexually reproducing individuals
- Not applicable on sibling/cryptic species
- Fossil record
- Incomplete speciation

END

Module# 193 Species Concepts

- **Recognition concept**: The organisms are supposed to share a 'specific mate recognition system', or SMRS.
- species are reproductive populations, and on the whole they are equivalent to biological species.
- However, there are two important differences.
- First, a population, in principle atleast, might vary geographically in how mates are recognized.
- Second, a single lineage might evolve a different SMRS.

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Species Concepts

- Cohesion species concept emphasizes the point that something infect hold the species together.
- It has been suggested that something other than sex might play this role.

Species Concepts

- Morphological species concept
- It is based on the morphological variations.
- Variations/different species/sub-species/sibling species

Module# 194 Worm Blooded vs Cold Blooded animals

END

Worm Blooded animals

- They maintain their body temperature
- Examples: Birds, Mammals

Cold Blooded animals

- Do not maintain their body temperature
- Examples: Amphibians, Fishes

Problems: Some cold blooded fishes, Lizards maintain body temperature and some worm blooded animals such as birds do not

Module# 195 Ecotherms, Endotherms & Heterotherm

Ectotherms

- Produce metabolic heat at low level and quickly exchange with environment.
- Absorb heat from surroundings to maintain body temperature
- Examples: Mostly invertebrates, Amphibians, Fishes etc

Enthotherm

- Generate own body heat for temperature regulate
- Examples: Birds, some

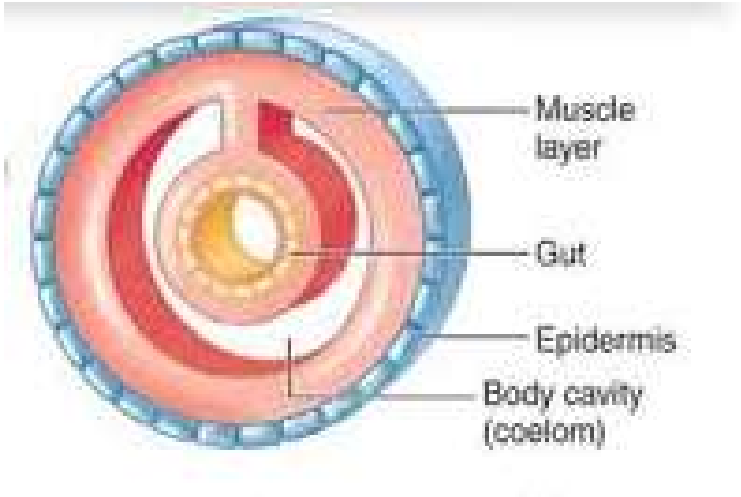
Ecotherms, Endotherms & Heterotherm

Heterotherms

- Capable of producing endothermic heat but generally do not use this to regulate their body temperature within a narrow range
- Examples: Bats, humming birds etc.

END

Module# 196 Coelomates, Acoelomates & Pseudocoelomates



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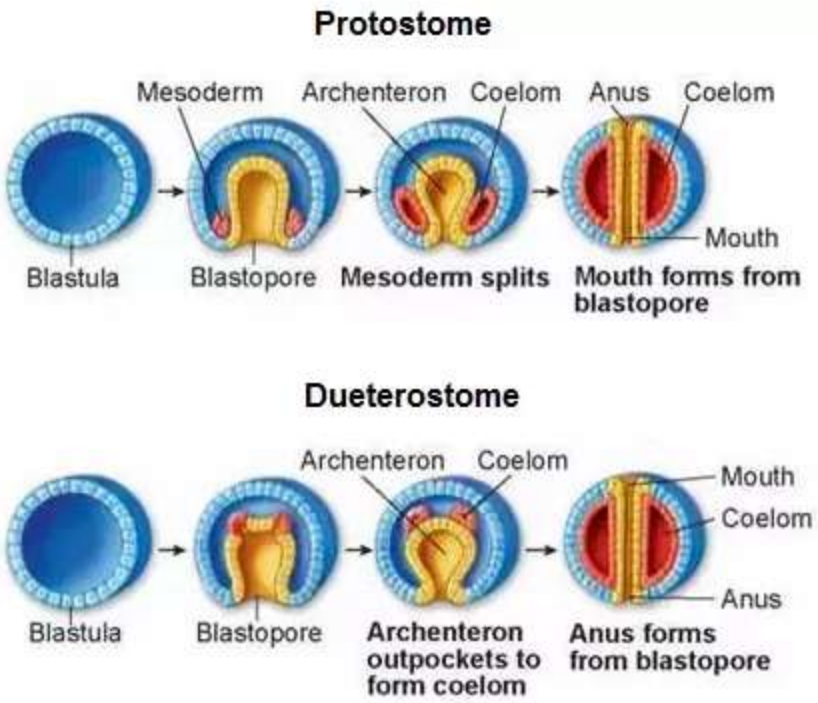
Coelom

- Body cavity between body wall and alimentary canal
- Schizocoel/enterocoel
- **Acoelomates:** Coelenterates, Platyhelminthes
- **Coelomates:** Mollusk, Annelids, Arthropod, Echinoderms, Vertebrates etc.
- **Pseudocoelomates:** Nematodes

Module# 197 Protostomes Vs Deuterostomes



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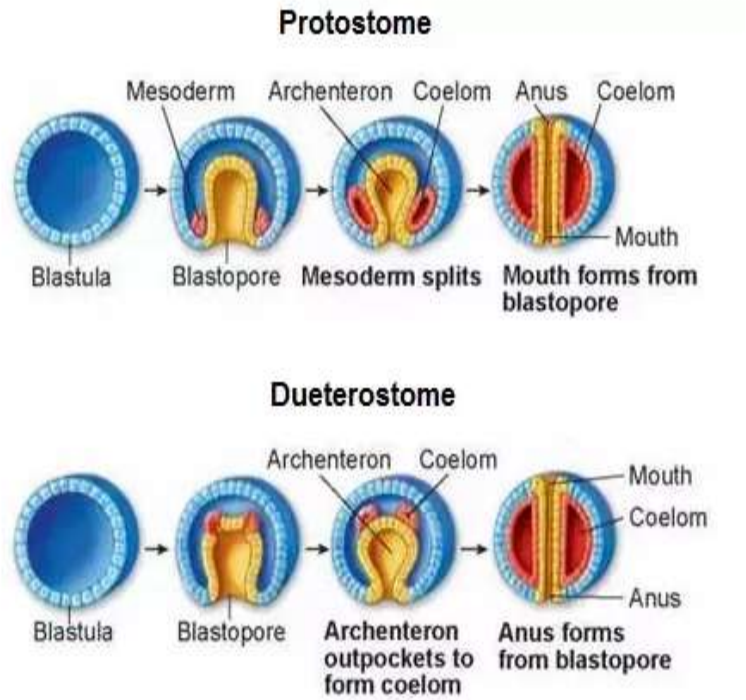
Protostomes

- Any member of the lower invertebrate phyla in which the mouth appears before the anus during development, cleavage is spiral and determinate and the coelom forms as a splitting of the mesoderm.
- Examples: Leeches, Insects, Spiders Crustaceans, Centipedes and millipedes, Earthworms Squid, Octopus, Snail and slugs, Bivalve

Protostomes Vs Deuterostomes



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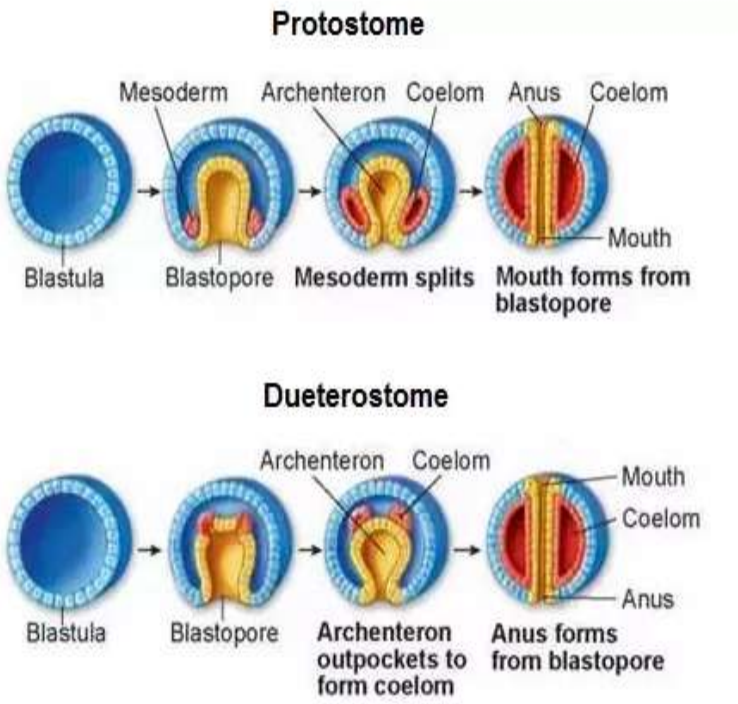
Deuterostomes

- In deuterostomes, the first cavity formed by the blastopore ends up as the organism's anus, while the mouth is formed secondarily on the opposite side.
- The animals whose embryonic development undergoes radial cleavage.

Protostomes Vs Deuterostomes



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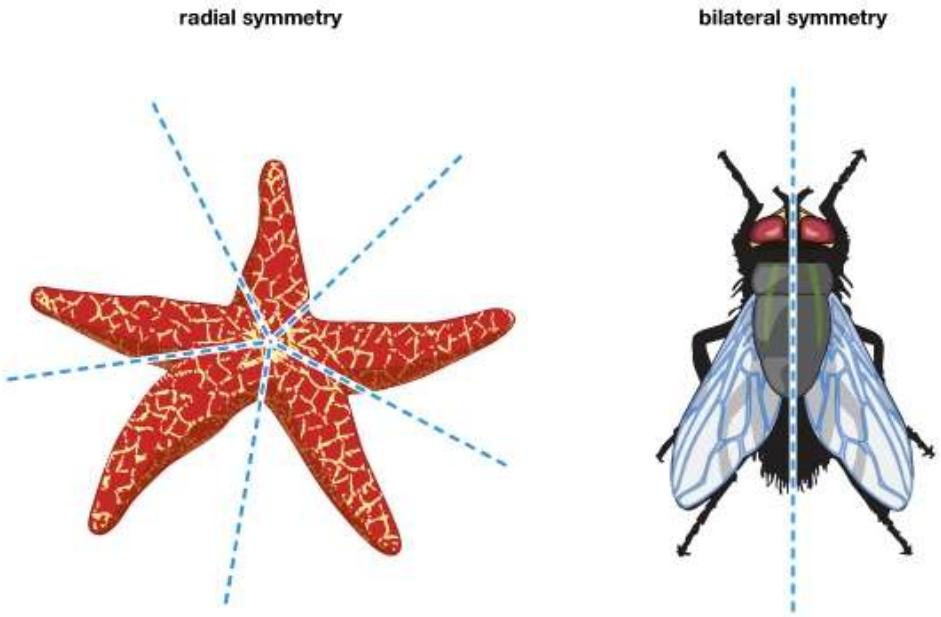
Examples of Deuterostomes

- Echinodermata
- Chordates include marine and terrestrial animals (mammals, reptiles, amphibians, frog, turtle, lizard, bat, tetrapods, Sloth, and jawless fish).
- Cephalochordata
- Vertebrata

Module# 198 Radial Symmetry Vs Bilateral Symmetry



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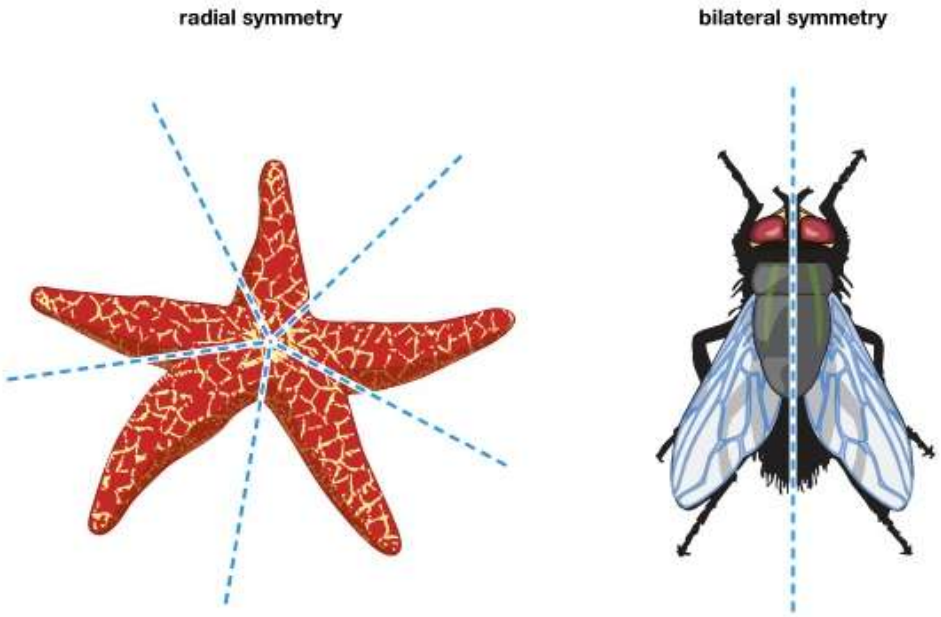
- The key difference between **radial and bilateral symmetry** is that **radial symmetry** generates identical body halves around the central axis whereas **bilateral symmetry** generates only two sides as left **and** right.

Radial Symmetry Vs Bilateral Symmetry



- Examples of radial symmetry includes starfish, hydra
- Examples of bilateral symmetry includes insects, vertebrates
- Activity
- Development

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Radial Symmetry Vs Bilateral Symmetry

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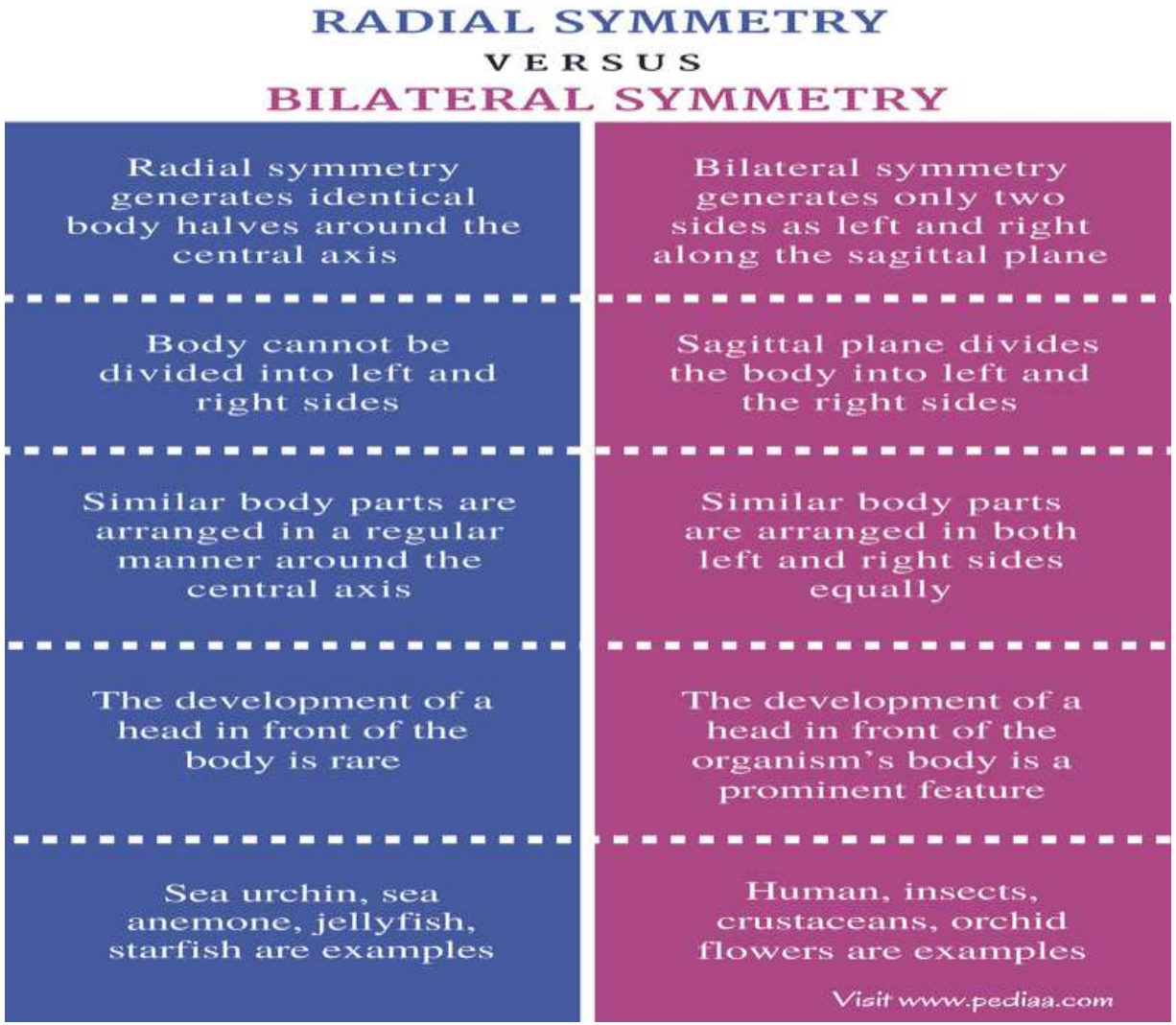


Module# 199 Chordates, Invertebrates Vs vertebrates

Chordates

- Five characteristic features of chordates present during some time of their life cycles are a **notochord**, a dorsal hollow tubular **nerve cord**, **pharyngeal slits**, endostyle/thyroid gland, and a **post-anal tail**.

Chordates, Invertebrates Vs vertebrates

Invertebrates

- Invertebrates are animals with no vertebral column or backbone.
- Invertebrates comprise approximately 97% of the animal kingdom.
- Most of them are soft-bodied and do not develop rigid internal skeletons.
- Many invertebrates do, however, possess hard exoskeletons to protect their bodies from their

Chordates, Invertebrates Vs vertebrates

Invertebrates

- Some common examples of invertebrates are snails, sponges, earthworms, squids, sea-stars, centipedes, butterflies, spiders, and jellyfish.

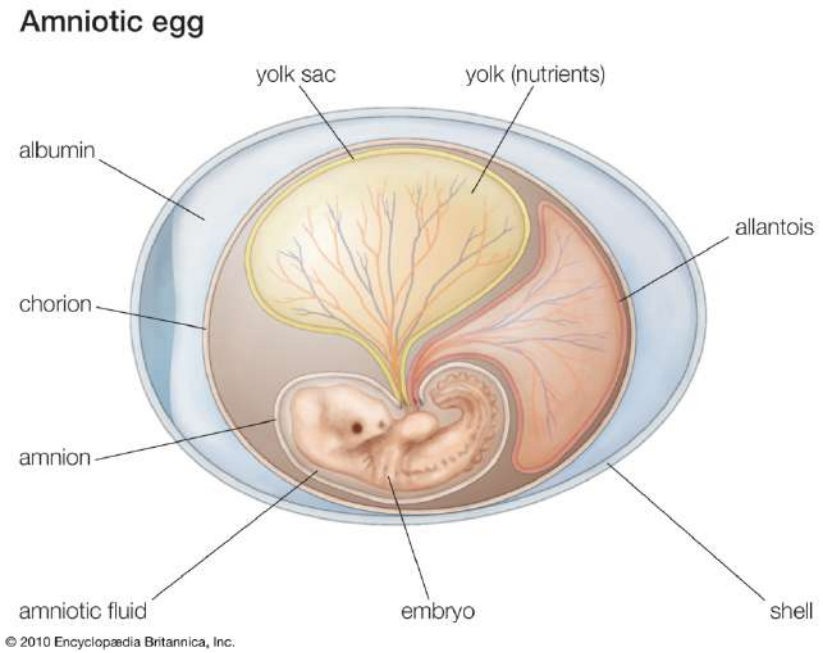
Chordates, Invertebrates Vs vertebrates

vertebrates

- Fish
- Amphibian
- Reptiles
- Birds
- Mammals

END

Module# 200 Amniotes Vs Anamniotes



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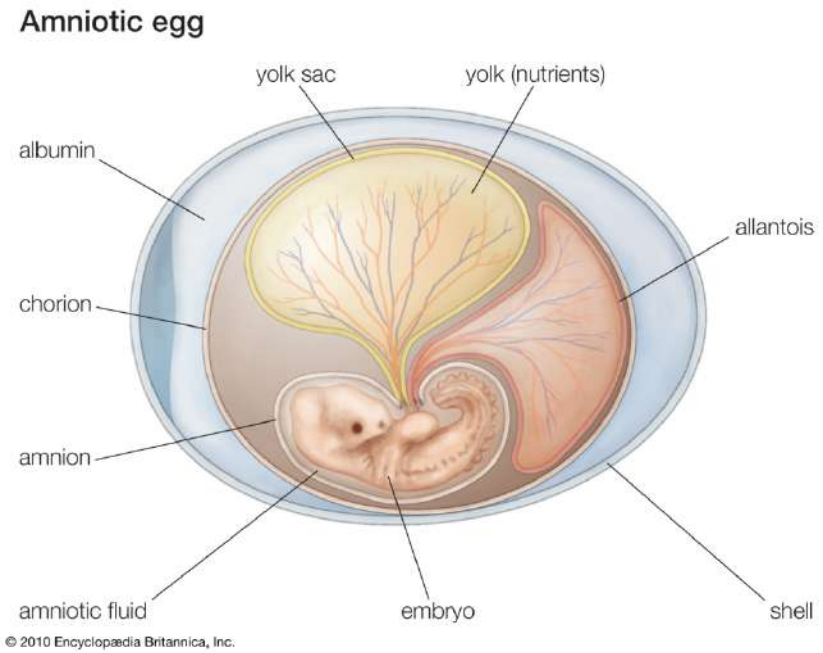
- **Amniotes** and **Anamniotes** are two vertebrates groups.
- **Amniotes** are higher vertebrates including reptiles, birds, and mammals
- **Anamniotes** are lower vertebrates including fishes and amphibians.

Amniotes Vs Anamniotes



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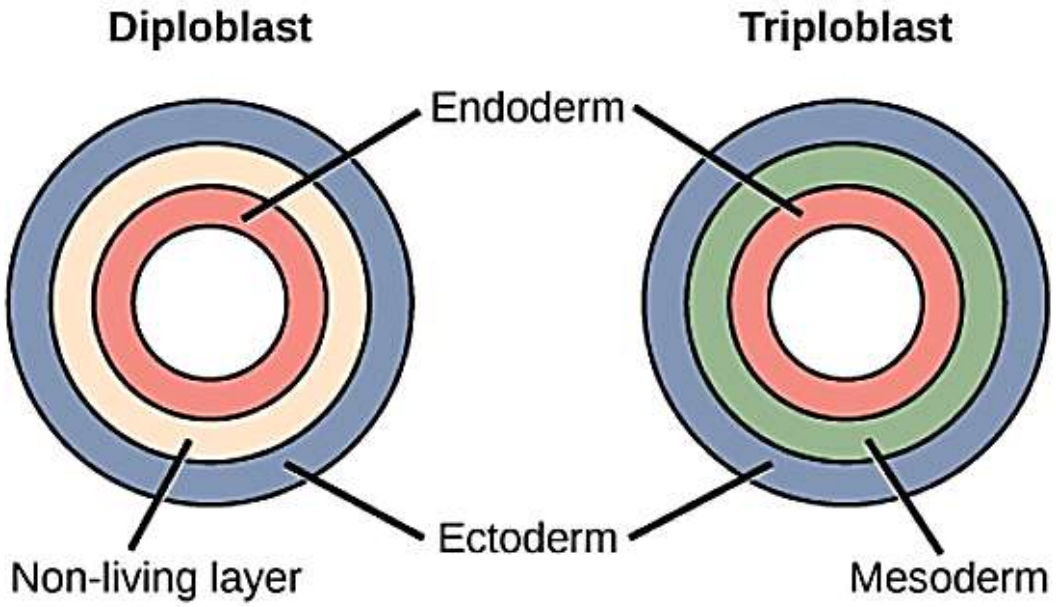
- The **anamniotes** are an informal group comprising the fishes and the amphibians, the so-called "lower vertebrates", which lay their eggs in water.
- They are distinguished from the **amniotes**, the "higher vertebrates" (reptiles, birds and mammals), which lay their eggs on land or retain the fertilized egg within the mother.

Module# 201 Diploblastic Vs Triploblastic



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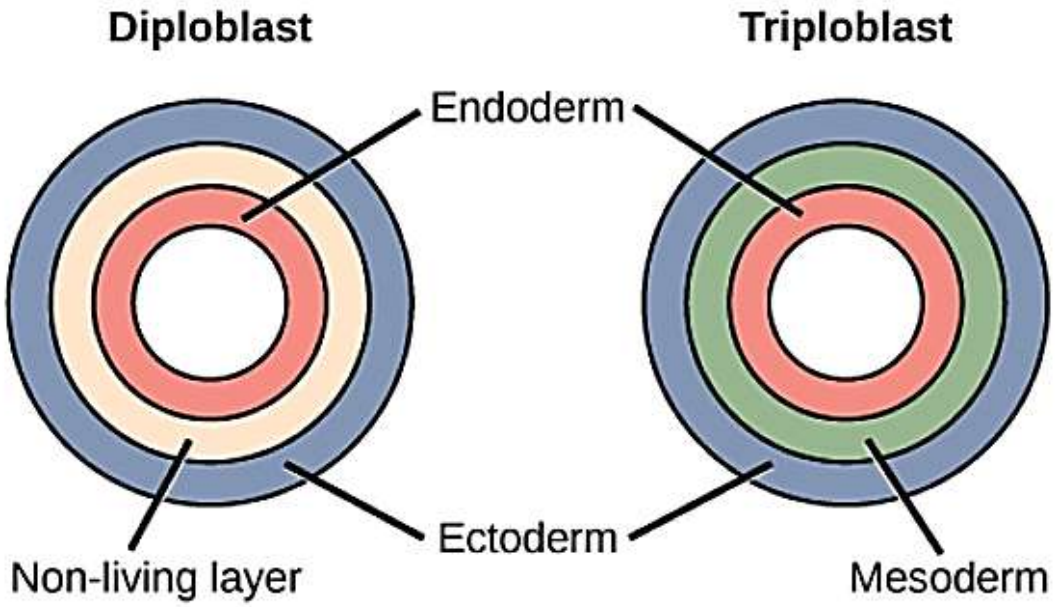
- Diploblastic organisms have only two primary germ layers in their blastula: endoderm and ectoderm.
- They lack the middle layer or the mesoderm.
- The inner layer, which is the endoderm, gives rise to tissues associated with gut and associated glands.
- In contrast, the outer layer, ectoderm, gives rise to covering tissues like epidermis.

Diploblastic Vs Triploblastic



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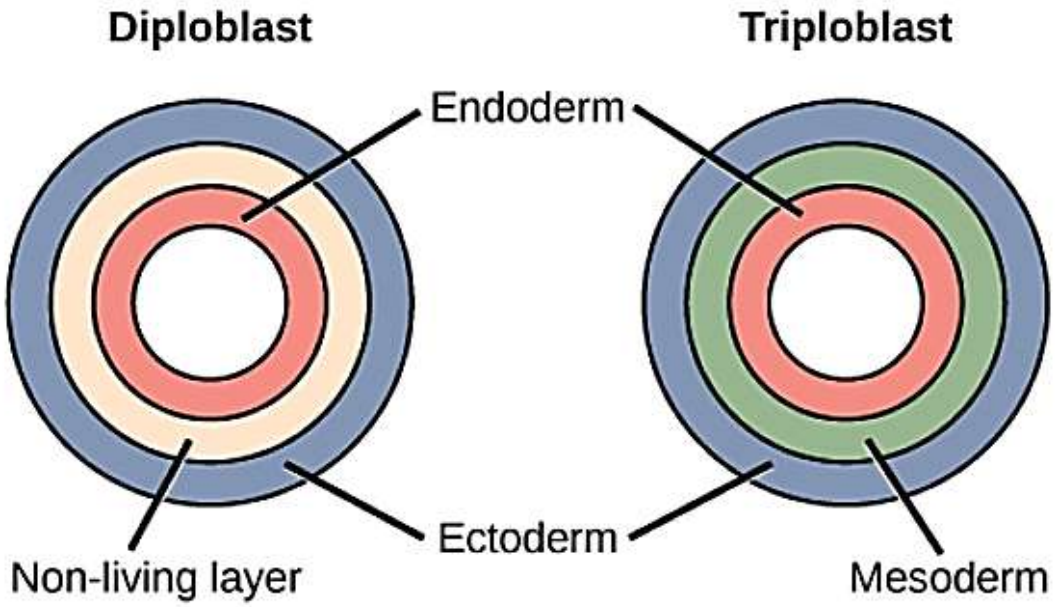
- Animals in phyla Cnidaria and Ctenophore belong to this group.
- Cnidarians include jellyfish, corals, sea pens, sea anemones, etc. and ctenophores include comb jellies.
- These simple, primitive metazoans basically lack body cavities and true organs.

Diploblastic Vs Triploblastic



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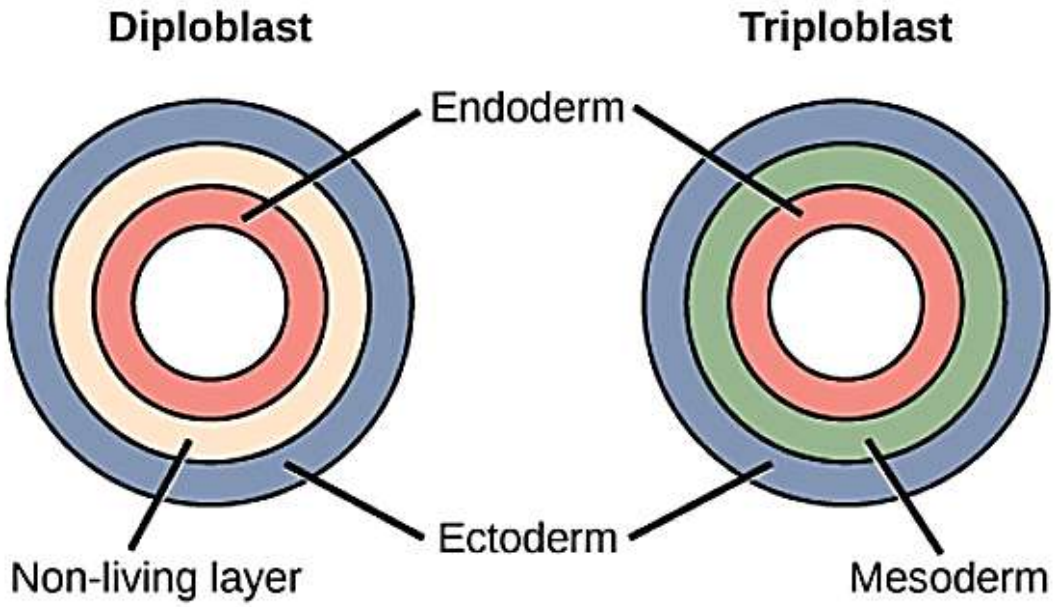
- Most metazoans develop three primary germ layers in their blastula; hence, they are referred to as triploblastic animals.
- The three primary germ layers are ectoderm, mesoderm, and endoderm.
- Ectoderm basically gives rise to the epidermis and may also give rise to sensory organs and parts of the nervous system.

Diploblastic Vs Triploblastic



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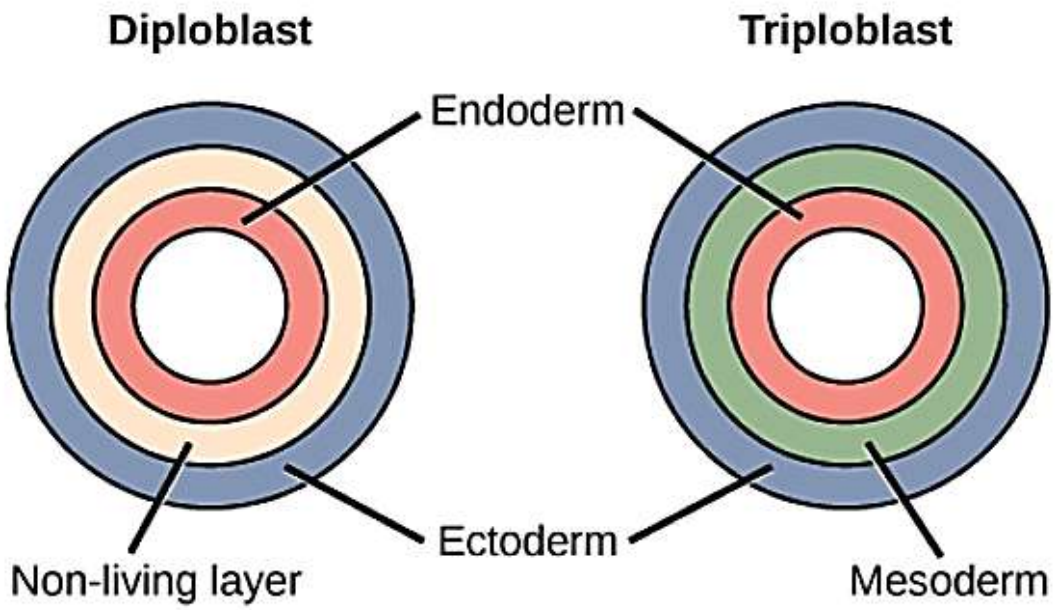
- Mesoderm mainly forms muscles, connective tissues, blood vessels, epithelial lining of interior cavities, certain excretory organs, and skeletal elements. Endoderm gives rise to parts of gastrointestinal tracts, respiratory tract, parts of endocrine glands and organs, and auditory system.

Diploblastic Vs Triploblastic



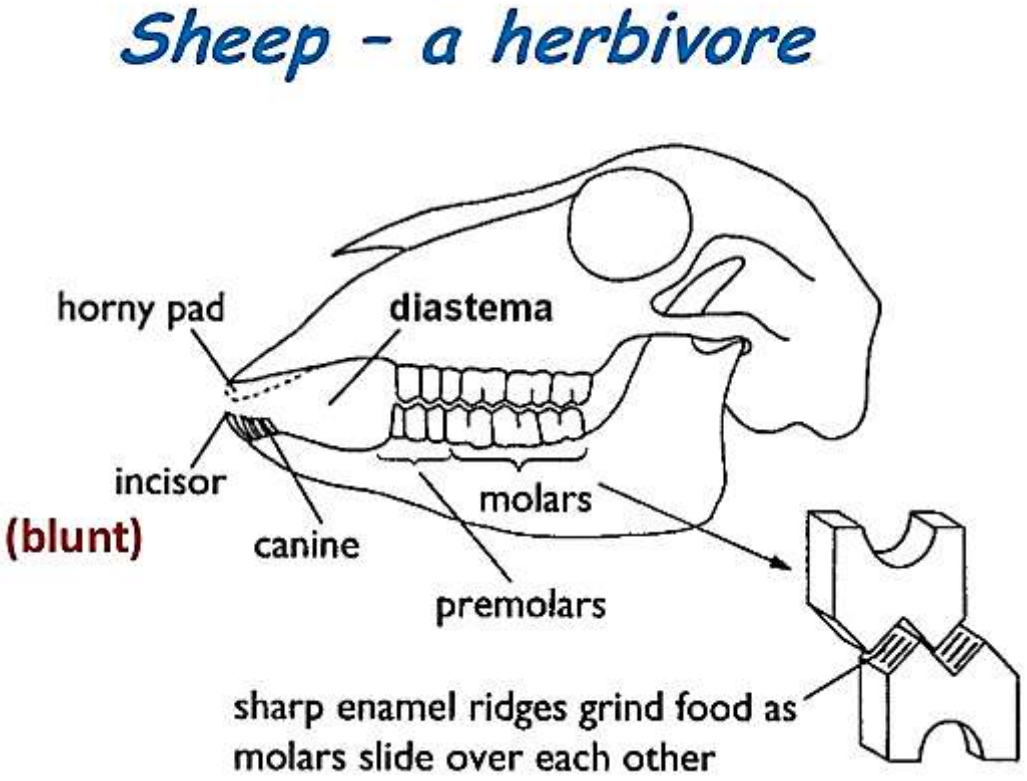
- Examples include arthropod, Echinoderms, Mollusks, vertebrate etc.

https://www.differencebetween.com/difference-between-diploblastic-and-vs-triploblastic/

Module# 202 Classification of Animals on the Basis of Nutrition

- Herbivores- **Animals** that only eat plant. ... Rabbits, Goat, sheep, grasshoppers etc
- Carnivores – **Animals** eat only other **animals**. ... dog, lion
- Omnivores- **Animals** that eat both plants and **animals**. ... Man, fox, Rats, crows
- Detrivores or Scavengers- **Animals** that eat the remains of died **animal**

Classification of Animals on the Basis of Nutrition



- Herbivores- Two important groups of herbivores are rodents (rat & squirrel) and ungulates.
- Rodents is the largest group of mammals.
- Ungulates are hoofed grassing animals such as horses, cattle and sheep.

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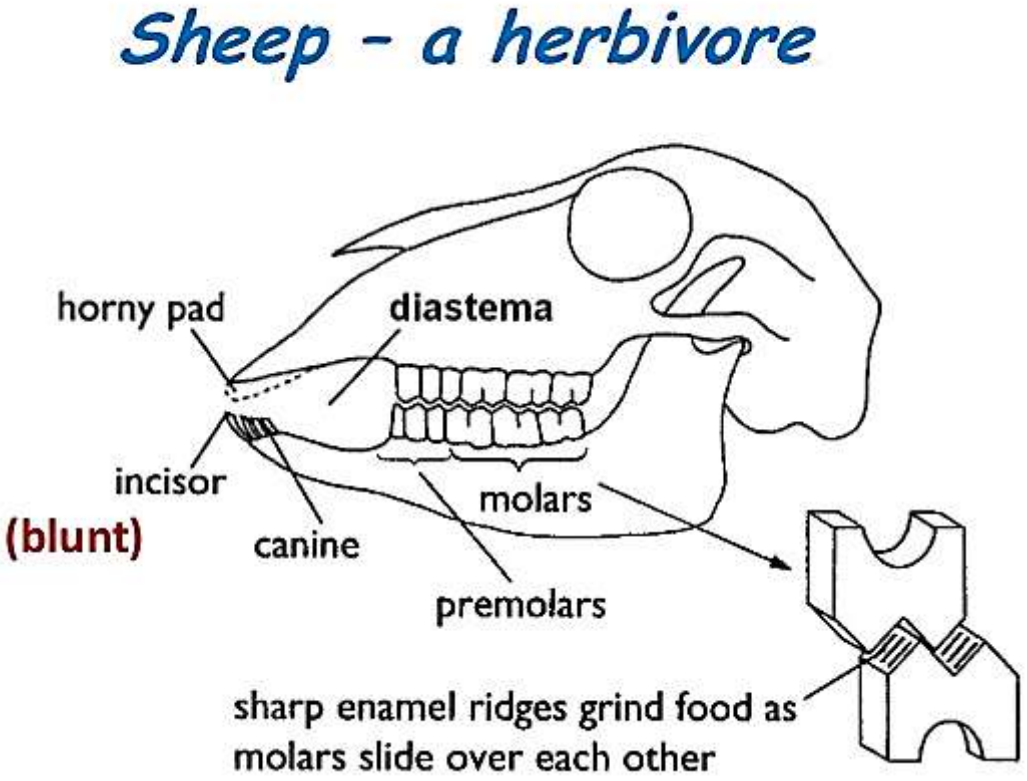
Classification of Animals on the Basis of Nutrition



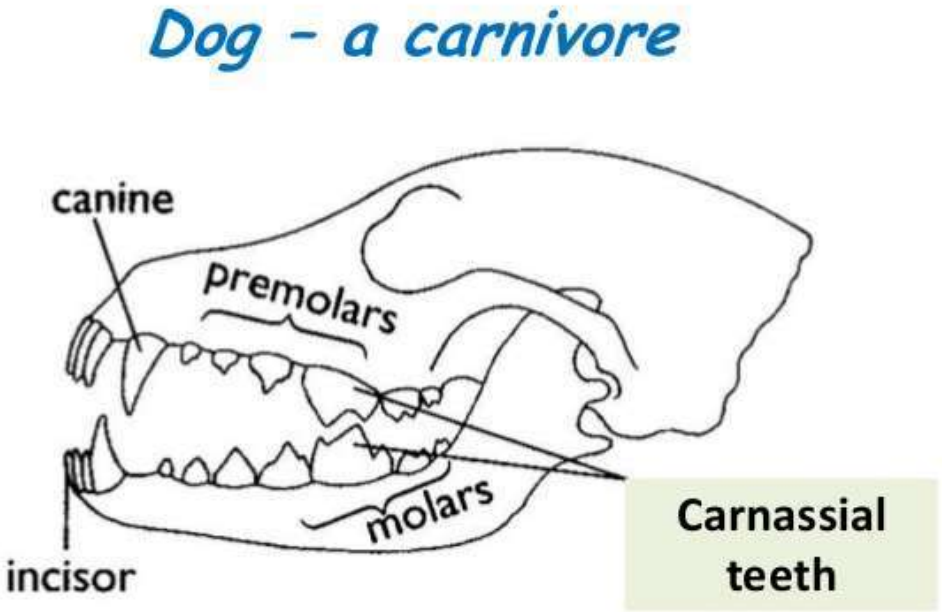
- Premolar and Molars have large grinding surfaces.
- Canines are missing
- In grassing and browsing herbivores i.e., dear and sheep, there are no upper incisors.

https://www.google.com/search?q=teeth+of+herbivores&sxsrf=ALeKk03sNAjhfjIvOBP9b61FVr4j_oOP2w:1612095117348&source=lnms&tbm=isch&sa=X&ved=2ahUKEwjr48CmksbuAhUFnVwKHa6BBLAQ_AUoAXoECBgQAw&biw=1360&bih=600#imgrc=o0oxWMDDU_sAKM

Module# 203 Principles of Systematics

Classification of Animals on the Basis of Nutrition

Classification of Animals on the Basis of Nutrition



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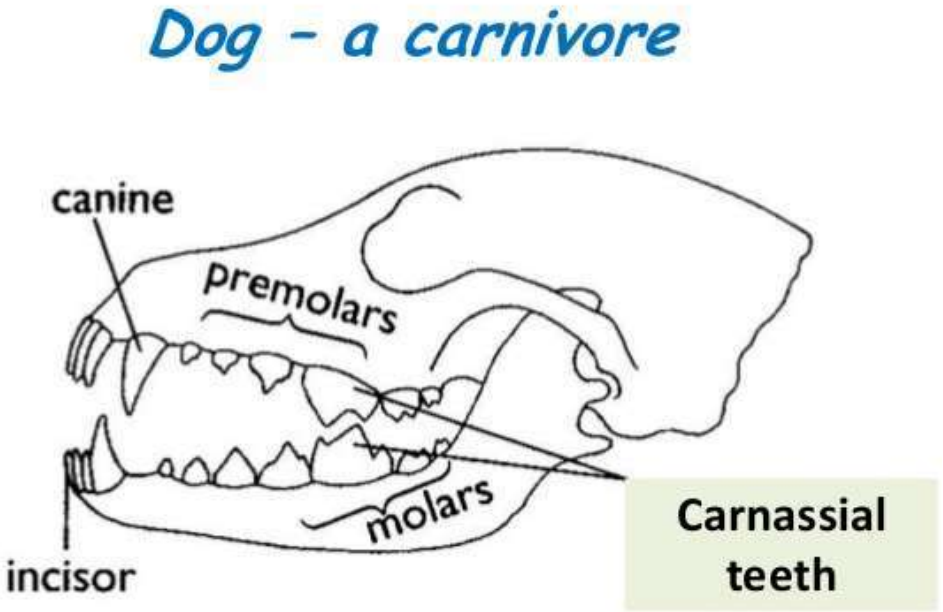
Carnivores

- The animals which feed on other animals are called carnivores.
- Have large canine teeth for catching and tearing the prey.
- Incisors, premolars and molars are all adapted for cutting flesh, cracking bones and reducing the chunks to sizes suitable for swallowing.

Classification of Animals on the Basis of Nutrition



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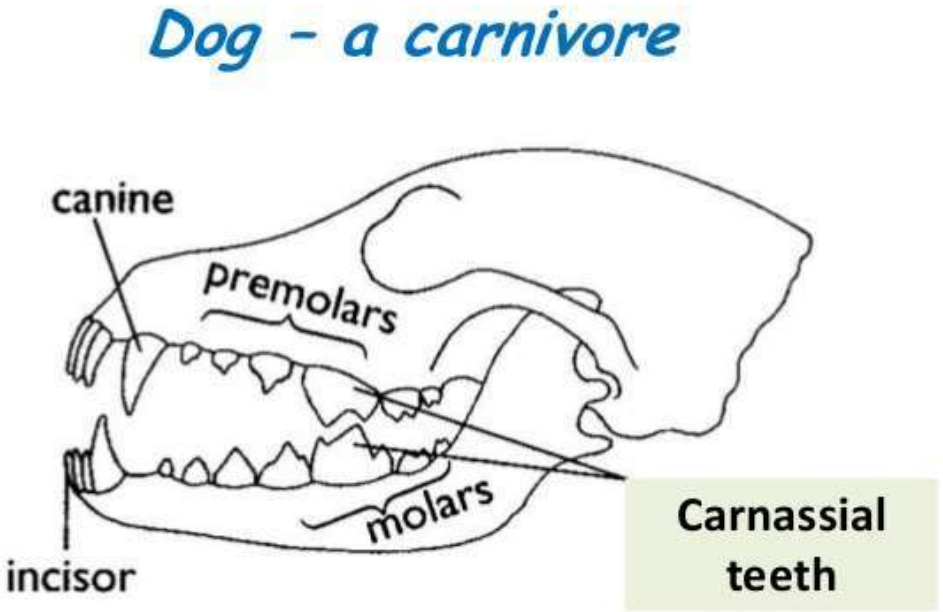
Predator

- Predator is an animal, which captures and readily kills live animals for its food.
- The animal which is eaten by predator is called prey.
- Prey-predator interaction helps in marinating ecosystem stable.
- A species in the area without its natural predator leads to disastrous results.

Classification of Animals on the Basis of Nutrition



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Omnivores

- Animals which eat both plants and animals.
- They are teethes structurally and functionally intermediate between the extremes of specialization attained by the teethes of herbivores and carnivores.
- Man, Cow,

Classification of Animals on the Basis of Nutrition

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Detrivores

- **Detritivores** (also known as detrivores, detritus feeders, or detritus eaters) are heterotrophs that obtain nutrients by consuming detritus (decomposing plant and animal parts as well as faeces).
- There are many kinds of invertebrates, vertebrates and plants that carry out coprophagy.

Classification of Animals on the Basis of Nutrition

- Detritivores are often invertebrate insects such as mites, beetles, butterflies and flies; mollusks such as slugs and snails; or soil-dwelling earthworms, millipedes and woodlice.

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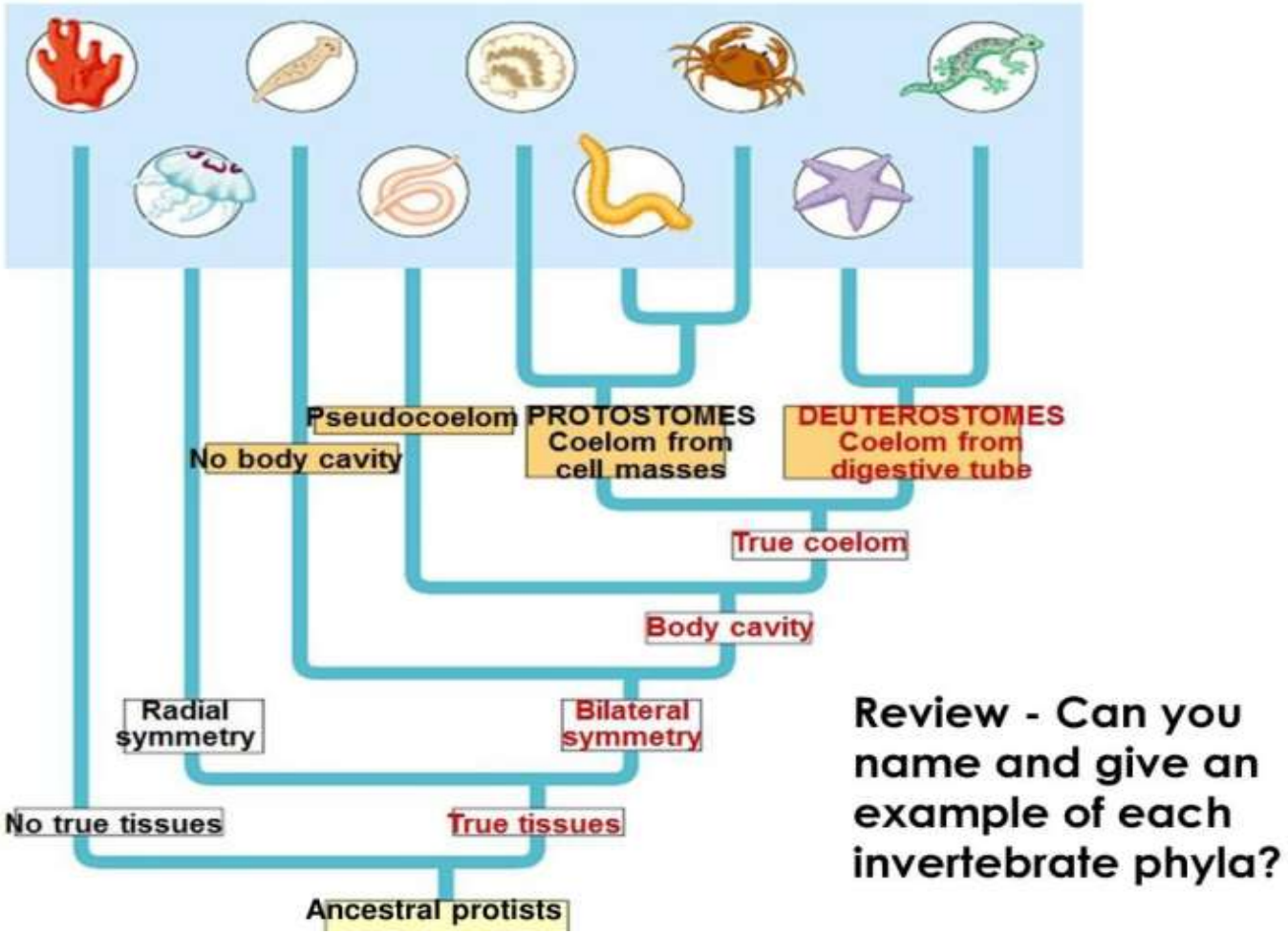
Module# 204 Invertebrates



Review - Can you name and give an example of each invertebrate phyla?

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Invertebrates

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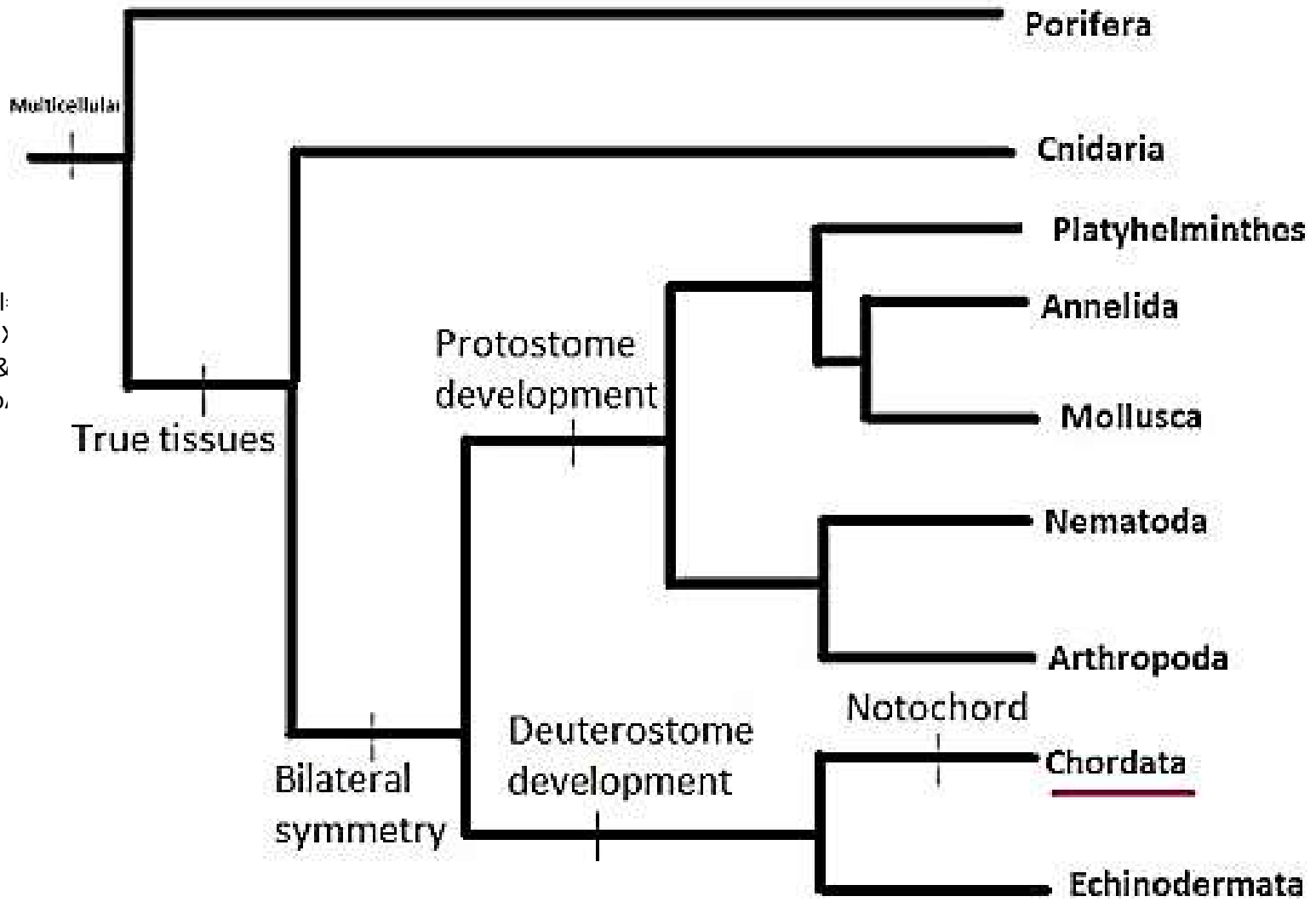

Module# 205 Phylum Porifera

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- Poriferans are commonly referred to as sponges. An early branching event in the history of animals separated the sponges from other metazoans.
- Fossil sponges are among the oldest known animal fossils, dating from the Late Precambrian.

Phylum Porifera

https://www.google.com/search?q=sponges&sxsrf=ALeKk016Jl1xaU5NvGLq9zHkKMni1CPzuA:1612107130711&source=lnms&tbm=isch&sa=X&ved=2ahUKEwitl_aGv8buAhUElFwKHXaaAMMQ_AUoAXoECBcQAw&biw=1360&bih=600

- The approximately 5,000 living sponge species are classified in the phylum Porifera, which is composed of three distinct groups, the Hexactinellida (glass sponges), the Demospongia, and the Calcariea (calcareous sponges).

Phylum Porifera

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- Sponges are characterized by the possession of a feeding system unique among animals. Poriferans don't have mouths; instead, they have tiny pores in their outer walls through which water is drawn.
- Cells in the sponge walls filter food from the water as the water is pumped through the body and out other larger openings.

Phylum Porifera

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- The cells of Poriferans are loosely organized.
- They are mostly found in marine water.
- Only a few are found in freshwater.
- They are either radially symmetrical or asymmetrical.
- Their body is usually cylindrical.
- The sclerob last secretes spicules while spongin fibres are secreted by spongioblasts.

Phylum Porifera

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- They have no organs in their body.
- They depict cellular grade of organization.
- The body comprises numerous pores known as Ostia and osculum.
- The central cavity is called spongocoel or atrium which opens to the outside through the osculum.

Phylum Porifera

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- They reproduce asexually by budding and fragmentation.
- The nutrition is holozoic.
- They have the power to regenerate the lost parts.
- The exchange of respiratory gases and nitrogenous wastes occurs by the process of diffusion.

Module# 206 Phylum Coelentrata

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- Coelentrata also known as Cnidaria.
- This name of cnidaria is due to single cavity Coelenteron.
- Cnidarian have tissue but don't have organ or organ system. That's why they posses tissue level of organization.
- They are Radially symmetry and Diploblastic.
- Radial Symmetry
- Diploblastic
- No separate body cavity.

Phylum Coelentrata

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- They have incomplete digestive system. It has a single aperture mouth but don't have anus.
- They respire through body surface by diffusion.
- They excrete through body surface by diffusion.
- Segmentations are absent in them.
- Tentacles are present in them which act as appendages.
- They are mostly marine, aquatic animals and free-living animals.

Phylum Coelentrata

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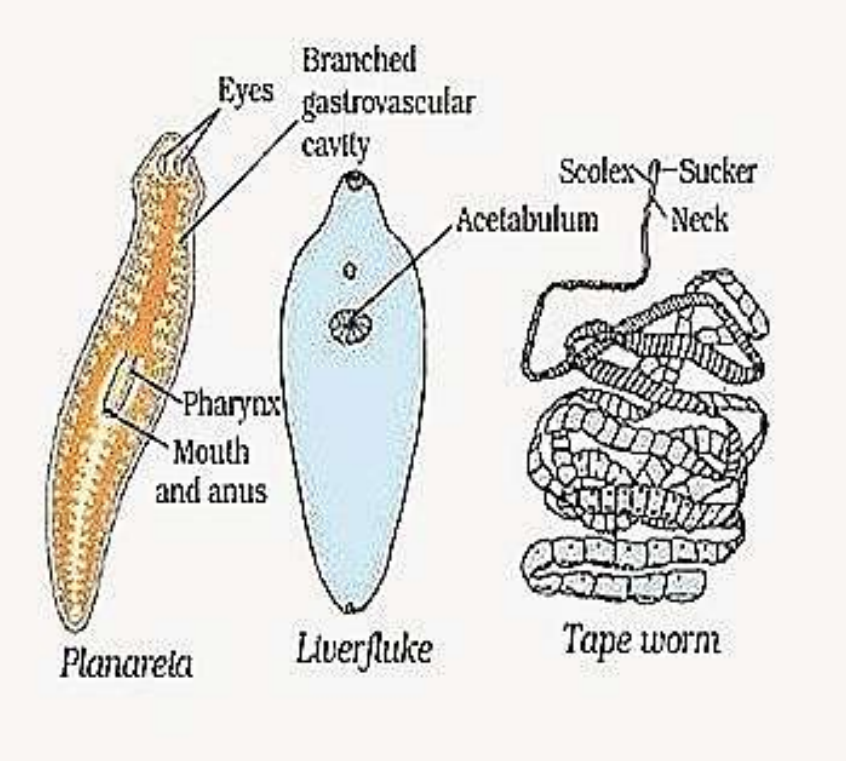
- They reproduce both sexually and asexually.
- Ex :- Sea-pan, Sea-fan, Brain coral etc.

Module# 207 Phylum Platyhelminthes



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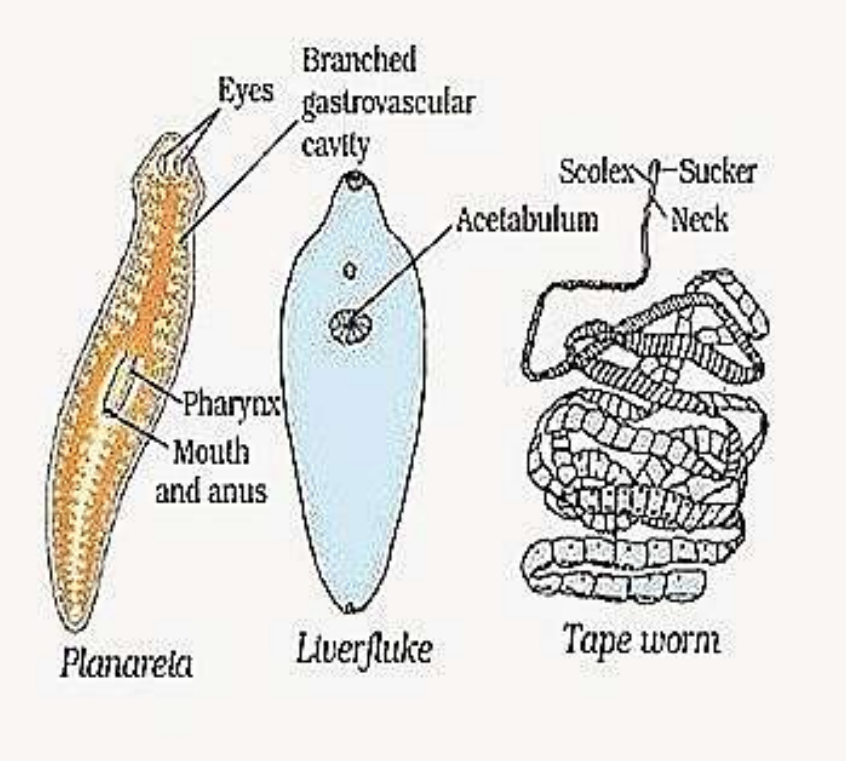
- Platys: flat; helminthes: worms)
- Kingdom: Animalia
- Habitat: mostly parasitic, few are free living in sea water or fresh water
- Grade of organization: Organ-system grade of organization
- Symmetry: Bilateral symmetry and Body is dorso-ventrally flattened

Phylum Platyhelminthes



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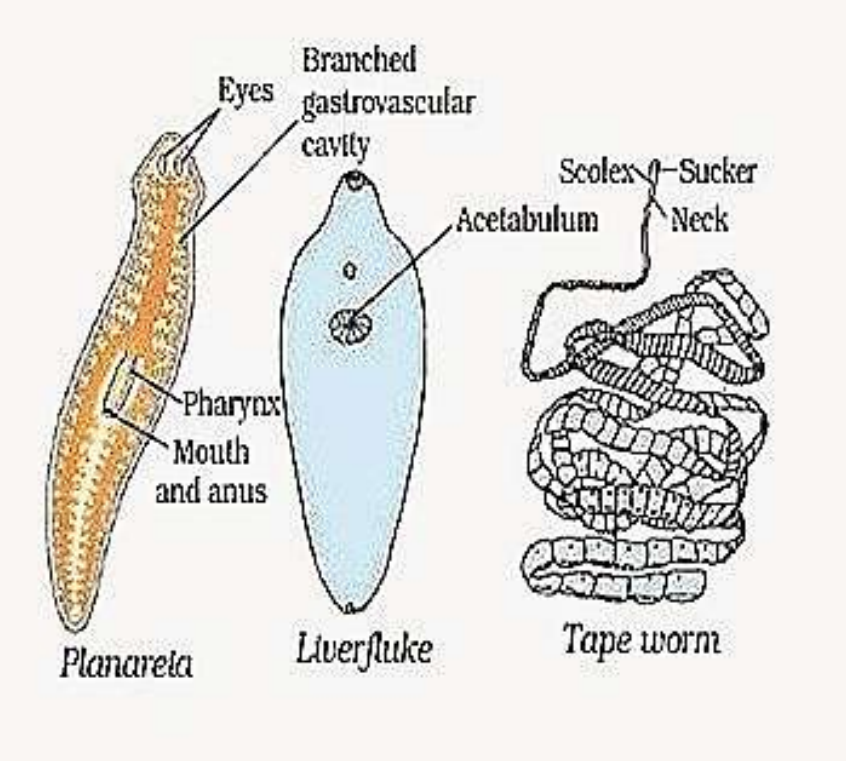
- Germ layer: Triploblastic.
- Ectoderm is very specialized with distinct epidermis
- Coelom: Absent (acoelomate).
- Digestive system: incomplete or absent
- No mouth (oral opening directly into pharynx).
- No true stomach structure (pharynx opens into a complex “intestinal” structure).

Phylum Platyhelminthes



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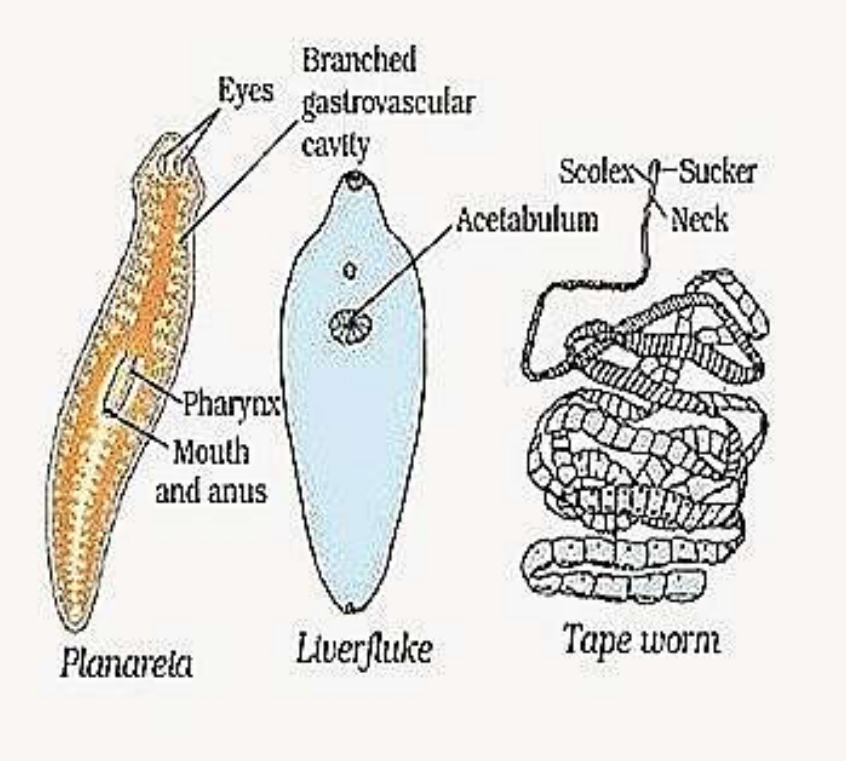
- In tapeworms, Direct absorption of soluble nutrients by cells and tissues.
- Excretory system: Protonephridia with Flame cell
- Respiration: by simple diffusion of gases through body surface
- Circulatory system: Absent

Phylum Platyhelminthes



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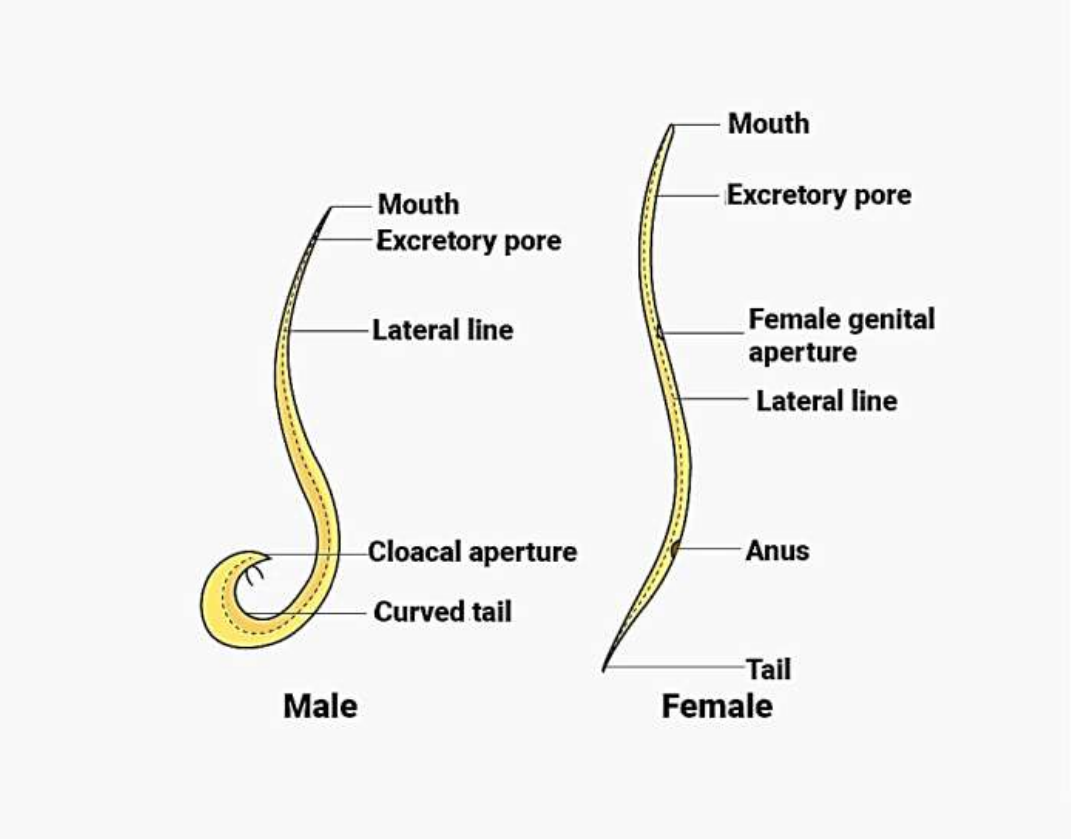
- Reproduction
- Sexual: by gamatic fusion in hermaphrodite species
- Asexual : by regeneration and fission
- Fertilization: internal.
- Life cycle is complex involving one or more hosts.
- Examples: Tapeworms, Planaria

Module# 208 Phylum Nematodes



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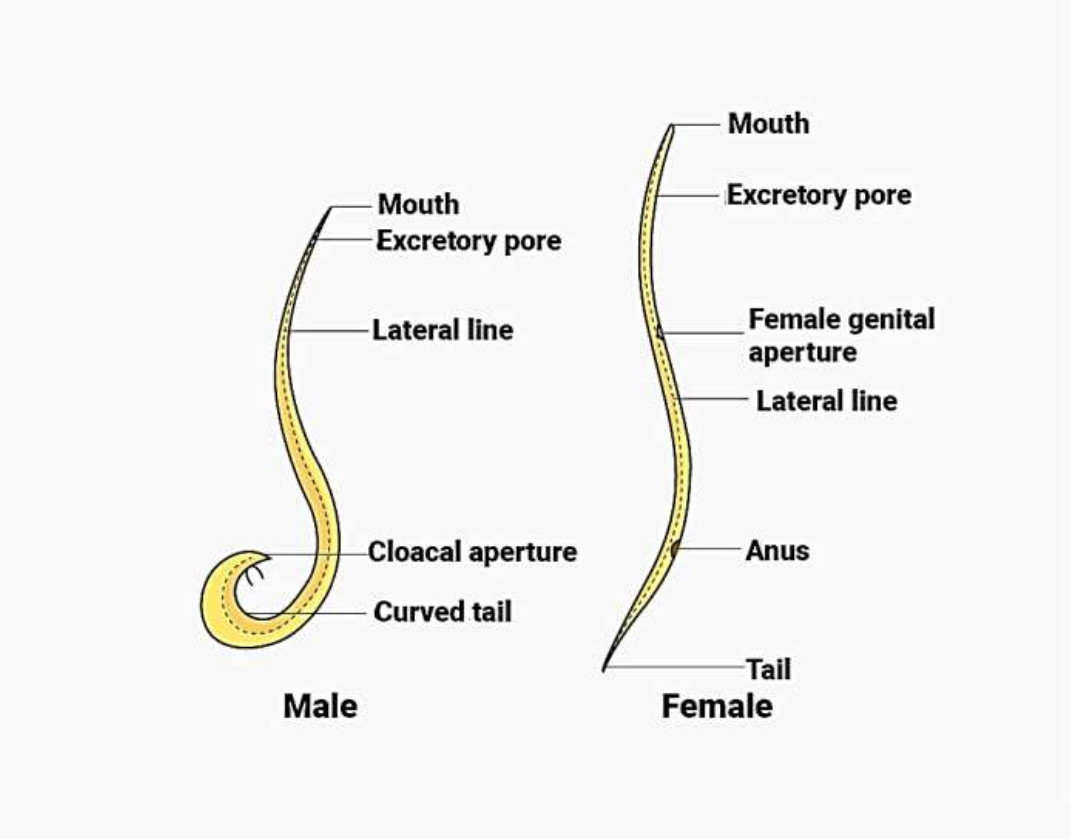
- Their body is bilaterally symmetrical and triploblastic.
- They are cylindrical in shape.
- They exhibit tissue level organization.
- Their body has a cavity or pseudocoelom.
- The alimentary canal is distinct, with the mouth and the anus.
- They are sexually dimorphic.

Phylum Nematodes



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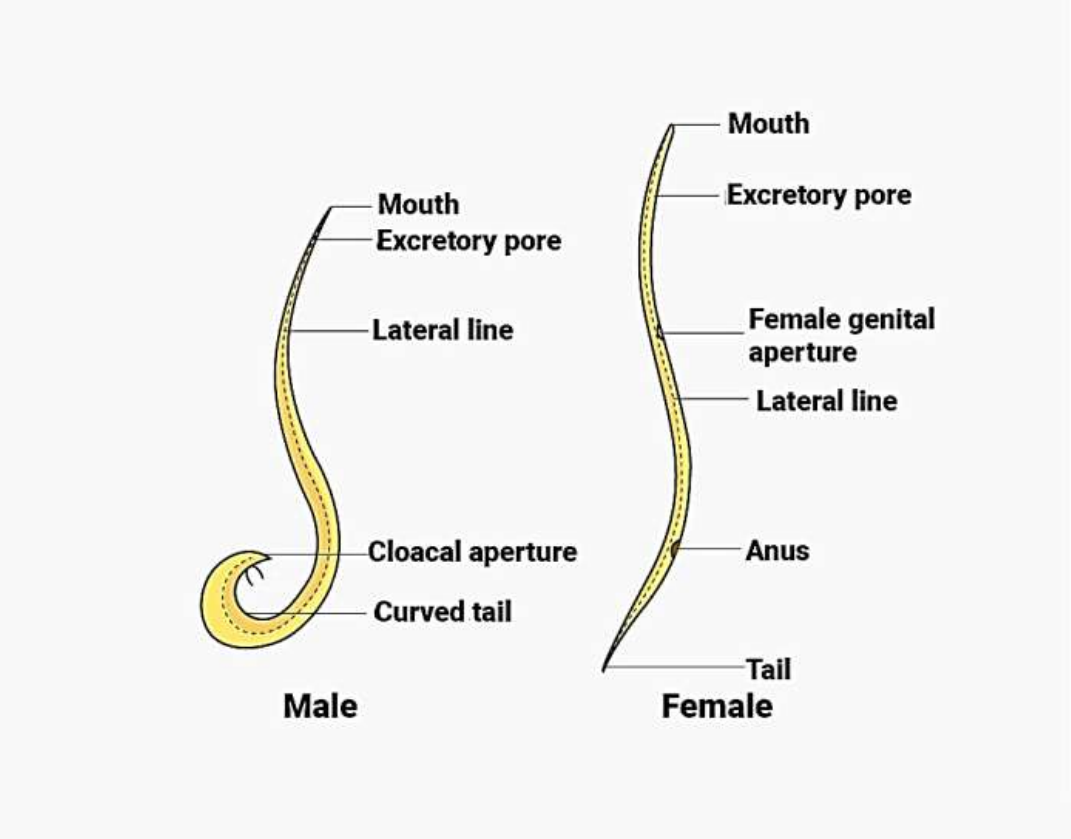
- They are devoid of the circulatory system and respiratory system.
- They are free-living or parasitic.
- Parasitic nematodes cause diseases in the host.
- Fertilization is internal and reproduction is sexual.
- Their cuticle moults periodically.
- The epidermis is synctical and contains dorsal or ventral nerve cords.

Phylum Nematodes



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- The body-wall muscles are longitudinal.
- They possess amoeboid sperm cells.
- They consist of chemosensory organs called aphids situated on the lips.

Module# 209 Phylum Annelida

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- Their body is bilaterally symmetrical
- They have a long and segmented body.
- **Annelids** are bilaterally symmetrical.
- They are triploblastic.
- Also, they exhibit organ system grade of organisation, showing organ differentiation

Phylum Annelida

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- The body is covered with a thin cuticle.
- They are coelomate.
- Annelids live in moist environments, moist soil, freshwater and marine water.
- They have parapodia and chitinous setae, used for locomotion.
- Their body appears red due to the presence of haemoglobin.

Phylum Annelida

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- Excretory and nervous systems are present.
- The digestive system is complete and developed.
- Respiration happens through the general body surface.
- These invertebrates have a true closed circulatory system.
- Sexes may be separate or united, wherein they are called hermaphrodites.

Phylum Annelida

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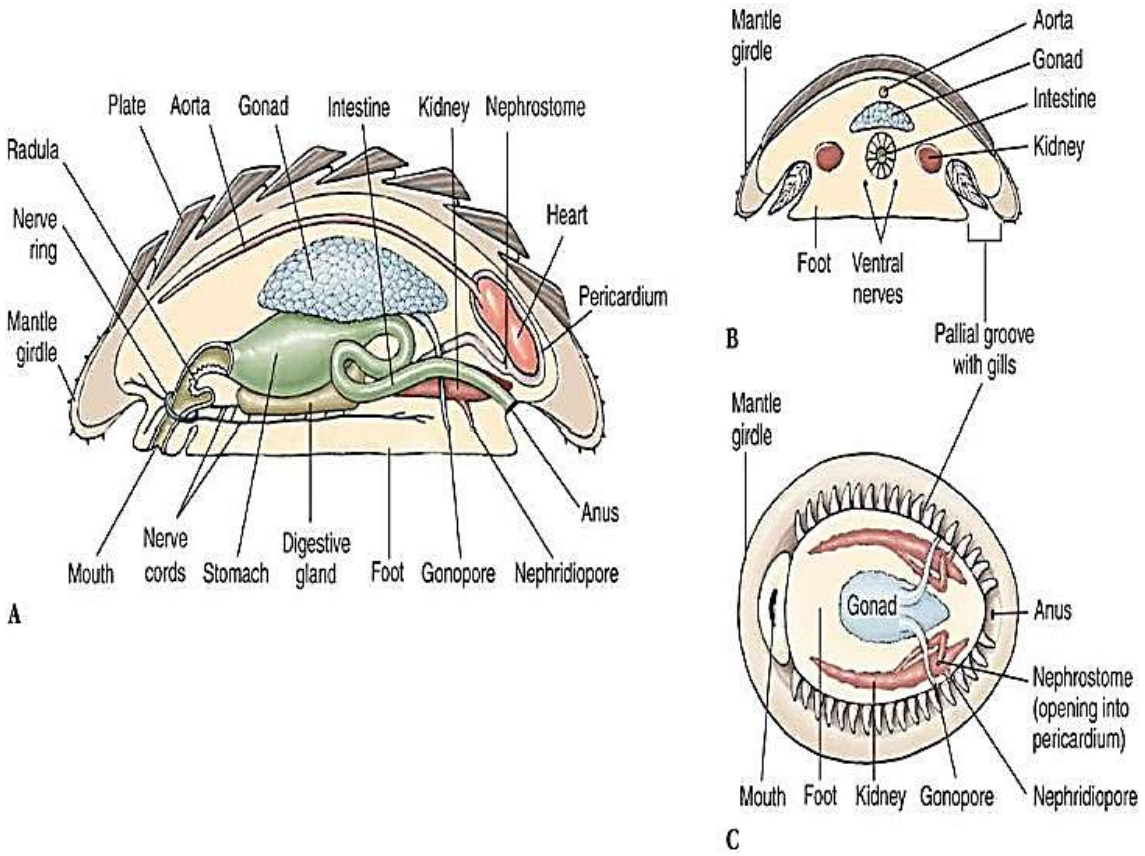
- Fertilization can be internal or external.
- Examples of Annelids
- Earthworm
- Leeches
- Polychaetes

Module# 210 Phylum Mollusca



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Phylum Mollusca

- They are mostly marine.
- They have segmented body with distinct head, muscular foot and visceral hump.
- They show bilateral symmetry.
- The shell may be external, internal or absent. It is secreted by mantle and made up of calcium carbonate.

Phylum Mollusca

Phylum Mollusca

- Coelom is greatly reduced. It is restricted to pericardial cavity and to small spaces within kidneys and gonads.
- They have complete digestive tract.
- They have open blood vascular system.
- Blood is blue due to copper containing blue respiratory pigment called haemocyanin.

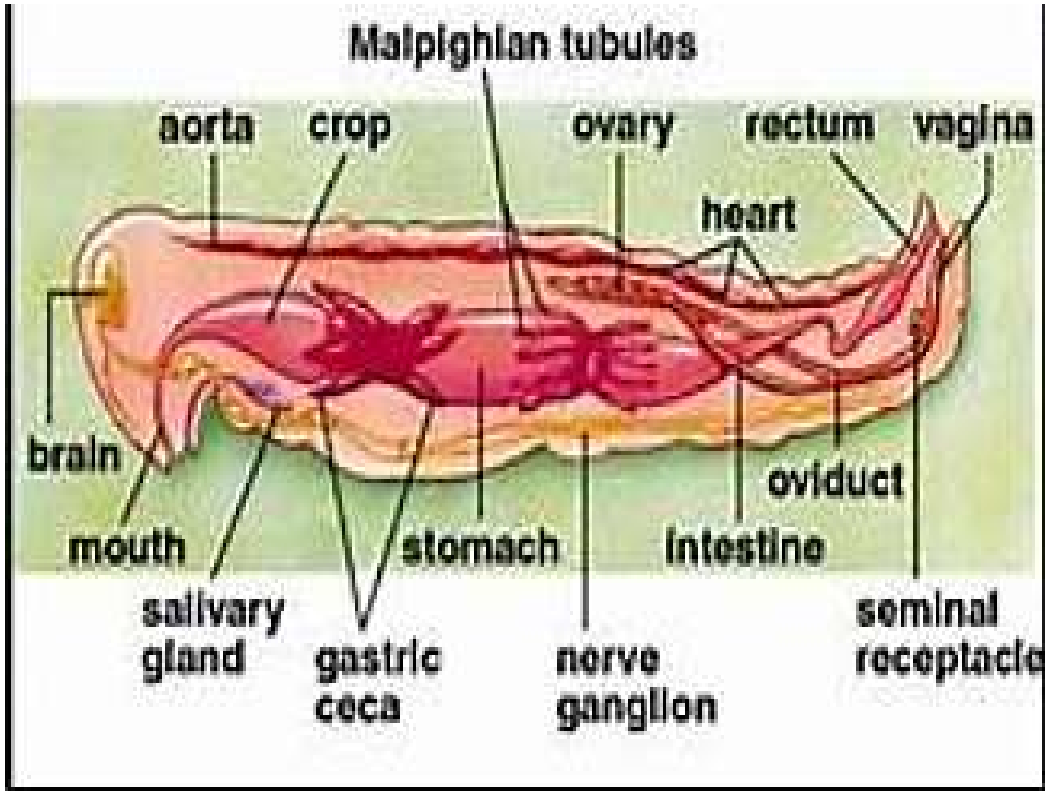
Phylum Mollusca

Phylum Mollusca

- Respirator organs are gills.
- Excretory organs are one or two pairs of sac-like kidneys.
- Osphradium is present in some molluscs for testing physical and chemical nature of water.
- The sexes are separated but some are hermaphrodite.
- Nervous system consists of cerebral, visceral, pleural and pedal ganglia.

END

Module# 211 Phylum Arthropoda



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Phylum Arthropoda

- Arthropods occur in all types of habitats i.e., marine, fresh water and terrestrial.
- There is an increasing tendency of cephalisation in arthropods.
- Their body is metamerically segment; composed of a linear series of segments or somites.

Phylum Arthropoda

Phylum Arthropoda

- Externally endoskeleton is divided into hardened segmental scletites. Internally the nervous system, muscular system and heart chambers are segmentally repeated.
- Metamerism is heteronomous.
- Appendages are joined. They act as lever systems providing mechanical advantages for locomotion.

Phylum Arthropoda

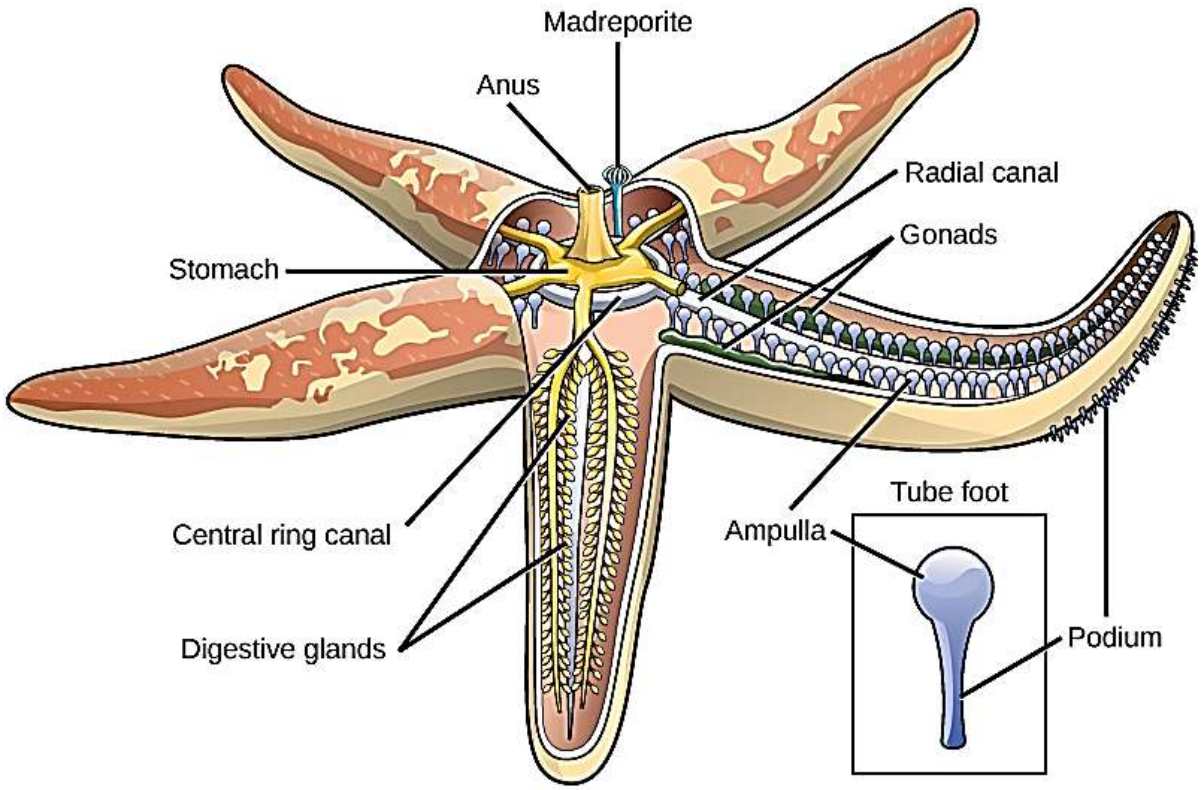
- Muscles of arthropods are striated
- Nervous system consists of nerve ring and a ganglionated double ventral nerve cord.
- Sense organs include compound eye, simple eyes, antennae, statocyst's etc.

Phylum Arthropoda

- Most arthropods are unisexual. Fertilization may be external or internal in aquatic arthropods but internal in all the terrestrial forms.
- Eggs are centrolecithal (yolk in center).
- Cleavage is meroblastic and superficial.

END

Module# 212 Phylum Echinodermata



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Phylum Echinodermata

- They are exclusively marine and are among the most common and widely distributed marine animals.
- Symmetry usually radial, nearly always pentamerous.
- Body shape globular, star-like, spherical, discoidal, or elongated.
- Water -vascular system of coelomic origin, including podia or tube feet for locomotion and usually with a madreporite.

Phylum Echinodermata

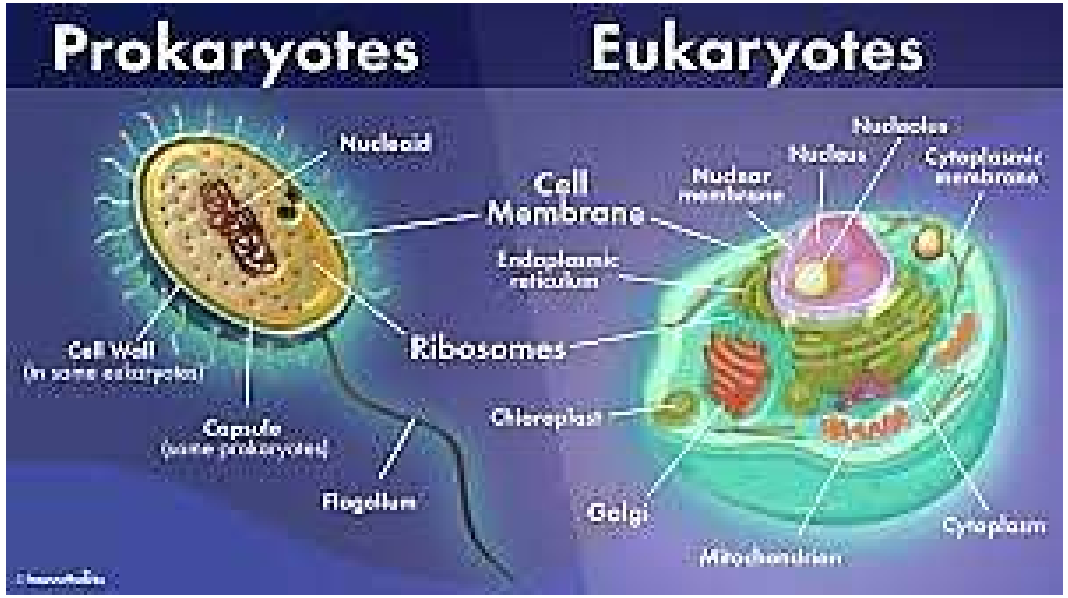
- Sexes are usually dioecious with few exceptions.
- Gonads large and single or multiple; fertilization external, while few echinoderms are viviparous.
- Regeneration of lost parts, a peculiarity.

Phylum Echinodermata

- A nervous system without a brain and with a circumoral ring and radial nerve.
- The excretory system is wanting.
- Respiratory organs include branchiae, tube-feet, respiratory tree, and bursae.
- Development is intermediate including characteristic larvae which undergo metamorphosis into the radially symmetrical adults.

END

Module# 213 Prokaryotes Vs Eukaryotes



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- **Prokaryotes** are organisms made up of cells that lack a cell nucleus or any membrane-encased organelles.
- **Eukaryotes** are organisms made up of cells that possess a membrane-bound nucleus that holds genetic material as well as membrane-bound organelles.

BAQIR HUSSAIN
BS ZOOLOGY
03336361244

Prokaryotes Vs Eukaryotes

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	Prokaryotes	Eukaryotes
DNA	DNA is naked	DNA bound to protein
	DNA is circular	DNA is linear
	Usually no introns	Usually has introns
Organelles	No nucleus	Has a nucleus
	No membrane-bound	Membrane-bound
	70S ribosomes	80S ribosomes
Reproduction	Binary fission	Mitosis and meiosis
	Single chromosome (haploid)	Chromosomes paired (diploid or more)
Average Size	Smaller (~1–5 µm)	Larger (~10–100 µm)

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BS ZOOLOGY
03336361244

Prokaryotes Vs Eukaryotes

https://www.technologynetworks.com/cell-science/articles/prokaryotes-vs-eukaryotes-what-are-the-key-differences-336095

Prokaryote vs eukaryote: key differences

	Prokaryote	Eukaryote
Nucleus	Absent	Present
Membrane-bound organelles	Absent	Present
Cell structure	Unicellular	Mostly multicellular; some unicellular
Cell size	Smaller (0.1-5 µm)	Larger (10-100 µm)
Complexity	Simpler	More complex
DNA Form	Circular	Linear
Examples	Bacteria, archaea	Animals, plants, fungi, protists

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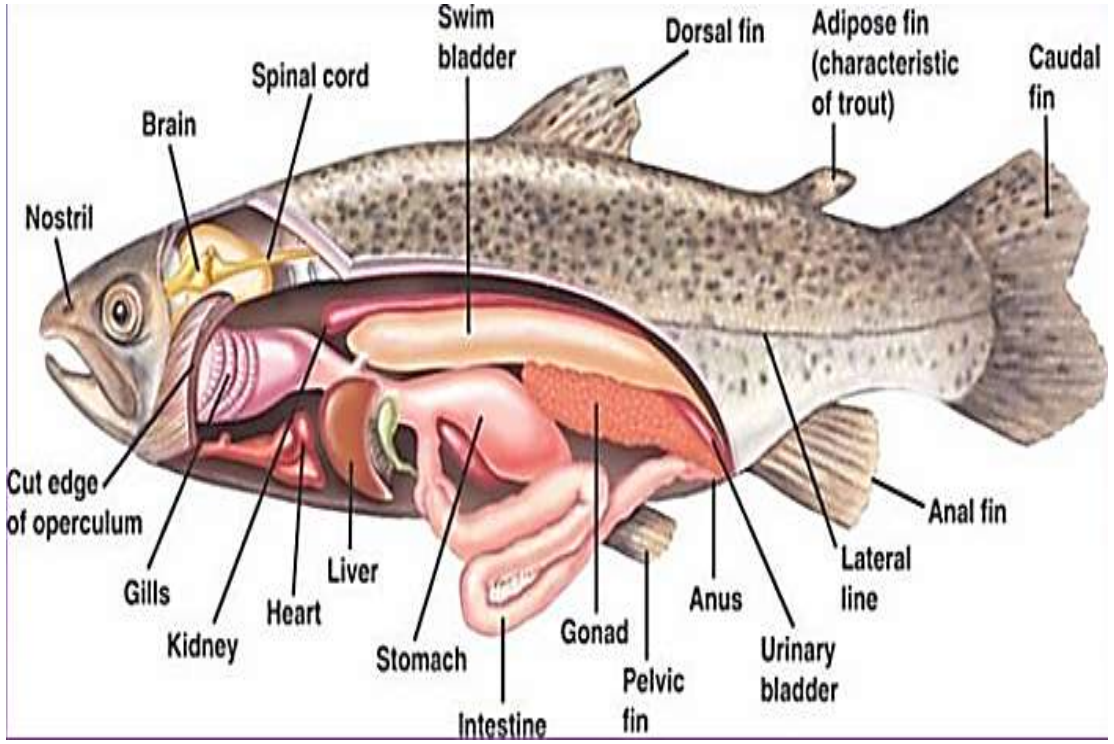
BS ZOOLOGY
03336361244

Module# 214 Class Pisces

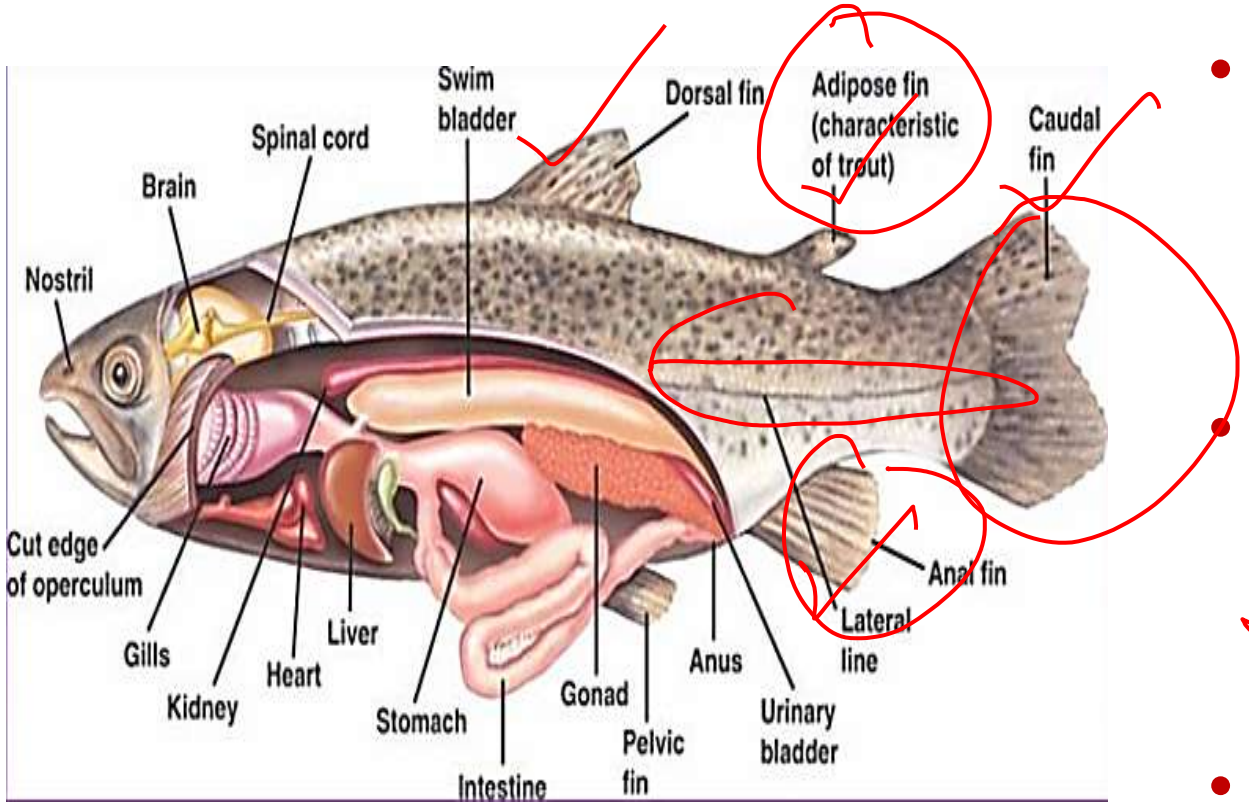


http://ourmarinespecies.com/wp-content/uploads/2018/12/fish_anatomy_9.jpg

Class Pisces

- They are found in fresh, marine, and brackish water.
- The body is usually streamlined. Some have a spindle-shaped or elongated body as well.
- Their body is distributed into a head, trunk and tail.
- They swim with the help of their tail.

Class Pisces



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- Paired and unpaired fins represent the appendages. These help the fish to balance while swimming.
- The lateral line system functions as a sensory organ to sense the disturbances in the nearby environment.
- The body is covered with thick-seated scales, which helps by providing protection to the internal organelles.

Class Pisces

- The gills help in respiration.
- Closed type blood circulation is observed.
- The internal skeleton is bony or cartilaginous.
- These are cold-blooded organisms.
- Swim bladder
- They may be herbivores or carnivores, oviparous or ovoviviparous.

Class Pisces

- The sexes are separate.
- Fertilization may be external or internal.
- They lack extra-embryonic membranes.
- The digestive system is well-developed.
- The nervous system comprises of the brain and ten pairs of the cranial nerves.
- Types

END

Module# 215 Class Amphibians

- They are the first cold blooded vertebrates from evolution point of view which came to the land.
- They occur in fresh water and moist land.
- Body is divisible into head and trunk. Tail may be present in some amphibians.
- The skin is smooth or rough having glands which keep it moist.

Class Amphibians

- They are usually without scales, but if present they are hidden beneath the skin (e. g. caecilians).
- Paired fins are absent. Unpaired fins may be present. Two pairs of limbs are used for locomotion except caecilions.
- The gills are present at least in the larval stage; some adult forms also carry external gills in addition to lungs (e.g., Proteus).

Class Amphibians

- Skull is dicondylic, i.e., with two occipital condyles for articulation with vertebral column.
- The respiratory organs are lungs, buccopharyngeal cavity, skin and gills.
- The heart is three chambered, having two auricles and one ventricle.
- Kidneys are mesonephric.

Class Amphibians

- Alimentary canal, urinary and reproductive tracts open into a common chamber called cloaca which opens outside through cloacal aperture.
- Ear consists of internal and middle ear. Tympanum (outer membrane) covers the middle ear. The eyes have eyelids. Nictitating membrane is well developed.

Class Amphibians

- Ten pairs of cranial nerves are present.
- Lateral line system is found during their development.
- Fertilization is external.

END

Module# 216 Class Reptilia

- Reptiles are the creeping and burrowing cold blooded vertebrates bearing epidermal scales.
- They are ectothermic (cold-blooded) and are found mostly in the warmer parts of the world.
- Skin is dry, rough and without glands, bearing epidermal scales or scutes.
- Snakes and lizards shed their scales as skin cast.

Class Reptilia

- They do not respire by means of gills. Respiration always takes place through lungs.
- Skull is monocondylic, i.e., with single occipital condyle.
- Except in snakes, there are two pairs of pentadactyl limbs, each with 5 digits bearing claws—tetrapodus pentadactyl type.

Class Reptilia

- Heart consists of two auricles and a partially divided ventricle. In crocodilians, heart is four chambered (two auricles and two ventricles).
- Twelve pairs of cranial nerves are present.
- Each ear consists of three parts; external, middle and internal. Snakes do not possess ears.
- The lateral line system is altogether absent.

Class Reptilia

- Tortoises feed almost entirely on vegetation.
- Some turtles are flesh eaters.
- All other reptiles are carnivorous/insectivorous.
- A typical cloaca is present.
- They are mostly oviparous.

END

Module# 217 Class Aves

- Birds are bipedal feathered and warm blooded (homoiothermous) animals.
- The hind-limbs are adapted for perching, walking or swimming, etc., and usually bear four, sometimes three and rarely two toes.
- Except uropygium gland (preen gland or oil gland), at the base of the tail, no skin gland is present. Ostrich and parrot lack oil gland.

Class Aves

- The upper and lower jaws are modified into beak, which lacks teeth.
- Legs are modified for walking, hopping, grasping, perching, wading and swimming. Legs bear homy epidermal scales.
- The alimentary canal has additional chambers, the crop and gizzard.

Class Aves

- The crop stores and softens the food; however, the gizzard helps in crushing and churning the food.
- Gall bladder is absent in some seed-eating birds (graminivorous) such as pigeons.
- Respiration is by lungs. The lungs are spongy and inelastic. Air sacs are connected to lungs for supplement respiration.

Class Aves

- Voice is produced by a special organ, the syrinx.
- The heart is four-chambered. Sinus venosus is absent.
- Renal portal system is very much reduced. Red blood corpuscles are nucleated, oval and biconvex.
- The kidneys are metanephric.
- Birds have 12 pairs of cranial nerves.

Class Aves

Class Aves

- Female has usually well developed single left ovary and oviduct. If right ovary and oviduct are present, they are vestigial (nonfunctional).
- Many birds show sexual dimorphism.
- All birds are oviparous.
- They show courtship, nest building, parental care, migration and territorial behaviour.

END

Module# 218 Class Mammalia

Class Mammalia

- These animals are warm blooded, hairy and have mammary or milk producing glands, (mammary glands).
- They are homoiothermous.
- Oil glands (sebaceous glands) and sweat glands (sudoriferous glands) are present in the skin.
- Teeth are of different types (heterodont) and are embedded in the sockets of jaws (the codont).

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Class Mammalia

- Except a few, mammals possess seven cervical (neck) vertebrae.
- The skull is dicondylic i.e., with two occipital condyles.
- Respiration is by lungs.
- They possess a muscular diaphragm dividing trunk into thorax and abdomen.
- The coelom is divided into four cavities.

Merged by BAQIR HUSSAIN BUZDAR

Class Mammalia

- A pericardial cavity lodging the heart, two pleural cavities each containing the lung and an abdominal cavity having the rest of viscera.
- The heart is four chambered. Sinus venosus is absent.
- The red blood corpuscles are without nucleus.
- Renal portal system is absent.

Merged by BAQIR HUSSAIN BUZDAR

Class Mammalia

- The brain has large cerebrum and cerebellum. Optic lobes are divided into four lobes called corpora quadrigemina.
- Corpus callosum connects the two cerebral hemispheres internally.
- 12 pairs of cranial nerves are present.

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Class Mammalia

- Each ear consists of three parts: external, middle and internal. Pinna is a part of external ear. Middle ear has 3 bony ear ossicles.
- Except egg laying mammals they are viviparous.
- They are dominant animals and are capable to learn because of their better developed brain.
- Types

END

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